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**MIDTERM EXAMINATION**

**MTH501- Linear Algebra (MCQ's File)**

**Question No: 1**

If for a linear transformation the equation T(x )= 0 has only the trivial solution then T is

► **one-to-one**
► onto

**Question No: 2**

Which one of the following is an matrix?

► $\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$
► $\begin{bmatrix} 1 & 0 & 1 \\ 0 & -3 & -3 \end{bmatrix}$
► $\begin{bmatrix} 1 & 0 \\ 2 & -3 \end{bmatrix}$
► $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$

**Question No: 3**

Let $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and let k be a scalar .A formula that relates det kA to k and det A is

► **det kA= k det A**

► det kA = det (k+A)

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► det k A = $k^2$ det A

► det kA = det A

**Question No:**

The equation x = p + t v describes a line

► through v parallel to p

► **through p parallel to v**
► through origin parallel to p

**Question No: 5**
Determine which of the following sets of vectors are linearly dependent.

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

►

$$v_1 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

►

$$v_1 = \begin{bmatrix} 5 \\ 2 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 10 \\ 4 \\ 6 \end{bmatrix}$$

►

**Question No: 6**

Every linear transformation is a matrix transformation

► **True**
► False

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**Question No: 7**

A null space is a vector space.

► **True**
► False

**Question No: 8**

If two row interchanges are made in succession, then the new determinant

► **equals to the old determinant**

► equals to -1 times the old determinant

**Question No: 9**

The determinant of A is the product of the pivots in any echelon form U of A , multiplied by $(-1)^r$ , Where r is

► the number of rows of A

► **the number of row interchanges made during row reduction from A to U**

► the number of rows of U

► the number of row interchanges made during row reduction U to A

**Question No: 10**

If A is invertible, then $\det(A)\det(A^{-1})=1$.

► **True**
► False

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**Question No: 11**

A square matrix $A=\left[a_{ij}\right]$ is lower triangular if and only if $a_{ij}=0$ for

- $i>j$
- $i<j$
- $i\leq j$
- $i=j$

**Question No: 12**

The product of upper triangular matrices is

- lower triangular matrix
- **upper triangular matrix**
- diagonal matrix

**Question No: 13**

The matrix multiplication is associative

- **True**
- False

**Question No: 14**

We can add the matrices of _____.

- **same order**
- same number of columns.
- same number of rows
- different order

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**Question No: 15**

By solving system of equations with iterative method, we stop the process when the entries in two successive iterations are _____.

- ► repeat
- ► large difference
- ► different
- ► **Same**

**Question No: 16**

Jacobi's Method is _____ converges to solution than Gauss Siedal Method.

- ► slow
- ► **fast**
- ► better

**Question No: 17**

A system of linear equations is said to be homogeneous if it can be written in the form _____.

- ► **AX = B**
- ► AX = 0
- ► AB = X
- ► $X = A^{-1}$

**Question No: 18**

The row reduction algorithm applies only to augmented matrices for a linear system.

- ► True
- ► **False**

**Question No: 19**

Whenever a system has no free variable, the solution set contains many solutions.

- ► True
- ► **False**

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**Question No: 20**

Which of the following is not a linear equation?

- ▶ $x_1 + 4x_2 + 1 = x_3$
- ▶ $x_1 = 1$
- ▶ $x_1 + 4x_2 - \sqrt{2}x_3 = \sqrt{4}$
- ▶ $x_1 + 4x_1x_2 - \sqrt{2}x_3 = \sqrt{4}$

**Question No: 21**

If a system of equations is solved using the Gauss-Seidel method, then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?
Select correct option:

All of its entries on the diagonal must be zero.
All of its entries below the diagonal must be zero.
All of its entries above the diagonal must be zero.
**All of its entries below and above the diagonal must be zero.**

**Question No: 22**

The determinant of a diagonal matrix is the product of the diagonal elements.
Select correct option:

- **TRUE**
- FALSE

**Question No: 23**

By using determinants, we can easily check that the solution of the given system of linear equation exits and it is unique.

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- FALSE
- **TRUE**

**Question No: 24**

A matrix A and its transpose have the same determinant.

- **TRUE**
- FALSE

**Question No: 25**

If both the Jacobi and Gauss-Seidel sequences converge for the solution of Ax=b, for any initial x(0), then which of the following is true about both the solutions?

- No solution
- **Unique solution**
- Different solutions
- Infinitely many solutions

**Question No: 26**

The value of the determinant of a square matrix remains unchanged if we multiply each element of a row or a column by some scalar.

- TRUE
- **FALSE**

**Question No: 27**

How many different permutations are there in the set of integers {1,2,3}?

- 2
- 4
- 6
- **8**

**Question No: 28**

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If A is n x n matrix and det (A) = 2 then det (5A) = _____.

- **10**
- 32
- 5
- 8

**Question No: 29**

Every vector space has at least two subspaces; one is itself and the second is:

- multiplication of vectors
- addition of vectors
- subspace {0}
- **scalar multiplication of vectors**

**Question No: 30**

If one row of A is multiplied by k to produce B, then which of the following condition is true?

Select correct option:

- det(AB) = (detA)(detB)
- **detB = k detA**
- detB = – detA
- detB = detA

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# MTH501-Linear Algebra

(Solved MCS's)

**LECTURE FROM (23 to 45)**

Junaidfazal08@gmail.com
Bc190202640@vu.edu.pk

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**JUNAID MALIK**
**0304-1659294**




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ALL answers are verified if found any mistake then Correct ACCORDINGLY

**1. Which statement about the General Least Square Method is true?**

- ❖ Solution obtained by this method is always Unique.
- ❖ This is a numerical method for the solution of system of Linear Equations.
- ❖ This method find an X that make Ax as close possible to the b.
- ❖ This method gives us exact solution of the System.

**2. Let v = (1, -2, 2, 0). The unit vector in the same direction as v is:**

- ❖ $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}, 0$
- ❖ $\frac{1}{3}, \frac{-2}{3}, \frac{2}{3}, 0$
- ❖ $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}, 0$
- ❖ $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}, \frac{1}{3}$

**3. Let u = (3, -2), v= (4, 5). For the weighted Euclidean inner product (u, v) = $4u_1v_1 + 5u_2v_2$**

(v, u) =

- ❖ 2
- ❖ -2
- ❖ 3
- ❖ -3

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**4. Let v = (0, 2, 2, 1). The unit vector in the same direction as v is**

- $0, \frac{2}{3}, \frac{2}{3}, \frac{1}{3}$
- $0, \frac{2}{3}, \frac{2}{3}, \frac{1}{3}$
- $0, \frac{1}{3}, \frac{2}{3}, \frac{2}{3}$
- $0, \frac{1}{3}, \frac{2}{3}, \frac{2}{3}$

**5. Let $I^3$ have the Eduction inner product. Then u = (2, 1, 3), V = (1, 7, k) are orthogonal for**

- K=9
- K= -3
- K= -9
- K= 3

**6. Let A ben x n Matrix whose entries are real. If $\lambda$ is an eigenvalue of A with x a corresponding eigenvector in $\mathbb{R}^n$ , then**

- $A\bar{x} = \lambda\bar{x}$
- $A\bar{x} = \lambda\bar{x}$
- $A\bar{x} = \lambda\bar{x}$
- $A\bar{x} = \lambda\bar{x}$

**7. Suppose that A = $\begin{bmatrix} 1.25 & -.75 \\ -.75 & \end{bmatrix}$ has eigenvalues 2 and 0.5. then 0rigin is a**

- Saddle point
- Repellor

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- Attractor

**8. Suppose that A = $\begin{bmatrix} 0.5 & 0.6 \\ -0.3 & 1.4 \end{bmatrix}$ has eigenvalues 0.8 and**

**1.1. then 0rigin is a**

- Saddle point
- Repellor
- Attractor

**9. if A is an m x n matrix with linearly independent column vector, then A can be factored as**

**A= QR**

**Where Q is an m X n matrix orthonormal column vector ,and R is an n x m.**

- Upper triangular matrix
- Invertible matrix
- Invertible lower triangular matrix
- invertible Upper triangular matrix

**10.The matrix equation $A^T A \hat{x} = A^T b$ represent a system of liner equation Commonly referred to as the**

- Normal equation for x
- Normal equation for X^
- Normal equation for A
- Normal equation for b

**11. By the best Approximation theorem , the distance from y to W is $\|y - \hat{Y}\|$ , where $\hat{Y}$=**

- Projw $\hat{Y}$
- $Proj_w y$

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- projy W

**12. ||u + v + w|| ||u|| + ||v|| +||w|| for all vectors u, v and w in an inner product space.**

- True
- False

**13. The dominate for the matrix A = $\begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ is**

- =1
- = -3
- = -1
- = 0

**14. A square matrix A I invertible if and only if and only is X = 0 is not an Eigen Value of A**

- True
- False

**15. A square matrix with orthogonal columns ----------matrix.**

- Is an orthogonal
- May be an orthogonal
- May not be an orthogonal
- Is not an orthogonal

**16. if two row are orthogonal , they are ---------- --.**

- Linearly independent
- Linearly Dependent

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**17. if X is orthogonal to both U and V, then must be ----------to u+v.**

- ❖ Orthogonal
- ❖ Orthonormal
- ❖ Perpendicular
- ❖ Parallel

**18. the given system** $\begin{matrix} 2x+3y=3 \\ 6x+9y=7 \end{matrix}$ **has**

- ❖ Unique solution
- ❖ Infinitely many solution
- ❖ No solution
- ❖ None of these

**19. Which statement about the matrix** $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 7 & 2 & 0 & 0 \\ 9 & 1 & 2 & 0 \\ 5 & 4 & 2 & -1 \end{bmatrix}$ **is false?**

- ❖ Eigenvalue 2 has algebraic multiplicity1
- ❖ Eigenvalue of the matrix 1, 2 and -1
- ❖ Characteristic polynomial of the matrix is $(1-\lambda)(2-\lambda)^2(-1-\lambda)$
- ❖ Eigenvalue -1 has multiplicity 1

**20 .if A =** $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ **is diagonalizable A has 2 district eigenvalues.**

- ❖ True
- ❖ False

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**21. A is diagonalizable if $A = PDP^{-1}$ where**

- D is any matrix and P is an invertible matrix
- D is a diagonal matrix and P is any matrix
- D is a diagonal matrix and P is invertible
- D is a invertible matrix and p is any matrix

**22.How many trems are there the algebraic expression $8X^2 + \sqrt{9x}$ x $25X^3$?**

- 4
- 3
- 2
- 1

**23. if two matrixes are added, then which of following should be true for them?**

- Both must have same order
- Both must have different order
- Both must be rectangular
- Both must be square

**24. if a matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 6 & 1 & 1 \end{bmatrix}$, then which of the following is true for A ?**

- It is a rectangular matrix
- It is row matrix
- It is singular matrix
- It is scalar matrix

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**25. if $v_1$, $v_2$ and $v_3$ are in $R^m$ then which of the following is equivalent to $[v_1, v_2, v_3]\begin{bmatrix}2\\-7\\5\end{bmatrix}$?**

- ❖ $2v_1 - 7v_2 + 5v_3$
- ❖ $5v_1 - 7v_2 + 2v_3$
- ❖ $5v_1 + 2v_2 - 7v_3$
- ❖ $2v_1 + 5v_2 - 7v_3$

***26.*** **if $(v_1, v_2, v_3)$ is linearly dependent set and $v_3 = cv_1$ $v_2$, (where 'c'is a scalar ), which option is true?**

- ❖ $v_1 \in \text{span}(v_1, v_2)$
- ❖ $v_3 \in \text{span}(v_1, v_2)$
- ❖ $v_2 \in \text{span}(v_1, v_2)$
- ❖ $v_3 \notin \text{span}(v_1, v_2)$

**27. if $A = \begin{bmatrix}1 & 2 & 3\\0 & 0 & 1\\0 & -1 & 1\end{bmatrix}$, then which of the following is true for the matrix A?**

- ❖ It is an invertible matrix
- ❖ It is a singular matrix
- ❖ It is a non – invertible matrix
- ❖ It is a rectangular matrix

**28. which of the following is true for the partitioned matrices A = (C D) and B= (E F), where Sub-**

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**matrices C and D have the same size As E and F respectively?**

- ❖ A + B =(CE DF)
- ❖ A + B = $\begin{bmatrix} C+E \\ D+F \end{bmatrix}$
- ❖ A + B = $[C+E \quad D+F]$
- ❖ A + B = $\begin{bmatrix} CE \\ DF \end{bmatrix}$

**29. If a matrix A is factorized into lower and upper triangular matrices, then which of the following is true for the matrix?**

- ❖ It is called an LU- procedure.
- ❖ It is called an LU –decomposition
- ❖ It is called an LU- matrices.
- ❖ It is called an LU- algorithm.

**30. if the matrix A = $\begin{bmatrix} 1 & 5 & 4 \\ 0 & 1 & 7 \\ 0 & 0 & 0 \end{bmatrix}$, then which of the following is true about it?**

- ❖ Its determinant is 0.
- ❖ Its determinant is 2.
- ❖ Its determinant is 4.
- ❖ Its determinant is 6.

**31. Let a set S is a basis of a vector space V, then which of the following is NOT true about it?**

- ❖ It is linearly dependent.
- ❖ Each element of S belong to V.

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- It spans V.
- It is linearly independent.

**32. if B =** $\begin{bmatrix}0\\2\end{bmatrix}, \begin{bmatrix}1\\0\end{bmatrix}$ **for** $R^2$ *and* sn $x^1 \in R^2$ has coordinate vector $[x^r]_B = \begin{bmatrix}1\\3\end{bmatrix}$, *then* which of the following is the value of $x^1$ ?

- $x^r = \begin{bmatrix}6\\1\end{bmatrix}$
- $x^r = \begin{bmatrix}2\\6\end{bmatrix}$
- $x^r = \begin{bmatrix}6\\1\end{bmatrix}$
- $x^r = \begin{bmatrix}1\\3\end{bmatrix}$

**33.if a set S={1, x, $x^2$} is a for $p_2$ and $[p^1]_s$ = (2, 4, 7 ) , then which of the following is the most appropriate option?**

- $P_2$= 2 – 4x+7 $x^2$
- $P_2$= 2 – 4x-7 $x^2$
- $P_2$= 2 + 4x+7 $x^2$
- $P_2$= 4x+7 $x^2$

**34. which of the following is the set of standard basis for $R^3$?**

- {(1, 1,0), (0, 1, 0), (1, 0, 1)}
- {(1, 0, 0), (0, 1, 1),(0, 0, 1)}
- {(1, 0, 0 ), (1, 1, 0), (0, 0, 1)}
- {(1, 0, 0), (0, 1, 0), (1, 1, 1)}

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**35. Consider the bases for $R^3$ given by B= { $^{r}b_1$ , $^{r}b_2$ } and C ={ $^{r}c_1$ , $^{r}c_2$ }; where $^{r}b_1 = \begin{bmatrix}1\\0\end{bmatrix}, {}^{r}b_2 = \begin{bmatrix}1\\1\end{bmatrix}$, $^{r}c_1 = \begin{bmatrix}5\\3\end{bmatrix}, {}^{r}c_2 \begin{bmatrix}2\\1\end{bmatrix}$, also assume that $p_{B\square c} = \begin{bmatrix}2 & -1\\3 & 1\end{bmatrix}$; then which of the following is the change of –coordinates matrix form B to C?**

- $p_{B\square c} = \begin{bmatrix}1 & 1\\3 & -2\end{bmatrix}$
- $p_{B\square c} = \begin{bmatrix}5 & -2\\3 & 1\end{bmatrix}$
- $p_{B\square c} = \begin{bmatrix}-8 & -3\\3 & \end{bmatrix}$

**36.if the general term of a typical signal is $(0.6)^k$ , then determine which of the following is the signal for k = -2 ?**

- $(0.6)^{-2} = 0$
- $(0.6)^{-2} = 0.6$
- $(0.6)^{-2} = (0.6)^2$
- $(0.6)^{-2} = 1/(0.6)^2$

**37. if the casorati matrix is not is not invertible , then which of the following is the most appropriate option regarding ding the associated signals?**

- The signals are linearly independent.
- The signals are linearly dependent.
- The signals may or may not dependent
- The signals may or may not independent

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**38. if $\{Y_k\}$= {…., 1, 0.7, 0, -0.7, 0, 0.7,1,0.7,0,….} and $0.35Y_{K+2}+ 0.6Y_{K+1}+ 0.42Y_K= Z_k$;**

**K=0**

**Then which of the following is the value of $Z_0$?**

- 0.840
- 0.049
- -0.770
- -1.139

**39.A system of linear equation is said to be homogenous if it can be written in the from----------**

- AX = B
- AX =0

**40.if AB = I = BA for matrices A, B and I, where I is an identity matrix , then**

- B is Inverse of A
- A is inverse of B
- $A^{(-1)}= B, B^{(-1)} = A$
- All of the above

**41. A square matrix A is said to be diagonal if A is similar to a matrix**

- Column matrix
- Zero matrix
- Diagonal matrix
- None of these

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**42. Let A be the matrix of order 2X3 and B be the matrix order 3X5, then which of the following is the order of the matrix AB?**

- 2x3
- 3x5
- 3x3
- 2x5

**43. Let 'Ax = 0' be a homogeneous liner system of 'n' equation and 'n' unknowns. Then, the coefficient matrix 'A' is invertible if and only if this system has ---- solution.**

- No
- Trivial
- Non- trivial
- Infinite many

**44. If X – 2 is a factor of the characteristic polynomial of matrix C then an eigenvalue of C is.**

- 2
- -2
- 1/2
- 0

**45.if □ +2 is a factor of the characteristic polynomial of matrix C. then which of the following is the eigenvalue of C?**

- 2
- -2

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- 1 / 2
- 0

**46. Let A and B be the square matrices. Then A and B are invertible with B = $A^{-1}$ and A = $B^{-1}$ if and only if AB = BA equals to a (an) ----matrix**

- Singular
- Square
- Identity
- Rectangular

**47. Let V be a five – dimensional vector space . and let S be a subset of V which spans V. Then S**

- Must be linearly dependent
- Must be a basis for V
- Must have infinitely many elements
- Must have at most five element.

**48. If U + V = U + W then**

- V+ W
- V ⍰w
- V = W
- None of the above

**49. if one of the eigenvalues of $[A]_{nxn}$ is zero , it implies----------**

- The solution to [A][X] =[C] a system of equation is unique
- The determinant of [A] is zero.

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- The solution to [A][X]= [0] system of equation is trivial
- The determinant of [A] is nonzero

**50.le a matrix A has both negative and positive Eigen values , so in this case origin behaves as a------- point**

- Saddle
- Critical

**51. If □ is an eigenvector of A , then every nonzero vector x such that Ax = □ x is called an -------- of A corresponding to---**

- Eigenvalue. □
- Eigenvector. □
- Eigenvalue. A
- Eigenvector. A

**52. Let A be n x n matrix, then A invertible if and only if**

- Det A is not Zero
- Det A is zero

**53. The invertible matrix theorem applies only to ---------matrices.**

- Rectangular
- Square
- Identity
- Scalar

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**54. A 3x 3 identity matrix have three and ---------- eigenvalues.**

- Same
- District

**55. An n x n matrix A is said to be diagonalizable if and only if A as n--------- eigenvectors.**

- Linearly dependent
- Linearly independent

**56. What is Eigen value?**

- A vector obtained from the coordinates
- A matrix determined from the algebraic equation
- A scalar associated with a given linear transformation
- It is the inverse of the transform

**57. A column replacement operation on A does not change the**

- Determinant
- Matrix
- Row
- Column

**58. Le a matrix A has both negative and positive eigen values, so in this case origin behaves as a---- point**

- Saddle
- Critical

**59. A null space is a vector space**

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- True
- False

**60. Each pair of eigenvalue and its corresponding eigenvector provides a solution of the equation X'= Ax which is called**

- Eigen solution
- Eigen function

**61. If A is invertible and b in $R^n$ be any vector. Then, we must have a matrix $A^{-1}$ b, which is a solution of ---**

- $A^{-1}$ =b
- $A^2$ = b
- $A^t$ = b
- Ax = b

**62. What is the maximum possible number of pivots in a 6 x 6 matrix?**

- 0
- 2
- 4
- 6

**63. If 3 is an eigenvalue of A and x is corresponding eigenvector, then what is the eigenvalue of $A^2$?**

- 12
- 9
- 6
- 3

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**64. The characteristics polynomial of 3 X 3 identity matrix is ------- if x is the eigen values of the given 3 x 3 identity matrix.**

- (x - 1)^3
- (x + 1)^3
- X^3
- (1 - x)^3

**65. A partitioned matrix 'A' is said to be block diagonal if the matrices on the main diagonal are square and all other position matrices are**

- Zero
- Unit
- NonZero Symmetric
- nonzero Skew Symmetric

**66. If A is an invertible square matrix then**

- $(A^T)^{-1} = (A^{-1})^T$
- $(A^T)^T=(A^{-1})^T$
- $(A^T)^{-1}=(A^{-1})^{-1}$
- None of these

**67.A blocked matrix in which block are repeated the down the diagonals of the matrix is called a ----- matrix.**

- Blocked Square
- Blocked diagonal- constant
- Blocked identity

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- Blocked rectangular

**68. Two equivalent vector must have the same initial point.**

- True
- False
- May be

**69. \lambda is an eigenvalue of a matrix A if and only if the equation (A - \lambdal)x= 0 has a------**

- Non- trivial solution
- Trivial solution

**70.which of following is the eigenvalue of the matrix?**

- 3
- 5
- 3.4
- 3.5

**71. the complex conjugate of a vector in $C^n$ is the vector x in $C^n$ whose entries the conjugates of the entries in x**

- Real
- Complex

**72. Multiplication of a partitioned matrix by a scalar is also computed-------**

- Row by Row
- Column by column
- Diagonal by diagonal

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- Block by block

**73.if the real part the eigenvalue is zero, then the trajectories from ------ around the origin.**

- Parabola
- Hyperbola
- Ellipse
- None of these

**74. A row interchange --------- the of the determinant.**

- Change
- Does not change

**75. For any subspace W of a vector space V, which one is not the axiom for subspace. 0 must be in W.**

- For all u, v in W and u – v must be in W.
- For all u, v in W and u.v must be in W.
- For any scalar k and u in W then k.u in W.

**76. Which one is not the axiom for vector space?**

- 0 + u = u
- 0.u = u
- 1.u = u
- u + v = v + u

**77. The Gauss-Seidel method is applicable to strictly diagonally dominant matrix.**

- TRUE
- FALSE

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**78. If a multiple of one row of a square matrix A is added to another row to produce a matrix B, then which of the following condition is true?**

- detB = detA
- detB = k detA
- detA detB = 0
- detA detB = detA

**79. Which of the following is the volume of the parallelepiped determined by the columns of A where A is a 3 x 3 matrix?**

- |det A|
- [A]
- det A
- A^(-1) ,that is inverse of A

**80. Determinant of a non-invertible(singular) matrix always**

- vanish
- unity
- non zero negative
- non zero positive

**81. Rank of a zero matrix of any order is**

- zero
- three
- four
- nine

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**82. can add the matrices of_____.**

► same order

► same number of columns.

► same number of rows

► different orde

**83. solving system of equations with iterative method, we stop the process when the entries in two successive iterations are_____.**

► repeat

► large difference

► different

**84. Jacobi's Method is_____converges to solution than Gauss Siedal Method.**

► slow

► fast

► better

**85.** is invertible, then det(A)det(A-1)=1.

► True

► False

**86. The determinant of A is the product of the pivots in any echelon form U of A , multiplied by (-1)r , Where r is**

► the number of rows of A

► the number of row interchanges made during row reduction from A to U

► the number of rows of U

► the number of row interchanges made during row reduction U to

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**87. If a system of equations is solved using the Gauss-Seidel method, then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?**

- All of its entries on the diagonal must be zero.
- All of its entries below the diagonal must be zero.
- All of its entries above the diagonal must be zero
- All of its entries below and above the diagonal must

**88. If M is a square matrix having two rows equal then which of the following about the determinant of the matrix is true?**

- det (M) is not equal to '1'
- det (M)=1
- det (M) is not equal to '0'
- det (M)=0

**89. if both the Jacobi and Gauss-Seidel sequences converge for the solution of Ax=b, for any initial x(0), then which of the following is true about both the solutions?**

- No solution
- Unique solution
- Different solutions

**90. Let t be any m x n matrix with orthonormal columns and v be any vector then ||t . v || =_____.**

- || v ||

- | v |
- V
- t . || v ||

**91. If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.**

- TRUE
- FALSE

**92. Every linear transformation is a matrix transformation.**

- True
- False

**93. All the lines those passes through origin are not the subspace of a plane.**

- FALSE
- TURE

**94. Why inverse of the matrix A= [1 2] is NOT possible?**

- Because it is a square matrix
- Because it is a zero matrix.
- Because it is an identity matrix.
- Because it is a rectangular matrix.

**95. Let W = {(1, y) such that y in R}. Is W a vector subspace of plane.**

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- YES
- NO

**96. If a system of equations is solved using the Jacobi's method , then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?**

- All of its entries on the diagonal must be zero.
- All of its entries above the diagonal must be zero.
- All of its entries below and above the diagonal must
- All of its entries below the diagonal must be zero.

**97. How many different permutations are there in the set of integers {1,2,3}?**

- 2
- 4
- 6
- 8

**98. Which one is the numerical method used for approximation of dominant eigenvalue of a matrix.**

- ► Power method
- ► Jacobi's method
- ► Guass Seidal method
- ► Gram Schmidt process

**99. The inverse of an invertible lower triangular matrix is.**

► lower triangular matrix

► upper triangular matrix
► diagonal matrix

**100. A is diagonalizable if $A = PDP^{-1}$ Where.**

► D is any matrix and P is an invertible matrix
► D is a diagonal matrix and P is any matrix
► D is a diagonal matrix and P is invertible matrix
► D is a invertible matrix and P is any matrix.

**101. The characteristic polynomial of a 5□5 matrix is, the eigenvalues are:**

□⁵ □ 4□⁴ □ 45□³ = 0

► 0,-5, 9
► 0,0,0,5,9
► 0,0,0,-5,9 (true)
► 0,0,5,-9

**102: A partitioned square matrix 'A' is said to the matrix if the matrices on the main diagonal square and all matrices above the main diagonal are zero.**

- Block lower triangular

**103: if there is a vector v= (2, 1, 0) then ||v|| is ……..**

- 5

**104: Each pair of eigenvalue and its corresponding eigenvectors provide a solution of equation.**

Eigen function

**105: which of the following is true for the matrix**

**Where $M_{11}$ .$M_{22}$ and $M_{33}$ are square sub – matrices, and O is Zero sub matrix?**

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It is block upper triangular matrix

**106: suppose that real solution $y_1$ and $y_2$ of $x^1 = Ax$ from a basis for the two- dimensional real vector space if $y_1$ and $y_2$ are ……..**

Linearly independent

**107: Let W be a subspace of R and $(u_1, u_2, \ldots\ldots u_p)$ is any orthogonal basis of W, then $y = c_1u_1 + c_2u_2 \ldots\ldots\ldots\ldots c_nu_n$ where**

$$Cj = \frac{y.uj}{uj.uj}$$

**108: Elementary row operation on a matrix do not affect the ….. relation among the column of the matrix**

Linear dependence

**109: Two vector are ……. if at least one of the vector is a multiple of the other**

Linearly dependent

**110: Electric circuit rotation is caused by sine and cosine function if there eigenvalues are complex and hence the origin is called….**

Spiral point

**111: Multiplication of a partitioned matrix by a scalar is also computed…..**

Column by column

**112: The invariable Matrix .Theorem applies only to……**

Square

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**113:Let A be a real 2 by 2 matrix with complex eign values □ = a – b + I,(b □) and associated eigenvectors vin $C^2$ then A PCP $^{-1}$ where p= ------**

P = [Revlmv]

**114: Let Ax = 0 be a homogeneous linear system of 'n' unknown , then the coefficient matrix 'A' is invertable if and only if this system has---------- solution**

Trivial

**115: the two vector are said to b equivalent if**

Same length and same direction

**116: if one of the eigenvalues of $[A]_{nxn}$ is zero .it implies**

The determined of [A] is zero

**117: let be a five – dimensional vector space. And let ba subset of consisting of five vectors.**

Mast be linearly dependent, but may or may not span V

**118: Suppose x, y, z are some vectors in an inner product space (v,<,>) such that <X,Y>=<X,Z> for all x □ V then Y=Z**

TRUE

**119: let U,V and W be vectors in $R^n$ ,then**

(U+ v), W=U.W+V.W

**Q120: if u and v non zero vector in either $R^2$or$R^3$ then by the law of cosines || u – vii$^2$……..**

$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos$

**121: If u +v – w, then**

V = w

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**122: Two vectors u and v in $R^n$ are orthogonal if…….**

u.v = o

**123:A vector whose length 1 is called ……..**

Unit vector

**124: A …… Matrix whit orthogonal columns is an orthogonal matrix.**

Square 1:

**125: if a square matrix has orthogonal columns then it also has …… rows.**

Orthogonal

**126: The norm of v is the non-negative scalar ||v|| defined by**

$||v|| = \sqrt{V}^2_1 + v_2^2 + \ldots\ldots\ldots\ldots v^2_n$

**127: the matrix A ^ T x A is invertable if and only if the columns of A are linearly independent**

True

**128: let V be a one Eigen vector then conjugate eigen vector is repented by.**

V

**129: Any finite dimensional inner product space an orthonormal basis.**

True

**130: the ……. vector is orthogonal to every vector in $R^n$.**

Zero

**131: Each pair of eigenvalue and this its corresponding eigenvector provides a solution of the**

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**equation $X^1$ = Ax which is called …….. of the differential equation.**

Eigen function

**132: if we divide a non – zero vector by its length we get a**

Unit vector

**133: A matrix A – (n\time n) has both positive and negative eigenvalues so in this case origin behave as a**

Saddle point

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# MTH501-Linear Algebra
# FINAL TERM MCQS
# Prepared by: JUNAID MALIK




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**ALL answers are verified if found any mistake then Correct ACCORDINGLY**

**1. Which statement about the General Least Square Method is true?**

- ❖Solution obtained by this method is always Unique.
- ❖This is a numerical method for the solution of system of Linear Equations.
- ❖This method find an X that make Ax as close possible to the b.
- ❖This method gives us exact solution of the System.

**2. Let v = (1, -2, 2, 0). The unit vector in the same direction as v is:**

- ❖ $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}, 0$
- ❖ $\frac{1}{3}, \frac{-2}{3}, \frac{2}{3}, 0$
- ❖ $\frac{-1}{3}, \frac{2}{3}, \frac{-2}{3}, 0$
- ❖ $\frac{1}{3}, \frac{-2}{3}, \frac{2}{3}, \frac{1}{3}$

**3. Let u = (3, -2), v= (4, 5). For the weighted Euclidean inner product (u, v) = $4u_1v_1 + 5\ u_2v_2$**

(v, u) =

- ❖2
- ❖-2
- ❖3
- ❖-3

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**4. Let v = (0, 2, 2, 1). The unit vector in the same direction as v is**

- $0, \frac{2}{3}, \frac{2}{3}, \frac{-1}{3}$
- $0, \frac{-2}{3}, \frac{2}{3}, \frac{-1}{3}$
- $0, \frac{1}{3}, \frac{2}{3}, \frac{2}{3}$
- $0, \frac{1}{3}, \frac{-2}{3}, \frac{2}{3}$

**5. Let $I^3$ have the Eduction inner product. Then u = (2, 1, 3), V = (1, 7, k) are orthogonal for**

- K=9
- K= -3
- K= -9
- K= 3

**6. Let A ben x n Matrix whose entries are real. If $\lambda$ is an eigenvalue of A with x a corresponding eigenvector in $\Sigma^n$ , then**

- $A\bar{x} = \lambda\bar{x}$
- $A\bar{x} = \overline{\lambda x}$
- $A\bar{x} = \bar{\lambda}x$
- $A\bar{x} = \lambda\bar{x}$

**7. Suppose that A =** $\begin{bmatrix} 1.25 & -.75 \\ -.75 & 1.25 \end{bmatrix}$ **has eigenvalues 2 and 0.5. then 0rigin is a**

- Saddle point
- Repellor

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- ❖Attractor

**8. Suppose that A =** $\begin{bmatrix} 0.5 & 0.6 \\ -0.3 & 1.4 \end{bmatrix}$ **has eigenvalues 0.8 and 1.1. then 0rigin is a**

- ❖Saddle point
- ❖Repellor
- ❖Attractor

**9. if A is an m x n matrix with linearly independent column vector, then A can be factored as**

**A= QR**

**Where Q is an m X n matrix orthonormal column vector ,and R is an n x m.**

- ❖Upper triangular matrix
- ❖Invertible matrix
- ❖Invertible lower triangular matrix
- ❖ invertible Upper triangular matrix

**10.The matrix equation $A^T A \hat{x} = A^T b$ represent a system of liner equation Commonly referred to as the**

- ❖Normal equation for x
- ❖Normal equation for X^
- ❖Normal equation for A
- ❖Normal equation for b

**11. By the best Approximation theorem , the distance from y to W is $||y - \hat{Y}||$ , where $\hat{Y}$=**

- ❖Projw $\hat{Y}$
- ❖$Proj_w y$

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❖projy W

**12. ||u + v + w|| ≤||u|| + ||v|| +||w|| for all vectors u, v and w in an inner product space.**

❖True

❖False

**13. The dominate for the matrix A =** $\begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ **is**

❖ $\lambda$ =1

❖ $\lambda$ = -3

❖ $\lambda$ = -1

❖ $\lambda$ = 0

**14. A square matrix A I invertible if and only if and only is X = 0 is not an Eigen Value of A**

❖True

❖False

**15. A square matrix with orthogonal columns ----------matrix.**

❖Is an orthogonal

❖May be an orthogonal

❖May not be an orthogonal

❖Is not an orthogonal

**16. if two row are orthogonal , they are ----------.**

❖Linearly independent

❖Linearly Dependent

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**17. if X is orthogonal to both U and V, then must be ----------to u+v.**

- ❖Orthogonal
- ❖Orthonormal
- ❖Perpendicular
- ❖Parallel

**18. the given system** $\begin{matrix} 2x+3y=3 \\ 6x+9y=7 \end{matrix}$ **has**

- ❖Unique solution
- ❖Infinitely many solution
- ❖No solution
- ❖None of these

**19. Which statement about the matrix** $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 7 & 2 & 0 & 0 \\ 9 & 1 & 2 & 0 \\ 5 & 4 & 2 & -1 \end{bmatrix}$ **is false?**

- ❖Eigenvalue 2 has algebraic multiplicity1
- ❖Eigenvalue of the matrix 1, 2 and -1
- ❖Characteristic polynomial of the matrix is $(1-\lambda)(2-\lambda)^2(-1-\lambda)$
- ❖Eigenvalue -1 has multiplicity 1

**20 .if A =** $\begin{bmatrix} \partial_{11} & \partial_{12} \\ \partial_{21} & \partial_{22} \end{bmatrix}$ **is diagonalizable A has 2 district eigenvalues.**

- ❖True
- ❖False

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**21. A is diagonalizable if $A = PDP^{-1}$ where**

- ❖ D is any matrix and P is an invertible matrix
- ❖D is a diagonal matrix and P is any matrix
- ❖D is a diagonal matrix and P is invertible
- ❖D is a invertible matrix and p is any matrix

**22.How many trems are there the algebraic expression $8X^2 + \sqrt{9x}$ x $25X^3$?**

- ❖4
- ❖3
- ❖2
- ❖1

**23. if two matrixes are added, then which of following should be true for them?**

- ❖Both must have same order
- ❖Both must have different order
- ❖Both must be rectangular
- ❖Both must be square

**24. if a matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 6 & 1 & 1 \end{bmatrix}$, then which of the following is true for A ?**

- ❖It is a rectangular matrix
- ❖It is row matrix
- ❖It is singular matrix
- ❖It is scalar matrix

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**25. if $^{i}v_{1}$, $^{i}v_{2}$, and $^{i}v_{3}$ are in $R^m$ then which of the following is equivalent to $[v_1, v_2, v_3]\begin{bmatrix}2\\-7\\5\end{bmatrix}$?**

- ❖ $2\,^{i}v_{1} - 7\,^{i}v_{2} + 5\,^{i}v_{3}$
- ❖ $5\,^{i}v_{1} - 7\,^{i}v_{2} + 2\,^{i}v_{3}$
- ❖ $5\,^{i}v_{1} + 2\,^{i}v_{2} - 7\,^{i}v_{3}$
- ❖ $2\,^{i}v_{1} + 5\,^{i}v_{2} - 7\,^{i}v_{3}$

**26. if ($^{r}v_{1}$, $^{r}v_{2}$, $^{r}v_{3}$) is linearly dependent set and $^{i}v_{1} \neq c\,^{i}v_{2}$ (where 'c'is a scalar ), which option is true?**

- ❖ $^{r}v_{1} \in \text{span}\,(^{r}v_{1}, ^{r}v_{2})$
- ❖ $^{r}v_{2} \in \text{span}\,(^{r}v_{1}, ^{r}v_{2})$
- ❖ $^{r}v_{3} \in \text{span}\,(^{r}v_{1}, ^{r}v_{2})$
- ❖ $^{r}v_{3} \notin \text{span}\,(^{r}v_{1}, ^{r}v_{2})$

**27. if $A = \begin{bmatrix}1 & 2 & 3\\0 & 0 & 1\\0 & -1 & 1\end{bmatrix}$, then which of the following is true for the matrix A?**

- ❖ It is an invertible matrix
- ❖ It is a singular matrix
- ❖ It is a non – invertible matrix
- ❖ It is a rectangular matrix

**28. which of the following is true for the partitioned matrices A = (C D) and B= (E F), where Sub-**

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**matrices C and D have the same size As E and F respectively?**

- ❖A + B =(CE DF)
- ❖A + B = $\begin{bmatrix} C+E \\ D+F \end{bmatrix}$
- ❖A + B = $[C+E \quad D+F]$
- ❖A + B = $\begin{bmatrix} CE \\ DF \end{bmatrix}$

**29. If a matrix A is factorized into lower and upper triangular matrices, then which of the following is true for the matrix?**

- ❖It is called an LU- procedure.
- ❖It is called an LU –decomposition
- ❖It is called an LU- matrices.
- ❖It is called an LU- algorithm.

**30. if the matrix A =** $\begin{bmatrix} 1 & 5 & 4 \\ 0 & 1 & 7 \\ 0 & 0 & 0 \end{bmatrix}$**, then which of the following is true about it?**

- ❖Its determinant is 0.
- ❖Its determinant is 2.
- ❖Its determinant is 4.
- ❖Its determinant is 6.

**31. Let a set S is a basis of a vector space V, then which of the following is NOT true about it?**

- ❖It is linearly dependent.
- ❖Each element of S belong to V.

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- It spans V.
- It is linearly independent.

**32. if B =** $\left\{\begin{pmatrix}0\\1\end{pmatrix},\begin{pmatrix}2\\0\end{pmatrix}\right\}$ for $R^2$ *and* sn $x^1 \in R^2$ has coordinate vector $\left[x^r\right]_B = \begin{pmatrix}1\\3\end{pmatrix}$, *then* which of the following is the value of $x^1$ ?

- $x^r = \begin{bmatrix}6\\1\end{bmatrix}$
- $x^r = \begin{bmatrix}2\\0\end{bmatrix}$
- $x^r = \begin{bmatrix}0\\1\end{bmatrix}$
- $x^r = \begin{bmatrix}1\\3\end{bmatrix}$

**33.if a set S={1, x, $x^2$} is a for $p_2$ and $[p^1]_s$ = (2, 4, 7 ) , then which of the following is the most appropriate option?**

- $P_2 = 2 - 4x + 7x^2$
- $P_2 = 2 - 4x - 7x^2$
- $P_2 = 2 + 4x + 7x^2$
- $P_2 = 4x + 7x^2$

**34. which of the following is the set of standard basis for $R^3$?**

- {(1, 1,0), (0, 1, 0), (1, 0, 1)}
- {(1, 0, 0), (0, 1, 1),(0, 0, 1)}
- {(1, 0, 0 ), (1, 1, 0), (0, 0, 1)}
- {(1, 0, 0), (0, 1, 0), (1, 1, 1)}

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**35. Consider the bases for $R^3$ given by B= $\{ {}^{r}b_1, {}^{r}b_2\}$ and C =$\{ {}^{r}c_1, {}^{r}c_2\}$; where ${}^{r}b_1=\begin{bmatrix}1\\0\end{bmatrix}, {}^{r}b_2=\begin{bmatrix}-1\\1\end{bmatrix}$, ${}^{r}c_1=\begin{bmatrix}-5\\3\end{bmatrix}, {}^{r}c_2\begin{bmatrix}-2\\1\end{bmatrix}$, also assume that $p_{B\leftarrow c}=\begin{bmatrix}-2 & -1\\3 & 1\end{bmatrix}$; then which of the following is the change of –coordinates matrix form B to C?**

- ❖ $p_{B\leftarrow c}=\begin{bmatrix}1 & 1\\-3 & -2\end{bmatrix}$
- ❖ $p_{B\leftarrow c}=\begin{bmatrix}1 & -1\\0 & 1\end{bmatrix}$
- ❖ $p_{B\leftarrow c}=\begin{bmatrix}-5 & -2\\3 & 1\end{bmatrix}$
- ❖ $p_{B\leftarrow c}=\begin{bmatrix}-8 & -3\\3 & 1\end{bmatrix}$

**36.if the general term of a typical signal is $(0.6)^k$ , then determine which of the following is the signal for k = -2 ?**

- ❖$(0.6)^{-2} = 0$
- ❖$(0.6)^{-2} =0.6$
- ❖$(0.6)^{-2}= (0.6)^2$
- ❖$(0.6)^{-2} = 1/(0.6)^2$

**37. if the casorati matrix is not is not invertible , then which of the following is the most appropriate option regarding ding the associated signals?**

- ❖The signals are linearly independent.
- ❖The signals are linearly dependent.
- ❖The signals may or may not dependent
- ❖ The signals may or may not independent

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**38. if $\{Y_k\}$= {…., 1, 0.7, 0, -0.7, 0, 0.7,1,0.7,0,….} and $0.35Y_{K+2}+ 0.6Y_{K+1}+ 0.42Y_K= Z_k$;**

$$\uparrow$$

$$K=0$$

**Then which of the following is the value of $Z_0$?**

- ❖0.840
- ❖0.049
- ❖-0.770
- ❖-1.139

**39.A system of linear equation is said to be homogenous if it can be written in the from----------**

- ❖AX = B
- ❖AX =0

**40.if AB = I = BA for matrices A, B and I, where I is an identity matrix , then**

- ❖ B is Inverse of A
- ❖ A is inverse of B
- ❖ $A^{(-1)}= B, B^{(-1)} = A$
- ❖ All of the above

**41. A square matrix A is said to be diagonal if A is similar to a matrix**

- ❖ Column matrix
- ❖ Zero matrix
- ❖ Diagonal matrix
- ❖ None of these

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**42. Let A be the matrix of order 2X3 and B be the matrix order 3X5, then which of the following is the order of the matrix AB?**

- 2x3
- 3x5
- 3x3
- 2x5

**43. Let 'Ax = 0' be a homogeneous liner system of 'n' equation and 'n' unknowns. Then, the coefficient matrix 'A' is invertible if and only if this system has ---- solution.**

- No
- Trivial
- Non- trivial
- Infinite many

**44. If X – 2 is a factor of the characteristic polynomial of matrix C then an eigenvalue of C is.**

- 2
- -2
- 1/2
- 0

**45.if $\lambda$ +2 is a factor of the characteristic polynomial of matrix C. then which of the following is the eigenvalue of C?**

- 2
- -2

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- 1 / 2
- 0

**46. Let A and B be the square matrices. Then A and B are invertible with $B = A^{-1}$ and $A = B^{-1}$ if and only if AB = BA equals to a (an)---- matrix**

- Singular
- Square
- Identity
- Rectangular

**47.Let V be a five – dimensional vector space . and let S be a subset of V which spans V. Then S**

- Must be linearly dependent
- Must be a basis for V
- Must have infinitely many elements
- Must have at most five element.

**48.If U + V = U + W then**

- V+ W
- V ≠ w
- V = W
- None of the above

**49. if one of the eigenvalues of $[A]_{nxn}$ is zero , it implies----------**

- The solution to [A][X] =[C] a system of equation is unique
- The determinant of [A] is zero.

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- The solution to [A][X]= [0] system of equation is trivial
- The determinant of [A] is nonzero

**50.le a matrix A has both negative and positive Eigen values , so in this case origin behaves as a -------point**

- Saddle
- Critical

**51. If $\lambda$ is an eigenvector of A , then every nonzero vector x such that Ax = $\lambda$ x is called an -------- of A corresponding to---**

- Eigenvalue. $\lambda$
- Eigenvector. $\lambda$
- Eigenvalue. A
- Eigenvector. A

**52. Let A be n x n matrix, then A invertible if and only if**

- Det A is not Zero
- Det A is zero

**53. The invertible matrix theorem applies only to ---------matrices.**

- Rectangular
- Square
- Identity
- Scalar

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**54. A 3x 3 identity matrix have three and ---------- eigenvalues.**

- Same
- District

**55. An n x n matrix A is said to be diagonalizable if and only if A as n--------- eigenvectors.**

- Linearly dependent
- Linearly independent

**56. What is Eigen value?**

- A vector obtained from the coordinates
- A matrix determined from the algebraic equation
- A scalar associated with a given linear transformation
- It is the inverse of the transform

**57. A column replacement operation on A does not change the**

- Determinant
- Matrix
- Row
- Column

**58. Le a matrix A has both negative and positive eigen values, so in this case origin behaves as a---- point**

- Saddle
- Critical

**59. A null space is a vector space**

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- True
- False

**60. Each pair of eigenvalue and its corresponding eigenvector provides a solution of the equation X'= Ax which is called**

- Eigen solution
- Eigen function

**61. If A is invertible and b in $R^n$ be any vector. Then, we must have a matrix $A^{-1}$ b, which is a solution of ---**

- $A^{-1}\ \chi = b$
- $A^2\ \chi = b$
- $A^t\ \chi = b$
- Ax = b

**62. What is the maximum possible number of pivots in a 6 x 6 matrix?**

- 0
- 2
- 4
- 6

**63. If 3 is an eigenvalue of A and x is corresponding eigenvector, then what is the eigenvalue of $A^2$?**

- 12
- 9
- 6
- 3

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**64. The characteristics polynomial of 3 X 3 identity matrix is ------- if x is the eigen values of the given 3 x 3 identity matrix.**

- (x - 1)^3
- (x + 1)^3
- X^3
- (1 - x)^3

**65. A partitioned matrix 'A' is said to be block diagonal if the matrices on the main diagonal are square and all other position matrices are**

- Zero
- Unit
- NonZero Symmetric
- nonzero Skew Symmetric

**66. If A is an invertible square matrix then**

- $(A^T)^{-1} = (A^{-1})^T$
- $(A^T)^T=(A^{-1})^T$
- $(A^T)^{-1}=(A^{-1})^{-1}$
- None of these

**67.A blocked matrix in which block are repeated the down the diagonals of the matrix is called a ----- matrix.**

- Blocked Square
- Blocked diagonal- constant
- Blocked identity

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- Blocked rectangular

**68. Two equivalent vector must have the same initial point.**

- True
- False
- May be

**69. \lambda is an eigenvalue of a matrix A if and only if the equation (A - \lambdal)x= 0 has a------**

- Non- trivial solution
- Trivial solution

**70.which of following is the eigenvalue of the matrix?**

- 3
- 5
- 3.4
- 3.5

**71. the complex conjugate of a vector in $C^n$ is the vector x in $C^n$ whose entries the conjugates of the entries in x**

- Real
- Complex

**72. Multiplication of a partitioned matrix by a scalar is also computed-------**

- Row by Row
- Column by column
- Diagonal by diagonal

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- Block by block

**73.if the real part the eigenvalue is zero, then the trajectories from ------ around the origin.**

- Parabola
- Hyperbola
- Ellipse
- None of these

**74. A row interchange --------- the of the determinant.**

- Change
- Does not change

**75. For any subspace W of a vector space V, which one is not the axiom for subspace. 0 must be in W.**

- For all u, v in W and u – v must be in W.
- For all u, v in W and u.v must be in W.
- For any scalar k and u in W then k.u in W.

**76. Which one is not the axiom for vector space?**

- 0 + u = u
- 0.u = u
- 1.u = u
- u + v = v + u

**77. The Gauss-Seidel method is applicable to strictly diagonally dominant matrix.**

- TRUE
- FALSE

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**78. If a multiple of one row of a square matrix A is added to another row to produce a matrix B, then which of the following condition is true?**

- detB = detA
- detB = k detA
- detA detB = 0
- detA detB = detA

**79. Which of the following is the volume of the parallelepiped determined by the columns of A where A is a 3 x 3 matrix?**

- |det A|
- [A]
- det A
- A^(-1) ,that is inverse of A

**80. Determinant of a non-invertible(singular) matrix always**

- vanish
- unity
- non zero negative
- non zero positive

**81. Rank of a zero matrix of any order is**

- zero
- three
- four
- nine

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**82. can add the matrices of _____.**

► same order

► same number of columns.

► same number of rows

► different orde

**83. solving system of equations with iterative method, we stop the process when the entries in two successive iterations are _____.**

► repeat

► large difference

► different

**84. Jacobi's Method is _____ converges to solution than Gauss Siedal Method.**

► slow

► fast

► better

85. is invertible, then det(A)det(A-1)=1.

► True

► False

**86. The determinant of A is the product of the pivots in any echelon form U of A , multiplied by (-1)r , Where r is**

► the number of rows of A

► the number of row interchanges made during row reduction from A to U

► the number of rows of U

► the number of row interchanges made during row reduction U to

**87. If a system of equations is solved using the Gauss-Seidel method, then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?**

- All of its entries on the diagonal must be zero.
- All of its entries below the diagonal must be zero.
- All of its entries above the diagonal must be zero
- All of its entries below and above the diagonal must

**88. If M is a square matrix having two rows equal then which of the following about the determinant of the matrix is true?**

- det (M) is not equal to '1'
- det (M)=1
- det (M) is not equal to '0'
- det (M)=0

**89. if both the Jacobi and Gauss-Seidel sequences converge for the solution of Ax=b, for any initial x(0), then which of the following is true about both the solutions?**

- No solution
- Unique solution
- Different solutions

**90. Let t be any m x n matrix with orthonormal columns and v be any vector then ||t . v || = _____.**

- || v ||

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- | v |
- V
- t . || v ||

**91. If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.**

- TRUE
- FALSE

**92. Every linear transformation is a matrix transformation.**

- True
- False

**93. All the lines those passes through origin are not the subspace of a plane.**

- FALSE
- TURE

**94. Why inverse of the matrix A= [1 2] is NOT possible?**

- Because it is a square matrix
- Because it is a zero matrix.
- Because it is an identity matrix.
- Because it is a rectangular matrix.

**95. Let W = {(1, y) such that y in R}. Is W a vector subspace of plane.**

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- YES
- NO

**96. If a system of equations is solved using the Jacobi's method , then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?**

- All of its entries on the diagonal must be zero.
- All of its entries above the diagonal must be zero.
- All of its entries below and above the diagonal must
- All of its entries below the diagonal must be zero.

**97. How many different permutations are there in the set of integers {1,2,3}?**

- 2
- 4
- 6
- 8

**98. Which one is the numerical method used for approximation of dominant eigenvalue of a matrix.**

- ► Power method
- ► Jacobi's method
- ► Guass Seidal method
- ► Gram Schmidt process

**99. The inverse of an invertible lower triangular matrix is.**

► lower triangular matrix

► upper triangular matrix
► diagonal matrix

**100. A is diagonalizable if $A = PDP^{-1}$ Where.**

► D is any matrix and P is an invertible matrix
► D is a diagonal matrix and P is any matrix
► D is a diagonal matrix and P is invertible matrix
► D is a invertible matrix and P is any matrix.

**101. The characteristic polynomial of a 5×5 matrix is, $\lambda^5 - 4\lambda^4 - 45\lambda^3 = 0$ the eigenvalues are:**

► 0,-5, 9
► 0,0,0,5,9
► 0,0,0,-5,9 (true)
► 0,0,5,-9

**102: A partitioned square matrix 'A' is said to the matrix if the matrices on the main diagonal square and all matrices above the main diagonal are zero.**

- Block lower triangular

**103: if there is a vector v= (2, 1, 0) then ||v|| is ……..**

- $\sqrt{5}$

**104: Each pair of eigenvalue and its corresponding eigenvectors provide a solution of equation.**

Eigen function

**105: which of the following is true for the matrix**

**Where $M_{11}$ .$M_{22}$ and $M_{33}$ are square sub – matrices, and O is Zero sub matrix?**

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It is block upper triangular matrix

**106: suppose that real solution $y_1$ and $y_2$ of $x^1 = Ax$ from a basis for the two- dimensional real vector space if $y_1$ and $y_2$ are ……..**

Linearly independent

**107: Let W be a subspace of R and ($u_1, u_2,.....u_p$) is any orthogonal basis of W, then $y = c_1u_1 + c_2u_2 \ldots\ldots c_nu_n$ where**

$$Cj = \frac{y.uj}{uj.uj}$$

**108: Elementary row operation on a matrix do not affect the ….. relation among the column of the matrix**

Linear dependence

**109: Two vector are …….. if at least one of the vector is a multiple of the other**

Linearly dependent

**110: Electric circuit rotation is caused by sine and cosine function if there eigenvalues are complex and hence the origin is called….**

Spiral point

**111: Multiplication of a partitioned matrix by a scalar is also computed…..**

Column by column

**112: The invariable Matrix .Theorem applies only to……**

Square

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**113:Let A be a real 2 by 2 matrix with complex eign values $\partial$ = a – b + I,(b $\neq$ ) and associated eigenvectors v in $C^2$ then A PCP $^{-1}$ where p= ------**

P = [Revlmv]

**114: Let Ax = 0 be a homogeneous linear system of 'n' unknown , then the coefficient matrix 'A' is invertable if and only if this system has ---------- solution**

Trivial

**115: the two vector are said to b equivalent if**

Same length and same direction

**116: if one of the eigenvalues of $[A]_{nxn}$ is zero .it implies**

The determined of [A] is zero

**117: let be a five – dimensional vector space. And let ba subset of consisting of five vectors.**

Mast be linearly dependent, but may or may not span V

**118: Suppose x, y, z are some vectors in an inner product space (v,<,>) such that <X,Y>=<X,Z> for all x ∑ V then Y=Z**

TRUE

**119: let U,V and W be vectors in $R^n$ ,then**

(U+ v), W=U.W+V.W

**Q120: if u and v non zero vector in either $R^2$or$R^3$ then by the law of cosines || u – vii$^2$……..**

$||u - v||^2 = ||u||^2 + ||v||^2 - 2\,||u||||v||\cos$

**121: If u +v – w, then**

V = w

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**122: Two vectors u and v in $R^n$ are orthogonal if…….**

u.v = o

**123:A vector whose length 1 is called ……..**

Unit vector

**124: A …… Matrix whit orthogonal columns is an orthogonal matrix.**

Square 1:

**125: if a square matrix has orthogonal columns then it also has …… rows.**

Orthogonal

**126: The norm of v is the non-negative scalar ||v|| defined by**

$||v|| = \sqrt{V}^2_1 + v_2^2 + \ldots\ldots\ldots\ldots v^2_n$

**127: the matrix A ^ T x A is invertable if and only if the columns of A are linearly independent**

True

**128: let V be a one Eigen vector then conjugate eigen vector is repented by.**

$V^{----}$

**129: Any finite dimensional inner product space an orthonormal basis.**

True

**130: the …….. vector is orthogonal to every vector in $R^n$.**

Zero

**131: Each pair of eigenvalue and this its corresponding eigenvector provides a solution of the**

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**equation $X^1 = Ax$ which is called …….. of the differential equation.**

Eigen function

**132: if we divide a non – zero vector by its length we get a**

Unit vector

**133: A matrix A – (n\time n) has both positive and negative eigenvalues so in this case origin behave as a**

Saddle point

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# Lecture 18

## Properties of Determinants

In this lecture, we will study the properties of the determinants. Some of them have already been discussed and you will be familiar with these. These properties become helpful, while computing the values of the determinants. The secret of determinants lies in how they change when row or column operations are performed.

**Theorem 3** (Row Operations)**:** Let ***A*** be a square matrix.

a. If a multiple of one row of *A* is added to another row, the resulting determinant will remain same.
b. If two rows of ***A*** are interchanged to produce ***B***, then *det* ***B*** = –*det* ***A***.
c. If one row of ***A*** is multiplied by *k* to produce ***B***, then *det* ***B*** = *k* . *det* ***A***.

The following examples show how to use Theorem 3 to find determinants efficiently.

a. If a multiple of one row of *A* is added to another row, the resulting determinant will remain same.

**Example**

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

*Multiplying 2nd row by non – zero scalar say 'k 'as*

$ka_{21} \quad ka_{22} \quad ka_{23}$ *– – – adding this in 1st row then 'A' becomes*

$$= \begin{vmatrix} a_{11}+ka_{21} & a_{12}+ka_{22} & a_{13}+ka_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \qquad R_1' \to R_1 + kR_2$$

If each element of any row(column) can be expressed as sum of two elements then the resulting determinant can be expressed as sum of two determinants, so in this case

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} ka_{21} & ka_{22} & ka_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + k \begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

By using property (c) of above theorem 3.

If any two rows or columns in a determinant are identical then value of this determinant is zero. So in this case $R_1 \equiv R_2$

$$\therefore \ \Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + k(0)$$

$$= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = A$$

b. If two rows of *A* are interchanged to produce *B*, then *det B = –det A.*

**Example 1**

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 1 & 1 \\ 0 & 8 & 9 \end{bmatrix}$$

$$\text{Now, } \det A = \begin{vmatrix} 1 & 2 & 3 \\ 5 & 1 & 1 \\ 0 & 8 & 9 \end{vmatrix} = 1(9-8) - 2(45-0) + 3(40-0) = 1 - 90 + 120 = 31$$

Now interchange column 1st with 2nd we get a new matrix, $B = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 5 & 1 \\ 8 & 0 & 9 \end{bmatrix}$

$$\det B = \begin{vmatrix} 2 & 1 & 3 \\ 1 & 5 & 1 \\ 8 & 0 & 9 \end{vmatrix} = 2(45-0) - 1(9-8) + 3(0-40) = 90 - 1 - 120 = -31$$

c. If one row of *A* is multiplied by *k* to produce *B*, then *det B = k . det A.*

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 0 & 1 \\ 0 & 8 & 9 \end{bmatrix}$$

$$|A| = 1(0-8) - 2(45-0) + 3(40-0)$$

$$= -8 - 90 + 120 = 22$$

*Multiplying $R_1$ by k, we get say*

$$B = \begin{bmatrix} 1k & 2k & 3k \\ 5 & 0 & 1 \\ 0 & 8 & 9 \end{bmatrix}$$

$$|B| = k(40-0) - 2k(45-0) + 3k(40-0)$$
$$= 40k - 90k + 120k = 22k$$
$$= k|A|$$

**Example 2**

Evaluate $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{bmatrix}$

**Solution**

$$\det A = \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -7 \\ 0 & -2 & -8 & -10 \\ 0 & -7 & -10 & -13 \end{vmatrix} \quad by\ R_2' \to R_2 + (-2)R_1, R_3' \to R_3 + (-3)R_1, R_4' \to R_4 + (-4)R_1$$

$$= \begin{vmatrix} -1 & -2 & -7 \\ -2 & -8 & -10 \\ -7 & -10 & -13 \end{vmatrix} \quad \text{expanding from Ist column}$$

$$= (-1)(-2)(-1)\begin{vmatrix} 1 & 2 & 7 \\ 1 & 4 & 5 \\ 7 & 10 & 13 \end{vmatrix} \quad taking\ (-1),(-2)\ and\ (-1)\ common\ \ from 1st, 2nd, 3rd\ rows$$

$$= (-2)\begin{vmatrix} 1 & 2 & 7 \\ 0 & 2 & -2 \\ 0 & -4 & -36 \end{vmatrix} \quad by\ R_2' \to R_2 + (-1)R_1, R_3' \to R_3 + (-7)R_1$$

$$= (-2)\begin{vmatrix} 2 & -2 \\ -4 & -36 \end{vmatrix} \quad \text{expanding by1st column}$$

$$= (-2)(2)(-4)\begin{vmatrix} 1 & -1 \\ 1 & 9 \end{vmatrix} \quad \text{taking 2 and (-4) common from1st and 2nd rows respectively.}$$

$$= 16\begin{vmatrix} 1 & -1 \\ 0 & 10 \end{vmatrix} \quad by\ R_2 + (-1)R_1$$
$$= 160$$

**Example 3** Evaluate the determinant of the matrix $A = \begin{bmatrix} 4 & 2 & 5 & 10 \\ 1 & 1 & 6 & 3 \\ 7 & 3 & 0 & 5 \\ 0 & 2 & 5 & 8 \end{bmatrix}$

**Solution**

$$\det A = \begin{vmatrix} 4 & 2 & 5 & 10 \\ 1 & 1 & 6 & 3 \\ 7 & 3 & 0 & 5 \\ 0 & 2 & 5 & 8 \end{vmatrix}$$

$$= -\begin{vmatrix} 1 & 1 & 6 & 3 \\ 4 & 2 & 5 & 10 \\ 7 & 3 & 0 & 5 \\ 0 & 2 & 5 & 8 \end{vmatrix}$$ interchanging $R_1$ and $R_2$ $(R_{12}')$

$$= -\begin{vmatrix} 1 & 1 & 6 & 3 \\ 0 & -2 & -19 & -2 \\ 0 & -4 & -42 & -16 \\ 0 & 2 & 5 & 8 \end{vmatrix}$$ By $R_2' \to R_2 + (-4)R_1, R_3' \to R_3 + (-7)R_1$

$$= -\begin{vmatrix} -2 & -19 & -2 \\ -4 & -42 & -16 \\ 2 & 5 & 8 \end{vmatrix}$$ expanding from 1st column

$$= (-1)^3 \begin{vmatrix} 2 & 19 & 2 \\ 4 & 42 & 16 \\ 2 & 5 & 8 \end{vmatrix}$$ taking (-1) as a common factor from $R_1$ and $R_2$

$$= -\begin{vmatrix} 2 & 19 & 2 \\ 4 & 42 & 16 \\ 2 & 5 & 8 \end{vmatrix}$$

$$= -2\begin{vmatrix} 1 & 19 & 2 \\ 2 & 42 & 16 \\ 1 & 5 & 8 \end{vmatrix}$$

$$= (-2)\begin{vmatrix} 1 & 19 & 2 \\ 0 & 4 & 12 \\ 0 & -14 & 6 \end{vmatrix}$$ By $R_2' \to R_2 + (-2)R_1, R_3' \to R_3 + (-1)R_1$

$$= (-2)\begin{vmatrix} 1 & 19 & 2 \\ 0 & 4 & 12 \\ 0 & -14 & 6 \end{vmatrix} \quad R_2 + (-2)R_1, R_3 + (-1)R_1$$

$$= -2\begin{vmatrix} 4 & 12 \\ -14 & 6 \end{vmatrix}$$ expand from Ist column

$$= -2(24+168) = -384$$

**Example 4** Without expansion, show that $\begin{vmatrix} x & a+x & b+c \\ x & b+x & c+a \\ x & c+x & a+b \end{vmatrix} = 0$

**Solution**

$$\begin{vmatrix} x & a+x & b+c \\ x & b+x & c+a \\ x & c+x & a+b \end{vmatrix}$$

$$= \begin{vmatrix} x & a+x-x & b+c \\ x & b+x-x & c+a \\ x & c+x-x & a+b \end{vmatrix} \quad By\ C_2' \to C_2 - C_1$$

$$= \begin{vmatrix} x & a & b+c \\ x & b & c+a \\ x & c & a+b \end{vmatrix}$$

*Taking 'x' common from* $C_1$

$$= x\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$$

$$= x\begin{vmatrix} 1 & a+b+c & b+c \\ 1 & b+c+a & c+a \\ 1 & c+a+b & a+b \end{vmatrix} \quad By\ C_2' \to C_2 + C_3$$

Now taking (a+b+c) common form $C_2$

$$= x(a+b+c)\begin{vmatrix} 1 & 1 & b+c \\ 1 & 1 & c+a \\ 1 & 1 & a+b \end{vmatrix}$$

$= 0$ as column Ist and 2nd are identical ($C_1 \equiv C_2$). So its value will be zero.

**Example 5** Evaluate $A=\begin{vmatrix} 2 & 3 & 1 & 0 & 1 \\ 1 & 1 & 3 & 1 & 2 \\ 2 & 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 1 & 2 \\ 4 & 1 & 1 & 0 & 0 \end{vmatrix}$

**Solution** Interchanging $R_1$ and $R_2$, we get

$$A=-\begin{vmatrix} 1 & 1 & 3 & 1 & 2 \\ 2 & 3 & 1 & 0 & 1 \\ 2 & 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 1 & 2 \\ 4 & 1 & 1 & 0 & 0 \end{vmatrix}$$

$$R_2' \to R_2 - 2R_1, R_3' \to R_3 - 2R_1, R_4' \to R_4 - 3R_1, R_5' \to R_5 - 4R_1$$

$$=-\begin{vmatrix} 1 & 1 & 3 & 1 & 2 \\ 0 & 1 & -5 & -2 & -3 \\ 0 & -1 & -4 & 1 & 0 \\ 0 & -1 & -8 & -2 & -4 \\ 0 & -3 & -11 & -4 & -8 \end{vmatrix}$$

expand from $C_1$

$$=-\begin{vmatrix} 1 & -5 & -2 & -3 \\ -1 & -4 & 1 & 0 \\ -1 & -8 & -2 & -4 \\ -3 & -11 & -4 & -8 \end{vmatrix}$$

$$R_2' \to R_2 + R_1, R_3' \to R_3 + R_1, R_4' \to R_4 + 3R_1$$

$$=-\begin{vmatrix} 1 & -5 & -2 & -3 \\ 0 & -9 & -1 & -3 \\ 0 & -13 & -4 & -7 \\ 0 & -26 & -10 & -17 \end{vmatrix}$$

expand from $C_1$

$$=-\begin{vmatrix} -9 & -1 & -3 \\ -13 & -4 & -7 \\ -26 & -10 & -17 \end{vmatrix}$$

taking (-1) common from Ist, 2nd and 3rd row

$$= \begin{vmatrix} 9 & 1 & 3 \\ 13 & 4 & 7 \\ 26 & 10 & 17 \end{vmatrix}$$

interchange Ist and 2nd Column($C_1'_2$)

$$= - \begin{vmatrix} 1 & 9 & 3 \\ 4 & 13 & 7 \\ 10 & 26 & 17 \end{vmatrix}$$

$C_2' \to C_2 - 9C_1, C_3' \to C_3 - 3C_1$

$$= - \begin{vmatrix} 1 & 0 & 0 \\ 4 & -23 & -5 \\ 10 & -64 & -13 \end{vmatrix}$$

expand from Ist row

$$= - \begin{vmatrix} -23 & -5 \\ -64 & -13 \end{vmatrix} = -(299 - 320) = 21$$

**An Algorithm to evaluate the determinant**

Algorithm means a sequence of a finite number of steps to get a desired result. The word Algorithm comes from the famous Muslim mathematician AL-Khwarizmi who invented the word algebra.

The step-by-step evaluation of det(***A***) of order ***n*** is obtained as follows:

**Step 1:** By an interchange of rows of ***A*** (and taking the resulting sign into account) bring a non zero entry to (1,1) the position (unless all the entries in the first column are zero in which case det ***A***=0).

**Step 2:** By adding suitable multiples of the first row to all the other rows, reduce the (n-1) entries, except (1,1) in the first column, to 0. Expand det(***A***) by its first column. Repeat this process or continue the following steps.

**Step 3:** Repeat step 1 and step 2 with the last remaining rows concentrating on the second column.

**Step 4:** Repeat step 1,step2 and step 3 with the remaining (n-2) rows, (n-3) rows and so on, until a triangular matrix is obtained.

**Step5:** Multiply all the diagonal entries of the resulting triangular matrix and then multiply it by its sign to get det(A)

**Example 6** Compute det $A$, where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$.

**Solution** *The strategy is to reduce* $A$ *to echelon form and then to use the fact that the determinant of a triangular matrix is the product of the diagonal entries.* The first two row replacements in column 1 do not change the determinant:

$$\det A = \begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix} \quad By\ R_2' \to R_2 + 2R_1, R_3' \to R_1 + R_3$$

An interchange of rows 2 and 3 ($R'_{23}$), it reverses the sign of the determinant, so

$$\det A = -\begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix} = -(1)(3)(-5) = 15$$

**Example 7** Compute det $A$, where

$$A = \begin{bmatrix} 2 & -8 & 6 & 8 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{bmatrix}.$$

**Solution** Taking'2' common from 1st row

$$\det A = 2\begin{vmatrix} 1 & -4 & 3 & 4 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{vmatrix}$$

$$\det A = 2\begin{vmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & -3 & 2 \end{vmatrix} \quad By\ R_2' \to R_2 - 3R_1, R_3' \to R_3 + 3R_1, R_4' \to R_4 - R_1$$

$$\det A = 2\begin{vmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix} \quad (By\ R_4' \to R_4 - \tfrac{1}{2}R_3)$$

$$= 2.\{(1)(3)(-6)(1)\} = -36$$

**Example 8** Show that $\begin{vmatrix} x & 2 & 2 & 2 \\ 2 & x & 2 & 2 \\ 2 & 2 & x & 2 \\ 2 & 2 & 2 & x \end{vmatrix} = (x+6)(x-2)^3$

**Solution**

$$\begin{vmatrix} x & 2 & 2 & 2 \\ 2 & x & 2 & 2 \\ 2 & 2 & x & 2 \\ 2 & 2 & 2 & x \end{vmatrix}$$

$$= \begin{vmatrix} x+6 & 2 & 2 & 2 \\ x+6 & x & 2 & 2 \\ x+6 & 2 & x & 2 \\ x+6 & 2 & 2 & x \end{vmatrix} \quad By\ \ C_1' \to C_1 + (C_2 + C_3 + C_4)$$

Taking (x+6) common from 1st column

$$= (x+6)\begin{vmatrix} 1 & 2 & 2 & 2 \\ 1 & x & 2 & 2 \\ 1 & 2 & x & 2 \\ 1 & 2 & 2 & x \end{vmatrix}$$

$$= (x+6)\begin{vmatrix} 1 & 2 & 2 & 2 \\ 0 & x-2 & 0 & 0 \\ 0 & 0 & x-2 & 0 \\ 0 & 0 & 0 & x-2 \end{vmatrix} \quad By\ R_2' \to R_2 - R_1, R_3' \to R_3 - R_1, R_4' \to R_4 - R_1$$

And this is the triangular matrix and its determinant is the prodcut of main diagonal's entries.

$$= (x+6)(x-2)^3$$

**Example 9** Compute det $A$, where $A = \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ -6 & 7 & -7 & 4 \\ -5 & -8 & 0 & 9 \end{bmatrix}$.

**Solution** $A = \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ -6 & 7 & -7 & 4 \\ -5 & -8 & 0 & 9 \end{bmatrix}$

$$\det A = \begin{vmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 5 & -3 & -6 \\ -5 & -8 & 0 & 9 \end{vmatrix} \quad R_3' \to R_3 + 2R_1$$

$= \mathbf{0}$ as $R_2 \equiv R_3$

**Example10** Compute det $A$, where

$$A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ -2 & -5 & 4 & -2 \end{bmatrix}$$

**Solution**

$$A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ -2 & -5 & 4 & -2 \end{bmatrix}$$

$$= \begin{vmatrix} 0 & 1 & 2 & -1 \\ 2 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -3 & 1 \end{vmatrix} \quad R_4' \to R_4 + R_2$$

$$= (-1) \begin{vmatrix} 2 & 1 & 2 & -1 \\ 0 & 5 & -7 & 3 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -3 & 1 \end{vmatrix} \quad By\ R_{12}'$$

Expanding from 1st row and 1st column

$$= -2\begin{vmatrix} 5 & -7 & 3 \\ 3 & 6 & 2 \\ 0 & -3 & 1 \end{vmatrix}$$

$$= (-2)\{5(6+6)-(-7)(3-0)+3(-9-0)\}$$

$$= 54$$

**Remarks**

Suppose that a square matrix ***A*** has been reduced to an echelon form ***U*** by row replacements and row interchanges.

*If there are r interchanges, then* $\det(A) = (-1)^r \det(U)$

Furthermore, all of the pivots are still visible in ***U*** (because they have not been scaled to ones). If ***A*** is invertible, then the pivots in ***U*** are on the diagonal (since ***A*** is row equivalent to the identity matrix). In this case, *det* ***U*** is the product of the pivots. If ***A*** is not invertible, then ***U*** has a row of zero and *det* ***U*** = *0*.

$$U = \begin{bmatrix} \bullet & \square & \square & \square \\ 0 & \bullet & \square & \square \\ 0 & 0 & \bullet & \square \\ 0 & 0 & 0 & \bullet \end{bmatrix} \qquad U = \begin{bmatrix} \bullet & \square & \square & \square \\ 0 & \bullet & \square & \square \\ 0 & 0 & \bullet & \square \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\det U \neq 0 \qquad \det U = 0$$

Thus we have the following formula

$$\det A = \begin{cases} (-1)^r .(\textit{Product of pivots in } U) & \textit{When A is invertible} \\ 0 & \textit{When A is not invertible} \end{cases} \qquad (1)$$

**Example**

**Case-01** For $2 \times 2$ invertible matrix

Reducing given $2 \times 2$ invertible matrix into Echelon form as follows;

$$A = \begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix}$$

By interchanging 1st and 2nd rows( $R'_{12}$)

$\square$ $\begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$ $\because$ one replacement of rows has occurred, $\therefore r = 1$

$\square$ $\begin{bmatrix} 3 & 2 \\ 0 & \frac{7}{3} \end{bmatrix}$ By $R_2' \to R_2 - \frac{4}{3}R_1$, we have desired row-echelon form $U = \begin{bmatrix} 3 & 2 \\ 0 & \frac{7}{3} \end{bmatrix}$.

Thus using the above formula as follows;

$$\det A = (-1)^r .(Product\ of\ \ pivots\ in U) = (-1)^1(3\cdot \frac{7}{3}) = -7$$

**Case-02** For $2\times 2$ non-invertible matrix

In this case say;

$$A=\begin{bmatrix}4 & 5\\ 8 & 10\end{bmatrix}$$

$$\begin{bmatrix}4 & 5\\ 0 & 0\end{bmatrix} \quad By\ R_2' \to R_2 - 2R_1, \text{desired row-echelon form is } U=\begin{bmatrix}4 & 5\\ 0 & 0\end{bmatrix}$$

Here no interchange of rows has occurred. So, $r = 0$ and

$$\therefore \det A = (-1)^r .(Product\ of\ \ pivots\ in U) = (-1)^0(4\cdot 0) = 0$$

**Theorem 5** If A is an $n\times n$ matrix, then *det* $\boldsymbol{A^T}$ = *det* $\boldsymbol{A}$.

**Example 11** If $A=\begin{bmatrix}1 & 4 & 1\\ 2 & 1 & 2\\ 3 & 1 & 3\end{bmatrix}$, find det($\boldsymbol{A}$) and det ($\boldsymbol{A^T}$)

$$\det A=\begin{vmatrix}1 & 4 & 1\\ 2 & 1 & 2\\ 3 & 1 & 3\end{vmatrix}=1(3-2)-4(6-6)+1(2-3)=1-0-1=0$$

Now

$$A^t=\begin{bmatrix}1 & 2 & 3\\ 4 & 1 & 1\\ 1 & 2 & 3\end{bmatrix}$$

$$\det A^t=\begin{vmatrix}1 & 2 & 3\\ 4 & 1 & 1\\ 1 & 2 & 3\end{vmatrix}=1(3-2)-2(12-1)+3(8-1)=1-22+21=0$$

**Remark**

Column operations are useful for both theoretical purposes and hand computations. However, for simplicity we'll perform only row operations in numerical calculations.

**Theorem 6 (Multiplicative Property)**

If $\boldsymbol{A}$ and $\boldsymbol{B}$ are $n\times n$ matrices, then $\det(AB) = (\det A)(\det B)$.

**Example 12** Verify Theorem 6 for $A=\begin{bmatrix}6 & 1\\ 3 & 2\end{bmatrix}$ *and* $B=\begin{bmatrix}4 & 3\\ 1 & 2\end{bmatrix}$

**Solution** $AB = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 14 & 13 \end{bmatrix}$

and $\det AB = 25.13 - 20.14 = 325 - 280 = 45$

Since *det* $\boldsymbol{A}$ = *9* and *det* $\boldsymbol{B}$ = *5*, $(\det A)(\det B) = 9.5 = 45 = \det AB$

**Remark**

$\det(\boldsymbol{A} + \boldsymbol{B}) \neq \det \boldsymbol{A} + \det \boldsymbol{B}$, in general.

For example,

If $A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & -3 \\ -1 & 5 \end{bmatrix}$. Then

$$A + B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \det(A + B) = 0$$

$$\det A + \det B = \begin{vmatrix} 2 & 3 \\ 1 & -5 \end{vmatrix} + \begin{vmatrix} -2 & -3 \\ -1 & 5 \end{vmatrix} = (-10 - 3) + (-10 - 3) = -26 \neq \det(A + B)$$

**Exercise**

Find the determinants in exercises 1 to 6 by row reduction to echelon form.

1. $\begin{vmatrix} 1 & 3 & 0 & 2 \\ -2 & -5 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{vmatrix}$

2. $\begin{vmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 2 & 5 & 4 & -3 \\ -3 & -7 & -5 & 2 \end{vmatrix}$

3. $\begin{vmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ -1 & 2 & 8 & 5 \\ 3 & -1 & -2 & 3 \end{vmatrix}$

4. $\begin{vmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ -2 & -6 & 2 & 3 & 9 \\ 3 & 7 & -3 & 8 & -7 \\ 3 & 5 & 5 & 2 & 7 \end{vmatrix}$

5. $\begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 8 & 6 & 1 \end{vmatrix}$

6. $\begin{vmatrix} 1 & 3 & 1 & 5 & 3 \\ -2 & -7 & 0 & -4 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{vmatrix}$

Combine the methods of row reduction and cofactor expansion to compute the determinants in exercises 7 and 8.

7. $\begin{vmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 9 \\ 4 & 10 & -4 & -1 \end{vmatrix}$

8. $\begin{vmatrix} 2 & 5 & 4 & 1 \\ 4 & 7 & 6 & 2 \\ 6 & -2 & -4 & 0 \\ -6 & 7 & 7 & 0 \end{vmatrix}$

9. Use determinant to find out whether the matrix is invertible $\begin{bmatrix} 2 & 0 & 0 & 8 \\ 1 & -7 & -5 & 0 \\ 3 & 8 & 6 & 0 \\ 0 & 7 & 5 & 4 \end{bmatrix}$

10. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be $3\times3$ matrices, with det $\boldsymbol{A}$ = 4 and det $\boldsymbol{B}$ = -3. Use properties of determinants to compute

(a) det $\boldsymbol{AB}$ (b) det $7\boldsymbol{A}$ (c) det $\boldsymbol{B^T}$ (d) det $\boldsymbol{A^T}$
(e) det $\boldsymbol{A^TA}$

11 Show that

(a) $\begin{vmatrix} a_1 & b_1 & a_1+b_1+c_1 \\ a_2 & b_2 & a_2+b_2+c_2 \\ a_3 & b_3 & a_3+b_3+c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

(b) $\begin{vmatrix} a_1+b_1 & a_1-b_1 & c_1 \\ a_2+b_2 & a_2-b_2 & c_2 \\ a_3+b_3 & a_3-b_3 & c_3 \end{vmatrix} = -2\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

12 Show that

(a) $\begin{vmatrix} a_1+b_1t & a_2+b_2t & a_3+b_3t \\ a_1t+b_1 & a_2t+b_2 & a_3t+b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = (1-t^2)\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

(b) $\begin{vmatrix} a_1 & b_1+ta_1 & c_1+rb_1+sa_1 \\ a_2 & b_2+ta_2 & c_2+rb_2+sa_2 \\ a_3 & b_3+ta_3 & c_3+rb_3+sa_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

9. Show that $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (y-x)(z-x)(z-y)$

# Lecture 19

## Cramer's Rule, Volume, and Linear Transformations

In this lecture, we shall apply the theory discussed in the last two lectures to obtain important theoretical formulae and a geometric interpretation of the determinant.

**Cramer's Rule** Cramer's rule is needed in a variety of theoretical calculations. For instance, it can be used to study how the solution of $\boldsymbol{Ax = b}$ is affected by changes in the entries of $\boldsymbol{b}$. However, the formula is inefficient for hand calculations, except for $2\times 2$ or perhaps $3\times 3$ matrices.

**Theorem 1 (Crammer's Rule)** Let $\boldsymbol{A}$ be an invertible $n\times n$ matrix. For any $\boldsymbol{b}$ in $\boldsymbol{R^n}$, the unique solution $x$ of $\boldsymbol{Ax = b}$ has entries given by

$$x_i = \frac{\det A_i(b)}{\det A}, \qquad i = 1, 2, ..., n \qquad (1)$$

**Example 1** Use Cramer's rule to solve the system

$$3x_1 - 2x_2 = 6$$
$$-5x_1 + 4x_2 = 8$$

**Solution** Write the system in matrix form, $\boldsymbol{Ax = b}$

$$\begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

*where*

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \& b = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$\det A = \begin{vmatrix} 3 & -2 \\ -5 & 4 \end{vmatrix} = 12 - 10 = 2$$

$$A_1(b) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, A_2(b) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

Since $\det \boldsymbol{A} = 2$, the system has a unique solution. By Cramer's rule,

$$x_1 = \frac{\det A_1(b)}{\det A} = \frac{24 + 16}{2} = 20$$
$$x_2 = \frac{\det A_2(b)}{\det A} = \frac{24 + 30}{2} = 27$$

**Example 2** Consider the following system in which $s$ is an unspecified parameter. Determine the values of $s$ for which the system has a unique solution and use Cramer's rule to describe the solution.

$$3sx_1 - 2x_2 = 4$$
$$-6x_1 + sx_2 = 1$$

**Solution** Here

$$A = \begin{bmatrix} 3s & -2 \\ -6 & s \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad A_1(b) = \begin{bmatrix} 4 & -2 \\ 1 & s \end{bmatrix}, \quad A_2(b) = \begin{bmatrix} 3s & 4 \\ -6 & 1 \end{bmatrix}$$

Since $\det A = 3s^2 - 12 = 3(s+2)(s-2)$

the system has a unique solution when

$\det A \neq 0$

$\Rightarrow 3(s+2)(s-2) \neq 0$

$\Rightarrow s^2 - 4 \neq 0$

$\Rightarrow s \neq \pm 2$

For such an $s$, the solution is $(x_1, x_2)$, where

$$x_1 = \frac{\det A_1(b)}{\det A} = \frac{4s+2}{3(s+2)(s-2)}, \qquad s \neq \pm 2$$

$$x_2 = \frac{\det A_2(b)}{\det A} = \frac{3s+24}{3(s+2)(s-2)} = \frac{s+8}{(s+2)(s-2)}, \qquad s \neq \pm 2$$

**Example 3** Solve, by Cramer's Rule, the system of equations

$$2x_1 - x_2 + 3x_3 = 1$$
$$x_1 + 2x_2 - x_3 = 2$$
$$3x_1 + 2x_2 + 2x_3 = 3$$

**Solution** Here $A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & -1 \\ 3 & 2 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $A_1 = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 2 & -1 \\ 3 & 2 & 2 \end{bmatrix}$

$$A_2 = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & -1 \\ 3 & 3 & 2 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 2 \\ 3 & 2 & 3 \end{bmatrix}$$

$$D = \det A = 2\cdot 6 - 1\cdot(-5) + 3(-4) = 5$$
$$D_1 = \det A_1 b = 1\cdot 6 + 2\cdot 8 + 3(-5) = 7$$
$$D_2 = \det A_2 b = 2\cdot(7) - 1\cdot(5) + 3(-3) = 0$$
$$D_3 = \det A_3 b = 2\cdot(2) + 1\cdot(-3) + 1(-4) = -3$$
$$So\ x_1 = \frac{D_1}{D} = \frac{7}{5},\quad x_2 = \frac{D_2}{D} = 0,\quad x_3 = \frac{D_3}{D} = -\frac{3}{5}$$

**Example 4** Use Cramer's Rule to solve.

$$x_1 + 2x_3 = 6$$
$$-3x_1 + 4x_2 + 6x_3 = 30$$
$$-x_1 - 2x_2 + 3x_3 = 8$$

**Solution**

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -3 & 4 & 6 \\ -1 & -2 & 3 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 30 \\ 8 \end{bmatrix}$$

$$\therefore A_1 = \begin{bmatrix} 6 & 0 & 2 \\ 30 & 4 & 6 \\ 8 & -2 & 3 \end{bmatrix}, A_2 = \begin{bmatrix} 1 & 6 & 2 \\ -3 & 30 & 6 \\ -1 & -8 & 3 \end{bmatrix} \quad A_3 = \begin{bmatrix} 1 & 0 & 6 \\ -3 & 4 & 30 \\ -1 & -2 & 8 \end{bmatrix}$$

Therefore,

$$x_1 = \frac{\det(A_1 b)}{\det(A)} = \frac{-40}{44} = \frac{-10}{11}, \qquad x_2 = \frac{\det(A_2 b)}{\det(A)} = \frac{72}{44} = \frac{18}{11}$$
$$x_3 = \frac{\det(A_3 b)}{\det(A)} = \frac{152}{44} = \frac{38}{11}$$

**Note** For any $n \times n$ matrix ***A*** and any ***b*** in $\boldsymbol{R^n}$, *let* $\boldsymbol{A_i(b)}$ be the matrix obtained from ***A*** by replacing ***i***th column by the vector ***b***.

$$A_i(b) = \begin{bmatrix} a_1 & \ldots & b & \ldots & a_n \end{bmatrix}$$
$$\uparrow$$
$$i^{th}\ column$$

**Formula for $A^{-1}$**

Cramer's rule leads easily to a general formula for the inverse of $n \times n$ matrix ***A***. The ***j****th* column of $\boldsymbol{A^{-1}}$ is a vector ***x*** that satisfies $\boldsymbol{Ax = e_j}$ where $\boldsymbol{e_j}$ is the ***j****th* column of the identity matrix, and the ***i****th* entry of ***x*** is the $(i, j)$-entry of $\boldsymbol{A^{-1}}$. By Cramer's rule,

$$\left\{(i,j) - entry\ of\ A^{-1}\right\} = x_{ij} = \frac{\det A_i(e_j)}{\det A} \qquad (2)$$

Recall that $A_{ji}$ denotes the submatrix of $A$ formed by deleting row $j$ and column $i$. A cofactor expansion down column $i$ of $A_i(e_j)$ shows that

$$\det A_i(e_j) = (-1)^{i+j} \det A_{ji} = C_{ji} \quad (3)$$

where $C_{ji}$ is a cofactor of $A$.

By (2), the $(i, j)$-entry of $A^{-1}$ is the cofactor $C_{ji}$ divided by $\det A$.

[Note that the subscripts on $C_{ji}$ are the reverse of $(i, j)$.] Thus

$$A^{-1} = \frac{1}{\det A}\begin{bmatrix} C_{11} & C_{21} & \dots & C_{n1} \\ C_{12} & C_{22} & \dots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \dots & C_{nn} \end{bmatrix} \quad (4)$$

The matrix of cofactors on the right side of (4) is called the **adjugate** (or **classical adjoint**) of $A$, denoted by adj $A$. (The term adjoint also has another meaning in advance texts on linear transformations.) The next theorem simply restates (4).

**Theorem 2 (An Inverse Formula)**

Let $A$ be an invertible $n \times n$ matrix, then $A^{-1} = \frac{1}{\det A} adj\, A$

**Example**

For the matrrix say

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix} \Rightarrow \det A = 10 - (-3) = 13$$

$\Rightarrow A^{-1}$ will also be a $2 \times 2$ matrix

As

$Aji =$ submatrix of $A$ formed by deleting row $j$ and column $i$

So in this case

$A_{11} =$ submatrix of $A$ formed by deleting row **1** and column **1** $= [5]$

$A_{12} =$ submatrix of $A$ formed by deleting row **1** and column **2** $= [-1]$

$A_{21} =$ submatrix of $A$ formed by deleting row **2** and column **1** $= [3]$

$A_{22} =$ submatrix of $A$ formed by deleting row **2** and column **2** $= [2]$

and

$$\det A_i(e_j) = (-1)^{i+j} \det(A_{ji}) = C_{ji}$$

where $e_j$ is the jth column of identity matrix $I_{n \times n}$

So in this case

$$C_{11} = \det A_{11}(e_1) = (-1)^{1+1} \det A_{11} = (+1)\det[5] = 5$$
$$C_{12} = \det A_{12}(e_1) = (-1)^{1+2} \det A_{12} = (-1)\det[-1] = (-1)(-1) = 1$$
$$C_{21} = \det A_1(e_2) = (-1)^{2+1} \det A_{21} = (-1)\det[3] = -3$$
$$C_{22} = \det A_2(e_2) = (-1)^{2+2} \det A_{22} = (+1)\det[2] = 2$$

By Cramer's rule,

$$\{(i,j)-entry\ of\ A^{-1}\} = x_{ij} = \frac{\det A_i(e_j)}{\det A} = \frac{C_{ji}}{\det A}$$

So for the current matrix;

$$\{(1,1)-entry\ of\ A^{-1}\} = x_{11} = \frac{\det A_1(e_1)}{\det A} = \frac{C_{11}}{\det A} = \frac{5}{13}$$
$$\{(1,2)-entry\ of\ A^{-1}\} = x_{12} = \frac{\det A_1(e_2)}{\det A} = \frac{C_{21}}{\det A} = \frac{-3}{13}$$
$$\{(2,1)-entry\ of\ A^{-1}\} = x_{21} = \frac{\det A_2(e_1)}{\det A} = \frac{C_{12}}{\det A} = \frac{1}{13}$$
$$\{(2,2)-entry\ of\ A^{-1}\} = x_{22} = \frac{\det A_2(e_2)}{\det A} = \frac{C_{22}}{\det A} = \frac{2}{13}$$

Hence by using equation # 4, we get

$$A^{-1} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} \frac{C_{11}}{\det A} & \frac{C_{21}}{\det A} \\ \frac{C_{12}}{\det A} & \frac{C_{22}}{\det A} \end{bmatrix} = \frac{1}{\det A}\begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix} = \begin{bmatrix} \frac{5}{13} & \frac{-3}{13} \\ \frac{1}{13} & \frac{2}{13} \end{bmatrix}$$

**Example 5** Find the inverse of the matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$.

**Solution** The nine cofactors are

$$C_{11} = +\begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2, \quad C_{12} = -\begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3, \quad C_{13} = +\begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5$$

$$C_{21} = -\begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14, \quad C_{22} = +\begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7, \quad C_{23} = -\begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -7$$

$$C_{31} = +\begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 4, \quad C_{32} = -\begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1, \quad C_{33} = +\begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$$

The adjoint matrix is the transpose of the matrix of cofactors. [For instance, $C_{12}$ goes in the (*2*, *1*) position.] Thus

$$adjA = \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$$

We could compute det $\boldsymbol{A}$ directly, but the following computation provides a check on the calculations above and produces det $\boldsymbol{A}$:

$$(adjA).A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 14 \end{bmatrix} = 14I$$

Since (adj $\boldsymbol{A}$) $\boldsymbol{A}$ = 14 $\boldsymbol{I}$, Theorem 2 shows that det $\boldsymbol{A}$ = 14 and

$$A^{-1} = \frac{1}{14}\begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} = \begin{bmatrix} -2/14 & 14/14 & 4/14 \\ 3/14 & -7/14 & 1/14 \\ 5/14 & -7/14 & -3/14 \end{bmatrix}$$

**Determinants as Area or Volume**

In the next application, we verify the geometric interpretation of determinants and we assume here that the usual Euclidean concepts of length, area, and volume are already understood for $\boldsymbol{R^2}$ and $\boldsymbol{R^3}$.

**Theorem 3** If $\boldsymbol{A}$ is a $2\times 2$ matrix, the area of the parallelogram determined by the columns of $A$ is $|\det A|$. If $\boldsymbol{A}$ is a $3\times 3$ matrix, the volume of the parallelepiped determined by the columns of $\boldsymbol{A}$ is $|\det A|$.

**Example 6** Calculate the area of the parallelogram determined by the points (-2, -2), (0, 3), (4, -1) and (6, 4).

**Solution**

Let A(-2,-2), B(0,3), C(4,-1) and D(6,4). Fixing one point say A(-2,-2) and find the adjacent lengths of parallelogram which are given by the column vectors as follows**;**

$$AB = \begin{bmatrix} 0-(-2) \\ 3-(-2) \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$AC = \begin{bmatrix} 4-(-2) \\ -1-(-2) \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

So the area of parallelogram ABCD determined by above column vectors

$$= \left|\det \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix}\right| = |2-30| = |-28| = 28$$

Now we translate the parallelogram ABCD to one having the origin as a vertex. For which we subtract the vertex (-2, -2) from each of the four vertices. The new parallelogram has the vertices say

A′=(-2-(-2),-2-(-2))=(0, 0)

B′=(0-(-2),3-(-2))=(2, 5)

C′=(4-(-2),-1-(-2))=(6,1)

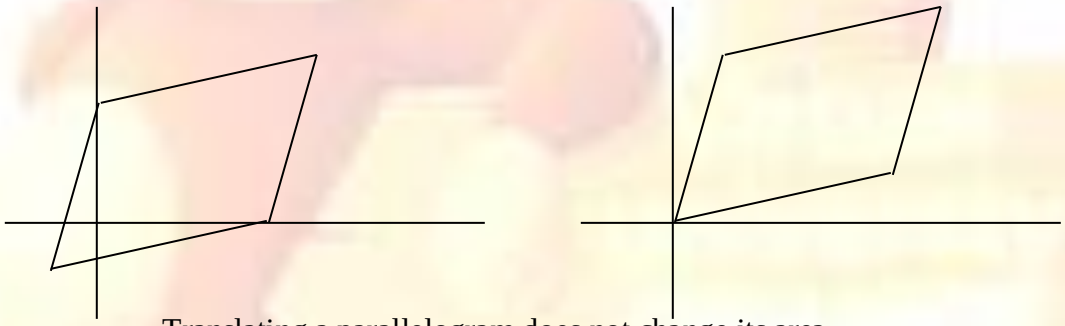
D′=(6-(-2),4-(-2))=(8, 6)

And fixing $A'(0,0)$ in this case, so

$$A'B' = \begin{bmatrix} 2-0 \\ 5-0 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$A'C' = \begin{bmatrix} 6-0 \\ 5-0 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$

See Fig below. The area of this parallelogram is also determined by the above columns vectors = $\left|\det \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix}\right| = |2-30| = |-28| = 28$



Translating a parallelogram does not change its area

**Linear Transformations**

Determinants can be used to describe an important geometric property of linear transformations in the plane and in $\boldsymbol{R^3}$. If ***T*** is a linear transformation and ***S*** is a set in the domain of ***T***, let ***T* (*S*)** denote the set of images of points in ***S***. We are interested in how the area (or volume) of ***T* (*S*)** compares with the area (or volume) of the original set ***S*.** For convenience, when ***S*** is a region bounded by a parallelogram, we also refer to ***S*** as a parallelogram.

**Theorem 4** Let $T: R^2 \to R^2$ be the linear transformation determined by a $2 \times 2$ matrix ***A***. If ***S*** is a parallelogram in $\boldsymbol{R^2}$, then

{area of ***T* (*S*)**} = |det***A***|. {area of ***S***}

If ***T*** is determined by a 3 x 3 matrix ***A***, and if ***S*** is a parallelepiped in $\boldsymbol{R^3}$, then

{volume of ***T* (*S*)**} = |det***A***|. {volume of ***S***}

**Example 7** Let $a$ and $b$ be positive numbers. Find the area of the region $E$ bounded by the ellipse whose equation is $\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} = 1$.

**Solution** We claim that $E$ is the image of the unit disk $D$ under the linear transformation **A:D→E** determined by the matrix $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, given as

$A\boldsymbol{u} = \boldsymbol{x}$ where $\boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in D$, $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in E$.

$$\Rightarrow \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Now $A\boldsymbol{u} = \boldsymbol{x} \Rightarrow \begin{bmatrix} au_1 \\ bu_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ then

$$\Rightarrow au_1 = x_1 \quad and \quad bu_2 = x_2$$

$$\Rightarrow u_1 = \frac{x_1}{a} \text{ and } u_2 = \frac{x_2}{b}$$

Since $u \in D$ (in the circular disk),it follows that the distance of $\boldsymbol{u}$ from origin will be less than unity i-e

$$\left(u_1^2 - 0\right) + \left(u_2^2 - 0\right) \le 1$$

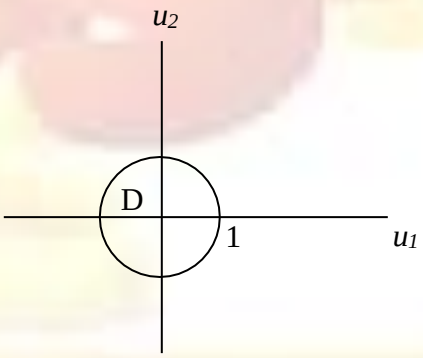
$$\Rightarrow \left(\frac{x_1}{a}\right)^2 + \left(\frac{x_2}{b}\right)^2 \le 1 \qquad \because u_1 = \frac{x_1}{a}, u_2 = \frac{x_2}{b}$$

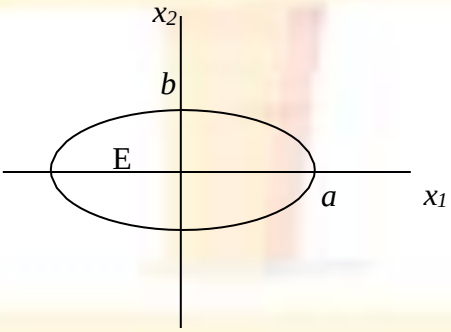
Hence by the generalization of theorem 4,

{area of ellipse} = {area of $\boldsymbol{A(D)}$} (*here* $T \equiv A$)

= |det $\boldsymbol{A}$|. {area of $\boldsymbol{D}$}

= $ab.\pi(1)^2 = \pi ab$



**Example 8** Let ***S*** be the parallelogram determined by the vectors $b_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, and let $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$. Compute the area of image of ***S*** under the mapping $x \to Ax$.

**Solution** The area of ***S*** is $\left| \det \begin{bmatrix} 1 & 5 \\ 3 & 1 \end{bmatrix} \right| = 14$, and det ***A*** = 2. By theorem 4, the area of image of ***S*** under the mapping $x \to Ax$ is |det ***A***|. {area of ***S***} = 2.14 = 28

**Exercises**

Use Cramer's Rule to compute the solutions of the systems in exercises 1 and 2.

1. $\begin{aligned} 2x_1 + x_2 &= 7 \\ -3x_1 + x_3 &= -8 \\ x_2 + 2x_3 &= -3 \end{aligned}$

2. $\begin{aligned} 2x_1 + x_2 + x_3 &= 4 \\ -x_1 + 2x_3 &= 2 \\ 3x_1 + x_2 + 3x_3 &= -2 \end{aligned}$

In exercises 3-6, determine the values of the parameter s for which the system has a unique solution, and describe the solution.

3. $\begin{aligned} 6sx_1 + 4x_2 &= 5 \\ 9x_1 + 2sx_2 &= -2 \end{aligned}$

4. $\begin{aligned} 3sx_1 - 5x_2 &= 3 \\ 9x_1 + 5sx_2 &= 2 \end{aligned}$

5. $\begin{aligned} sx_1 - 2sx_2 &= -1 \\ 3x_1 + 6sx_2 &= 4 \end{aligned}$

6. $\begin{aligned} 2sx_1 + x_2 &= 1 \\ 3sx_1 + 6sx_2 &= 2 \end{aligned}$

In exercises 7 and 8, compute the adjoint of the given matrix, and then find the inverse of the matrix.

7. $\begin{bmatrix} 3 & 5 & 4 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$

8. $\begin{bmatrix} 3 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 3 & 2 \end{bmatrix}$

In exercises 9 and 10, find the area of the parallelogram whose vertices are listed.

9. *(0, 0), (5, 2), (6, 4), (11, 6)*

10. *(-1, 0), (0, 5), (1, -4), (2, 1)*

11. Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at *(1, 0, -2), (1, 2, 4), (7, 1, 0)*.

12. Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at *(1, 4, 0), (-2, -5, 2), (-1, 2, -1)*.

13. Let $\boldsymbol{S}$ be the parallelogram determined by the vectors $\boldsymbol{b_1} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ and $\boldsymbol{b_2} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$, and let $\boldsymbol{A} = \begin{bmatrix} 6 & -2 \\ -3 & 2 \end{bmatrix}$. Compute the area of the image of $\boldsymbol{S}$ under the mapping $\boldsymbol{x} \rightarrow \boldsymbol{Ax}$.

14. Let $\boldsymbol{S}$ be the parallelogram determined by the vectors $\boldsymbol{b_1} = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$ and $\boldsymbol{b_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and let $\boldsymbol{A} = \begin{bmatrix} 7 & 2 \\ 1 & 1 \end{bmatrix}$. Compute the area of the image of $\boldsymbol{S}$ under the mapping $\boldsymbol{x} \rightarrow \boldsymbol{Ax}$.

15. Let $\boldsymbol{T}: \boldsymbol{R^3} \rightarrow \boldsymbol{R^3}$ be the linear transformation determined by the matrix $\boldsymbol{A} = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$, where a, b, c are positive numbers. Let $\boldsymbol{S}$ be the unit ball, whose bounding surface has the equation $x_1^2 + x_2^2 + x_3^2 = 1$.

a. Show that $\boldsymbol{T}(\boldsymbol{S})$ is bounded by the ellipsoid with the equation $\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} + \frac{x_3^2}{c^2} = 1$.

b. Use the fact that the volume of the unit ball is $4\pi/3$ to determine the volume of the region bounded by the ellipsoid in part (a).

# Lecture 20

## Vector Spaces and Subspaces

**Case Example**

The space shuttle's control systems are absolutely critical for flight. Because the shuttle is an unstable airframe, it requires constant computer monitoring during atmospheric flight. The flight control system sends a stream of commands to aerodynamic control surfaces.

Mathematically, the input and output signals to an engineering system are functions. It is important in applications that these functions can be added, and multiplied by scalars. These two operations on functions have algebraic properties that are completely analogous to the operation of adding vectors in $\boldsymbol{R^n}$ and multiplying a vector by a scalar, as we shall see in the lectures 20 and 27. For this reason, the set of all possible inputs (functions) is called a *vector space*. The mathematical foundation for systems engineering rests on vector spaces of functions, and we need to extend the theory of vectors in $\boldsymbol{R^n}$ to include such functions. Later on, we will see how other vector spaces arise in engineering, physics, and statistics.

**Definition** Let $\boldsymbol{V}$ be an arbitrary nonempty set of objects on which two operations are defined, addition and multiplication by scalars (numbers). If the following axioms are satisfied by all objects $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ in $\boldsymbol{V}$ and all scalars $k$ and $l$, then we call $\boldsymbol{V}$ a vector space.

**Axioms of Vector Space**

**1. Closure Property** For any two vectors $\boldsymbol{u}$ & $\boldsymbol{v} \in \boldsymbol{V}$, implies $\boldsymbol{u} + \boldsymbol{v} \in \boldsymbol{V}$

**2. Commutative Property** For any two vectors $\boldsymbol{u}$ & $\boldsymbol{v} \in \boldsymbol{V}$, implies $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$

**3. Associative Property** For any three vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \boldsymbol{V}$, $\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$

**4. Additive Identity** For any vector $\boldsymbol{u} \in \boldsymbol{V}$, there exist a zero vector $\boldsymbol{0}$ such that
$\boldsymbol{0} + \boldsymbol{u} = \boldsymbol{u} + \boldsymbol{0} = \boldsymbol{u}$

**5. Additive Inverse** For each vector $\boldsymbol{u} \in \boldsymbol{V}$, there exist a vector $-\boldsymbol{u}$ in $\boldsymbol{V}$ such that
$-\boldsymbol{u} + \boldsymbol{u} = \boldsymbol{0} = \boldsymbol{u} + (-\boldsymbol{u})$

**6. Scalar Multiplication** For any scalar $k$ and a vector $\boldsymbol{u} \in \boldsymbol{V}$ implies k $\boldsymbol{u} \in \boldsymbol{V}$

**7. Distributive Law** For any scalar $k$ if $\boldsymbol{u}$ & $\boldsymbol{v} \in \boldsymbol{V}$, then $k(\boldsymbol{u} + \boldsymbol{v}) = k\,\boldsymbol{u} + k\,\boldsymbol{v}$

**8.** For scalars $m, n$ and for any vector $\boldsymbol{u} \in \boldsymbol{V}$, $(m + n)\,\boldsymbol{u} = m\,\boldsymbol{u} + n\,\boldsymbol{u}$

**9.** For scalars $m, n$ and for any vector $\boldsymbol{u} \in \boldsymbol{V}$, $m\,(n\,\boldsymbol{u}) = (m\,n)\,\boldsymbol{u} = n\,(m\,\boldsymbol{u})$

**10.** For any vector $\boldsymbol{u} \in \boldsymbol{V}$, $1\boldsymbol{u} = \boldsymbol{u}$ where $1$ is the multiplicative identity of real numbers.

**Examples of vector spaces** The following examples will specify a non empty set $\boldsymbol{V}$ and two operations: addition and scalar multiplication; then we shall verify that the ten vector space axioms are satisfied.

**Example 1** Show that the set of all ordered n-tuple $\boldsymbol{R^n}$ is a vector space under the standard operations of addition and scalar multiplication.

**Solution**

(i) **Closure Property:**

Suppose that $\boldsymbol{u} = (u_1, u_2, \ldots, u_n)$ and $\boldsymbol{v} = (v_1, v_2, \ldots, v_n) \in \boldsymbol{R^n}$

Then by definition, $\boldsymbol{u} + \boldsymbol{v} = (u_1, u_2, \ldots, u_n) + (v_1, v_2, \ldots, v_n)$

$= (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n) \in \boldsymbol{R^n}$ (By closure property)

Therefore, $\boldsymbol{R^n}$ is closed under addition.

(ii) **Commutative Property**

Suppose that $\boldsymbol{u} = (u_1, u_2, \ldots, u_n)$ and $\boldsymbol{v} = (v_1, v_2, \ldots, v_n) \in \boldsymbol{R^n}$

Now $\boldsymbol{u} + \boldsymbol{v} = (u_1, u_2, \ldots, u_n) + (v_1, v_2, \ldots, v_n)$

$= (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$ (By closure property)

$= (v_1 + u_1, v_2 + u_2, \ldots, v_n + u_n)$ (By commutative law of real numbers)

$= (v_1, v_2, \ldots, v_n) + (u_1, u_2, \ldots, u_n)$ (By closure property)

$= \boldsymbol{v} + \boldsymbol{u}$

Therefore, $\boldsymbol{R^n}$ is commutative under addition.

(iii) **Associative Property**

Suppose that $\boldsymbol{u} = (u_1, u_2, \ldots, u_n)$, $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)$ *and* $\boldsymbol{w} = (w_1, w_2, \ldots, w_n) \in \boldsymbol{R^n}$

Now $(\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w} = [(u_1, u_2, \ldots, u_n) + (v_1, v_2, \ldots, v_n)] + (w_1, w_2, \ldots, w_n)$

$= (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n) + (w_1, w_2, \ldots, w_n)$ (By closure property)

$= ((u_1 + v_1) + w_1, (u_2 + v_2) + w_2, \ldots, (u_n + v_n) + w_n))$ (By closure property)

$= (u_1 + (v_1 + w_1), u_2 + (v_2 + w_2), \ldots, u_n + (v_n + w_n))$ (By associative law of real numbers)

$= (u_1, u_2, \ldots, u_n) + (v_1 + w_1, v_2 + w_2, \ldots, v_n + w_n)$ (By closure property)

$= (u_1, u_2, \ldots, u_n) + [(v_1, v_2, \ldots, v_n) + (w_1, w_2, \ldots, w_n)]$ (By closure property)

$= \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$

Hence $\boldsymbol{R^n}$ is associative under addition.

(iv) **Additive Identity**

Suppose $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R^n}$. There exists $\boldsymbol{0} = (0, 0, \ldots, 0) \in \boldsymbol{R^n}$ such that
$\boldsymbol{0} + \boldsymbol{u} = (0, 0, \ldots, 0) + (u_1, u_2, \ldots, u_n)$

$= (0 + u_1, 0 + u_2, \ldots, 0 + u_n)$ (By closure property)

$= (u_1, u_2, \ldots, u_n) = \boldsymbol{u}$ (Existence of identity of real numbers)

Similarly, $\boldsymbol{u} + \boldsymbol{0} = \boldsymbol{u}$

Hence $\boldsymbol{0} = (0, 0, \ldots, 0)$ is the additive identity for $\boldsymbol{R^n}$.

(v) **Additive Inverse**

Suppose $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R^n}$. There exists $-\boldsymbol{u} = (-u_1, -u_2, \ldots, -u_n) \in \boldsymbol{R^n}$

Such that $\boldsymbol{u} + (-\boldsymbol{u}) = (u_1, u_2, \ldots, u_n) + (-u_1, -u_2, \ldots, -u_n)$

$= (u_1 + (-u_1), u_2 + (-u_2), \ldots, u_n + (-u_n))$ (By closure property)

$= (0, 0, \ldots, 0) = \boldsymbol{0}$

Similarly, $(-\boldsymbol{u}) + \boldsymbol{u} = \boldsymbol{0}$

Hence the inverse of each element of $\boldsymbol{R^n}$ exists in $\boldsymbol{R^n}$.

(vi) **Scalar Multiplication**

If $k$ is any scalar and $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R^n}$.

Then by definition, $k\,\boldsymbol{u} = k\,(u_1, u_2, \ldots, u_n) = (k\,u_1, k\,u_2, \ldots, k\,u_n) \in \boldsymbol{R^n}$
(By closure property)

(vii) **Distributive Law**

Suppose $k$ is any scalar and $\boldsymbol{u} = (u_1, u_2, \ldots, u_n)$, $\boldsymbol{v} = (v_1, v_2, \ldots, v_n) \in \boldsymbol{R^n}$

Now $k\,(\boldsymbol{u} + \boldsymbol{v}) = k\,[(u_1, u_2, \ldots, u_n) + (v_1, v_2, \ldots, v_n)]$

$= k\,(u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$ (By closure property)

$= (k\,(u_1 + v_1), k\,(u_2 + v_2), \ldots, k\,(u_n + v_n))$ (By scalar multiplication)

$= (k\,u_1 + k\,v_1, k\,u_2 + k\,v_2, \ldots, k\,u_n + k\,v_n)$ (By Distributive Law)

$= (k\,u_1, k\,u_2, \ldots, k\,u_n) + (k\,v_1, k\,v_2, \ldots, k\,v_n)$ (By closure property)

$= k\,(u_1, u_2, \ldots, u_n) + k\,(v_1, v_2, \ldots, v_n)$ (By scalar multiplication)

$= k\,\boldsymbol{u} + k\,\boldsymbol{v}$

(viii) Suppose $k$ and $l$ be any scalars and $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R}^n$

Then $(k + l)\,\boldsymbol{u} = (k + l)\,(u_1, u_2, \ldots, u_n)$

$= ((k + l)u_1, (k + l)u_2, \ldots, (k + l)u_n)$ (By scalar multiplication)

$= (k\,u_1 + l\,u_1, k\,u_2 + l\,u_2, \ldots, k\,u_n + l\,u_n)$ (By Distributive Law)

$= (k\,u_1, k\,u_2, \ldots, k\,u_n) + (l\,u_1, l\,u_2, \ldots, l\,u_n)$ (By closure property)

$= k\,(u_1, u_2, \ldots, u_n) + l\,(u_1, u_2, \ldots, u_n)$ (By scalar multiplication)

$= k\,\boldsymbol{u} + l\,\boldsymbol{u}$

(ix) Suppose $k$ and $l$ be any scalars and $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R}^n$

Then $k\,(l\,\boldsymbol{u}) = k\,[l\,(u_1, u_2, \ldots, u_n)]$

$= k\,(l\,u_1, l\,u_2, \ldots, l\,u_n)$ (By scalar multiplication)

$= (k\,(l\,u_1), k\,(l\,u_2), \ldots, k\,(l\,u_n))$ (By scalar multiplication)

$= ((k\,l)u_1, (k\,l)u_2, \ldots, (k\,l)u_n)$ (By associative law)

$= (k\,l)\,(u_1, u_2, \ldots, u_n)$ (By scalar multiplication)

$= (k\,l)\,\boldsymbol{u}$

(x) Suppose $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \boldsymbol{R}^n$

Then $1\,\boldsymbol{u} = 1\,(u_1, u_2, \ldots, u_n)$

$= (1u_1, 1u_2, \ldots, 1u_n)$ (By scalar multiplication)

$= (u_1, u_2, \ldots, u_n) = \boldsymbol{u}$ (Existence of identity in scalrs)

Hence, $\boldsymbol{R^n}$ is the real vector space with the standard operations of addition and scalar multiplication.

**Note** The three most important special cases of $\boldsymbol{R^n}$ are $\boldsymbol{R}$ (the real numbers), $\boldsymbol{R^2}$ (the vectors in the plane), and $\boldsymbol{R^3}$ (the vectors in 3-space).

**Example 2** Show that the set $\boldsymbol{V}$ of all 2x2 matrices with real entries is a vector space if vector addition is defined to be matrix addition and vector scalar multiplication is defined to be matrix scalar multiplication.

**Solution** Suppose that $\boldsymbol{u} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}, \boldsymbol{v} = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} \in \boldsymbol{V}$

and $k$ and $l$ be two any scalars.

(i) **Closure property** To prove axiom (i), we must show that $\boldsymbol{u} + \boldsymbol{v}$ is an object in $\boldsymbol{V}$: that is , we must show that $\boldsymbol{u} + \boldsymbol{v}$ is a 2x2 matrix. But this is clear from the definition of matrix addition, since $\boldsymbol{u}+\boldsymbol{v} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = \begin{bmatrix} u_{11}+v_{11} & u_{12}+v_{12} \\ u_{21}+v_{21} & u_{22}+v_{22} \end{bmatrix}$

(By closure property)

(ii) **Commutative property** Now it is very easy to verify the Axiom (ii)

$\boldsymbol{u}+\boldsymbol{v} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = \begin{bmatrix} u_{11}+v_{11} & u_{12}+v_{12} \\ u_{21}+v_{21} & u_{22}+v_{22} \end{bmatrix}$ (By closure property)

$= \begin{bmatrix} v_{11}+u_{11} & v_{12}+u_{12} \\ v_{21}+u_{21} & v_{22}+u_{22} \end{bmatrix}$ (Commutative property of real numbers)

$= \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} + \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \boldsymbol{v}+\boldsymbol{u}$

(iii) **Associative property** $(\boldsymbol{u}+\boldsymbol{v})+\boldsymbol{w} = \left( \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} \right) + \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$

$= \begin{bmatrix} u_{11}+v_{11} & u_{12}+v_{12} \\ u_{21}+v_{21} & u_{22}+v_{22} \end{bmatrix} + \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$ (By closure property)

$= \begin{bmatrix} (u_{11}+v_{11})+w_{11} & (u_{12}+v_{12})+w_{12} \\ (u_{21}+v_{21})+w_{21} & (u_{22}+v_{22})+w_{22} \end{bmatrix}$

$= \begin{bmatrix} u_{11}+(v_{11}+w_{11}) & u_{12}+(v_{12}+w_{12}) \\ u_{21}+(v_{21}+w_{21}) & u_{22}+(v_{22}+w_{22}) \end{bmatrix}$ (By associative property of real numbers)

$$= \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11}+w_{11} & v_{12}+w_{12} \\ v_{21}+w_{21} & v_{22}+w_{22} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \left( \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} + \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} \right) = \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$$

Therefore, ***V*** is associative under '+'.

(iv) **Additive Identity** Now to prove the axiom (iv), we must find an object $\boldsymbol{0}$ in ***V*** such that $\boldsymbol{0} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{0} = \boldsymbol{v}$ for all ***u*** in ***V***. This can be done by defining $\boldsymbol{0} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

$$\boldsymbol{0} + \boldsymbol{u} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} 0+u_{11} & 0+u_{12} \\ 0+u_{21} & 0+u_{22} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \boldsymbol{u}$$

and similarly $\boldsymbol{u} + \boldsymbol{0} = \boldsymbol{u}$.

(v) **Additive Inverse** Now to prove the axiom (v) we must show that each object ***u*** in ***V*** has a negative $-\boldsymbol{u}$ such that $\boldsymbol{u} + (-\boldsymbol{u}) = \boldsymbol{0} = (-\boldsymbol{u}) + \boldsymbol{0}$. Defining the negative of ***u*** to be

$$-\boldsymbol{u} = \begin{bmatrix} -u_{11} & -u_{12} \\ -u_{21} & -u_{22} \end{bmatrix}.$$

$$\boldsymbol{u} + (-\boldsymbol{u}) = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} -u_{11} & -u_{12} \\ -u_{21} & -u_{22} \end{bmatrix} = \begin{bmatrix} u_{11}+(-u_{11}) & u_{12}+(-u_{12}) \\ u_{21}+(-u_{21}) & u_{22}+(-u_{22}) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \boldsymbol{0}$$

Similarly, $(-\boldsymbol{u}) + \boldsymbol{u} = \boldsymbol{0}$

(vi) **Scalar Multiplication**

Axiom (vi) also holds because for any real number *k* we have

$$k\boldsymbol{u} = k \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} ku_{11} & ku_{12} \\ ku_{21} & ku_{22} \end{bmatrix}$$ (By closure property)

so that k ***u*** is a 2x2 matrix and consequently is an object in ***V***.

(vii) **Distributive Law**

$$k(\boldsymbol{u} + \boldsymbol{v}) = k\left( \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} \right)$$

$$= k\begin{bmatrix} u_{11}+v_{11} & u_{12}+v_{12} \\ u_{21}+v_{21} & u_{22}+v_{22} \end{bmatrix} = \begin{bmatrix} k(u_{11}+v_{11}) & k(u_{12}+v_{12}) \\ k(u_{21}+v_{21}) & k(u_{22}+v_{22}) \end{bmatrix}$$

$$= \begin{bmatrix} ku_{11}+kv_{11} & ku_{12}+kv_{12} \\ ku_{21}+kv_{21} & ku_{22}+kv_{22} \end{bmatrix} = \begin{bmatrix} ku_{11} & ku_{12} \\ ku_{21} & ku_{22} \end{bmatrix} + \begin{bmatrix} kv_{11} & kv_{12} \\ kv_{21} & kv_{22} \end{bmatrix}$$

$$= k\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + k\begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = k\boldsymbol{u} + k\boldsymbol{v}$$

(viii) $(k+l)\boldsymbol{u} = (k+l)\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} (k+l)u_{11} & (k+l)u_{12} \\ (k+l)u_{21} & (k+l)u_{22} \end{bmatrix}$

$$= \begin{bmatrix} ku_{11}+lu_{11} & ku_{12}+lu_{12} \\ ku_{21}+lu_{21} & ku_{22}+lu_{22} \end{bmatrix} = \begin{bmatrix} ku_{11} & ku_{12} \\ ku_{21} & ku_{22} \end{bmatrix} + \begin{bmatrix} lu_{11} & lu_{12} \\ lu_{21} & lu_{22} \end{bmatrix}$$

$$= k\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + l\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = k\boldsymbol{u} + l\boldsymbol{u}$$

(ix) $k(l\boldsymbol{u}) = k\left(l\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}\right) = k\begin{bmatrix} lu_{11} & lu_{12} \\ lu_{21} & lu_{22} \end{bmatrix}$

$$= \begin{bmatrix} k(lu_{11}) & k(lu_{12}) \\ k(lu_{21}) & k(lu_{22}) \end{bmatrix} = \begin{bmatrix} (kl)u_{11} & (kl)u_{12} \\ (kl)u_{21} & (kl)u_{22} \end{bmatrix} = (kl)\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = (kl)\boldsymbol{u}$$

(x) Finally axiom (x) is a simple computation

$$1\boldsymbol{u} = 1\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} 1u_{11} & 1u_{12} \\ 1u_{21} & 1u_{22} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \boldsymbol{u}$$

Hence the set of all 2x2 matrices with real entries is vector space under matrix addition and matrix scalar multiplication.

**Note** Example 2 is a special case of a more general class of vector spaces. The arguments in that example can be adapted to show that a set $\boldsymbol{V}$ of all $\boldsymbol{m} \times \boldsymbol{n}$ matrices with real entries, together with the operations of matrix addition and scalar multiplication, is a vector space.

**Example 3** Let $V$ be the set of all real-valued functions defined on the entire real line$(-\infty,\infty)$. If $f, g \in V$, then $f + g$ is a function defined by

$$(f+g)(x) = f(x) + g(x), \text{ for all } x \in R.$$

The product of a scalar $a \in R$ and a function $f$ in $V$ is defined by

$$(a f)(x) = a f(x), \text{ for all } x \in R.$$

**Solution**

(i) **Closure Property** If $f, g \in V$, then by definition

$(f+g)(x) = f(x) + g(x) \in V$. Therefore, $V$ is closed under addition.

(ii) **Commutative Property** If $f$ and $g$ are in $V$, then for all $x \in R$

$(f + g)(x) = f(x) + g(x)$ (By definition)

$= g(x) + f(x)$ (By commutative property)

$= (g + f)(x)$ (By definition)

So that $f + g = g + f$

(iii) **Associative Property** If $f, g$ and $h$ are in $V$, then for all $x \in R$

$((f + g) + h)(x) = (f + g)(x) + h(x)$ (By definition)

$= (f(x) + g(x)) + h(x)$ (By definition)

$= f(x) + (g(x) + h(x))$ (By associative property)

$= f(x) + (g + h)(x)$ (By definition)

$= (f + (g + h))(x)$

And so $(f + g) + h = f + (g + h)$

(iv) **Additive Identity** The additive identity of $V$ is the zero function defined by $0(x) = 0$, for all $x \in R$ because $(0 + f)(x) = 0(x) + f(x)$ (By definition)

$= 0 + f(x) = f(x)$ (Existence of identity)

i.e. $0 + f = f$. Similarly, $f + 0 = f$.

(v) **Additive Inverse** The additive inverse of a function $f$ in $V$ is $(-1) f = -f \in V$ because

$(f + (-f))(x) = f(x) + (-f)(x)$ (By definition)

$= f(x) - f(x)$ (By definition)

$= 0$ (Existence of inverse)

i.e. $\boldsymbol{f} + (-\boldsymbol{f}) = \boldsymbol{0}$. Similarly, $(-\boldsymbol{f}) + \boldsymbol{f} = \boldsymbol{0}$.

(vi) **Scalar Multiplication** If $\boldsymbol{f}$ is in $\boldsymbol{V}$ and $a$ is in $R$, then by definition $(a\,\boldsymbol{f})(x) = a\,\boldsymbol{f}(x) \in \boldsymbol{V}$.

(vii) **Distributive Law** If $\boldsymbol{f}, \boldsymbol{g}$ are in $\boldsymbol{V}$ and $a \in R$, then
$(a(\boldsymbol{f} + \boldsymbol{g}))(x) = a(\boldsymbol{f} + \boldsymbol{g})(x) = a(\boldsymbol{f}(x) + \boldsymbol{g}(x)) = a\,\boldsymbol{f}(x) + a\,\boldsymbol{g}(x)$

$= (a\,\boldsymbol{f})(x) + (a\,\boldsymbol{g})(x) = (a\,\boldsymbol{f} + a\,\boldsymbol{g})(x)$ and, therefore, $a(\boldsymbol{f} + \boldsymbol{g}) = a\,\boldsymbol{f} + a\,\boldsymbol{g}$

(viii) Let $a, b$ in $R$ and $\boldsymbol{f} \in \boldsymbol{V}$, then

$((a + b)\,\boldsymbol{f})(x) = (a + b)\,\boldsymbol{f}(x) = a\,\boldsymbol{f}(x) + b\,\boldsymbol{f}(x) = (a\,\boldsymbol{f})(x) + (b\,\boldsymbol{f})(x) = (a\,\boldsymbol{f} + b\,\boldsymbol{f})(x)$

Thus $(a + b)\,\boldsymbol{f} = a\,\boldsymbol{f} + b\,\boldsymbol{f}$

(ix) $a(b\,\boldsymbol{f})(x) = a(b\,\boldsymbol{f}(x)) = (a\,b)\,\boldsymbol{f}(x)$ showing that $a(b\,\boldsymbol{f}) = (a\,b)\,\boldsymbol{f}$

(x) $(1.\boldsymbol{f})(x) = 1\,\boldsymbol{f}(x) = \boldsymbol{f}(x)$ (Existence of identity)

And so $1.\boldsymbol{f} = \boldsymbol{f}$

Hence $\boldsymbol{V}$ is a real vector space.

**Example 4** If $\boldsymbol{p}(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$

and $\boldsymbol{q}(x) = b_0 + b_1 x + b_2 x^2 + \ldots + b_n x^n$

We define

$$\boldsymbol{p}(x) + \boldsymbol{q}(x) = (a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n) + (b_0 + b_1 x + b_2 x^2 + \ldots + b_n x^n)$$

$$= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \ldots + (a_n + b_n)x^n$$ and for any scalar $k$,

$$k\boldsymbol{p}(x) = k(a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n) = ka_0 + ka_1 x + ka_2 x^2 + \ldots + k\,a_n x^n$$

Clearly the given polynomial is a vector space under the addition and scalar multiplication.

**Example 5** **(The Zero Vector Space)** Let $\boldsymbol{V}$ consists of a single object, which we define by $\boldsymbol{0}$ and $\boldsymbol{0} + \boldsymbol{0} = \boldsymbol{0}$ and k $\boldsymbol{0} = \boldsymbol{0}$ for all scalars $k$. It is easy to check that all the vector space axioms are satisfied. We call $\boldsymbol{V} = \{0\}$ as the zero vector space.

**Example 6** (Every plane through the origin is a vector space)

Let $\boldsymbol{V}$ be any plane through the origin in $\boldsymbol{R^3}$. We shall show that the points in $\boldsymbol{V}$ form a vector space under a standard addition and scalar multiplication operations for vectors in $\boldsymbol{R^3}$.

From example1, we know that $\boldsymbol{R^3}$ itself is a vector space under these operations. Thus, Axioms 2, 3, 7, 8, 9 and 10 hold for all points in $\boldsymbol{R^3}$ and consequently for all points in the plane $\boldsymbol{V}$. We therefore need only show that Axioms 1, 4, 5 and 6 are satisfied.

Since the plane is passing through the origin, it has an equation of the form

$$a\,x + b\,y + c\,z = 0 \qquad (1)$$

Thus, if $\boldsymbol{u} = (u_1, u_2, u_3)$ and $\boldsymbol{v} = (v_1, v_2, v_3)$ are points in $\boldsymbol{V}$, then

$$a\,u_1 + b\,u_2 + c\,u_3 = 0 \text{ and } a\,v_1 + b\,v_2 + c\,v_3 = 0.$$

Adding these equations gives $a\,(u_1 + v_1) + b\,(u_2 + v_2) + c\,(u_3 + v_3) = 0$

This equality tell us that the coordinates of the point

$$\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3)$$

satisfies (1); thus, $\boldsymbol{u} + \boldsymbol{v}$ lies in plane $\boldsymbol{V}$. This proves that the Axiom 1 is satisfied.

There exists $\boldsymbol{0} = (0, 0, 0)$ such that $a\,(0) + b\,(0) + c\,(0) = 0$. Therefore, Axiom 4 is satisfied.

Multiplying $a\,u_1 + b\,u_2 + c\,u_3 = 0$ through by $k$ gives

$$a\,(ku_1) + b\,(ku_2) + c\,(ku_3) = 0$$

Thus, $(ku_1, k\,u_2, k\,u_3) = k\,(u_1, u_2, u_3) = k\,\boldsymbol{u} \in \boldsymbol{V}$. Hence, Axiom 6 is satisfied.

We shall prove the axiom 5 is satisfied. Multiplying $a\,u_1 + b\,u_2 + c\,u_3 = 0$ through by $-1$ gives $a\,(-1u_1) + b\,(-1u_2) + c\,(-1u_3) = 0$

Thus, $(-u_1, -u_2, -u_3) = -\,(u_1, u_2, u_3) = -\boldsymbol{u} \in \mathbf{V}$. This establishes Axiom 5.

**Example 7** (A set that is not a vector space)

Let $\boldsymbol{V}=\boldsymbol{R^2}$ and define addition and scalar multiplication operation as follows. If $\boldsymbol{u} = (u_1, u_2)$ *and* $\boldsymbol{v} = (v_1, v_2)$ then define

$\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1, u_2 + v_2)$ and if $k$ is any real number then define $k\boldsymbol{u} = (ku_1, 0)$.

For any vector $\boldsymbol{u} \in \boldsymbol{V}$, $1\boldsymbol{u} = 1(u_1, u_2) = (1\,u_1, 0) = (u_1, 0) \neq \boldsymbol{u}$ where $1$ is the multiplicative identity of real numbers. Therefore, the axiom 10 is not satisfied.

Hence, $\boldsymbol{V}=\boldsymbol{R^2}$ is not a vector space.

**Theorem 1** Let $\boldsymbol{V}$ be a vector space, $\boldsymbol{u}$ a vector in $\boldsymbol{V}$, and $k$ is a scalar, then

(i) $0\boldsymbol{u} = \boldsymbol{0}$ (ii) $k\boldsymbol{0} = \boldsymbol{0}$

(iii) $(-1)\,\boldsymbol{u} = -\boldsymbol{u}$ (iv) If $k\,\boldsymbol{u} = \boldsymbol{0}$ then $k = 0$ or $\boldsymbol{u} = \boldsymbol{0}$

**Definition** A subset $\boldsymbol{W}$ of a vector space $\boldsymbol{V}$ is called a subspace of $\boldsymbol{V}$ if $\boldsymbol{W}$ itself a vector space under the addition and scalar multiplication defined on $\boldsymbol{V}$.

**Note** If $\boldsymbol{W}$ is a part of a larger set $\boldsymbol{V}$ that is already known to be a vector space, then certain axioms need not be verified for $\boldsymbol{W}$ because they are "inherited" from $\boldsymbol{V}$. For example, there is no need to check that $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$ (Axiom 2) for $\boldsymbol{W}$ because this holds for all vectors in $\boldsymbol{V}$ and consequently for all vectors in $\boldsymbol{W}$. Other Axioms are inherited by $\boldsymbol{W}$ from $\boldsymbol{V}$ are 3, 7, 8, 9, and 10. Thus, to show that a set $\boldsymbol{W}$ is a subspace of a vector space $\boldsymbol{V}$, we need only verify Axioms 1, 4, 5 and 6. The following theorem shows that even Axioms 4 and 5 can be omitted.

**Theorem 2** If $\boldsymbol{W}$ is a set of one or more vectors from a vector space $\boldsymbol{V}$, then $\boldsymbol{W}$ is subspace of $\boldsymbol{V}$ if and only if the following conditions hold.

(a) If $\boldsymbol{u}$ and $\boldsymbol{v}$ are vectors in $\boldsymbol{W}$, then $\boldsymbol{u} + \boldsymbol{v}$ is in $\boldsymbol{W}$
(b) If $k$ is any scalar and $\boldsymbol{u}$ is any vector in $\boldsymbol{W}$, then $k\,\boldsymbol{u}$ is in $\boldsymbol{W}$.

**Proof** If $\boldsymbol{W}$ is a subspace of $\boldsymbol{V}$, then all the vector space axioms are satisfied; in particular, Axioms 1 and 6 hold. But these are precisely conditions (a) and (b).

Conversely, assume conditions (a) and (b) hold. Since these conditions are vector space Axioms 1 and 6, we need only show that $\boldsymbol{W}$ satisfies the remaining 8 axioms. The vectors in $\boldsymbol{W}$ automatically satisfy axioms 2, 3, 7, 8, 9, and 10 since they are satisfied by all vectors in $\boldsymbol{V}$. Therefore, to complete the proof, we need only to verify that vectors in $\boldsymbol{W}$ satisfy axioms 4 and 5.

Let $\boldsymbol{u}$ be any vector in $\boldsymbol{W}$. By condition (b), $k\,\boldsymbol{u}$ is in $\boldsymbol{W}$ for every scalar $k$. Setting $k = 0$, it follows from theorem 1 that $0\,\boldsymbol{u} = \boldsymbol{0}$ is in $\boldsymbol{W}$, and setting $k = -1$, it follows that $(-1)\,\boldsymbol{u} = -\boldsymbol{u}$ is in $\boldsymbol{W}$. □

**Remark**

**(1)** The theorem states that $\boldsymbol{W}$ is a subspace of $\boldsymbol{V}$ if and only if $\boldsymbol{W}$ is closed under addition and closed under scalar multiplication.

**(2)** Every vector space has at least two subspaces, itself and the subspace $\{\boldsymbol{0}\}$ consisting only of the zero vector. Thus the subspace $\{\boldsymbol{0}\}$ is called the zero subspace.

**Example 8** Let $\boldsymbol{W}$ be the subset of $\boldsymbol{R^3}$ consisting of the all the vectors of the form $(a, b, 0)$, where $a$ and $b$ are real numbers. To check if $\boldsymbol{W}$ is subspace of $\boldsymbol{R^3}$, we first see that axiom 1 and 6 of a vector space holds.

Let $\boldsymbol{u} = (a_1, b_1, 0)$ *and* $\boldsymbol{v} = (a_2, b_2, 0)$ be vectors in $\boldsymbol{W}$ then $\boldsymbol{u} + \boldsymbol{v} = (a_1, b_1, 0) + (a_2, b_2, 0) = (a_1 + a_2, b_1 + b_2, 0)$ is in $\boldsymbol{W}$. Since the third component is zero. Also $c$ is scalar, and then $c\boldsymbol{u} = c(a_1, b_1, 0) = (ca_1, cb_1, 0)$ is in $\boldsymbol{W}$. Therefore the I$^{st}$ and 6$^{th}$ axioms of the vector space holds. We can also verify the other axioms of vector space. Hence $\boldsymbol{W}$ is a subspace.

**Example 9** Consider the set $\boldsymbol{W}$ consisting of all 2x3 matrices of the form $\begin{bmatrix} a & b & 0 \\ 0 & c & d \end{bmatrix}$, Where $a$, $b$, $c$ and $d$ are arbitrary real numbers. Show that the $\boldsymbol{W}$ is a subspace $\boldsymbol{M_{2x3}}$.

**Solution** Consider $\boldsymbol{u} = \begin{bmatrix} a_1 & b_1 & 0 \\ 0 & c_1 & d_1 \end{bmatrix}, \boldsymbol{v} = \begin{bmatrix} a_2 & b_2 & 0 \\ 0 & c_2 & d_2 \end{bmatrix}$ in $\boldsymbol{W}$

Then $\boldsymbol{u} + \boldsymbol{v} = \begin{bmatrix} a_1 & b_1 & 0 \\ 0 & c_1 & d_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 & 0 \\ 0 & c_2 & d_2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 & b_1 + b_2 & 0 \\ 0 & c_1 + c_2 & d_1 + d_2 \end{bmatrix}$ is in $\boldsymbol{W}$.

So that the (a) part of the theorem is satisfied. Also $k$ is a scalar, and then $k\boldsymbol{u} = \begin{bmatrix} kd_1 & kb_1 & 0 \\ 0 & kc_1 & kd_1 \end{bmatrix}$ is in $\boldsymbol{W}$. So the (b) part of the above theorem is also satisfied.

Hence $\boldsymbol{W}$ is a subspace of $\boldsymbol{M_{2x3}}$.

**Note** Let $\boldsymbol{V}$ is a vector space then every subset of $\boldsymbol{V}$ is not necessary a subspace of $\boldsymbol{V}$. For example, let $\boldsymbol{V} = \boldsymbol{R^2}$ then any line in $\boldsymbol{R^2}$ not passing through origin is not a subspace of $\boldsymbol{R^2}$. Similarly, a plane in $\boldsymbol{R^3}$ not passing through the origin is not a subspace of $\boldsymbol{R^3}$.

**Example 10** Let $\boldsymbol{W}$ be the subset of $\boldsymbol{R^3}$ consisting of all vectors of the form $(a, b, 1)$, where $a$, $b$ are any real numbers. To check whether property (a) and (b) of the above theorem holds. Let $\boldsymbol{u} = (a_1, b_1, 1)$ *and* $\boldsymbol{v} = (a_2, b_2, 1)$ be vectors in $\boldsymbol{W}$.

Then $\boldsymbol{u} + \boldsymbol{v} = (a_1, b_1, 1) + (a_2, b_2, 1) = (a_1 + a_2, b_1 + b_2, 1 + 1)$ which is not in $\boldsymbol{W}$ because the third component 2 is not 1. As the Ist property does not hold therefore, the given set of vectors is not a vector space.

**Example 11** Which of the following are subspaces of $\boldsymbol{R^3}$

(i) All vectors of the form $(a, 0, 0)$
(ii) All vectors of the form $(a, 1, 1)$
(iii) All vectors of the form $(a, b, c)$, where $b = a + c$
(iv) All vectors of the form $(a, b, c)$, where $b = a + c + 1$

**Solution** Let $\boldsymbol{W}$ is the set of all vectors of the form $(a, 0, 0)$.

(i) Suppose $\boldsymbol{u} = (u_1, 0, 0)$ and $\boldsymbol{v} = (v_1, 0, 0)$ are in $\boldsymbol{W}$.

Then $\boldsymbol{u} + \boldsymbol{v} = (u_1, 0, 0) + (v_1, 0, 0) = (u_1 + v_1, 0, 0)$ which is of the form $(a, 0, 0)$.

Therefore, $\boldsymbol{u} + \boldsymbol{v} \in \boldsymbol{W}$

If $k$ is any scalar and $\boldsymbol{u} = (u_1, 0, 0)$ is any vector in $\boldsymbol{W}$, then $k\,\boldsymbol{u} = k\,(u_1, 0, 0) = (k\,u_1, 0, 0)$
which is of the form $(a, 0, 0)$. Therefore, $k\,\boldsymbol{u} \in \boldsymbol{W}$. Hence $\boldsymbol{W}$ is the subspace of $\boldsymbol{R^3}$.

(ii) Let $\boldsymbol{W}$ is the set of all vectors of the form $(a, 1, 1)$.

Suppose $\boldsymbol{u} = (u_1, 1, 1)$ and $\boldsymbol{v} = (v_1, 1, 1)$ are in $\boldsymbol{W}$. Then $\boldsymbol{u} + \boldsymbol{v} = (u_1, 1, 1) + (v_1, 1, 1) = (u_1 + v_1, 2, 2)$ which is not of the form $(a, 1, 1)$. Therefore, $\boldsymbol{u} + \boldsymbol{v} \notin \boldsymbol{W}$. Hence $\boldsymbol{W}$ is not the subspace of $\boldsymbol{R^3}$.

(iii) Suppose $\boldsymbol{W}$ is the set of all vectors of the form $(a, b, c)$, where $b = a + c$

Suppose $\boldsymbol{u} = (u_1, u_1 + u_3, u_3)$ and $\boldsymbol{v} = (v_1, v_1 + v_3, v_3)$ are in $\boldsymbol{W}$.

Then $\boldsymbol{u} + \boldsymbol{v} = (u_1, u_1 + u_3, u_3) + (v_1, v_1 + v_3, v_3)$
$= (u_1 + v_1, u_1 + u_3 + v_1 + v_3, u_3 + v_3)$
$= (u_1 + v_1, (u_1 + v_1) + (u_3 + v_3), u_3 + v_3)$, which is of the form $(a, a + c, c)$.

Therefore, $\boldsymbol{u} + \boldsymbol{v} \in \boldsymbol{W}$

If $k$ is any scalar and $\boldsymbol{u} = (u_1, u_1 + u_3, u_3)$ is any vector in $\boldsymbol{W}$, then
$k\,\boldsymbol{u} = k\,(u_1, u_1 + u_3, u_3) = (k\,u_1, k\,(u_1 + u_3), k\,u_3)$ (By definition)
$= (k\,u_1, k\,u_1 + k\,u_3, k\,u_3)$ (By Distributive Law)
Which is of the form $(a, a + c, c)$. Therefore, $k\,\boldsymbol{u} \in \boldsymbol{W}$. Hence $\boldsymbol{W}$ is the subspace of $\boldsymbol{R^3}$.

(iv) Let $\boldsymbol{W}$ is the set of all vectors of the form $(a, b, c)$, where $b = a + c + 1$
Suppose $\boldsymbol{u} = (u_1, u_1 + u_3 + 1, u_3)$ and $\boldsymbol{v} = (v_1, v_1 + v_3 + 1, v_3)$ are in $\boldsymbol{W}$.
Then $\boldsymbol{u} + \boldsymbol{v} = (u_1, u_1 + u_3 + 1, u_3) + (v_1, v_1 + v_3 + 1, v_3)$
$= (u_1 + v_1, u_1 + u_3 + 1 + v_1 + v_3 + 1, u_3 + v_3)$
$= (u_1 + v_1, (u_1 + v_1) + (u_3 + v_3) + 2, u_3 + v_3)$
Which is not of the form $(a, a + c + 1, c)$. Therefore, $\boldsymbol{u} + \boldsymbol{v} \notin \boldsymbol{W}$. Hence $\boldsymbol{W}$ is not the subspace of $\boldsymbol{R^3}$.

**Example 12** Determine which of the following are subspaces of $\boldsymbol{P_3}$.
(i) All polynomials $a_0 + a_1\,\boldsymbol{x} + a_2\,\boldsymbol{x^2} + a_3\,\boldsymbol{x^3}$ for which $a_0 = 0$
(ii) All polynomials $a_0 + a_1\,\boldsymbol{x} + a_2\,\boldsymbol{x^2} + a_3\,\boldsymbol{x^3}$ for which $a_0 + a_1 + a_2 + a_3 = 0$
(iii) All polynomials $a_0 + a_1\,\boldsymbol{x} + a_2\,\boldsymbol{x^2} + a_3\,\boldsymbol{x^3}$ for which $a_0$, $a_1$, $a_2$, and $a_3$ are integers
(iv) All polynomials of the form $a_0 + a_1\,\boldsymbol{x}$, where $a_0$ and $a_1$ are real numbers.

**Solution** (i) Let $W$ is the set of all polynomials $a_0 + a_1 x + a_2 x^2 + a_3 x^3$ for which $a_0 = 0$.
Suppose that $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where $c_0 = 0$) and $v = b_0 + b_1 x + b_2 x^2 + b_3 x^3$ (where $b_0 = 0$) are in $W$. Then $u + v = (c_0 + c_1x + c_2x^2 + c_3x^3) + (b_0 + b_1x + b_2x^2 + b_3x^3) = (c_0 + b_0) + (c_1 + b_1)x + (c_2 + b_2)x^2 + (c_3 + b_3) x^3$, where $c_0 + b_0 = 0$.
Therefore, $u + v \in W$

If $k$ is any scalar and $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where $c_0 = 0$) is any vector in $W$.
Then $k\,u = k\,(c_0 + c_1x + c_2x^2 + c_3x^3) = (kc_0) + (kc_1)\,x + (kc_2)\,x^2 + (kc_3)\,x^3$ where $k\,c_0 = 0$. Therefore, $k\,u \in W$. Hence $W$ is the subspace of $P_3$.

(ii) Let $W$ is the set of all polynomials $a_0 + a_1x + a_2x^2 + a_3x^3$ for which $a_0+a_1+a_2+a_3 = 0$.
Suppose that $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where $c_0 + c_1 + c_2 + c_3 = 0$) and $v = b_0 + b_1 x + b_2 x^2 + b_3 x^3$ (where $b_0 + b_1 + b_2 + b_3 = 0$) are in $W$.

Now
$u + v = (c_0 + c_1 x + c_2 x^2 + c_3 x^3) + (b_0 + b_1 x + b_2 x^2 + b_3 x^3)$

$= (c_0 + b_0) + (c_1 + b_1)\,x + (c_2 + b_2)\,x^2 + (c_3 + b_3)\,x^3$

Where $(c_0+b_0) + (c_1+b_1) + (c_2+b_2) + (c_3+b_3) = (c_0+c_1+c_2+c_3) + (b_0+b_1+b_2+b_3) = 0 + 0 = 0$. Therefore, $u + v \in W$

If $k$ is any scalar and $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where $c_0 + c_1 + c_2 + c_3 = 0$) is any vector in $W$. Then $k\,u = k\,(c_0 + c_1x + c_2x^2 + c_3x^3) = (kc_0) + (kc_1)\,x + (kc_2)\,x^2 + (kc_3)\,x^3$
Where $(k\,c_0) + (k\,c_1) + (k\,c_2) + (k\,c_3) = k(c_0 + c_1 + c_2 + c_3) = k.0 = 0$

Therefore, $k\,u \in W$. Hence $W$ is the subspace of $P_3$.

(iii) Let $W$ is the set of all polynomials $a_0 + a_1 x + a_2 x^2 + a_3 x^3$ for which $a_0$, $a_1$, $a_2$, and $a_3$ are integers.
Suppose that the vectors $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where $c_0$, $c_1$, $c_2$, and $c_3$ are integers) and $v = b_0 + b_1 x + b_2 x^2 + b_3 x^3$ (where $b_0$, $b_1$, $b_2$, and $b_3$ are integers) are in $W$.
Now
$u + v = (c_0 + c_1 x + c_2 x^2 + c_3 x^3) + (b_0 + b_1 x + b_2 x^2 + b_3 x^3)$
$= (c_0 + b_0) + (c_1 + b_1)\,x + (c_2 + b_2)\,x^2 + (c_3 + b_3)\,x^3$, where
$(c_0 + b_0)$, $(c_1 + b_1)$, $(c_2 + b_2)$, and $(c_3 + b_3)$ are integers (integers are closed under addition). Therefore, $u + v \in W$

If $k$ is any scalar and $u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ (where $c_0$, $c_1$, $c_2$, and $c_3$ are integers) is any vector in $W$. Then $k\,u = k\,(c_0 + c_1x + c_2x^2 + c_3x^3 = (kc_0) + (kc_1)\,x + (kc_2)\,x^2 + (kc_3)\,x^3$, where $(k\,c_0)$, $(k\,c_1)$, $(k\,c_2)$, and $(k\,c_3)$ are not integers (product of real number and integer). Therefore, $k\,u \notin W$. Hence, $W$ is not the subspace of $P_3$.

(iv) Let $\boldsymbol{W}$ is the set of all polynomials of the form $a_0 + a_1 \boldsymbol{x}$, where $a_0$ and $a_1$ are real numbers. Suppose that $\boldsymbol{u} = c_0 + c_1 \boldsymbol{x}$ (where $c_0$ and $c_1$ are real numbers) and $\boldsymbol{v} = b_0 + b_1 \boldsymbol{x}$ (where $b_0$ and $b_1$ are real numbers) are in $\boldsymbol{W}$.

Then $\boldsymbol{u} + \boldsymbol{v} = (c_0 + c_1 \boldsymbol{x}) + (b_0 + b_1 \boldsymbol{x}) = (c_0 + b_0) + (c_1 + b_1) \boldsymbol{x}$
Where $(c_0 + b_0)$ and $(c_1 + b_1)$ are real numbers.
Therefore, $\boldsymbol{u} + \boldsymbol{v} \in \boldsymbol{W}$

If $k$ is any scalar and $\boldsymbol{u} = c_0 + c_1 \boldsymbol{x}$ (where $c_0$ and $c_1$ are real numbers) is any vector in $\boldsymbol{W}$.
Then $k\,\boldsymbol{u} = k\,(c_0 + c_1 \boldsymbol{x}) = (k\,c_0) + (k\,c_1) \boldsymbol{x}$
Where $(k\,c_0)$ and $(k\,c_1)$ are real numbers.
Therefore, $k\,\boldsymbol{u} \in \boldsymbol{W}$. Hence $\boldsymbol{W}$ is the subspace of $\boldsymbol{P_3}$.

**Example 13** Determine which of the following are subspaces of $\boldsymbol{M_{22}}$.

(i) All matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a + b + c + d = 0$

(ii) All 2 x 2 matrices $\boldsymbol{A}$ such that det $(\boldsymbol{A}) = 0$

(iii) All the matrices of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$

**Solution** Let $\boldsymbol{W}$ is the set of all matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a + b + c + d = 0$.

**(i)** Suppose $\boldsymbol{u} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$ (where $e + f + g + h = 0$) and $\boldsymbol{v} = \begin{bmatrix} l & m \\ n & p \end{bmatrix}$

(Where $l + m + n + p = 0$) are in $\boldsymbol{W}$.

Then $\boldsymbol{u} + \boldsymbol{v} = \begin{bmatrix} e & f \\ g & h \end{bmatrix} + \begin{bmatrix} l & m \\ n & p \end{bmatrix} = \begin{bmatrix} e+l & f+m \\ g+n & h+p \end{bmatrix}$ (By definition)

Where $(e + l) + (f + m) + (g + n) + (h + p)$
$= (e + f + g + h) + (l + m + n + p) = 0 + 0 = 0$
Therefore, $\boldsymbol{u} + \boldsymbol{v} \in \boldsymbol{W}$

If $k$ is any scalar and $\boldsymbol{u} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$ (where $e + f + g + h = 0$) is any vector in $\boldsymbol{W}$.

Then $k\,\boldsymbol{u} = k\begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ke & kf \\ kg & kh \end{bmatrix}$ (by definition)

Where $k\,e + k\,f + k\,g + k\,h = k\,(e + f + g + h) = k\,0 = 0$
Hence, $k\,\boldsymbol{u} \in \boldsymbol{W}$. Therefore, $\boldsymbol{W}$ is subspace of $\boldsymbol{M_{22}}$.

**(ii)** Let $\boldsymbol{W}$ is the set of all 2 x 2 matrices $\boldsymbol{A}$ such that det $(\boldsymbol{A}) = 0$

Suppose $\boldsymbol{u} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$ (Where det $(\boldsymbol{u}) = e\,h - f\,g = 0$) and $\boldsymbol{v} = \begin{bmatrix} l & m \\ n & p \end{bmatrix}$

(Where det $(v) = l\,p - m\,n = 0$) are in $W$.

Then $u + v = \begin{bmatrix} e & f \\ g & h \end{bmatrix} + \begin{bmatrix} l & m \\ n & p \end{bmatrix} = \begin{bmatrix} e+l & f+m \\ g+n & h+p \end{bmatrix}$ (By definition)

Where det $(u + v) = (e + l)(h + p) - (f + m)(g + n)$

$= e\,h + e\,p + l\,h + l\,p - f\,g - f\,n - m\,g - m\,n$

$= (e\,h - f\,g) + (l\,p - m\,n) + e\,p + l\,h - f\,n - m\,g = e\,p + l\,h - f\,n - m\,g \neq 0$

Therefore, $u + v \notin W$. Therefore, $W$ is not subspace of $M_{22}$.

**(iii)** Let $W$ is the set of all matrices of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$

Suppose $u = \begin{bmatrix} e & f \\ 0 & g \end{bmatrix}$ and $v = \begin{bmatrix} l & m \\ 0 & n \end{bmatrix}$ are in $W$.

Then $u + v = \begin{bmatrix} e & f \\ 0 & g \end{bmatrix} + \begin{bmatrix} l & m \\ 0 & n \end{bmatrix} = \begin{bmatrix} e+l & f+m \\ 0 & g+n \end{bmatrix}$ (By definition)

Which is of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$. Therefore, $u + v \in W$

If $k$ is any scalar and $u = \begin{bmatrix} e & f \\ 0 & g \end{bmatrix}$ is any vector in $W$.

Then $k\,u = k\begin{bmatrix} e & f \\ 0 & g \end{bmatrix} = \begin{bmatrix} ke & kf \\ 0 & kg \end{bmatrix}$ (By definition)

Which is of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$. Hence, $k\,u \in W$

Therefore, $W$ is subspace of $M_{22}$.

**Example 14** Determine which of the following are subspaces of the space $F(-\infty,\infty)$.

(i) All $f$ such that $f(x) \leq 0$ for all $x$ (ii) all $f$ such that $f(0) = 0$

(iii) All $f$ such that $f(0) = 2$ (iv) all constant functions

(v) All $f$ of the form $k_1 + k_2 \sin x$, where $k_1$ and $k_2$ are real numbers

(vi) All everywhere differentiable functions that satisfy $f' + 2f = 0$.

**Solution** **(i)** Let $W$ is the set of all $f$ such that $f(x) \leq 0$ for all $x$.

Suppose $g$ and $h$ are the vectors in $W$. Then $g(x) \leq 0$ for all $x$ and $h(x) \leq 0$ for all $x$.

Now $(g + h)(x) = g(x) + h(x) \leq 0$. Therefore, $g + h \in W$

If $k$ is any scalar and $g$ is any vector in $W$. Then $g(x) \leq 0$ for all $x$

Now $(k\,g)(x) = k\,g(x)$, which is greater than $0$ for negative real values of $k$.

$\therefore kg \notin W \quad \forall\ k < 0$.

Hence $W$ is not the subspace of $F(-\infty,\infty)$.

**(ii)** Let $W$ is the set of all $f$ such that $f(0) = 0$.
Suppose $g$ and $h$ are the vectors in $W$. Then $g(0) = 0$ and $h(0) = 0$
Now $(g + h)(0) = g(0) + h(0) = 0 + 0 = 0$. Therefore, $g + h \in W$
If $k$ is any scalar and $g$ is any vector in $W$. Then $g(0) = 0$
Now $(k\,g)(0) = k\,g(0) = k\,0 = 0$. $\therefore kg \in W$ . Hence $W$ is the subspace of $F(-\infty,\infty)$.

**(iii)** Let $W$ is the set of all $f$ such that $f(0) = 2$
Suppose $g$ and $h$ are the vectors in $W$. Then $g(0) = 2$ and $h(0) = 2$
Now $(g + h)(0) = g(0) + h(0) = 2 + 2 \neq 2 \therefore kg \notin W$ . Hence $W$ is not the subspace of $F(-\infty,\infty)$.

**(iv)** Let $W$ is the set of all constant functions. Suppose $g$ and $h$ are the vectors in $W$.
Then $g(x) = a$ and $h(x) = b$, where $a$ and $b$ are constants.
Now $(g + h)(x) = g(x) + h(x) = a + b$, which is constant. Therefore, $g + h \in W$
If $k$ is any scalar and $g$ is any vector in $W$. Then $g(x) = a$, where $a$ is any constant.
Now $(k\,g)(x) = k\,g(x) = k\,a$, which is a constant. $\therefore kg \in W$ . Hence $W$ is the subspace of $F(-\infty,\infty)$.

**(v)** Let $W$ is the set of all $f$ of the form $k_1 + k_2 \sin x$, where $k_1$ and $k_2$ are real numbers
Suppose $g$ and $h$ are the vectors in $W$. Then $g(x) = m_1 + m_2 \sin x$ and $h(x) = n_1 + n_2 \sin x$, where $m_1$, $m_2$, $n_1$and $n_2$ are real numbers.

Now $(g + h)(x) = g(x) + h(x) = [m_1 + m_2 \sin x] + [n_1 + n_2 \sin x] = (m_1 + n_1) + (m_2 + n_2) \sin x$
Which is of the form $k_1 + k_2 \sin x$. Therefore, $g + h \in W$

If $k$ is any scalar and $g$ is any vector in $W$. Then $g(x) = m_1 + m_2 \sin x$, where $m_1$ and $m_2$ are any real numbers.

Now $(k\,g)(x) = k\,g(x) = k\,[m_1 + m_2 \sin x] = (k\,m_1) + (k\,m_2) \sin x$
Which is of the form $k_1 + k_2 \sin x$. $\therefore kg \in W$ . Hence $W$ is the subspace of $F(-\infty,\infty)$.

**(vi)** Let $W$ is the set of all everywhere differentiable functions that satisfy $f' + 2f = 0$.
Suppose $g$ and $h$ are the vectors in $W$. Then $g' + 2g = 0$ and $h' + 2h = 0$
Now $(g + h)' + 2(g + h) = g' + h' + 2(g + h) = (g' + 2g) + (h' + 2h) = 0 + 0 = 0$
Therefore, $g + h \in W$
If $k$ is any scalar and $g$ is any vector in $W$. Then $g' + 2g = 0$
Now $(kg)' + 2(kg) = kg' + 2kg = k(g' + 2g) = k.0 = 0$
$\therefore kg \in W$ . Hence $W$ is the subspace of $F(-\infty,\infty)$.

**Remark** Let $n$ be a non-negative integer, and let $\boldsymbol{P_n}$ be the set of real valued function of the form $\boldsymbol{p(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n}$ where $a_0, a_1, a_2, \ldots, a_n$ *are* real numbers, then $\boldsymbol{P_n}$ is a subspace $\boldsymbol{F(-\infty, \infty)}$.

**Example 15** Show that the invertible n x n matrices do not form a subspace of $\boldsymbol{M_{n\,x\,n}}$.

**Solution** Let $\boldsymbol{W}$ is the set of invertible matrices in $\boldsymbol{M_{n\,x\,n}}$. This set fails to be a subspace on both counts- it is closed under neither scalar multiplication nor addition.

For example consider invertible matrices $\boldsymbol{W} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, \boldsymbol{V} = \begin{bmatrix} -1 & 2 \\ -2 & 5 \end{bmatrix}$ in $\boldsymbol{M_{n\,x\,n}}$.

The matrix $\boldsymbol{0.U}$ is a 2x2 zero matrix, hence is not invertible; and the matrix $\boldsymbol{U + V}$ has a column of zeros, hence is not invertible.

**Theorem** If $\boldsymbol{Ax = 0}$ is a homogeneous linear system of $m$ equations in $n$ unknowns, then the set of solution vectors is a subspace of $\boldsymbol{R^n}$.

**Example 16** Consider the linear systems

*(a)* $\begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 3 & -6 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ *(b)* $\begin{bmatrix} 1 & -2 & 3 \\ -3 & 7 & -8 \\ -2 & 4 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

*(c)* $\begin{bmatrix} 1 & -2 & 3 \\ -3 & 7 & -8 \\ 4 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ *(d)* $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Each of the systems has three unknowns, so the solutions form subspaces of $\boldsymbol{R^3}$. Geometrically, this means that each solution space must be a line through origin, a plane through origin, the origin only, or all of $\boldsymbol{R^3}$.

**Solution** *(a)* The solutions are $x = 2s - 3t, y = s, z = t$. From which it follows that $x = 2y - 3z$ *or* $x - 2y + 3z = 0$.
This is the equation of the plane through the origin with $\boldsymbol{n} = (1, -2, 3)$ as a normal vector.

**(b)** The solutions are $x = -5t, y = -t, z = t$, which are parametric equations for the line through the origin parallel to the vector $\boldsymbol{v} = (-5, -1, 1)$.

**(c)** The solution is $x = 0, y = 0, z = 0$ so the solution space is the origin only, that is $\{\boldsymbol{0}\}$.

**(d)** The solutions are $x = r, y = s, z = t$. where $r$, $s$ and $t$ have arbitrary values, so the solution space is all $\boldsymbol{R^3}$.

**A Subspace Spanned by a Set**: The next example illustrates one of the most common ways of describing a subspace. We know that the term linear combination refers to any

sum of scalar multiples of vectors, and Span $\{v_1, \ldots, v_p\}$ denotes the set of all vectors that can be written as linear combinations of $v_1, \ldots, v_p$.

**Example 17** Given $v_1$ and $v_2$ in a vector space $V$, let $H$ = Span $\{v_1, v_2\}$. Show that $H$ is a subspace of $V$.

**Solution** The zero vector is in $H$, since $0 = 0v_1 + 0v_2$. To show that $H$ is closed under vector addition, take two arbitrary vectors in $H$, say,

$$u = s_1v_1 + s_2v_2 \quad \text{and} \quad w = t_1v_1 + t_2v_2$$

By Axioms 2, 3 and 8 for the vector space $V$.

$$u + w = (s_1v_1 + s_2v_2) + (t_1v_1 + t_2v_2)$$
$$= (s_1 + t_1)v_1 + (s_2 + t_2)v_2$$

So $u + w$ is in $H$. Furthermore, if $c$ is any scalar, then by Axioms 7 and 9,

$$cu = c(s_1v_1 + s_2v_2) = (cs_1)v_1 + (cs_2)v_2$$

Which shows that $cu$ is in $H$ and $H$ is closed under scalar multiplication.

Thus $H$ is a subspace of $V$. □

Later on we will prove that every nonzero subspace of $R^3$, other than $R^3$ itself, is either Span $\{v_1, v_2\}$ for some linearly independent $v_1$ and $v_2$ or Span $\{v\}$ for $v \neq 0$. In the first case the subspace is a plane through the origin and in the second case a line through the origin. (See Figure below) It is helpful to keep these geometric pictures in mind, even for an abstract vector space.

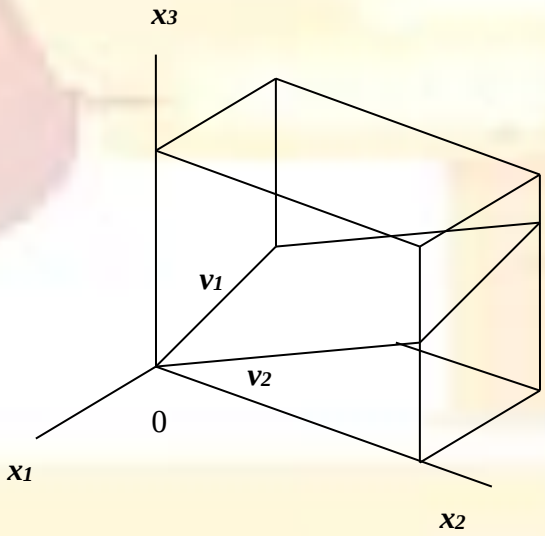



**Figure 9 – An example of a subspace**

The argument in Example 17 can easily be generalized to prove the following theorem.

**Theorem 3** If $v_1, \ldots, v_p$ are in a vector space $V$, then Span $\{v_1, \ldots, v_p\}$ is a subspace of $V$.

We call Span $\{v_1, \ldots, v_p\}$ the subspace spanned (or generated) by $\{v_1, \ldots, v_p\}$. Given any subspace $H$ of $V$, a spanning (or generating) set for $H$ is a set $\{v_1, \ldots, v_p\}$ in $H$ such that $H$ = Span $\{v_1, \ldots, v_p\}$.

**Proof**

The zero vector is in $H$, since $0 = 0v_1 + 0v_2 + \ldots + 0v_n = \sum_{j=0}^{j=n} 0V_j = 0\left(\sum_{j=0}^{j=n} V_j\right) = 0$

To show that $H$ is closed under vector addition, take two arbitrary vectors in $H$, say,

$$u = s_1v_1 + s_2v_2 + \ldots + t_nv_n = \sum_{i=0}^{i=n} s_iV_i$$

and

$$w = t_1v_1 + t_2v_2 \ldots + t_nv_n = \sum_{k=0}^{k=n} t_kV_k$$

By Axioms 2, 3 and 8 for the vector space $V$.

$$u + w = \sum_{i=0}^{i=n} s_iV_i + \sum_{k=0}^{k=n} t_kV_k = (s_1v_1 + s_2v_2 + \ldots + s_nv_n) + (t_1v_1 + t_2v_2 \ldots + t_nv_n)$$

$$= (s_1 + t_1)v_1 + (s_2 + t_2)v_2 + \ldots + (s_n + t_n)v_n = \sum_{p=0}^{p=n} (s_p + t_p)V_p$$

So $u + w$ is in $H$. Furthermore, if $c$ is any scalar, then by Axioms 7 and 9,

$$cu = c(s_1v_1 + s_2v_2 + \ldots + s_nv_n) = (cs_1)v_1 + (cs_2)v_2 + \ldots + (cs_n)v_n = \sum_{r=0}^{r=n} cs_rV_r .$$

Which shows that $cu$ is in $H$ and $H$ is closed under scalar multiplication.

Thus $H$ is a subspace of $V$. □

**Example 18** Let $H$ be the set of all vectors of the form $(a - 3b, b - a, a, b)$, where $a$ and $b$ are arbitrary scalars. That is, let $H = \{(a - 3b, b - a, a, b): a \text{ and } b \text{ in } R\}$. Show that $H$ is a subspace of $R^4$.

**Solution** Write the vectors in $H$ as column vectors. Then an arbitrary vector in $H$ has the form

$$\begin{bmatrix} a-3b \\ b-a \\ a \\ b \end{bmatrix} = a\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$v_1$ $v_2$

This calculation shows that $\boldsymbol{H}$ = Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$, where $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ are the vectors indicated above. Thus $\boldsymbol{H}$ is a subspace of $\boldsymbol{R^4}$ by Theorem 3. □

Example18 illustrates a useful technique of expressing a subspace $\boldsymbol{H}$ as the set of linear combinations of some small collection of vectors. If $\boldsymbol{H}$ = Span $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$, we can think of the vectors $\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}$ in the spanning set as "handles" that allow us to hold on to the subspace $\boldsymbol{H}$. *Calculations with the infinitely many vectors in* $\boldsymbol{H}$ *are often reduced to operations with the finite number of vectors in the spanning set.*



**Exercises**

In exercises 1-13 a set of objects is given together with operations of addition and scalar multiplication. Determine which sets are vector spaces under the given operations. For those that are not, list all axioms that fail to hold.

1. The set of all triples of real numbers *(x, y, z)* with the operations
$$(x,y,z)+(x',y',z') = (x+x',y+y',z+z') \text{ and } k(x,y,z) = (kx,y,z)$$

2. The set of all triples of real numbers *(x, y, z)* with the operations
$$(x,y,z)+(x',y',z') = (x+x',y+y',z+z') \text{ and } k(x,y,z) = (0,0,0)$$

3. The set of all pairs of real numbers *(x, y)* with the operations
$$(x,y)+(x',y') = (x+x',y+y') \text{ and } k(x,y) = (2kx,2ky)$$

4. The set of all pairs of real numbers of the form *(x, 0)* with the standard operations on $\boldsymbol{R^2}$.

5. The set of all pairs of real numbers of the form *(x, y)*, where $x \geq 0$, with the standard operations on $\boldsymbol{R^2}$.

6. The set of all n-tuples of real numbers of the form *(x, x, …, x)* with the standard operations on $\boldsymbol{R^n}$.

7. The set of all pairs of real numbers *(x, y)* with the operations.
$$(x,y)+(x',y') = (x+x'+1,y+y'+1) \text{ and } k(x,y) = (kx,ky)$$

8. The set of all 2x2 matrices of the form $\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix}$ with matrix addition and scalar multiplication.

9. The set of all 2x2 matrices of the form $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ with matrix addition and scalar multiplication.

10. The set of all pairs of real numbers of the form *(1, x)* with the operations
$$(1,y)+(1,y') = (1,y+y') \text{ and } k(1,y) = (1,ky)$$

11. The set of polynomials of the form $\boldsymbol{a} + b\boldsymbol{x}$ with the operations
$$(\boldsymbol{a_0}+a_1\boldsymbol{x})+(\boldsymbol{b_0}+b_1\boldsymbol{x}) = (\boldsymbol{a_0}+\boldsymbol{b_0})+(a_1+b_1)\boldsymbol{x} \text{ and } k(\boldsymbol{a_0}+a_1\boldsymbol{x}) = (k\boldsymbol{a_0})+(ka_1)\boldsymbol{x}$$

12. The set of all positive real numbers with operations $\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{xy}$ and $k\boldsymbol{x} = \boldsymbol{x}^k$

13. The set of all real numbers *(x, y)* with operations

$$(x, y)+(x', y')=(xx', yy') \text{ and } k(x, y)=(kx, ky)$$

14. Determine which of the following are subspaces of $\boldsymbol{M_{nn}}$.
(a) all $n$ x $n$ matrices $\boldsymbol{A}$ such that tr $(\boldsymbol{A}) = 0$
(b) all $n$ x $n$ matrices $\boldsymbol{A}$ such that $\boldsymbol{A}^T = -\boldsymbol{A}$
(c) all $n$ x $n$ matrices $\boldsymbol{A}$ such that the linear system $\boldsymbol{Ax} = \boldsymbol{0}$ has only the trivial solution
(d) all $n$ x $n$ matrices $\boldsymbol{A}$ such that $\boldsymbol{AB} = \boldsymbol{BA}$ for a fixed $n$ x $n$ matrix $\boldsymbol{B}$

15. Determine whether the solution space of the system $\boldsymbol{Ax} = \boldsymbol{0}$ is a line through the origin, a plane through the origin, or the origin only. If it is a plane, find an equation for it; if it is a line, find parametric equations for it.

(a) $\boldsymbol{A}=\begin{bmatrix} -1 & 1 & 1 \\ 3 & -1 & 0 \\ 2 & -4 & -5 \end{bmatrix}$ (b) $\boldsymbol{A}=\begin{bmatrix} 1 & -2 & 3 \\ -3 & 6 & 9 \\ -2 & 4 & -6 \end{bmatrix}$

(c) $\boldsymbol{A}=\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ (d) $\boldsymbol{A}=\begin{bmatrix} 1 & 2 & -6 \\ 1 & 4 & 4 \\ 3 & 10 & 6 \end{bmatrix}$

16. Determine if the set "all polynomial in $\boldsymbol{p_n}$ such that $\boldsymbol{p(0) = 0}$" is a subspace of $\boldsymbol{P_n}$ for an appropriate value of $n$. Justify your answer.

17. Let $\boldsymbol{H}$ be the set of all vectors of the form $\begin{bmatrix} s \\ 3s \\ 2s \end{bmatrix}$. Find a vector $\boldsymbol{v}$ in $\boldsymbol{R^3}$ such that $\boldsymbol{H}$ = Span $\{\boldsymbol{v}\}$. Why does this show that $\boldsymbol{H}$ is a subspace of $\boldsymbol{R^3}$?

18. Let $\boldsymbol{W}$ be the set of all vectors of the form $\begin{bmatrix} 5b+2c \\ b \\ c \end{bmatrix}$, where $b$ and $c$ are arbitrary.

Find vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ such that $\boldsymbol{W}$ = Span $\{\boldsymbol{u}, \boldsymbol{v}\}$. Why does this show that $\boldsymbol{W}$ is a subspace of $\boldsymbol{R^3}$?

19. Let $\boldsymbol{W}$ be the set of all vectors of the form $\begin{bmatrix} s+3t \\ s-t \\ 2s-t \\ 4t \end{bmatrix}$. Show that $\boldsymbol{W}$ is a subspace of $\boldsymbol{R^4}$.

20. Let $\boldsymbol{v_1} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \boldsymbol{v_2} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \boldsymbol{v_3} = \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$, and $\boldsymbol{w} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$.

(a) Is $\boldsymbol{w}$ in $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$? How many vectors are in $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$?
(b) How many vectors are in Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$?
(c) Is $\boldsymbol{w}$ in the subspace spanned by $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$? Why?

In exercises 21 and 22, let $\boldsymbol{W}$ be the set of all vectors of the form shown, where $a$, $b$ and $c$ represent arbitrary real numbers. In each case, either find a set $\boldsymbol{S}$ of vectors that spans $\boldsymbol{W}$ or give an example to show that $\boldsymbol{W}$ is not a vector space.

21. $\begin{bmatrix} 3a+b \\ 4 \\ a-5b \end{bmatrix}$

22. $\begin{bmatrix} a-b \\ b-c \\ c-a \\ b \end{bmatrix}$

23. Show that $\boldsymbol{w}$ is in the subspace of $\boldsymbol{R^4}$ spanned by $\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}$, where

$$\boldsymbol{w} = \begin{bmatrix} -9 \\ 7 \\ 4 \\ 8 \end{bmatrix}, \boldsymbol{v_1} = \begin{bmatrix} 7 \\ -4 \\ -2 \\ 9 \end{bmatrix}, \boldsymbol{v_2} = \begin{bmatrix} -4 \\ 5 \\ -1 \\ -7 \end{bmatrix}, \boldsymbol{v_3} = \begin{bmatrix} -9 \\ 4 \\ 4 \\ -7 \end{bmatrix}$$

24. Determine if $\boldsymbol{y}$ is in the subspace of $\boldsymbol{R^4}$ spanned by the columns of $\boldsymbol{A}$, where

$$\boldsymbol{y} = \begin{bmatrix} 6 \\ 7 \\ 1 \\ -4 \end{bmatrix}, \boldsymbol{A} = \begin{bmatrix} 5 & -5 & -9 \\ 8 & 8 & -6 \\ -5 & -9 & 3 \\ 3 & -2 & -7 \end{bmatrix}$$

# Lecture 21

## Null Spaces, Column Spaces, and Linear Transformations

Subspaces arise in as set of all solutions to a system of homogenous linear equations as the set of all linear combinations of certain specified vectors. In this lecture, we compare and contrast these two descriptions of subspaces, allowing us to practice using the concept of a subspace. In applications of linear algebra, subspaces of $\boldsymbol{R^n}$ usually arise in one of two ways:

- as the set of all solutions to a system of homogeneous linear equations or
- as the set of all linear combinations of certain specified vectors.

Our work here will provide us with a deeper understanding of the relationships between the solutions of a linear system of equations and properties of its coefficient matrix.

### Null Space of a Matrix

Consider the following system of homogeneous equations:

$$\begin{aligned} x_1 - 3x_2 - 2x_3 &= 0 \\ -5x_1 + 9x_2 + x_3 &= 0 \end{aligned} \qquad (1)$$

In matrix form, this system is written as $\boldsymbol{Ax = 0}$, where

$$\boldsymbol{A} = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix} \qquad (2)$$

Recall that the set of all $\boldsymbol{x}$ that satisfy (1) is called the solution set of the system (1). Often it is convenient to relate this set directly to the matrix $\boldsymbol{A}$ and the equation $\boldsymbol{Ax = 0}$. We call the set of $\boldsymbol{x}$ that satisfy $\boldsymbol{Ax = 0}$ the **null space** of the matrix $\boldsymbol{A}$. The reason for this name is that if matrix A is viewed as a linear operator that maps points of some vector space V into itself, it can be viewed as mapping all the elements of this solution space of AX = 0 into the null element "0". Thus the null space N of A is that subspace of all vectors in V which are imaged into the null element “0" by the matrix A.

### NULL SPACE

**Definition** The **null space** of an $m \times n$ matrix $\boldsymbol{A}$, written as Nul $\boldsymbol{A}$, is the set of all solutions to the homogeneous equation $\boldsymbol{Ax = 0}$. In set notation,

Nul $\boldsymbol{A}$ = $\{\boldsymbol{x}$: $\boldsymbol{x}$ is in $\boldsymbol{R^n}$ and $\boldsymbol{Ax = 0}\}$

*OR*

$$Nul(A) = \{x / \forall x \in \square, Ax = 0\}$$

A more dynamic description of Nul $\boldsymbol{A}$ is the set of all $\boldsymbol{x}$ in $\boldsymbol{R^n}$ that are mapped into the zero vector of $\boldsymbol{R^m}$ via the linear transformation $\boldsymbol{x} \rightarrow \boldsymbol{Ax}$ , where A is a matrix of transformation. See Figure1

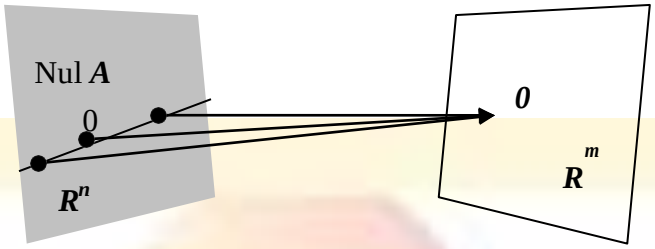



**Figure 1**

**Example 1** Let $A = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix}$ and let $\boldsymbol{u} = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$. Determine if $\boldsymbol{u} \in \text{Nul } \boldsymbol{A}$.

**Solution** To test if $\boldsymbol{u}$ satisfies $\boldsymbol{Au} = \boldsymbol{0}$, simply compute

$$\boldsymbol{Au} = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 5-9+4 \\ -25+27-2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$ Thus $\boldsymbol{u}$ is in Nul $\boldsymbol{A}$.

**Example** Determine the null space of the following matrix:

$$A = \begin{pmatrix} 4 & 0 \\ -8 & 20 \end{pmatrix}$$

**Solution** To find the null space of $A$ we need to solve the following system of equations:

$$\begin{pmatrix} 4 & 0 \\ -8 & 20 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x_1 + 0x_2 \\ -8x_1 + 20x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow 4x_1 + 0x_2 = 0 \qquad \Rightarrow x_1 = 0$$

*and* $$\Rightarrow -8x_1 + 20x_2 = 0 \qquad \Rightarrow x_2 = 0$$

We can find Null space of a matrix with two ways i.e. with matrices or with system of linear equations. We have given this in both matrix form and (here first we convert the matrix into system of equations) equation form. In equation form it is easy to see that by solving these equations together the only solution is $x_1 = x_2 = 0$. In terms of vectors from

$\mathbb{R}^2$ the solution consists of the single vector $\{0\}$ and hence the null space of $A$ is $\{0\}$.

**Activity** Determine the null space of the following matrices:

**1.** $0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

**2.** $M = \begin{pmatrix} 1 & -5 \\ -5 & 25 \end{pmatrix}$

In earlier (previous) lectures, we developed the technique of elementary row operations to solve a linear system. We know that performing elementary row operations on an augmented matrix does not change the solution set of the corresponding linear system ***Ax=0***. Therefore, we can say that it does not change the null space of ***A***. We state this result as a theorem:

**Theorem 1** Elementary row operations do not change the null space of a matrix.

**Or**

Null space N(A) of a matrix A can not be changed (always same) by changing the matrix with elementary row operations.

**Example** Determine the null space of the following matrix using the elementary row operations: (Taking the matrix from the above Example)

$$A = \begin{pmatrix} 4 & 0 \\ -8 & 20 \end{pmatrix}$$

**Solution** First we transform the matrix to the reduced row echelon form:

$$\begin{pmatrix} 4 & 0 \\ -8 & 20 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ -8 & 20 \end{pmatrix} \quad \frac{1}{4}R_1$$

$$\sim \begin{pmatrix} 1 & 0 \\ 0 & 20 \end{pmatrix} \quad R_2 + 8R_1$$

$$\sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \frac{1}{20}R_2$$

which corresponds to the system

$$x_1 = 0$$
$$x_2 = 0$$

Since every column in the coefficient part of the matrix has a leading entry that means our system has the **trivial solution only**:

$$x_1 = 0$$
$$x_2 = 0$$

This means **the null space consists only of the zero vector**.

We can observe and compare both the above examples which show the same result.

**Theorem 2** The null space of an $m \times n$ matrix $A$ is a subspace of $R^n$. Equivalently, the set of all solutions to a system $Ax = 0$ of $m$ homogeneous linear equations in $n$ unknowns is a subspace of $R^n$.

Or simply, the null space is the space of all the vectors of a Matrix A of any order those are mapped (assign) onto zero vector in the space $R^n$ *(i.e. Ax = 0)*.

**Proof** We know that the subspace of $A$ consists of all the solution to the system $\boldsymbol{Ax = 0}$.

First, we should point out that the zero vector, $\mathbf{0}$, in $\boldsymbol{R^n}$ will be a solution to this system and so we know that the null space is not empty. This is a good thing since a vector space (subspace or not) must contain at least one element.

Now we know that the null space is not empty. Consider $\boldsymbol{u}$, $\boldsymbol{v}$ be two any vectors (elements) (in) from the null space and let $c$ be any scalar. We just need to show that the sum ($\boldsymbol{u+v}$) and scalar multiple ($c.\boldsymbol{u}$) of these are also in the null space.

Certainly Nul $\boldsymbol{A}$ is a subset of $\boldsymbol{R^n}$ because $\boldsymbol{A}$ has $n$ columns. To show that Nul(A) is the subspace, we have to check three conditions whether they are satisfied or not. If Nul(A) satisfies the all three condition, we say Nul(A) is a subspace otherwise not.

First, zero vector "0" must be in the space and subspace. If zero vector does not in the space we can not say that is a vector space (generally, we use space for vector space).

And we know that zero vector maps on zero vector so 0 is in Nul(A). Now choose any vectors $u$, $v$ from Null space and using definition of Null space (i.e. Ax=0)

$$\boldsymbol{Au = 0} \text{ and } \boldsymbol{Av = 0}$$

Now the other two conditions are vector addition and scalar multiplication. For this we proceed as follow:

Let start with vector addition:

To show that $\boldsymbol{u + v}$ is in Nul $\boldsymbol{A}$, we must show that $\boldsymbol{A\ (u + v) = 0}$. Using the property of matrix multiplication, we find that

$$\boldsymbol{A\ (u + v) = Au + Av = 0 + 0 = 0}$$

Thus $\boldsymbol{u + v}$ is in Nul $\boldsymbol{A}$, and Nul $\boldsymbol{A}$ is closed under vector addition.

For Matrix multiplication, consider any scalar , say $c$,

$$\mathbf{A}\ (c\boldsymbol{u}) = c\ (\boldsymbol{Au}) = c\ (\mathbf{0}) = \mathbf{0}$$

which shows that $c\boldsymbol{u}$ is in Nul $\boldsymbol{A}$. Thus Nul $\boldsymbol{A}$ is a subspace of $\boldsymbol{R^n}$.

**Example 2** The set $\boldsymbol{H}$, of all vectors in $\boldsymbol{R^4}$ whose coordinates $a$, $b$, $c$, $d$ satisfy the equations

$$a - 2b + 5c = d$$
$$c - a = b$$

is a subspace of $\boldsymbol{R^4}$.

**Solution** Since $a - 2b + 5c = d$

$c - a = b$

By rearranging the equations, we get

$$a - 2b + 5c - d = 0$$
$$-a - b + c = 0$$

We see that $\boldsymbol{H}$ is the set of all solutions of the above system of homogeneous linear equations.
Therefore from the Theorem 2, $\boldsymbol{H}$ is a subspace of $\boldsymbol{R^4}$.

It is important that the linear equations defining the set $\boldsymbol{H}$ are homogeneous. Otherwise, the set of solutions will definitely not be a subspace (because the zero-vector (origin) is not a solution of a non- homogeneous system), geometrically means that a line that not passes through origin can not be a subspace, because subspace must hold the zero vector (origin). Also, in some cases, the set of solutions could be empty. In this case, we can not find any solution of a system of linear equations, geometrically says that lines are parallel or not intersecting.
If the null space having more than one vector, geometrically means that the lines intersect more than one point and must passes through origin (zero vector) .

**An Explicit Description of Nul *A***

There is no obvious relation between vectors in Nul $\boldsymbol{A}$ and the entries in $\boldsymbol{A}$. We say that Nul $\boldsymbol{A}$ is defined implicitly, because it is defined by a condition that must be checked. No explicit list or description of the elements in Nul $\boldsymbol{A}$ is given. However, when we solve the equation $\boldsymbol{Ax} = \boldsymbol{0}$, we obtain an explicit description of Nul $\boldsymbol{A}$.

**Example 3** Find a spanning set for the null space of the matrix

$$\boldsymbol{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

**Solution** The first step is to find the general solution of $\boldsymbol{Ax} = \boldsymbol{0}$ in terms of free variables.
After transforming the augmented matrix $[\boldsymbol{A} \quad \boldsymbol{0}]$ to the reduced row echelon form and we get;

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

which corresponds to the system

$$x_1 - 2x_2 - x_4 + 3x_5 = 0$$
$$x_3 + 2x_4 - 2x_5 = 0$$
$$0 = 0$$

The general solution is

$x_1 = 2x_2 + x_4 - 3x_5$

$x_2 =$ *free variable*

$x_3 = -2x_4 + 2x_5$

$x_4 =$ *free variable*

$x_5 =$ *free variable*

Next, decompose the vector giving the general solution into a linear combination of vectors where the weights are the free variables. That is,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\uparrow \qquad \uparrow \qquad \uparrow$$

$$\boldsymbol{u} \qquad \boldsymbol{v} \qquad \boldsymbol{w}$$

$$= x_2\boldsymbol{u} + x_4\boldsymbol{v} + x_5\boldsymbol{w} \qquad (3)$$

Every linear combination of ***u, v*** and ***w*** is an element of Nul ***A***. Thus ***{u, v, w}*** is a spanning set for Nul ***A***.

Two points should be made about the solution in Example 3 that apply to all problems of this type. We will use these facts later.

1. The spanning set produced by the method in Example 3 is automatically linearly independent because the free variables are the weights on the spanning vectors. For instance, look at the 2nd, 4th and 5th entries in the solution vector in (3) and note that $x_2\boldsymbol{u} + x_4\boldsymbol{v} + x_5\boldsymbol{w}$ can be *0* only if the weights $x_2$, $x_4$ and $x_5$ are all zero.
2. When Nul *A* contains nonzero vector, the number of vectors in the spanning set for Nul *A* equals the number of free variables in the equation $Ax = 0$.

**Example 4** Find a spanning set for the null space of $\boldsymbol{A} = \begin{bmatrix} 1 & -3 & 2 & 2 & 1 \\ 0 & 3 & 6 & 0 & -3 \\ 2 & -3 & -2 & 4 & 4 \\ 3 & -6 & 0 & 6 & 5 \\ -2 & 9 & 2 & -4 & -5 \end{bmatrix}$.

**Solution** The null space of ***A*** is the solution space of the homogeneous system

$$x_1 - 3x_2 + 2x_3 + 2x_4 + x_5 = 0$$
$$0x_1 + 3x_2 + 6x_3 + 0x_4 - 3x_5 = 0$$
$$2x_1 - 3x_2 - 2x_3 + 4x_4 + 4x_5 = 0$$
$$3x_1 - 6x_2 + 0x_3 + 6x_4 + 5x_5 = 0$$
$$-2x_1 + 9x_2 + 2x_3 - 4x_4 - 5x_5 = 0$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 3 & 6 & 0 & -3 & 0 \\ 2 & -3 & -2 & 4 & 4 & 0 \\ 3 & -6 & 0 & 6 & 5 & 0 \\ -2 & 9 & 2 & -4 & -5 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 3 & 6 & 0 & -3 & 0 \\ 0 & 3 & -6 & 0 & 2 & 0 \\ 0 & 3 & -6 & 0 & 2 & 0 \\ 0 & 3 & 6 & 0 & -3 & 0 \end{bmatrix} \begin{matrix} \\ \\ -3R_1 + R_4 \\ 2R_1 + R_5 \end{matrix} \quad -2R_1 + R_3$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 3 & -6 & 0 & 2 & 0 \\ 0 & 3 & -6 & 0 & 2 & 0 \\ 0 & 3 & 6 & 0 & -3 & 0 \end{bmatrix} \quad (1/3)R_2$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & -12 & 0 & 5 & 0 \\ 0 & 0 & -12 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ -3R_2 + R_4 \\ -3R_2 + R_5 \end{matrix} \quad -3R_2 + R_3$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -5/12 & 0 \\ 0 & 0 & -12 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (-1/12)R_3$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -5/12 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad 12R_3 + R_4$$

$$\begin{bmatrix} 1 & -3 & 0 & 2 & 11/6 & 0 \\ 0 & 1 & 0 & 0 & -1/6 & 0 \\ 0 & 0 & 1 & 0 & -5/12 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ -2R_3 + R_2 \\ \\ -2R_3 + R_1 \\ \end{matrix}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 2 & 4/3 & 0 \\ 0 & 1 & 0 & 0 & -1/6 & 0 \\ 0 & 0 & 1 & 0 & -5/12 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \quad 3R_2 + R_1$$

The reduced row echelon form of the augmented matrix corresponds to the system

$$\begin{aligned} 1x_1 + \quad & 2x_4 + (4/3)x_5 = 0 \\ 1x_2 + \quad & (-1/6)x_5 = 0 \\ 1x_3 + \quad & (-5/12)x_5 = 0 . \\ & 0 = 0 \\ & 0 = 0 \end{aligned}$$

No equation of this system has a form zero = nonzero; Therefore, the system is **consistent**. The system has **infinitely many solutions**:

$x_1 = -2x_4 + (-4/3)x_5 \quad x_2 = +(1/6)x_5 \quad x_3 = +(5/12)x_5$

$x_4$ = arbitrary $\quad x_5$ = arbitrary

The solution can be written in the vector form:

$$\boldsymbol{c}_4 = (-2,0,0,1,0) \qquad \boldsymbol{c}_5 = (-4/3,1/6,5/12,0,1)$$

Therefore $\{(-2,0,0,1,0), (-4/3,1/6,5/12,0,1)\}$ is a spanning set for Null space of $\boldsymbol{A}$.

**Activity:** Find an explicit description of Nul $A$ where:

**1.** $A = \begin{pmatrix} 3 & 5 & 5 & 3 & 9 \\ 5 & 1 & 1 & 0 & 3 \end{pmatrix}$

**2.** $A = \begin{pmatrix} 4 & 1 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$

**The Column Space of a Matrix** Another important subspace associated with a matrix is its column space. Unlike the null space, the column space is defined explicitly via linear combinations.

**Definition (Column Space)** The column space of an $m \times n$ matrix $\boldsymbol{A}$, written as Col $\boldsymbol{A}$, is the set of all linear combinations of the columns of $\boldsymbol{A}$. If $\boldsymbol{A} = [\boldsymbol{a_1} \ \ldots \ \boldsymbol{a_n}]$, then

Col $\boldsymbol{A}$ = *Span* $\{\boldsymbol{a_1}, \ldots, \boldsymbol{a_n}\}$

Since Span $\{\boldsymbol{a_1}, \ldots, \boldsymbol{a_n}\}$ is a subspace, by Theorem of lecture 20 i.e. if $v_1,...,v_p$ are in a vector space $V$, then Span $\{v_1,...,v_p\}$ is a subspace of $V$.

The column space of a matrix is that subspace spanned by the columns of the matrix (columns viewed as vectors). It is that space defined by all linear combinations of the column of the matrix.

**Example**, in the given matrix,

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 1 & 4 \\ 3 & 1 & 5 \\ 4 & 1 & 6 \end{pmatrix}$$

The column space Col$\boldsymbol{A}$ is all the linear combination of the first (1, 2, 3, 4), the second (1, 1, 1, 1) and the third column ( 3, 4, 5, 6). That is, Col$\boldsymbol{A}$ = { a·(1, 2, 3, 4) + b·(1, 1, 1, 1) + c·( 3, 4, 5, 6) }. In general, **the column space** Col$\boldsymbol{A}$ **contains all the linear combinations of columns of A**.

The next theorem follows from the definition of Col $\boldsymbol{A}$ and the fact that the columns of $\boldsymbol{A}$ are in $\boldsymbol{R^m}$.

**Theorem 3** The column space of an $m \times n$ matrix $A$ is a subspace of $R^m$.

Note that a typical vector in Col $\boldsymbol{A}$ can be written as $\boldsymbol{Ax}$ for some $\boldsymbol{x}$ because the notation $\boldsymbol{Ax}$ stands for a linear combination of the columns of $\boldsymbol{A}$. That is,

Col $\boldsymbol{A}$ = $\{\boldsymbol{b}: \boldsymbol{b} = \boldsymbol{Ax}$ for some $\boldsymbol{x}$ in $\boldsymbol{R^n}\}$

The notation $\boldsymbol{Ax}$ for vectors in Col $\boldsymbol{A}$ also shows that Col $\boldsymbol{A}$ is the range of the linear transformation $\boldsymbol{x} \rightarrow \boldsymbol{Ax}$.

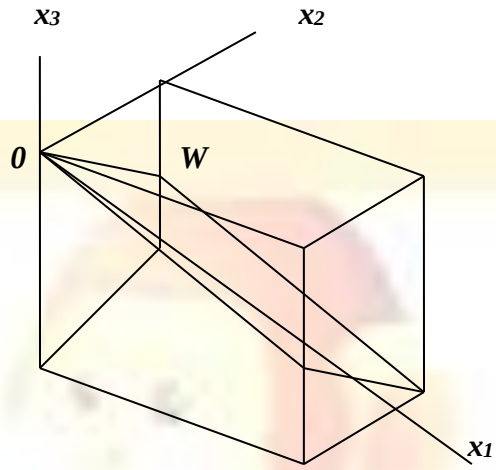
**Example 6** Find a matrix $\boldsymbol{A}$ such that $\boldsymbol{W}$ = Col $\boldsymbol{A}$. $\boldsymbol{W} = \left\{ \begin{bmatrix} 6a - b \\ a + b \\ -7a \end{bmatrix} : a,b \text{ in } R \right\}$

**Solution** First, write $\boldsymbol{W}$ as a set of linear combinations.

$$\boldsymbol{W} = \left\{ a \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} : a,b \text{ in } R \right\} = \text{Span} \left\{ \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Second, use the vectors in the spanning set as the columns of $\boldsymbol{A}$. Let $\boldsymbol{A} = \begin{bmatrix} 6 & -1 \\ 1 & 1 \\ -7 & 0 \end{bmatrix}$.

Then $\boldsymbol{W}$ = Col $\boldsymbol{A}$, as desired.




We know that the columns of $A$ span $R^m$ if and only if the equation $Ax = b$ has a solution for each $b$. We can restate this fact as follows:

The column space of an $m \times n$ matrix $A$ is all of $R^m$ if and only if the equation $Ax = b$ has a solution for each $b$ in $R^m$.

**Theorem 4** A system of linear equations $Ax = b$ is consistent if and only if $b$ in the column space of $A$.

**Example 6** A vector $b$ in the column space of $A$. Let $Ax = b$ is the linear system

$$\begin{bmatrix} -1 & 3 & 2 \\ 1 & 2 & -3 \\ 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

. Show that $b$ is in the column space of $A$, and express $b$ as a linear combination of the column vectors of $A$.

**Solution** Augmented Matrix is given by

$$\begin{bmatrix} -1 & 3 & 2 & 1 \\ 1 & 2 & -3 & -9 \\ 2 & 1 & -2 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & -2 & -1 \\ 0 & 5 & -1 & -8 \\ 0 & 7 & 2 & -1 \end{bmatrix} \begin{matrix} -1R_1 \\ -1R_1 + R_2 \\ -2R_1 + R_3 \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & -2 & -1 \\ 0 & 1 & -1/5 & -8/5 \\ 0 & 0 & 17/5 & 51/5 \end{bmatrix} \begin{matrix} 1/5R_2 \\ -7R_2 + R_3 \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 0 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{matrix} (5/17)R_3 \\ (1/5)R_3 + R_2 \\ 2R_3 + R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} 3R_2 + R_1$$

$\Rightarrow x_1 = 2, x_2 = -1, x_3 = 3$ . Since the system is consistent, ***b*** is in the column space of ***A***.

Moreover, $$2\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 3\begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

**Example** Determine whether **b** is in the column space of **A** and if so, express **b** as a linear combination of the column vectors of **A**:

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{pmatrix} : \; b = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$

**Solution**

The coefficient matrix $Ax = b$ is:

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$

The augmented matrix for the linear system that corresponds to the matrix equation $Ax = b$ is:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 3 & 2 \end{array}\right)$$

We reduce this matrix to the Reduced Row Echelon Form:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 3 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & -1 & -1 & 1 \\ 2 & 1 & 3 & 2 \end{array}\right) \; R_2 + (-1)R_1$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & -1 & -1 & 4 \end{array}\right) \; R_3 + (-2)R_1$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 4 \end{array}\right) \; (-1)R_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & -1 \\ \hline 0 & 0 & 0 & 3 \end{array}\right) \quad R_3 + R_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & -1 \\ \hline 0 & 0 & 0 & 1 \end{array}\right) \quad \frac{1}{3} R_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right) \quad R_2 + R_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right) \quad R_1 + R_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right) \quad R_1 + (-1) R_2$$

The new system for the equation $Ax = b$ is

$$\begin{aligned} x_1 + x_3 &= 0 \\ x_2 + x_3 &= 0 \\ 0 &= 1 \end{aligned}$$

Equation $0 = 1$ cannot be solved, therefore, the system has **no solution** (i.e. the system is **inconsistent**).

Since the equation $A\mathbf{x} = \mathbf{b}$ has no solution, therefore **b** is not in the column space of ***A***.

**Activity** Determine whether **b** is in the column space of A and if so, express **b** as a linear combination of the column vectors of **A**:

**1.**

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 9 & 3 & 1 \\ 1 & 1 & 1 \end{pmatrix}; \quad b = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$$

**2.** $$A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & -1 & -1 \end{pmatrix}; \quad b = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

**3.** $A=\begin{pmatrix} 1 & 1 & -2 & 1 \\ 0 & 2 & 0 & 1 \\ 1 & 1 & 1 & -3 \\ 0 & 2 & 2 & 1 \end{pmatrix}$; $b=\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$

**Theorem 5** If $x_0$ denotes any single solution of a consistent linear system $Ax=b$ and if $\boldsymbol{v_1, v_2, v_3, ..., v_k}$ form the solution space of the homogeneous system $Ax=0$, then every solution of $Ax=b$ can be expressed in the form $\boldsymbol{x = x_0 + c_1 v_1 + c_2 v_2 + ... + c_k v_k}$ and, conversely, for all choices of scalars $c_1, c_2, c_3, ..., c_k$, the vector $x$ is a solution of $Ax=b$.

General and Particular Solutions: The vector $x_0$ is called a particular solution of $Ax=b$ .The expression $x_0 + c_1 v_1 + c_2 v_2 + \ldots + c_k v_k$ is called the general solution of $Ax=b$ , and the expression $c_1 v_1 + c_2 v_2 + \ldots + c_k v_k$ is called the general solution of $Ax=0$.

**Example 7** Find the vector form of the general solution of the given linear system $\boldsymbol{Ax = b}$; then use that result to find the vector form of the general solution of $\boldsymbol{Ax=0}$.

$$x_1 + 3x_2 - 2x_3 + 2x_5 = 0$$
$$2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1$$
$$5x_3 + 10x_4 + 15x_6 = 5$$
$$2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6$$

**Solution** We solve the non-homogeneous linear system. The augmented matrix of this system is given by

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & -2 & 0 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{bmatrix} \begin{matrix} \\ -2R_1 + R_2 \\ \\ -2R_1 + R_4 \end{matrix}$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{bmatrix} \begin{matrix} \\ -1R_2 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 2 \end{bmatrix} \begin{matrix} \\ \\ -4R_2 + R_4 \end{matrix} \quad -5R_2 + R_3$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 6 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad R_{34}$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (1/6)R_3$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad -3R_3 + R_2$$

$$\begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad 2R_2 + R_1$$

The reduced row echelon form of the augmented matrix corresponds to the system

$$\begin{aligned} 1x_1 + 3x_2 + \quad 4x_4 + 2x_5 &= 0 \\ 1x_3 + 2x_4 &= 0 \\ 1x_6 &= (1/3) \\ 0 &= 0 \end{aligned}$$

No equation of this system has a form zero = nonzero; Therefore, the system is **consistent**. The system has **infinitely many solutions**:

$$x_1 = -3x_2 - 4x_4 - 2x_5 \quad x_2 = r \quad x_3 = -2x_4$$
$$x_4 = s \quad x_5 = t \quad x_6 = 1/3$$

$$x_1 = -3r - 4s - 2t \quad x_2 = r \quad x_3 = -2s$$
$$x_4 = s \quad x_5 = t \quad x_6 = \frac{1}{3}$$

This result can be written in vector form as

$$\begin{bmatrix} x \\ x_2^1 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -3r-4s-2t \\ r \\ -2s \\ s \\ t \\ 1 \\ \overline{3} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ \overline{3} \end{bmatrix} + r\begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s\begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \qquad \text{(A)}$$

which is the general solution of the given system. The vector $\boldsymbol{x_0}$ in (A) is a particular solution of the given system; the linear combination $r\begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s\begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ in (A) is the general solution of the homogeneous system.

**Activity:**

**1.** Suppose that $x_1 = -1,\ x_2 = 2,\ x_3 = 4,\ x_4 = -3$ is a solution of a non-homogenous linear system $Ax = b$ and that the solution set of the homogenous system $Ax = 0$ is given by this formula:

$x_1 = -3r + 4s,$

$x_2 = r - s,$

$x_3 = r,$

$x_4 = s$

**(a)** Find the vector form of the general solution of $Ax = 0$.

**(b)** Find the vector form of the general solution of $Ax = 0$.

Find the vector form of the general solution of the following linear system $\boldsymbol{Ax} = \boldsymbol{b}$; then use that result to find the vector form of the general solution of $\boldsymbol{Ax}=\boldsymbol{0}$:

**2.**
$$\begin{aligned} x_1 - 2x_2 &= 1 \\ 3x_1 - 9x_2 &= 2 \end{aligned}$$

**3.**
$$\begin{aligned} x_1 + 2x_2 - 3x_3 + x_4 &= 3 \\ -3x_1 - x_2 + 3x_3 + x_4 &= -1 \\ -x_1 + 3x_2 - x_3 + 2x_4 &= 2 \\ 4x_1 - 5x_2 - 3x_4 &= -5 \end{aligned}$$

**The Contrast between Nul *A* and Col *A***

It is natural to wonder how the null space and column space of a matrix are related. In fact, the two spaces are quite dissimilar. Nevertheless, a surprising connection between the null space and column space will emerge later.

**Example 8** Let $\boldsymbol{A} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$

(a) If the column space of $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$, what is $k$?
(b) If the null space of $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$, what is $k$?

**Solution**

(a) The columns of $\boldsymbol{A}$ each have three entries, so Col $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$, where $k = 3$.
(b) A vector $\boldsymbol{x}$ such that $\boldsymbol{Ax}$ is defined must have four entries, so Nul $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$, where $k = 4$.

When a matrix is not square, as in Example 8, the vectors in Nul $\boldsymbol{A}$ and Col $\boldsymbol{A}$ live in entirely different "universes". For example, we have discussed no algebraic operations that connect vectors in $\boldsymbol{R}^3$ with vectors in $\boldsymbol{R}^4$. Thus we are not likely to find any relation between individual vectors in Nul $\boldsymbol{A}$ and Col $\boldsymbol{A}$.

**Example 9** If $\boldsymbol{A} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$, find a nonzero vector in Col $\boldsymbol{A}$ and a nonzero vector in Nul $\boldsymbol{A}$

**Solution** It is easy to find a vector in Col $\boldsymbol{A}$. Any column of $\boldsymbol{A}$ will do, say, $\begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$. To find a nonzero vector in Nul $\boldsymbol{A}$, we have to do some work. We row reduce the augmented matrix $[\boldsymbol{A} \quad \boldsymbol{0}]$ to obtain $[\boldsymbol{A} \quad \boldsymbol{0}] \sim \begin{bmatrix} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$. Thus if $\boldsymbol{x}$ satisfies $\boldsymbol{Ax} = \boldsymbol{0}$, then $x_1 = -9x_3, x_2 = 5x_3, x_4 = 0$, and $x_3$ is free. Assigning a nonzero value to $x_3$ (say), $x_3 = 1$, we obtain a vector in Nul $\boldsymbol{A}$, namely, $\boldsymbol{x} = (-9, 5, 1, 0)$.

**Example 10** With $\boldsymbol{A} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$, let $\boldsymbol{u} = \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix}$ and $\boldsymbol{v} = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}$.

(a) Determine if $\boldsymbol{u}$ is in Nul $\boldsymbol{A}$. Could $\boldsymbol{u}$ be in Col $\boldsymbol{A}$?

(b) Determine if $\boldsymbol{v}$ is in Col $\boldsymbol{A}$. Could $\boldsymbol{v}$ be in Nul $\boldsymbol{A}$?

**Solution** (a) An explicit description of Nul $\boldsymbol{A}$ is not needed here. Simply compute the product

$$\boldsymbol{Au} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Obviously, $\boldsymbol{u}$ is not a solution of $\boldsymbol{Ax = 0}$, so $\boldsymbol{u}$ is not in Nul $\boldsymbol{A}$.
Also, with four entries, $\boldsymbol{u}$ could not possibly be in Col $\boldsymbol{A}$, since Col $\boldsymbol{A}$ is a subspace of $\boldsymbol{R^3}$.
(b) Reduce $[\boldsymbol{A} \quad \boldsymbol{v}]$ to an echelon form:

$$[\boldsymbol{A} \quad \boldsymbol{v}] = \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ -2 & -5 & 7 & 3 & -1 \\ 3 & 7 & -8 & 6 & 3 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ 0 & 1 & -5 & -4 & 2 \\ 0 & 0 & 0 & 17 & 1 \end{bmatrix}$$

At this point, it is clear that the equation $\boldsymbol{Ax = v}$ is consistent, so $\boldsymbol{v}$ is in Col $\boldsymbol{A}$. With only three entries, $\boldsymbol{v}$ could not possibly be in Nul $\boldsymbol{A}$, since Nul $\boldsymbol{A}$ is a subspace of $\boldsymbol{R^4}$.

The following table summarizes what we have learned about Nul $\boldsymbol{A}$ and Col $\boldsymbol{A}$.

| | |
|---|---|
| 1. **Nul $A$ is a subspace of $R^n$.** | 1. **Col $A$ is a subspace of $R^m$.** |
| 2. **Nul $A$ is implicitly defined; i.e. we are given only a condition ($Ax = 0$) that vectors in Nul $A$ must satisfy.** | 2. **Col $A$ is explicitly defined; that is, we are told how to build vectors in Col $A$.** |
| 3. **It takes time to find vectors in Nul $A$. Row operations on $[A \quad 0]$ are required.** | 3. **It is easy to find vectors in Col $A$ The columns of $A$ are displayed; others are formed from them.** |
| 4. **There is no obvious relation between Nul $A$ and the entries in $A$.** | 4. **There is an obvious relation between Col $A$ and the entries in $A$, since each column of $A$ is in Col $A$.** |
| 5. **A typical vector v in Nul $A$ has the property that $Av = 0$.** | 5. **A typical vector $v$ in Col $A$ has the property that the equation $Ax = v$ is consistent.** |
| 6. **Given a specific vector $v$, it is easy to tell if $v$ is in Nul $A$. Just compute $Av$.** | 6. **Given a specific vector $v$, it may take time to tell if $v$ is in Col $A$. Row operations on $[A \quad v]$ are required.** |
| 7. **Nul $A = \{0\}$ if and only if the equation $Ax = 0$ has only the trivial solution.** | 7. **Col $A = R^m$ if and only if the equation $Ax = b$ has a solution for every $b$ in $R^m$.** |
| 8. **Nul $A = \{0\}$ if and only if the linear transformation $x \rightarrow Ax$ is one-to-one.** | 8. **Col $A = R^m$ if and only if the linear transformation $x \rightarrow Ax$ maps $R^n$ onto $R^m$.** |

**Kernel and Range of A Linear Transformation**

Subspaces of vector spaces other than $\boldsymbol{R^n}$ are often described in terms of a linear transformation instead of a matrix. To make this precise, we generalize the definition given earlier in Segment I.

**Definition** A **linear transformation** $\boldsymbol{T}$ from a vector space $\boldsymbol{V}$ into a vector space $\boldsymbol{W}$ is a rule that assigns to each vector $\boldsymbol{x}$ in $\boldsymbol{V}$ a unique vector $\boldsymbol{T}(\boldsymbol{x})$ in $\boldsymbol{W}$, such that

(i) $\boldsymbol{T}(\boldsymbol{u}+\boldsymbol{v}) = \boldsymbol{T}(\boldsymbol{u}) + \boldsymbol{T}(\boldsymbol{v})$ for all $\boldsymbol{u}, \boldsymbol{v}$ in $\boldsymbol{V}$, and

(ii) $\boldsymbol{T}(c\boldsymbol{u}) = c\,\boldsymbol{T}(\boldsymbol{u})$ for all $\boldsymbol{u}$ in $\boldsymbol{V}$ and all scalars $c$.

The **kernel** (or **null space**) of such a $\boldsymbol{T}$ is the set of all $\boldsymbol{u}$ in $\boldsymbol{V}$ such that $\boldsymbol{T}(\boldsymbol{u}) = \boldsymbol{0}$ (the zero vector in $\boldsymbol{W}$). The **range** of $\boldsymbol{T}$ is the set of all vectors in $\boldsymbol{W}$ of the form $\boldsymbol{T}(\boldsymbol{x})$ for some $\boldsymbol{x}$ in $\boldsymbol{V}$. If $\boldsymbol{T}$ happens to arise as a matrix transformation, say, $\boldsymbol{T}(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$ for some matrix $\boldsymbol{A}$ – then the kernel and the range of $\boldsymbol{T}$ are just the null space and the column space of $\boldsymbol{A}$, as defined earlier. So if $\boldsymbol{T}(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$, col $\boldsymbol{A}$ = range of $\boldsymbol{T}$.

**Definition** If $T : V \to W$ is a linear transformation, then the set of vectors in $V$ that $\boldsymbol{T}$ maps into $\boldsymbol{0}$ is called the **kernel** of $\boldsymbol{T}$; it is denoted by $\boldsymbol{ker(T)}$. The set of all vectors in $\boldsymbol{W}$ that are images under $\boldsymbol{T}$ of at least one vector in $\boldsymbol{V}$ is called the **range** of $\boldsymbol{T}$; it is denoted by $\boldsymbol{R(T)}$.

**Example** If $T_A : \mathbb{R}^n \to \mathbb{R}^m$ is multiplication by the $m \times n$ matrix A, then from the above definition; the kernel of $T_A$ is the null space of $\boldsymbol{A}$ and the range of $T_A$ is the column space of $\boldsymbol{A}$.

**Remarks** The kernel of $\boldsymbol{T}$ is a subspace of $\boldsymbol{V}$ and the range of $\boldsymbol{T}$ is a subspace of $\boldsymbol{W}$.

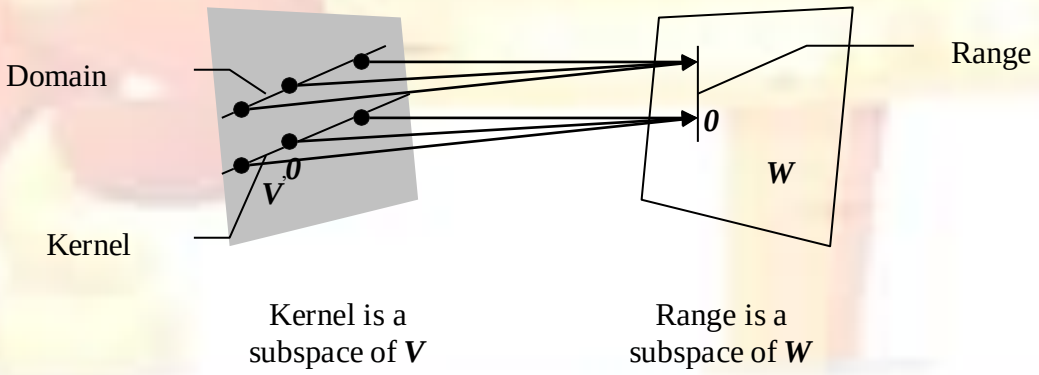



**Figure 2** Subspaces associated with a linear transformation.

In applications, a subspace usually arises as either the kernel or the range of an appropriate linear transformation. For instance, the set of all solutions of a homogeneous linear differential equation turns out to be the kernel of a linear transformation. Typically, such a linear transformation is described in terms of one or more derivatives of a

function. To explain this in any detail would take us too far a field at this point. So we present only two examples. The first explains why the operation of differentiation is a linear transformation.

**Example 11** Let $\boldsymbol{V}$ be the vector space of all real-valued functions $\boldsymbol{f}$ defined on an interval $[a, b]$ with the property that they are differentiable and their derivatives are continuous functions on $[a, b]$. Let $\boldsymbol{W}$ be the vector space of all continuous functions on $[a, b]$ and let $\boldsymbol{D}:\boldsymbol{V}\to\boldsymbol{W}$ be the transformation that changes $\boldsymbol{f}$ in $\boldsymbol{V}$ into its derivative $\boldsymbol{f}'$. In calculus, two simple differentiation rules are

$$\boldsymbol{D}(\boldsymbol{f}+\boldsymbol{g})=\boldsymbol{D}(\boldsymbol{f})+\boldsymbol{D}(\boldsymbol{g}) \text{ and } \boldsymbol{D}(c\boldsymbol{f})=c\boldsymbol{D}(\boldsymbol{f})$$

That is, $\boldsymbol{D}$ is a linear transformation. It can be shown that the kernel of $\boldsymbol{D}$ is the set of constant functions of $[a, b]$ and the range of $\boldsymbol{D}$ is the set $\boldsymbol{W}$ of all continuous functions on $[a, b]$.

**Example 12** The differential equation $y''+wy=0$ (4)
where $w$ is a constant, is used to describe a variety of physical systems, such as the vibration of a weighted spring, the movement of a pendulum and the voltage in an inductance – capacitance electrical circuit. The set of solutions of (4) is precisely the kernel of the linear transformation that maps a function $\boldsymbol{y}=\boldsymbol{f}(\boldsymbol{t})$ into the function $\boldsymbol{f}''(\boldsymbol{t})+w\boldsymbol{f}(\boldsymbol{t})$. Finding an explicit description of this vector space is a problem in differential equations.

**Example 13** Let $\boldsymbol{W}=\left\{\begin{bmatrix}a\\b\\c\end{bmatrix}: a-3b-c=0\right\}$. Show that $W$ is a subspace of $R^3$ in different ways.

**Solution** First method: $W$ is a subspace of $R^3$ by Theorem 2 because $W$ is the set of all solutions to a system of homogeneous linear equations (where the system has only one equation). Equivalently, $W$ is the null space of the *1x3* matrix $\boldsymbol{A}=[1\ \ -3\ \ -1]$.

Second method: Solve the equation $a - 3b - c = 0$ for the leading variable a in terms of the free variables $b$ and $c$.

Any solution has the form $\begin{bmatrix}3b+c\\b\\c\end{bmatrix}$, where $b$ and $c$ are arbitrary, and

$$\begin{bmatrix}3b+c\\b\\c\end{bmatrix}=b\begin{bmatrix}3\\1\\0\end{bmatrix}+c\begin{bmatrix}1\\0\\1\end{bmatrix}$$

$\boldsymbol{v}_1$ $\boldsymbol{v}_2$

This calculation shows that $W$ = Span$\{v_1, v_2\}$. Thus $W$ is a subspace of $R^3$ by Theorem i.e. if $v_1,...,v_p$ are in a vector space$V$, then Span $\{v_1,...,v_p\}$ is a subspace of$V$. We could also solve the equation $a - 3b - c = 0$ for $b$ or $c$ and get alternative descriptions of $W$ as a set of linear combinations of two vectors.

**Example 14** Let $\boldsymbol{A} = \begin{bmatrix} 7 & -3 & 5 \\ -4 & 1 & -5 \\ -5 & 2 & -4 \end{bmatrix}$, $\boldsymbol{v} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$, *and* $\boldsymbol{W} = \begin{bmatrix} 7 \\ 6 \\ -3 \end{bmatrix}$

Suppose you know that the equations $\boldsymbol{Ax} = \boldsymbol{v}$ and $\boldsymbol{Ax} = \boldsymbol{w}$ are both consistent. What can you say about the equation $\boldsymbol{Ax} = \boldsymbol{v} + \boldsymbol{w}$?

**Solution** Both $\boldsymbol{v}$ and $\boldsymbol{w}$ are in Col $\boldsymbol{A}$. Since Col $\boldsymbol{A}$ is a vector space, $\boldsymbol{v} + \boldsymbol{w}$ must be in Col $\boldsymbol{A}$. That is, the equation $\boldsymbol{Ax} = \boldsymbol{v} + \boldsymbol{w}$ is consistent.

**Activity**

1. Let $\boldsymbol{V}$ and $\boldsymbol{W}$ be any two vector spaces. The mapping $T: V \to W$ such that $\boldsymbol{T}(\boldsymbol{v}) = \boldsymbol{0}$ for every $\boldsymbol{v}$ in $\boldsymbol{V}$ is a linear transformation called the **zero transformation**. Find the kernel and range of the zero transformation.

2. Let $\boldsymbol{V}$ be any vector space. The mapping $I: V \to V$ defined by $\boldsymbol{I(v)} = \boldsymbol{v}$ is called the **identity operator** on $\boldsymbol{V}$. Find the kernel and range of the identity operator.

**Exercises**

1. Determine if $\boldsymbol{w} = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}$ is in Nul $\boldsymbol{A}$, where $\boldsymbol{A} = \begin{bmatrix} 5 & 21 & 19 \\ 13 & 23 & 2 \\ 8 & 14 & 1 \end{bmatrix}$.

In exercises 2 and 3, find an explicit description of Nul $\boldsymbol{A}$, by listing vectors that span the null space.

2. $\begin{bmatrix} 1 & 3 & 5 & 0 \\ 0 & 1 & 4 & -2 \end{bmatrix}$

3. $\begin{bmatrix} 1 & -2 & 0 & 4 & 0 \\ 0 & 0 & 1 & -9 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

In exercises 4-7, either use an appropriate theorem to show that the given set, $\boldsymbol{W}$ is a vector space, or find a specific example to the contrary.

4. $\left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} : a+b+c = 2 \right\}$

5. $\left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} : \begin{matrix} a-2b = 4c \\ 2a = c+3d \end{matrix} \right\}$

6. $\left\{ \begin{bmatrix} b-2d \\ 5+d \\ b+3d \\ d \end{bmatrix} : b,d \text{ real} \right\}$

7. $\left\{ \begin{bmatrix} -a+2b \\ a-2b \\ 3a-6b \end{bmatrix} : a,b \text{ real} \right\}$

In exercises 8 and 9, find $\boldsymbol{A}$ such that the given set is Col $\boldsymbol{A}$.

8. $\left\{ \begin{bmatrix} 2s+3t \\ r+s-2t \\ 4r+s \\ 3r-s-t \end{bmatrix} : r,s,t \text{ real} \right\}$

9. $\left\{ \begin{bmatrix} b-c \\ 2b+c+d \\ 5c-4d \\ d \end{bmatrix} : b,c,d \text{ real} \right\}$

For the matrices in exercises 10-13, (a) find $k$ such that Nul $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$, and (b) find $k$ such that Col $\boldsymbol{A}$ is a subspace of $\boldsymbol{R}^k$.

10. $A = \begin{bmatrix} 2 & -6 \\ -1 & 3 \\ -4 & 12 \\ 3 & -9 \end{bmatrix}$

11. $A = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 0 & -5 \\ 0 & -5 & 7 \\ -5 & 7 & -2 \end{bmatrix}$

12. $A = \begin{bmatrix} 4 & 5 & -2 & 6 & 0 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix}$

13. $A = \begin{bmatrix} 1 & -3 & 9 & 0 & -5 \end{bmatrix}$

14. Let $A = \begin{bmatrix} -6 & 12 \\ -3 & 6 \end{bmatrix}$ and $w = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Determine if $w$ is in Col $A$. Is $w$ in Nul $A$?

15. Let $A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$ and $w = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. Determine if $w$ is in Col $A$. Is $w$ in Nul $A$?

16. Define $T$: $P_2 \rightarrow R^2$ by $T(p) = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix}$. For instance, if $p(t) = 3 + 5t + 7t^2$, then $T(p) = \begin{bmatrix} 3 \\ 15 \end{bmatrix}$.

a. Show that $T$ is a linear transformation.
b. Find a polynomial $p$ in $P_2$ that spans the kernel of $T$, and describe the range of $T$.

17. Define a linear transformation $T$: $P_2 \rightarrow R^2$ by $T(p) = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix}$. Find polynomials $p_1$ and $p_2$ in $P_2$ that span the kernel of $T$, and describe the range of $T$.

18. Let $M_{2x2}$ be the vector space of all 2x2 matrices, and define $T$: $M_{2x2} \rightarrow M_{2x2}$ by $T(A) = A + A^T$, where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

(a) Show that $T$ is a linear transformation.
(b) Let $B$ be any element of $M_{2x2}$ such that $B^T = B$. Find an $A$ in $M_{2x2}$ such that $T(A) = B$.
(c) Show that the range of $T$ is the set of $B$ in $M_{2x2}$ with the property that $B^T = B$.
(d) Describe the kernel of $T$.

19. Determine whether $w$ is in the column space of $A$, the null space of $A$, or both, where

(a) $\boldsymbol{w} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -3 \end{bmatrix}, \boldsymbol{A} = \begin{bmatrix} 7 & 6 & -4 & 1 \\ -5 & -1 & 0 & -2 \\ 9 & -11 & 7 & -3 \\ 19 & -9 & 7 & 1 \end{bmatrix}$ (b) $\boldsymbol{w} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \boldsymbol{A} = \begin{bmatrix} -8 & 5 & -2 & 0 \\ -5 & 2 & 1 & -2 \\ 10 & -8 & 6 & -3 \\ 3 & -2 & 1 & 0 \end{bmatrix}$

20. Let $\boldsymbol{a_1}, \ldots, \boldsymbol{a_5}$ denote the columns of the matrix $\boldsymbol{A}$, where

$$\boldsymbol{A} = \begin{bmatrix} 5 & 1 & 2 & 2 & 0 \\ 3 & 3 & 2 & -1 & -12 \\ 8 & 4 & 4 & -5 & 12 \\ 2 & 1 & 1 & 0 & -2 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} \boldsymbol{a_1} & \boldsymbol{a_2} & \boldsymbol{a_4} \end{bmatrix}$$

(a) Explain why $\boldsymbol{a_3}$ and $\boldsymbol{a_5}$ are in the column space of $\boldsymbol{B}$

(b) Find a set of vectors that spans Nul $\boldsymbol{A}$

(c) Let $\boldsymbol{T}$: $\boldsymbol{R^5} \rightarrow \boldsymbol{R^4}$ be defined by $\boldsymbol{T}(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$. Explain why $\boldsymbol{T}$ is neither one-to-one nor onto.

# Lecture 22

## Linearly Independent Sets; Bases

First we revise some definitions and theorems from the Vector Space:

**Definition** Let $V$ be an arbitrary nonempty set of objects on which two operations are defined, addition and multiplication by scalars.

If the following axioms are satisfied by all objects $u, v, w$ in $V$ and all scalars $l$ and $m$, then we call $V$ a vector space.

**Axioms of Vector Space**

For any set of vectors $u, v, w$ in V *and scalars l, m, n:*

1. $u + v$ is in $V$
2. $u + v = v + u$
3. $u + (v + w) = (u + v) + w$
4. There exist a zero vector 0 such that
   $0 + u = u + 0 = u$
5. There exist a vector $-u$ in $V$ such that
   $-u + u = 0 = u + (-u)$
6. $(l\ u)$ is in $V$
7. $l\ (u + v) = l\ u + l\ v$
8. $m\ (n\ u) = (m\ n)\ u = n\ (m\ u)$
9. $(l + m)\ u = I\ u + m\ u$
10. $1u = u$ where 1 is the multiplicative identity

**Definition** A subset $W$ of a vector space $V$ is called a subspace of $V$ if $W$ itself is a vector space under the addition and scalar multiplication defined on $V$.

**Theorem** If $W$ is a set of one or more vectors from a vector space $V$, then $W$ is subspace of $V$ if and only if the following conditions hold:

(a) If $u$ and $v$ are vectors in $W$, then $u + v$ is in $W$
(b) If $k$ is any scalar and $u$ is any vector in $W$, then $k\ u$ is in $W$.

**Definition** The null space of an $m \times n$ matrix $A$ *(Nul A)* is the set of all solutions of the hom equation $Ax = 0$

$Nul\ A = \{x: x\ is\ in\ R^n\ and\ Ax = 0\}$

**Definition** The column space of an m x n matrix $A$ *(Col A)* is the set of all linear combinations of the columns of $A$.

If $A = [a_1 \ \ldots \ a_n]$,
then
$Col\ A = Span\ \{ a_1, \ldots , a_n \}$

Since we know that a set of vectors $S = \{v_1, v_2, v_3, ...v_p\}$ spans a given vector space $\boldsymbol{V}$ if every vector in $\boldsymbol{V}$ is expressible as a linear combination of the vectors in $\boldsymbol{S}$. In general there may be more than one way to express a vector in $\boldsymbol{V}$ as linear combination of vectors in a spanning set. We shall study conditions under which each vector in $\boldsymbol{V}$ is expressible as a linear combination of the spanning vectors in exactly one way. Spanning sets with this property play a fundamental role in the study of vector spaces.
In this Lecture, we shall identify and study the subspace $\boldsymbol{H}$ as "efficiently" as possible. The key idea is that of linear independence, defined as in $\boldsymbol{R^n}$.

**Definition** An ***indexed set*** of vectors $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$ in $\boldsymbol{V}$ is said to be **linearly independent** if the vector equation

$$c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + \ldots + c_p\boldsymbol{v_p} = 0 \qquad (1)$$

has only the trivial solution, i.e. $c_1 = 0, \ldots, c_p = 0$.
The set $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$ is said to be **linearly dependent** if (1) has a nontrivial solution, that is, if there are some weights, $c_1, \ldots, c_p$, not all zero, such that (1) holds. In such a case, (1) is called a **linear dependence relation** among $\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}$. Alternatively, to say that the $\boldsymbol{v}$'s are linearly dependent is to say that the zero vector **0** can be expressed as a *nontrivial linear combination* of the $\boldsymbol{v}$'s.

If the trivial solution is the only solution to this equation then the vectors in the set are called **linearly independent** and the set is called a **linearly independent set**. If there is another solution then the vectors in the set are called **linearly dependent** and the set is called a **linearly dependent set**.

Just as in $\boldsymbol{R^n}$, a set containing a single vector $\boldsymbol{v}$ is linearly independent if and only if $\boldsymbol{v} \neq \boldsymbol{0}$. Also, a set of two vectors is linearly dependent if and only if one of the vectors is a multiple of the other. And any set containing the zero-vector is linearly dependent.

Determining whether a set of vectors $a_1, a_2, a_3, ...a_n$ is linearly independent is easy when one of the vectors is **0**: if, say, $a_1 = 0$, then we have a simple solution to $x_1a_1 + x_2a_2 + x_3a_3 + \ldots + x_na_n = 0$ given by choosing $x_1$ to be any nonzero value and putting all the other $x$'s equal to 0. Consequently, *if a set of vectors contains the zero vector, it must always be linearly dependent*. Equivalently, *any set of linearly independent vectors cannot contain the zero vector*.

Another situation in which it is easy to determine linear independence is when there are more vectors in the set than entries in the vectors. If $n > m$, then the $n$ vectors $a_1, a_2, a_3, ...a_n$ in $\boldsymbol{R^m}$ are columns of an $m \times n$ matrix $A$. The vector equation $x_1a_1 + x_2a_2 + x_3a_3 + \ldots + x_na_n = 0$ is equivalent to the matrix equation $A\mathbf{x} = \mathbf{0}$ whose corresponding linear system has more variables than equations. Thus there must be at least one free variable in the solution, meaning that there are nontrivial solutions

to $x_1a_1 + x_2a_2 + x_3a_3 + \ldots + x_na_n = 0$: If $n > m$, then the set $\{a_1, a_2, a_3, \ldots a_n\}$ of vectors in $\boldsymbol{R^m}$ *must* be linearly dependent.

When $n$ is small we have a clear geometric picture of the relation amongst linearly independent vectors. For instance, the case $n = 1$ produces the equation $x_1a_1 = 0$, and as long as $a_1 \neq 0$, we only have the trivial solution $x_1 = 0$. *A single nonzero vector always forms a linearly independent set*.

When $n = 2$, the equation takes the form $x_1a_1 + x_2a_2 = 0$. If this were a linear dependence relation, then one of the $x$'s, say $x_1$, would have to be nonzero. Then we could solve the equation for $a_1$ and obtain a relation indicating that $a_1$ is a scalar multiple of $a_2$. Conversely, if one of the vectors is a scalar multiple of the other, we can express this in the form $x_1a_1 + x_2a_2 = 0$. Thus, *a set of two nonzero vectors is linearly dependent if and only if they are scalar multiples of each other*.

**Example** **(linearly independent set)**

Show that the following vectors are linearly independent:

$$v_1 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

**Solution** Let there exist scalars $c_1, c_2, c_3$ in $\boldsymbol{R}$ such that

$$c_1v_1 + c_3v_2 + c_3v_3 = 0$$

Therefore,

$$\Rightarrow c_1\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} + c_2\begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} + c_3\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -2c_1 \\ c_1 \\ c_1 \end{bmatrix} + \begin{bmatrix} 2c_2 \\ c_2 \\ -2c_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ c_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -2c_1 + 2c_2 \\ c_1 + c_2 \\ c_1 - 2c_2 + c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The above can be written as:

$-2c_1 + 2c_2 = 0$ .....(1) $\Rightarrow -c_1 + c_2 = 0$.....(4) (*dividing by* 2 *onboth sides of* (1))

$c_1 + c_2 = 0$.....(2)

$c_1 - 2c_2 + c_3 = 0$.....(3)

*Solving* (2) *and* (4) *implies*:

$$\begin{array}{l} c_1+c_2=0 \\ -c_1+c_2=0 \\ \hline 0+2c_2=0 \\ \Rightarrow c_2=0 \end{array}$$

*Solving* (2) *implies*: $c_1+0=0 \Rightarrow c_1=0$

*Solving* (3) *implies*: $0+0+c_3=0 \Rightarrow c_3=0$

$\Rightarrow c_1=c_2=c_3=0$ ; *scalars* $c_1, c_2, c_3 \in R$ *are all zero*

$\therefore$ The system has trivial solution.

*Hence the given vectors* $v_1, v_2, v_3$ *are linearly independent.*

**Example** **(linearly dependent set)**

If $v_1=\{2,-1,0,3\}, v_2=\{1,2,5,-1\}$ *and* $v_3=\{7,-1,5,8\}$ , then the set of vectors $S=\{v_1,v_2,v_3\}$ is linearly dependent, since $3v_1+v_2-v_3=0$

**Example** **(linearly dependent set)**

The polynomials $p_1=-x+1,\ p_2=-2x^2+3x+5,$ *and* $p_3=-x^2+3x+1$ form a linearly dependent set in $p_2$ since $3p_1-p_2+2p_3=0$.

**Note** The linearly independent or linearly dependent sets can also be determined using the Echelon Form or the Reduced Row Echelon Form methods.

**Theorem 1** An indexed set $\{ v_1, \ldots, v_p \}$ of two or more vectors, with $v_1 \neq 0$, is linearly dependent if and only if some $v_j$ (with $j>1$) is a linear combination of the preceding vectors, $v_1, \ldots, v_{j-1}$.

The main difference between linear dependence in $R^n$ and in a general vector space is that when the vectors are not n – tuples, the homogeneous equation (1) usually cannot be written as a system of *n* linear equations. That is, the vectors cannot be made into the columns of a matrix $A$ in order to study the equation $Ax = 0$. We must rely instead on the definition of linear dependence and on Theorem 1.

**Example 1** Let $p_1(t) = 1$, $p_2(t) = t$ and $p_3(t) = 4 - t$. Then $\{ p_1, p_2, p_3 \}$ is linearly dependent in $P$ because $p_3 = 4p_1 - p_2$.

**Example 2** The set $\{Sin\ t, Cos\ t\}$ is linearly independent in $C[0, 1]$ because *Sin t* and *Cos t* are not multiples of one another as vectors in $C[0, 1]$. That is, there is no scalar *c* such that *Cos t* = *c*. *Sin t* for all *t* in *[0, 1]*. (Look at the graphs of *Sin t* and *Cos t*.) However, *{Sin t Cos t, Sin 2t}* is linearly dependent because of he identity:
*Sin 2t = 2 Sin t Cos t*, for all *t*.

**Useful results**

- A set containing the zero vector is linearly dependent.
- A set of two vectors is linearly dependent if and only if one is a multiple of the other.
- A set containing one nonzeoro vector is linearly independent. i.e. consider the set containing one nonzeoro vector $\{v_1\}$ so $\{v_1\}$ is linearly independent when $v_1 \neq 0$.
- A set of two vectors is linearly independent if and only if neither of the vectors is a multiple of the other.

**Activity** Determine whether the following sets of vectors are linearly independent or linearly dependent:

1. $i=(1,0,0,0),\ j=(0,1,0,0), k=(0,0,0,1)$ in $\mathbb{R}^4$.
2. $v_1=(2,0,-1),\ v_2=(-3,-2,-5),\ v_3=(-6,1,-1),\ v_4=(-7,0,2)$ in $\mathbb{R}^3$.
3. $i=(1,0,0,...,0),\ j=(0,1,0,...,0), k=(0,0,0,...,1)$ in $\mathbb{R}^m$.
4. $3x^2+3x+1,\ 4x^2+x,\ 3x^2+6x+5,\ -x^2+2x+7$ in $p_2$

**Definition** Let ***H*** be a subspace of a vector space ***V***. An indexed set of vectors ***B*** = ***{b1, ..., bp}*** in **V** is a **basis** for ***H*** if

(i) ***B*** is a linearly independent set, and
(ii) the subspace spanned by ***B*** coincides with ***H***; that is,
***H*** = *Span* ***{b1,...,bp }***

The definition of a basis applies to the case when ***H*** = ***V***, because any vector space is a subspace of itself. Thus a basis of ***V*** is a linearly independent set that spans ***V***. Observe that when $H \neq V$, condition (ii) includes the requirement that each of the vectors ***b1,...,bp*** must belong to ***H***, because *Span* ***{ b1,...,bp }*** contains ***b1, ...,bp***, as we saw in lecture 21.

**Example 3** Let *A* be an invertible $n \times n$ matrix – say, *A* = ***[a1 ... an]***. Then the columns of *A* form a basis for $\mathbb{R}^n$ because they are linearly independent and they span $\mathbb{R}^n$, by the Invertible Matrix Theorem.

**Example 4** Let ***e1, ..., en*** be the columns of the $n \times n$ identity matrix, $I_n$. That is,

$$\mathbf{e}_1=\begin{bmatrix}1\\0\\ \vdots \\0\end{bmatrix},\quad \mathbf{e}_2=\begin{bmatrix}0\\1\\ \vdots \\0\end{bmatrix},\quad \ldots\quad \mathbf{e}_n=\begin{bmatrix}0\\0\\ \vdots \\1\end{bmatrix}$$

The set ***{e1, ..., en}*** is called the standard basis for $\mathbb{R}^n$ (Fig. 1).

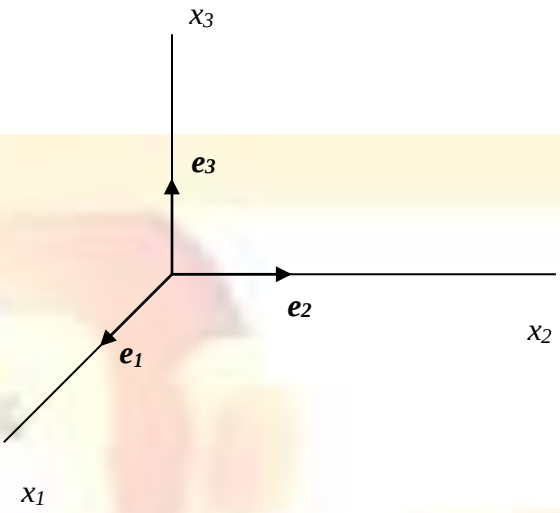



**Figure 1 - The standard basis for $R^3$**

**Example 5** Let $\boldsymbol{v}_1 = \begin{bmatrix} 3 \\ 0 \\ -6 \end{bmatrix}, \boldsymbol{v}_2 = \begin{bmatrix} -4 \\ 1 \\ 7 \end{bmatrix}$, and $\boldsymbol{v}_3 = \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$. Determine if $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ is a basis for $\boldsymbol{R^3}$.

**Solution** Since there are exactly three vectors here in $\boldsymbol{R^3}$, we can use one of any methods to determine whether they are basis for $\mathbb{R}^3$ or not. For this, let solve with help of matrices. First form a matrix of vectors i.e. matrix $A = [\boldsymbol{v}_1 \quad \boldsymbol{v}_2 \quad \boldsymbol{v}_3]$. If this matrix is invertible (i.e. $|A| \neq 0$ determinant should be non zero).
For instance, a simple computation shows that $\det A = 6 \neq 0$. Thus $A$ is invertible. As in example 3, the columns of $A$ form a basis for $\boldsymbol{R^3}$.

**Example 6** Let $\boldsymbol{S} = \{\boldsymbol{1}, \boldsymbol{t}, \boldsymbol{t^2}, \ldots, \boldsymbol{t^n}\}$. Verify that $\boldsymbol{S}$ is a basis for $\boldsymbol{P_n}$. This basis is called the **standard basis** for $\boldsymbol{P_n}$.

**Solution** Certainly $\boldsymbol{S}$ spans $\boldsymbol{P_n}$. To show that $\boldsymbol{S}$ is linearly independent, suppose that $c_0, \ldots, c_n$ satisfy

$$c_0\boldsymbol{.1} + c_1\boldsymbol{t} + c_2\boldsymbol{t^2} + \ldots\ldots + c_n\boldsymbol{t^n} = \boldsymbol{0}\ \boldsymbol{(t)} \qquad (2)$$

This equality means that the polynomial on the left has the same values as the zero polynomial on the right. A fundamental theorem in algebra says that the only polynomial in $\boldsymbol{P_n}$ with more than $n$ zeros is the zero polynomial. That is, (2) holds for all **t** only if $c_0 = \ldots = c_n = 0$. This proves that $S$ is linearly independent and hence is a basis for $\boldsymbol{P_n}$. See Figure 2.

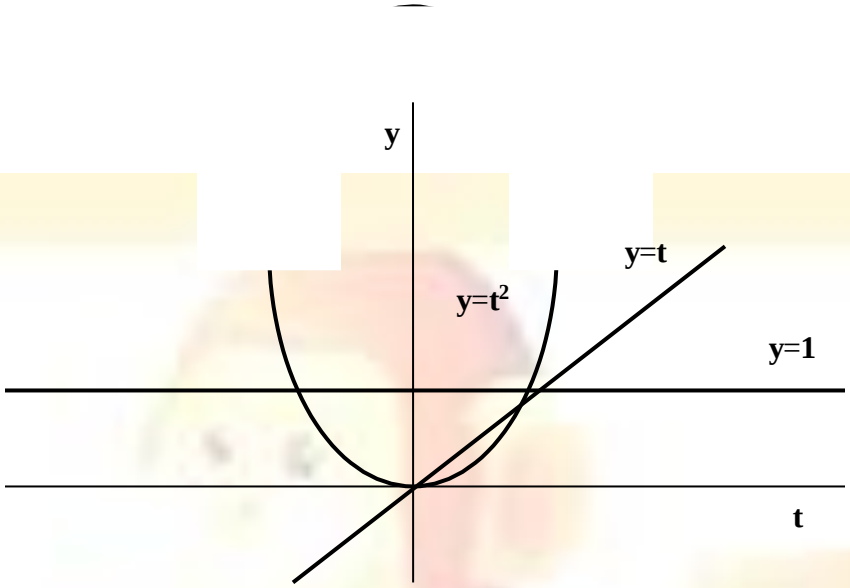



**Figure 2 – The standard basis for $P_2$**

Problems involving linear independence and spanning in $\boldsymbol{P_n}$ are handled best by a technique to be discussed later.

**Example 7** Check whether the set of vectors *{(2, -3, 1), (4, 1, 1), (0, -7, 1)}* is basis for $\boldsymbol{R^3}$?

**Solution** The set $\boldsymbol{S} = \{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ of vectors in $\boldsymbol{R^3}$ spans $\boldsymbol{V} = \boldsymbol{R^3}$ if

$$c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + c_3\boldsymbol{v_3} = d_1\boldsymbol{w_1} + d_2\boldsymbol{w_2} + d_3\boldsymbol{w_3} \qquad (*)$$

with $\boldsymbol{w_1} = (1,0,0)$, $\boldsymbol{w_2} = (0,1,0)$ , $\boldsymbol{w_3} = (0,0,1)$ has at least one solution for every set of values of the coefficients $d_1, d_2, d_3$. Otherwise (i.e., if no solution exists for at least some values of $d_1, d_2, d_3$), $\boldsymbol{S}$ does not span $\boldsymbol{V}$. With our vectors $\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}$, (*) becomes

$$c_1(2,-3,1) + c_2(4,1,1) + c_3(0,-7,1) = d_1(1,0,0) + d_2(0,1,0) + d_3(0,0,1)$$

Rearranging the left hand side yields

$$\begin{aligned} 2c_1 + 4c_2 + 0c_3 &= 1d_1 + 0d_2 + 0d_3 \\ -3c_1 + 1c_2 - 7c_3 &= 0d_1 + 1d_2 + 0d_3 \qquad (A) \\ 1c_1 + 1c_2 + 1c_3 &= 0d_1 + 0d_2 + 1d_3 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} 2 & 4 & 0 \\ -3 & 1 & -7 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

We now find the determinant of coefficient matrix $\begin{bmatrix} 2 & 4 & 0 \\ -3 & 1 & -7 \\ 1 & 1 & 1 \end{bmatrix}$ to determine whether the system is consistent (so that $\boldsymbol{S}$ spans $\boldsymbol{V}$), or inconsistent ($\boldsymbol{S}$ does not span $\boldsymbol{V}$).

Now $\det \begin{bmatrix} 2 & 4 & 0 \\ -3 & 1 & -7 \\ 1 & 1 & 1 \end{bmatrix} = 2(8) - 4(4) + 0 = 0$

Therefore, the system (A) is inconsistent, and, consequently, the set $\boldsymbol{S}$ does not span the space $\boldsymbol{V}$.

**Example 8** Check whether the set of vectors
$\{-4 + 1\,t + 3\,t^2,\ 6 + 5\,t + 2\,t^2,\ 8 + 4\,t + 1\,t^2\}$ is a basis for $\boldsymbol{P_2}$?

**Solution** The set $\boldsymbol{S} = \{p_1(t), p_2(t), p_3(t)\}$ of vectors in $\boldsymbol{P_2}$ spans $\boldsymbol{V} = \boldsymbol{P_2}$ if

$$c_1\,p_1(t) + c_2\,p_2(t) + c_3\,p_3(t) = d_1\,q_1(t) + d_2\,q_2(t) + d_3\,q_3(t) \qquad (*)$$

with $q_1(t) = 1 + 0\,t + 0\,t^2$, $q_2(t) = 0 + 1\,t + 0\,t^2$, $q_3(t) = 0 + 0\,t + 1\,t^2$ has at least one solution for every set of values of the coefficients $d_1, d_2, d_3$. Otherwise (i.e., if no solution exists for at least some values of $d_1, d_2, d_3$), $\boldsymbol{S}$ does not span $\boldsymbol{V}$. With our vectors $p_1(t), p_2(t), p_3(t)$, (*) becomes:

$$c_1(-4 + 1\,t + 3\,t^2) + c_2(6 + 5\,t + 2\,t^2) + c_3(8 + 4\,t + 1\,t^2) =$$
$$d_1(1 + 0\,t + 0\,t^2) + d_2(0 + 1\,t + 0\,t^2) + d_3(0 + 0\,t + 1\,t^2)$$

Rearranging the left hand side yields

$$(-4c_1 + 6c_2 + 8c_3)1 + (1c_1 + 5c_2 + 4c_3)\,t + (3c_1 + 2c_2 + 1c_3)\,t^2 =$$
$$(1d_1 + 0d_2 + 0d_3)1 + (0d_1 + 1d_2 + 0d_3)\,t + (0d_1 + 0d_2 + 1d_3)\,t^2$$

In order for the equality above to hold for all values of t, the coefficients corresponding to the same power of **t** on both sides of the equation must be equal. This yields the following system of equations:

$$\begin{aligned} -4c_1 + 6c_2 + 8c_3 &= 1d_1 + 0d_2 + 0d_3 \\ 1c_1 + 5c_2 + 4c_3 &= 0d_1 + 1d_2 + 0d_3 \\ 3c_1 + 2c_2 + 1c_3 &= 0d_1 + 0d_2 + 1d_3 \end{aligned} \qquad (A)$$

$$\Rightarrow \begin{bmatrix} -4 & 6 & 8 \\ 1 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

We now find the determinant of coefficient matrix $\begin{bmatrix} -4 & 6 & 8 \\ 1 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$ to determine whether the system is consistent (so that $\boldsymbol{S}$ spans $\boldsymbol{V}$), or inconsistent ($\boldsymbol{S}$ does not span $\boldsymbol{V}$).

Now $\det \begin{bmatrix} -4 & 6 & 8 \\ 1 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} = -26 \neq 0$. Therefore, the system (A) is consistent, and, consequently, the set $\boldsymbol{S}$ spans the space $\boldsymbol{V}$.

The set $\boldsymbol{S} = \{p_1(t), p_2(t), p_3(t)\}$ of vectors in $\boldsymbol{P_2}$ is linearly independent if the only solution of

$$c_1\,p_1(t) + c_2\,p_2(t) + c_3\,p_3(t) = 0 \qquad (**)$$

is $c_1, c_2, c_3 = 0$. In this case, the set $\boldsymbol{S}$ forms a basis for span $\boldsymbol{S}$. Otherwise (i.e., if a solution with at least some nonzero values exists), $\boldsymbol{S}$ is linearly dependent. With our vectors $\boldsymbol{p_1(t)}, \boldsymbol{p_2(t)}, \boldsymbol{p_3(t)}$, (2) becomes: $c_1(-4 + 1\,t + 3\,t^2) + c_2(6 + 5\,t + 2\,t^2) + c_3(8 + 4\,t + 1\,t^2) = 0$ Rearranging the left hand side yields

$$(-4\,c_1 + 6\,c_2 + 8\,c_3)1 + (1\,c_1 + 5\,c_2 + 4\,c_3)\,t + (3\,c_1 + 2\,c_2 + 1\,c_3)\,t^2 = 0$$

This yields the following homogeneous system of equations:

$$\begin{aligned} -4c_1 + 6c_2 + 8c_3 &= 0 \\ 1c_1 + 5c_2 + 4c_3 &= 0 \\ 3c_1 + 2c_2 + 1c_3 &= 0 \end{aligned} \Rightarrow \begin{bmatrix} -4 & 6 & 8 \\ 1 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

As $det \begin{bmatrix} -4 & 6 & 8 \\ 1 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} = -26 \neq 0$. Therefore the set $\boldsymbol{S} = \{\boldsymbol{p_1(t)}, \boldsymbol{p_2(t)}, \boldsymbol{p_3(t)}\}$ is linearly independent. Consequently, the set $\boldsymbol{S}$ forms a basis for span $\boldsymbol{S}$.

**Example 9** The set $\boldsymbol{S} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ is a basis for the vector space $\boldsymbol{V}$ of all 2 x 2 matrices.

**Solution** To verify that $\boldsymbol{S}$ is linearly independent, we form a linear combination of the vectors in $\boldsymbol{S}$ and set it equal to zero:

$$c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

This gives $\begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, which implies that $c_1 = c_2 = c_3 = c_4 = 0$. Hence $\boldsymbol{S}$ is linearly independent.

To verify that $\boldsymbol{S}$ spans $\boldsymbol{V}$ we take any vector $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ in $\boldsymbol{V}$ and we must find scalars $c_1, c_2,$ $c_3$, and $c_4$ such that

$$c_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

We find that $c_1 = a$, $c_2 = b$, $c_3 = c$, and $c_4 = d$ so that $\boldsymbol{S}$ spans $\boldsymbol{V}$.

The basis $\boldsymbol{S}$ in this example is called the standard basis for $\boldsymbol{M_{22}}$. More generally, the standard basis for $\boldsymbol{M_{mn}}$ consists of $mn$ different matrices with a single $1$ and zeros for the remaining entries

**Example 10** Show that the set of vectors

$$\left\{ \begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} \right\}$$

is a basis for the vector space $V$ of all 2 x 2 matrices (i.e. $\boldsymbol{M_{22}}$).

**Solution** The set $S = \{v_1, v_2, v_3, v_4\}$ of vectors in $M_{22}$ spans $V = M_{22}$ if

$$c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 = d_1 w_1 + d_2 w_2 + d_3 w_3 + d_4 w_4 \quad (*)$$

with $w_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, w_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, w_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, w_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

has at least one solution for every set of values of the coefficients $d_1, d_2, d_3, d_4$. Otherwise (i.e., if no solution exists for at least some values of $d_1, d_2, d_3, d_4$), $S$ does not span $V$. With our vectors $v_1, v_2, v_3, v_4$, (*) becomes:

$$c_1 \begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix} + c_2 \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix} + c_4 \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$$

$$= d_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + d_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + d_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Rearranging the left hand side yields

$$\begin{bmatrix} 3c_1 + 0c_2 + 0c_3 + 1c_4 & 6c_1 - 1c_2 - 8c_3 + 0c_4 \\ 3c_1 - 1c_2 - 12c_3 - 1c_4 & -6c_1 + 0c_2 - 4c_3 + 2c_4 \end{bmatrix} =$$

$$\begin{bmatrix} 1d_1 + 0d_2 + 0d_3 + 0d_4 & 0d_1 + 1d_2 + 0d_3 + 0d_4 \\ 0d_1 + 0d_2 + 1d_3 + 0d_4 & 0d_1 + 0d_2 + 0d_3 + 1d_4 \end{bmatrix}$$

The matrix equation above is equivalent to the following system of equations

$$\begin{aligned} 3c_1 + 0c_2 + 0c_3 + 1c_4 &= 1d_1 + 0d_2 + 0d_3 + 0d_4 \\ 6c_1 - 1c_2 - 8c_3 + 0c_4 &= 0d_1 + 1d_2 + 0d_3 + 0d_4 \\ 3c_1 - 1c_2 - 12c_3 - 1c_4 &= 0d_1 + 0d_2 + 1d_3 + 0d_4 \\ -6c_1 + 0c_2 - 4c_3 + 2c_4 &= 0d_1 + 0d_2 + 0d_3 + 1d_4 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} 3 & 0 & 0 & 1 \\ 6 & -1 & -8 & 0 \\ 3 & -1 & -12 & -1 \\ -6 & 0 & -4 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{bmatrix}$$

We now find the determinant of coefficient matrix $A = \begin{bmatrix} 3 & 0 & 0 & 1 \\ 6 & -1 & -8 & 0 \\ 3 & -1 & -12 & -1 \\ -6 & 0 & -4 & 2 \end{bmatrix}$ to determine whether the system is consistent (so that $S$ spans $V$), or inconsistent ($S$ does not span $V$).

Now $det\ (A) = 48 \neq 0$. Therefore, the system (A) is consistent, and, consequently, the set $\boldsymbol{S}$ spans the space $\boldsymbol{V}$.

Now, the set $\boldsymbol{S} = \{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}, \boldsymbol{v_4}\}$ of vectors in $\boldsymbol{M_{22}}$ is linearly independent if the only solution of $c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + c_3\boldsymbol{v_3} + c_4\boldsymbol{v_4} = 0$ is $c_1, c_2, c_3, c_4 = 0$. In this case the set $\boldsymbol{S}$ forms a basis for span $\boldsymbol{S}$. Otherwise (i.e., if a solution with at least some nonzero values exists), $\boldsymbol{S}$ is linearly dependent. With our vectors $\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}, \boldsymbol{v_4}$, we have

$$c_1\begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix} + c_2\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} + c_3\begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix} + c_4\begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Rearranging the left hand side yields

$$\begin{bmatrix} 3c_1 + 0c_2 + 0c_3 + 1c_4 & 6c_1 - 1c_2 - 8c_3 + 0c_4 \\ 3c_1 - 1c_2 - 12c_3 - 1c_4 & -6c_1 + 0c_2 - 4c_3 + 2c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

The matrix equation above is equivalent to the following homogeneous equation.

$$\begin{bmatrix} 3 & 0 & 0 & 1 \\ 6 & -1 & -8 & 0 \\ 3 & -1 & -12 & -1 \\ -6 & 0 & -4 & 2 \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

As $\det(A) = 48 \neq 0$

Therefore the set $\boldsymbol{S} = \{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}, \boldsymbol{v_4}\}$ is linearly independent. Consequently, the set $\boldsymbol{S}$ forms a basis for span $\boldsymbol{S}$.

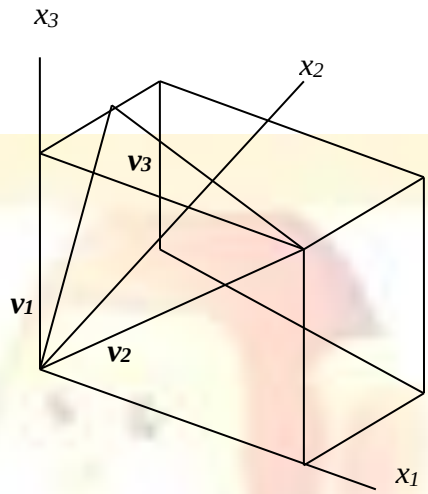
**Example 11** Let $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -2 \\ -3 \end{bmatrix}$, $\boldsymbol{v}_2 = \begin{bmatrix} -3 \\ 5 \\ 7 \end{bmatrix}$, $\boldsymbol{v}_3 = \begin{bmatrix} -4 \\ 5 \\ 6 \end{bmatrix}$, and $\boldsymbol{H} = Span\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$.

Note that $\boldsymbol{v_3} = 5\boldsymbol{v_1} + 3\boldsymbol{v_2}$ and show that Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ = Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$. Then find a basis for the subspace $\boldsymbol{H}$.

**Solution**

Every vector in Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ belongs to $\boldsymbol{H}$ because

$$c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} = c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + 0\boldsymbol{v_3}$$




Now let $\boldsymbol{x}$ be any vector in $\boldsymbol{H}$ – say, $\boldsymbol{x} = c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + c_3\boldsymbol{v_3}$. Since $\boldsymbol{v_3} = 5\boldsymbol{v_1} + 3\boldsymbol{v_2}$, we may substitute

$$\boldsymbol{x} = c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + c_3\,(5\boldsymbol{v_1} + 3\boldsymbol{v_2})$$
$$= (c_1 + 5c_3)\,\boldsymbol{v_1} + (c_2 + 3c_3)\,\boldsymbol{v_2}$$

Thus $\boldsymbol{x}$ is in Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$, so every vector in $\boldsymbol{H}$ already belongs to Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$. We conclude that $\boldsymbol{H}$ and *Span* $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ are actually the same set of vectors. It follows that $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is a basis of $\boldsymbol{H}$ since $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is obviously linearly independent.

**Activity** Show that the following set of vectors is basis for $\mathbb{R}^3$ :

**1.**

$$v_1 = (1, 0, 0),\ v_2 = (0, 2, 1),\ v_1 = (3, 0, 1)$$

**2.**

$$v_1 = (1, 2, 3),\ v_2 = (0, 1, 1),\ v_1 = (0, 1, 3)$$

**The Spanning Set Theorem**

As we will see, a basis is an "efficient" spanning set that contains no unnecessary vectors. In fact, a basis can be constructed from a spanning set by discarding unneeded vectors.

**Theorem 2** **(The Spanning Set Theorem)** Let $\boldsymbol{S} = \{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$ be a set in $\boldsymbol{V}$ and let $\boldsymbol{H}$ = Span $\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_p}\}$.

- **a.** If one of the vectors in $\boldsymbol{S}$ – say, $\boldsymbol{v_k}$ – is a linear combination of the remaining vectors in $\boldsymbol{S}$, then the set formed from $\boldsymbol{S}$ by removing $\boldsymbol{v_k}$ still spans $\boldsymbol{H}$.
- **b.** If $\boldsymbol{H} \neq \{\boldsymbol{0}\}$, some subset of $\boldsymbol{S}$ is a basis for $\boldsymbol{H}$.

Since we know that span is the set of all linear combinations of some set of vectors and basis is a set of linearly independent vectors whose span is the entire vector space. The spanning set is a set of vectors whose span is the entire vector space. "The Spanning set theorem" is that a spanning set of vectors always contains a subset that is a basis.

**Remark** Let $V = R^m$ and let $S = \{v_1, v_2, \ldots, v_n\}$ be a set of nonzero vectors in $V$.

**Procedure**

The procedure for finding a subset of $S$ that is a basis for $W$ = span $S$ is as follows:

**Step 1** Write the Equation,

$$c_1v_1 + c_2v_2 + \ldots + c_n v_n = 0 \qquad (3)$$

**Step 2** Construct the augmented matrix associated with the homogeneous system of Equation (1) and transforms it to reduced row echelon form.

**Step 3** The vectors corresponding to the columns containing the leading 1's form a basis for $W$ = span $S$.

Thus if $S = \{v_1, v_2, \ldots, v_6\}$ and the leading 1's occur in columns *1, 3,* and *4*, then $\{ v_1 , v_3 , v_4\}$ is a basis for span $S$.

**Note** In step 2 of the procedure above, it is sufficient to transform the augmented matrix to row echelon form.

**Example 12** Let $S = \{v_1, v_2, v_3, v_4, v_5\}$ be a set of vectors in $R^4$, where $v_1 = (1,2,-2,1)$, $v_2 = (-3,0,-4,3)$, $v_3 = (2,1,1,-1)$, $v_4 = (-3,3,-9,6)$, and $v_5 = (9,3,7,-6)$. Find a subset of $S$ that is a basis for $W$ = span $S$.

**Solution** Step 1 Form Equation (3),

$c_1 (1,2, -2,1) + c_2(-3,0,-4,3) + c_3(2,1,1,-1) + c_4(-3,3,-9,6) + c_5(9,3.7,-6) = (0,0,0,0)$.

Step 2 Equating corresponding components, we obtain the homogeneous system

$$\begin{aligned} c_1 - 3c_2 + 2c_3 - 3c_4 + 9c_5 &= 0 \\ 2c_1 + c_3 + 3c_4 + 3c_5 &= 0 \\ -2c_1 - 4c_2 + c_3 - 9c_4 + 7c_5 &= 0 \\ c_1 + 3c_2 - c_3 + 6c_4 - 6c_5 &= 0 \end{aligned}$$

The reduced row echelon form of the associated augmented matrix is

$$\begin{bmatrix} 1 & 0 & ½ & 3/2 & 3/2 & : 0 \\ 0 & 1 & -1/2 & 3/2 & -5/2 & : 0 \\ 0 & 0 & 0 & 0 & 0 & : 0 \\ 0 & 0 & 0 & 0 & 0 & : 0 \end{bmatrix}$$

Step 3 The leading 1's appear in columns 1 and 2, so $\{v_1, v_2\}$ is a basis for $W$ = span $S$.

**Two Views of a Basis When the Spanning Set Theorem is used, the deletion of vectors from a spanning set must stop when the set becomes linearly independent**. If an additional vector is deleted, it will not be a linear combination of the remaining vectors and hence the smaller set will no longer span ***V***. Thus a basis is a spanning set that is as small as possible.

A basis is also a linearly independent set that is as large as possible. If ***S*** is a basis for ***V***, and if ***S*** is enlarged by one vector – say, ***w*** – from ***V***, then the new set cannot be linearly independent, because *S* spans *V*, and *w* is therefore a linear combination of the elements in ***S***.

**Example 13** The following three sets in $\boldsymbol{R^3}$ show how a linearly independent set can be enlarged to a basis and how further enlargement destroys the linear independence of the set. Also, a spanning set can be shrunk to a basis, but further shrinking destroys the spanning property.

$$\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix}\right\} \qquad \left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix},\begin{bmatrix}4\\5\\6\end{bmatrix}\right\} \qquad \left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix},\begin{bmatrix}4\\5\\6\end{bmatrix},\begin{bmatrix}7\\8\\9\end{bmatrix}\right\}$$

Linearly independent but does not span $\boldsymbol{R^3}$ — A basis for $\boldsymbol{R^3}$ — Spans $\boldsymbol{R^3}$ but is linearly dependent

**Example 14** Let $\boldsymbol{v}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}$, $\boldsymbol{v}_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}$, and $\boldsymbol{H} = \left\{\begin{bmatrix}s\\s\\0\end{bmatrix} : s \text{ in } R\right\}$. then every vector in ***H*** is a linear combination of $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ because $\begin{bmatrix}s\\s\\0\end{bmatrix} = s\begin{bmatrix}1\\0\\0\end{bmatrix} + s\begin{bmatrix}0\\1\\0\end{bmatrix}$. Is $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ a basis for ***H***?

**Solution** Neither $\boldsymbol{v_1}$ nor $\boldsymbol{v_2}$ is in ***H***, so $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ cannot a basis for ***H***. In fact, $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is a basis for the plane of all vectors of the form $(c_1, c_2, 0)$, but ***H*** is only a line.

**Activity** Find a Basis for the subspace W in $\mathbb{R}^3$ spanned by the following sets of vectors:

1. $v_1 = (1,0,2), v_2 = (3,2,1), v_3 = (1,0,6), v_4 = (3,2,1)$
2. $v_1 = (1,2,2), v_2 = (3,2,1), v_3 = (1,1,7), v_4 = (7,6,4)$

**Exercises**

Determine which set in exercises 1-4 are bases for $R^2$ or $R^3$. Of the sets that are not bases, determine which one are linearly independent and which ones span $R^2$ or $R^3$. Justify your answers.

1. $\begin{bmatrix}1\\0\\-2\end{bmatrix}, \begin{bmatrix}3\\2\\-4\end{bmatrix}, \begin{bmatrix}-3\\-5\\1\end{bmatrix}$

2. $\begin{bmatrix}1\\-3\\0\end{bmatrix}, \begin{bmatrix}-2\\9\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\-3\\5\end{bmatrix}$

3. $\begin{bmatrix}1\\2\\-3\end{bmatrix}, \begin{bmatrix}-4\\-5\\6\end{bmatrix}$

4. $\begin{bmatrix}1\\-4\\3\end{bmatrix}, \begin{bmatrix}0\\3\\-1\end{bmatrix}, \begin{bmatrix}3\\-5\\4\end{bmatrix}, \begin{bmatrix}0\\2\\-2\end{bmatrix}$

5. Find a basis for the set of vectors in $R^3$ in the plane $x + 2y + z = 0$.

6. Find a basis for the set of vectors in $R^2$ on the line $y = 5x$.

7. Suppose $R^4 = \text{Span}\{v_1, v_2, v_3, v_4\}$. Explain why $\{v_1, v_2, v_3, v_4\}$ is a basis for $R^4$.

8. Explain why the following sets of vectors are not bases for the indicated vector spaces. (Solve this problem by inspection).
(a) $u_1 = (1, 2)$, $u_2 = (0, 3)$, $u_3 = (2, 7)$ for $R^2$
(b) $u_1 = (-1, 3, 2)$, $u_2 = (6, 1, 1)$ for $R^3$
(c) $p_1 = 1 + x + x^2$, $p_2 = x - 1$ for $P_2$
(d) $A = \begin{bmatrix}1 & 1\\2 & 3\end{bmatrix}, B = \begin{bmatrix}6 & 0\\-1 & 4\end{bmatrix}, C = \begin{bmatrix}3 & 0\\1 & 7\end{bmatrix}, D = \begin{bmatrix}5 & 1\\4 & 2\end{bmatrix}, E = \begin{bmatrix}7 & 1\\2 & 9\end{bmatrix}$ for $M_{22}$

9. Which of the following sets of vectors are bases for $R^2$?
(a) (2, 1), (3, 0) (b) (4, 1), (-7, -8) (c) (0, 0), (1, 3) (d) (3, 9), (-4, -12)

10. Let $V$ be the space spanned by $v_1 = \text{Cos}^2 x$, $v_2 = \text{Sin}^2 x$, $v_3 = \cos 2x$.
(a) Show that $S = \{v_1, v_2, v_3\}$ is not a basis for $V$ (b) Find a basis for $V$

In exercises 11-13, determine a basis for the solution space of the system.

11.
$$\begin{aligned} x_1 + x_2 - x_3 &= 0\\ -2x_1 - x_2 + 2x_3 &= 0\\ -x_1 + x_3 &= 0 \end{aligned}$$

12.
$$\begin{aligned} 2x_1 + x_2 + 3x_3 &= 0\\ x_1 + 5x_3 &= 0\\ x_2 + x_3 &= 0 \end{aligned}$$

13.
$$x + y + z = 0$$
$$3x + 2y - 2z = 0$$
$$4x + 3y - z = 0$$
$$6x + 5y + z = 0$$

14. Determine bases for the following subspace of $\boldsymbol{R^3}$

(a) the plane $3x - 2y + 5z = 0$ (b) the plane $x - y = 0$

(c) the line $x = 2t$, $y = -t$, $z = 4t$ (d) all vectors of the form (a, b, c), where $b = a + c$

15. Find a standard basis vector that can be added to the set $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ to produce a basis for $\boldsymbol{R^3}$.

(a) $\boldsymbol{v_1} = (-1, 2, 3)$, $\boldsymbol{v_2} = (1, -2, -2)$ (b) $\boldsymbol{v_1} = (1, -1, 0)$, $\boldsymbol{v_2} = (3, 1, -2)$

16. Find a standard basis vector that can be added to the set $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ to produce a basis for $\boldsymbol{R^4}$.

$\boldsymbol{v_1} = (1, -4, 2, -3)$, $\boldsymbol{v_2} = (-3, 8, -4, 6)$

# Lecture No.23
## Coordinate System

**OBJECTIVES**

The objectives of the lecture are to learn about:

- Unique representation theorem.
- Coordinate of the element of a vector space relative to the basis B.
- Some examples in which B- coordinate vector is uniquely determined using basis of a vector space.
- Graphical interpretation of coordinates.
- Coordinate Mapping

**Theorem**

Let $B=\{b_1,b_2,...,b_n\}$ be a basis for a vector space V. Then for each $x$ in $V$, there exist a unique set of scalars $c_1,c_2,...,c_n$ such that

$$x=c_1b_1,c_2b_2,...,c_nb_n \quad \text{.....(1)}$$

**Proof**

Since $B$ is a basis for a vector space $V$, then by definition of basis every element of $V$ can be written as a linear combination of basis vectors. That is if $x\in V$, then $x=c_1b_1,c_2b_2,...,c_nb_n$. Now, we show that this representation for $x$ is unique.
For this, suppose that we have two representations for $x$.
i.e.

$$x=c_1b_1,c_2b_2,...,c_nb_n \quad \text{.....(2)}$$

and

$$x=d_1b_1,d_2b_2,...,d_nb_n \quad \text{.....(3)}$$

We will show that the coefficients are actually equal. To do this, subtracting (3) from (2), we have

$$0=(c_1-d_1)b_1+(c_2-d_2)b_2+...+(c_n-d_n)b_n.$$

Since $B$ is a basis, it is linearly independent set. Thus the coefficients in the last linear combination must all be zero. That is

$$c_1=d_1,...,c_n=d_n.$$

Thus the representation for $x$ is unique.

**Definition (B-Coordinate of $x$)**

Suppose that the set $B=\{b_1,b_2,...,b_n\}$ is a basis for $V$ and $x$ is in $V$. The coordinates of $x$ relative to basis $B$ (**or the *B*-coordinate of** $x$) are the weights $c_1,c_2,...,c_n$ such that

$$x=c_1b_1,c_2b_2,...,c_nb_n.$$

**Note**

If $c_1, c_2, ..., c_n$ are the ***B***- coordinates of $x$, then the vector in $R^n$, $[x]_B = \begin{bmatrix} c_1 \\ c_2 \\ \cdot \\ \cdot \\ \cdot \\ c_n \end{bmatrix}$ is the coordinate vector of $x$ (relative to ***B)*** or ***B***- coordinates of $x$.

**Example 1**

Consider a basis $B = \{b_1, b_2\}$ for $R^2$, where $b_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Suppose an $x$ in $R^2$ has the coordinate vector $[x]_B = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$. Find $x$

**Solution**

Using above definition $x$ is uniquely determined using coordinate vector and the basis. That is

$$\begin{aligned} x &= c_1 b_1, c_2 b_2 \\ &= (-2)b_1 + (3)b_2 \\ &= (-2)\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} -2 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ 6 \end{bmatrix} \\ x &= \begin{bmatrix} 1 \\ 6 \end{bmatrix} \end{aligned}$$

**Example 2**

Let $\boldsymbol{S} = \{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ be the basis for $\boldsymbol{R^3}$, where $\boldsymbol{v_1}$ = (1, 2, 1), $\boldsymbol{v_2}$ = (2, 9, 0), and $\boldsymbol{v_3}$ = (3, 3, 4).

(a) Find the coordinates vector of $\boldsymbol{v}$ = (5, -1, 9) with respect to ***S***.

(b) Find the vector $\boldsymbol{v}$ in $\boldsymbol{R^3}$ whose coordinate vector with respect to the basis ***S*** is

$[\boldsymbol{v}]_S$ = (-1, 3, 2)

**Solution**

Since **S** is a basis for $\mathbf{R^3}$, Thus

$x = c_1 v_1 + c_2 v_2 + c_3 v_3$.

Further

$(5,-1,9) = c_1(1,2,1) + c_2(2,9,0) + c_3(3,3,4)$ .....(A)

To find the coordinate vector of $\boldsymbol{v}$, we have to find scalars $c_1, c_2, c_3$.

For this equating corresponding components in (A) gives

$c_1 + 2c_2 + 3c_3 = 5$ (1)

$2c_1 + 9c_2 + 3c_3 = -1$ (2)

$c_1 + 4c_3 = 9$ (3)

*Now find values of* $c_1, c_2$ *and* $c_3$ *from these equations.*

*From equation (3)*

$c_1 = 9 - 4c_3$

*Put this value of* $c_1$ *in equations* (1) *and* (2)

$9 - 4c_3 + 2c_2 + 3c_3 = 5$

$2c_2 - c_3 = -4$ (4)

*and*

$2(9 - 4c_3) + 9c_2 + 3c_3 = -1$

$18 - 8c_3 + 9c_2 + 3c_3 = -1$

$9c_2 - 5c_3 = -19$ (5)

*Multiply equation* (4) *by 5*

$10c_2 - 5c_3 = -20$

*Subtract equation* (5) *from above equation*

$10c_2 - 5c_3 = -20$

$\pm 9c_2 \mp 5c_3 = \mp 19$

---

$c_2 = -1$

*Put value of* $c_2$ *in equation* (4) *to get* $c_3$

$2(-1) - c_3 = -4$

$-2 - c_3 = -4$

$c_3 = 4 - 2 = 2$

*Put value of* $c_3$ *in equation* (3) *to get* $c_1$

$c_1 + 4(2) = 9$

$c_1 = 9 - 8 = 1$

Thus, we obtain $c_1 = 1, c_2 = -1, c_3 = 2$

Therefore, $[\boldsymbol{v}]_S = (1, -1, 2)$

Using the definition of coordinate vector, we have

$$v = c_1v_1 + c_2v_2 + c_3v_3$$
$$= (-1)v_1 + 3v_2 + 2v_3$$
$$= (-1)(1,2,1) + 3(2,9,0) + 2(3,3,4)$$
$$= (-1+6+6, -2+27+6, -1+0+8)$$
$$= (11,31,7)$$

Therefore

$$v = (11,31,7)$$

**Example 3**

Find the coordinates vector of the polynomial $\boldsymbol{p} = \boldsymbol{a_0} + a_1\boldsymbol{x} + a_2\boldsymbol{x^2}$ relative to the basis $\boldsymbol{S} = \{\boldsymbol{1}, \boldsymbol{x}, \boldsymbol{x^2}\}$ for $\boldsymbol{p_2}$.

**Solution**

To find the coordinator vector of the polynomial $p$ , we write it as a linear combination of the basis set $S$ . That is

$$a_0 + a_1x + a_2x^2 = c_1(1) + c_2(x) + c_3(x^2)$$
$$\Rightarrow c_1 = a_0, c_2 = a_1, c_3 = a_2$$

Therefore

$$[p]_s = (a_0, a_1, a_2)$$

**Example 4**

Find the coordinates vector of the polynomial $\boldsymbol{p} = \boldsymbol{5} - \boldsymbol{4x} + \boldsymbol{3x^2}$ relative to the basis $\boldsymbol{S} = \{\boldsymbol{1}, \boldsymbol{x}, \boldsymbol{x^2}\}$ for $\boldsymbol{p_2}$.

**Solution**

To find the coordinator vector of the polynomial $p$ , we write it as a linear combination of the basis set $S$ . That is

$$5 - 4x + 3x^2 = c_1(1) + c_2(x) + c_3(x^2)$$
$$\Rightarrow c_1 = 5, c_2 = -4, c_3 = 3$$

Therefore

$$[p]_s = (5, -4, 3)$$

**Example 5**

Find the coordinate vector of A relative to the basis S = {A1, A2, A3, A4}

$$\boldsymbol{A} = \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}; \boldsymbol{A}_1 = \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}; \boldsymbol{A_2} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}; \boldsymbol{A_3} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}; \boldsymbol{A_4} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

**Solution**

To find the coordinator vector of $A$ , we write it as a linear combination of the basis set $S$ . That is

$\boldsymbol{A} = c_1 \boldsymbol{A_1} + c_2 \boldsymbol{A_2} + c_3 \boldsymbol{A_3} + c_4 \boldsymbol{A_4}$

$$\begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix} = c_1 \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -c_1 & c_1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} c_2 & c_2 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ c_3 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & c_4 \end{bmatrix}$$

$$= \begin{bmatrix} -c_1 + c_2 + 0 + 0 & c_1 + c_2 + 0 + 0 \\ 0 + 0 + c_3 + 0 & 0 + 0 + 0 + c_4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} -c_1 + c_2 & c_1 + c_2 \\ c_3 & c_4 \end{bmatrix}$$

$$-c_1 + c_2 = 2 \quad (1)$$

$$c_1 + c_2 = 0 \quad (2)$$

$$c_3 = -1 \quad (3)$$

$$c_4 = 3 \quad (4)$$

Adding (1) and (2), gives

$2c_2 = 2 \Rightarrow c_2 = 1$

Putting the value of $c_2$ in (2) to get $c_1$, $c_1 = -1$

So $c_1 = -1, c_2 = 1, c_3 = -1, c_4 = 3$

Therefore, $[\boldsymbol{v}]_S = (-1, 1, -1, 3)$

**Graphical Interpretation of Coordinates**

A coordinate system on a set consists of a one-to-one mapping of the points in the set into $\boldsymbol{R^n}$. For example, ordinary graph paper provides a coordinate system for the plane when one selects perpendicular axes and a unit of measurement on each axis. Figure 1 shows the standard basis $\{\boldsymbol{e_1}, \boldsymbol{e_2}\}$, the vectors $\boldsymbol{b_1}$ (= $\boldsymbol{e_1}$) and $\boldsymbol{b_2}$ from Example 1, that is,

$\boldsymbol{b}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\boldsymbol{b}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Vector $\boldsymbol{x} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$, the coordinates 1 and 6 give the location of $\boldsymbol{x}$ relative to the standard basis: 1 unit in the $\boldsymbol{e_1}$ direction and 6 units in the $\boldsymbol{e_2}$ direction.

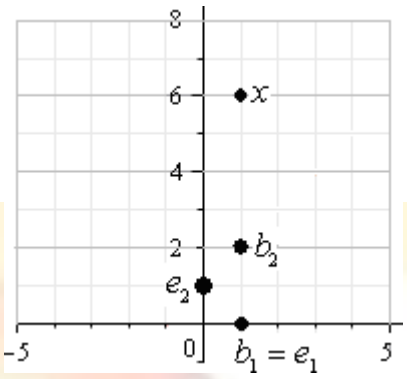



**Figure 1**

Figure 2 shows the vectors $\boldsymbol{b_1}$, $\boldsymbol{b_2}$, and $\boldsymbol{x}$ from Figure 1. (Geometrically, the three vectors lie on a vertical line in both figures.) However, the standard coordinate grid was erased and replaced by a grid especially adapted to the basis $\boldsymbol{B}$ in Example 1. The coordinate vector $[\boldsymbol{x}]_B = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ gives the location of $\boldsymbol{x}$ on this new coordinate system: – 2 units in the $\boldsymbol{b_1}$ direction and *3* units in the $\boldsymbol{b_2}$ direction.

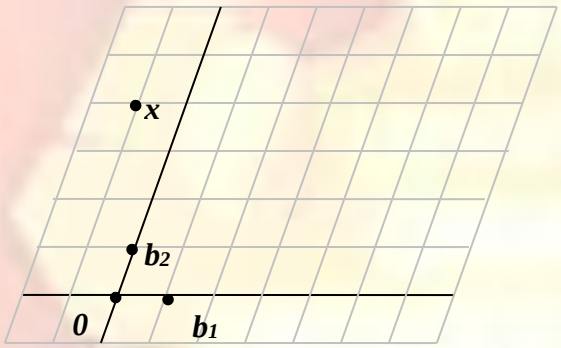
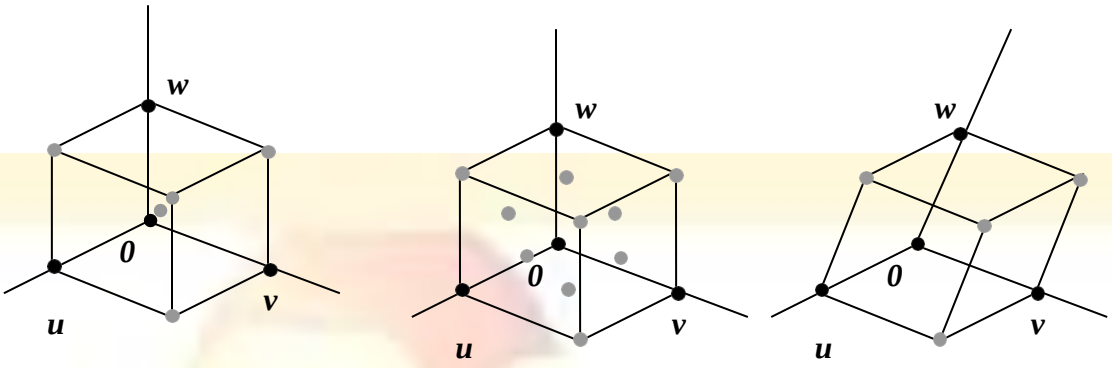



Figure 2

**Example 6**

In crystallography, the description of a crystal lattice is aided by choosing a basis $\{\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}\}$ for $\boldsymbol{R^3}$ that corresponds to three adjacent edges of one "unit cell" of the crystal. An entire lattice is constructed by stacking together many copies of one cell. There are fourteen basic types of unit cells; three are displayed in Figure 3.




(a) Body centered cubic

(b) Face centered orthorhombic

(c) Simple monoclinic

**Figure 3 – Examples of unit cells**

The coordinates of atoms within the crystal are given relative to the basis for the lattice. For instance, $\begin{bmatrix} 1/2 \\ 1/2 \\ 1 \end{bmatrix}$ identifies the top face-centered atom in the cell in Figure 3(b).

**Coordinates in $R^n$** When a basis $\boldsymbol{B}$ for $R^n$ is fixed, the $\boldsymbol{B}$-coordinate vector of a specified $\boldsymbol{x}$ is easily found, as in the next example.

**Example 7** Let $\boldsymbol{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \boldsymbol{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \boldsymbol{x} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, and $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2\}$.

Find the coordinate vector $[\boldsymbol{x}]_{\boldsymbol{B}}$ of $\boldsymbol{x}$ relative to $\boldsymbol{B}$.

**Solution** The $\boldsymbol{B}$ – coordinates $c_1$ , $c_2$ of $\boldsymbol{x}$ satisfy

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$\boldsymbol{b}_1$ $\boldsymbol{b}_2$ $\boldsymbol{x}$

*or*

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \qquad (3)$$

$\boldsymbol{b}_1$ $\boldsymbol{b}_2$ $\boldsymbol{x}$

Now, inverse of matrix $\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ -1 & 2 \\ \frac{}{3} & \frac{}{3} \end{bmatrix}$

From equation (3) we get

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{-1}{3} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{3}(4) + \frac{1}{3}(5) \\ \frac{-1}{3}(4) + \frac{2}{3}(5) \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Thus, $c_1 = 3$, $c_2 = 2$.
(Equation (3) can also be solved by row operations on an augmented matrix. Try it yourself )
Thus $\boldsymbol{x} = 3\boldsymbol{b}_1 + 2\boldsymbol{b}_2$ and $[\boldsymbol{x}]_{\boldsymbol{B}} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

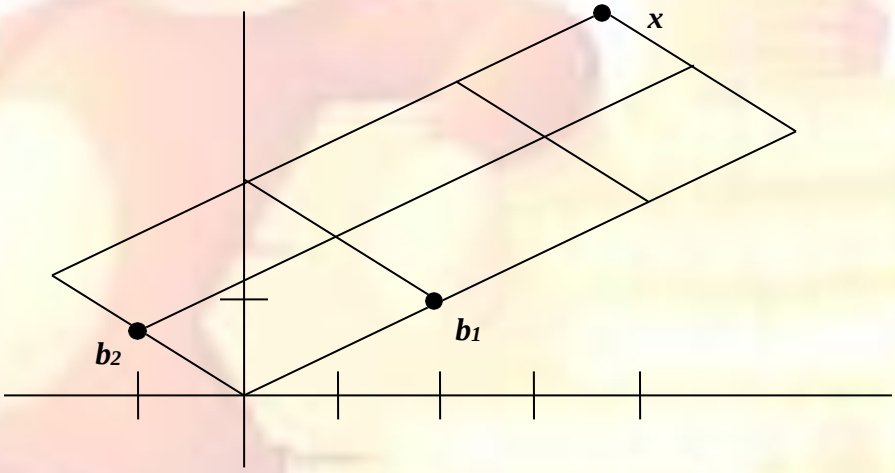



**Figure 4 – The *B*-coordinate vector of *x* is (3,2)**

The matrix in (3) changes the $\boldsymbol{B}$-coordinates of a vector $\boldsymbol{x}$ into the standard coordinates for $\boldsymbol{x}$. An analogous change of coordinates can be carried out in $\boldsymbol{R}^n$ for a basis $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_n\}$.

Let $\boldsymbol{P}_B = [\boldsymbol{b}_1 \quad \boldsymbol{b}_2 \quad \ldots \quad \boldsymbol{b}_n]$

Then the vector equation $\boldsymbol{x} = c_1\boldsymbol{b}_1 + c_2\boldsymbol{b}_2 + \ldots + c_n\boldsymbol{b}_n$

is equivalent to $\boldsymbol{x} = \boldsymbol{P}_B[\boldsymbol{x}]_B$ (4)

We call $\boldsymbol{P}_B$ the **change-of-coordinates matrix** from $\boldsymbol{B}$ to the standard basis in $\boldsymbol{R}^n$.
Left-multiplication by $\boldsymbol{P}_B$ transforms the coordinate vector $[\boldsymbol{x}]_B$ into $\boldsymbol{x}$. The change-of-coordinates equation (4) is important and will be needed at several points in next lectures.

Since the columns of $\boldsymbol{P_B}$ form a basis for $\boldsymbol{R^n}$, $\boldsymbol{P_B}$ is invertible (by the Invertible Matrix Theorem). Left-multiplication by $\boldsymbol{P_B^{-1}}$ converts $\boldsymbol{x}$ into its $\boldsymbol{B}$-coordinate vector:

$$\boldsymbol{P_B^{-1}x = [x]_B}$$

The correspondence $\boldsymbol{x \to [x]_B}$ produced here by $\boldsymbol{P_B^{-1}}$, is the coordinate mapping mentioned earlier. Since $\boldsymbol{P_B^{-1}}$ is an invertible matrix, the coordinate mapping is a one-to-one linear transformation from $\boldsymbol{R^n}$ onto $\boldsymbol{R^n}$, by the Invertible Matrix Theorem. (See also Theorem 3 in lecture 10) This property of the coordinate mapping is also true in a general vector space that has a basis, as we shall see.

**The Coordinate Mapping** Choosing a basis $\boldsymbol{B} = \{\boldsymbol{b_1}, \boldsymbol{b_2}, \ldots, \boldsymbol{b_n}\}$ for a vector space $\boldsymbol{V}$ introduces a coordinate system in $\boldsymbol{V}$. The coordinate mapping $\boldsymbol{x \to [x]_B}$ connects the possibly unfamiliar space $\boldsymbol{V}$ to the familiar space $\boldsymbol{R^n}$. See Figure 5. Points in $\boldsymbol{V}$ can now be identified by their new "names".

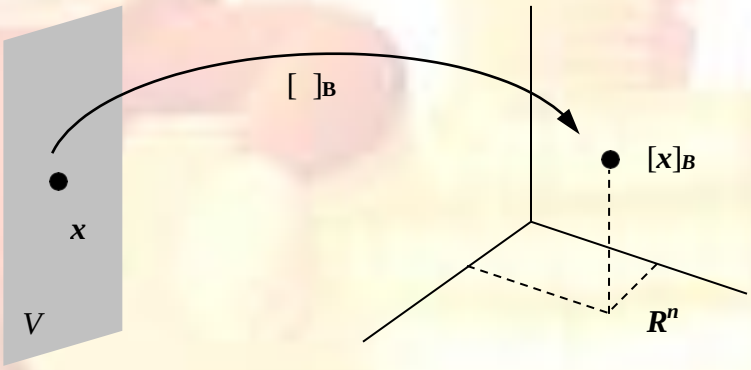



**Figure 5 – The coordinate mapping from $\boldsymbol{V}$ onto $\boldsymbol{R^n}$**

**Theorem 2** Let $\boldsymbol{B} = \{\boldsymbol{b_1}, \boldsymbol{b_2}, \ldots, \boldsymbol{b_n}\}$ be a basis for a vector space $\boldsymbol{V}$. Then the coordinate mapping $\boldsymbol{x \to [x]_B}$ is a one-to-one linear transformation from $\boldsymbol{V}$ onto $\boldsymbol{R^n}$.

**Proof** Take two typical vectors in $\boldsymbol{V}$, say

$$\boldsymbol{u} = c_1\boldsymbol{b_1} + c_2\boldsymbol{b_2} + \ldots + c_n\boldsymbol{b_n}$$

$$\boldsymbol{w} = d_1\boldsymbol{b_1} + d_2\boldsymbol{b_2} + \ldots + d_n\boldsymbol{b_n}$$

Then, using vector operations, $\boldsymbol{u+w} = (c_1 + d_1)\boldsymbol{b_1} + (c_2 + d_2)\boldsymbol{b_2} + \ldots + (c_n + d_n)\boldsymbol{b_n}$

It follows that $[\boldsymbol{u+w}]_{\boldsymbol{B}} = \begin{bmatrix} c_1 + d_1 \\ \vdots \\ c_n + d_n \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} + \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix} = [\boldsymbol{u}]_{\boldsymbol{B}} + [\boldsymbol{w}]_{\boldsymbol{B}}$

Thus the coordinate mapping preserves addition. If $r$ is any scalar, then

$$r\boldsymbol{u} = r(c_1\boldsymbol{b_1} + c_2\boldsymbol{b_2} + \cdots + c_n\boldsymbol{b_n}) = (rc_1)\boldsymbol{b_1} + (rc_2)\boldsymbol{b_2} + \cdots + (rc_n)\boldsymbol{b_n}$$

So $$[r\boldsymbol{u}]_{\boldsymbol{B}} = \begin{bmatrix} rc_1 \\ \vdots \\ rc_n \end{bmatrix} = r\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = r[\boldsymbol{u}]_{\boldsymbol{B}}$$

Thus the coordinate mapping also preserves scalar multiplication and hence is a linear transformation. It can be verified that the coordinate mapping is one-to-one and maps $\boldsymbol{V}$ onto $\boldsymbol{R^n}$.

The linearity of the coordinate mapping extends to linear combinations, just as in lecture 9. If $\boldsymbol{u_1}, \boldsymbol{u_2}, \ldots, \boldsymbol{u_p}$ are in $\boldsymbol{V}$ and if $c_1, c_2, \ldots, c_p$ are scalars, then

$$[c_1\boldsymbol{u_1} + c_2\boldsymbol{u_2} + \ldots + c_p\boldsymbol{u_p}]_{\boldsymbol{B}} = c_1[\boldsymbol{u_1}]_{\boldsymbol{B}} + c_2[\boldsymbol{u_2}]_{\boldsymbol{B}} + \ldots + c_p[\boldsymbol{u_p}]_{\boldsymbol{B}} \quad (5)$$

In words, (5) says that the $\boldsymbol{B}$-coordinate vector of a linear combination of $\boldsymbol{u_1}, \boldsymbol{u_2}, \ldots, \boldsymbol{u_p}$ is the same linear combination of their coordinate vectors.

The coordinate mapping in Theorem 2 is an important example of an isomorphism from $\boldsymbol{V}$ onto $\boldsymbol{R^n}$. In general, a one-to-one linear transformation from a vector space $\boldsymbol{V}$ onto a vector space $\boldsymbol{W}$ is called an **isomorphism** from $\boldsymbol{V}$ onto $\boldsymbol{W}$ (iso from the Greek for "the same", and morph from the Greek for "form" or "structure"). The notation and terminology for $\boldsymbol{V}$ and $\boldsymbol{W}$ may differ, but the two spaces are indistinguishable as vector spaces. Every vector space calculation in $\boldsymbol{V}$ is accurately reproduced in $\boldsymbol{W}$, and vice versa.

**Example 8** Let $\boldsymbol{B}$ be the standard basis of the space $\boldsymbol{P_3}$ of polynomials; that is, let $\boldsymbol{B} = \{\boldsymbol{1}, \boldsymbol{t}, \boldsymbol{t^2}, \boldsymbol{t^3}\}$. A typical element $\boldsymbol{p}$ of $\boldsymbol{P_3}$ has the form $\boldsymbol{p}(\boldsymbol{t}) = \boldsymbol{a_0} + a_1\boldsymbol{t} + a_2\boldsymbol{t^2} + a_3\boldsymbol{t^3}$
Since $\boldsymbol{p}$ is already displayed as a linear combination of the standard basis vectors, we conclude that $[\boldsymbol{p}]_{\boldsymbol{B}} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$. Thus the coordinate mapping $\boldsymbol{p} \to [\boldsymbol{p}]_{\boldsymbol{B}}$ is an isomorphism from $\boldsymbol{P_3}$ onto $\boldsymbol{R^4}$. All vector space operations in $\boldsymbol{P_3}$ correspond to operations in $\boldsymbol{R^4}$.

If we think of $\boldsymbol{P_3}$ and $\boldsymbol{R^4}$ as displays on two computer screens that are connected via the coordinate mapping, then every vector space operation in $\boldsymbol{P_3}$ on one screen is exactly duplicated by a corresponding vector operation in $\boldsymbol{R^4}$ on the other screen. The vectors on the $\boldsymbol{P_3}$ screen look different from those on the $\boldsymbol{R^4}$ screen, but they "act" as vectors in exactly the same way. See Figure 6.

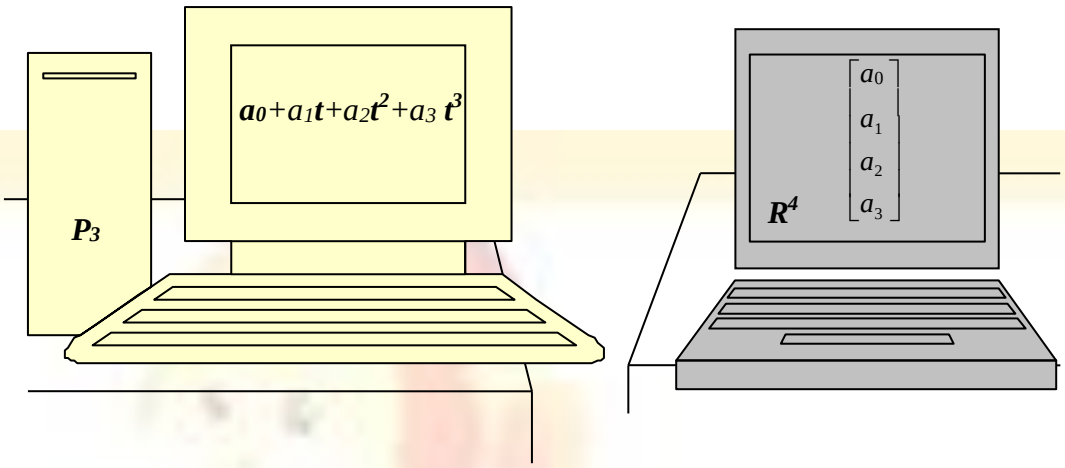



**Figure 6 – The space $P_3$ is isomorphic to $R^4$**

**Example 9** Use coordinate vector to verify that the polynomials $\mathbf{1} + 2\mathbf{t}^2$, $\mathbf{4} + \mathbf{t} + 5\mathbf{t}^2$ and $\mathbf{3} + 2\mathbf{t}$ are linearly dependent in $\mathbf{P}_2$.

**Solution** The coordinate mapping from Example 8 produces the coordinate vectors $(1, 0, 2)$, $(4, 1, 5)$ and $(3, 2, 0)$, respectively. Writing these vectors as the columns of a matrix $\mathbf{A}$, we can determine their independence by row reducing the augmented matrix for $\mathbf{Ax} = \mathbf{0}$:

$$\begin{bmatrix} 1 & 4 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 2 & 5 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The columns of $\mathbf{A}$ are linearly dependent, so the corresponding polynomials are linearly dependent. In fact, it is easy to check that column 3 of $\mathbf{A}$ is 2 times column 2 minus 5 times column 1. The corresponding relation for the polynomials is

$$\mathbf{3} + 2\mathbf{t} = 2(\mathbf{4} + \mathbf{t} + 5\mathbf{t}^2) - 5(\mathbf{1} + 2\mathbf{t}^2)$$

**Example 10** Let $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$, and $\mathbf{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$. Then $\mathbf{B}$ is a basis for $\mathbf{H} = Span\{\mathbf{v}_1, \mathbf{v}_2\}$. Determine if $\mathbf{x}$ is in $\mathbf{H}$ and if it is, find the coordinate vector of $\mathbf{x}$ relative to $\mathbf{B}$.

**Solution** If $\mathbf{x}$ is in $\mathbf{H}$, then the following vector equation is consistent.

$$c_1 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

The scalars, $c_1$ and $c_2$, if they exist, are the $\mathbf{B}$ – coordinates of $\mathbf{x}$.

Using row operations, we obtain $\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$.

Thus $c_1 = 2$, $c_2 = 3$ and $[\boldsymbol{x}]_{\boldsymbol{B}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. The coordinate system on $\boldsymbol{H}$ determined by $\boldsymbol{B}$ is shown in Figure 7.

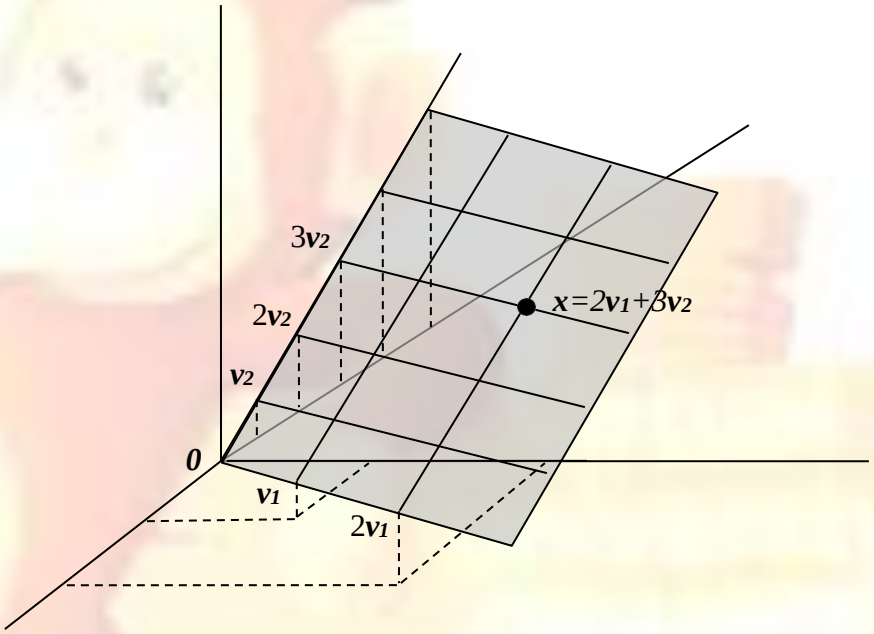



**Figure 7 – A coordinate system on a plane $H$ in $R^3$**

If a different basis for $\boldsymbol{H}$ were chosen, would the associated coordinate system also make $\boldsymbol{H}$ isomorphic to $\boldsymbol{R}^2$? Surely, this must be true. We shall prove it in the next lecture.

**Example 11** Let $\boldsymbol{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\boldsymbol{b}_2 = \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix}$, $\boldsymbol{b}_3 = \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix}$, and $\boldsymbol{x} = \begin{bmatrix} -8 \\ 2 \\ 3 \end{bmatrix}$.

a. Show that the set $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3\}$ is a basis of $\boldsymbol{R}^3$.
b. Find the change-of-coordinates matrix from $\boldsymbol{B}$ to the standard basis.
c. Write the equation that relates $\boldsymbol{x}$ in $\boldsymbol{R}^3$ to $[\boldsymbol{x}]_{\boldsymbol{B}}$.
d. Find $[\boldsymbol{x}]_{\boldsymbol{B}}$, for the $\boldsymbol{x}$ given above.

**Solution**

a. It is evident that the matrix $\boldsymbol{P}_{\boldsymbol{B}} = [\boldsymbol{b}_1 \ \ \boldsymbol{b}_2 \ \ \boldsymbol{b}_3]$ is row equivalent to the identity matrix. By the Invertible Matrix Theorem, $\boldsymbol{P}_{\boldsymbol{B}}$ is invertible and its columns form a basis for $\boldsymbol{R}^3$.

b. From part (a), the change-of-coordinates matrix is $\boldsymbol{P}_{\boldsymbol{B}} = \begin{bmatrix} 1 & -3 & 3 \\ 0 & 4 & -6 \\ 0 & 0 & 3 \end{bmatrix}$.

c. $\boldsymbol{x} = \boldsymbol{P}_{\boldsymbol{B}}[\boldsymbol{x}]_{\boldsymbol{B}}$.

d. To solve part (c), it is probably easier to row reduce an augmented matrix instead of computing $\boldsymbol{P}_{\boldsymbol{B}}^{-1}$. We have

$$\begin{bmatrix} 1 & -3 & 3 & -8 \\ 0 & 4 & -6 & 2 \\ 0 & 0 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$\boldsymbol{P}_{\boldsymbol{B}}$ $\quad$ $\boldsymbol{x}$ $\quad$ $\boldsymbol{I}$ $\quad$ $[\boldsymbol{x}]_{\boldsymbol{B}}$

Hence $[\boldsymbol{x}]_{\boldsymbol{B}} = \begin{bmatrix} -5 \\ 2 \\ 1 \end{bmatrix}$

**Example 12** The set $\boldsymbol{B} = \{\boldsymbol{1} + \boldsymbol{t}, \boldsymbol{1} + \boldsymbol{t}^2, \boldsymbol{t} + \boldsymbol{t}^2\}$ is a basis for $\boldsymbol{P}_2$. Find the coordinate vector of $\boldsymbol{p}(\boldsymbol{t}) = \boldsymbol{6} + 3\boldsymbol{t} - \boldsymbol{t}^2$ relative to $\boldsymbol{B}$.

**Solution** The coordinates of $\boldsymbol{p}(\boldsymbol{t}) = \boldsymbol{6} + 3\boldsymbol{t} - \boldsymbol{t}^2$ with respect to $\boldsymbol{B}$ satisfy

$$c_1(\boldsymbol{1} + \boldsymbol{t}) + c_2(\boldsymbol{1} + \boldsymbol{t}^2) + c_3(\boldsymbol{t} + \boldsymbol{t}^2) = \boldsymbol{6} + 3\boldsymbol{t} - \boldsymbol{t}^2$$

$$c_1 + c_1 t + c_2 + c_2 t^2 + c_3 t + c_3 t^2 = 6 + 3t - t^2$$

$$c_1 + c_2 + c_1 t + c_3 t + c_2 t^2 + c_3 t^2 = 6 + 3t - t^2$$

$$c_1 + c_2 + (c_1 + c_3)t + (c_2 + c_3)t^2 = 6 + 3t - t^2$$

Equating coefficients of like powers of $\boldsymbol{t}$, we have

$c_1 + c_2 = 6$ ---------- (1)

$c_1 + c_3 = 3$ ---------- (2)

$c_2 + c_3 = -1$ ---------- (3)

*Subtract equation* (2) *from* (1) *we get*

$c_2 - c_3 = 6 - 3 = 3$

*Add this equation with equation* (3)

$2c_2 = -1 + 3 = 2$

$\Rightarrow c_2 = 1$

*Put value of* $c_2$ *in equation* (3)

$1 + c_3 = -1$

$\Rightarrow c_3 = -2$

*From equation* (1) *we have*

$c_1 + c_2 = 6$

$c_1 = 6 - 1 = 5$

Solving, we find that $c_1 = 5$, $c_2 = 1$, $c_3 = -2$, and $[\boldsymbol{p}]_{\boldsymbol{B}} = \begin{bmatrix} 5 \\ 1 \\ -2 \end{bmatrix}$.

**Exercises**

In exercises 1 and 2, find the vector $\boldsymbol{x}$ determined by the given coordinate vector $[\boldsymbol{x}]_{\boldsymbol{B}}$ and the given basis $\boldsymbol{B}$.

1. $\boldsymbol{B} = \left\{ \begin{bmatrix} 3 \\ -5 \end{bmatrix}, \begin{bmatrix} -4 \\ 6 \end{bmatrix} \right\}, [\boldsymbol{x}]_{\boldsymbol{B}} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$

2. $\boldsymbol{B} = \left\{ \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ -2 \end{bmatrix}, \begin{bmatrix} 4 \\ -7 \\ 0 \end{bmatrix} \right\}, [\boldsymbol{x}]_{\boldsymbol{B}} = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$

In exercises 3-6, find the coordinate vector $[\boldsymbol{x}]_{\boldsymbol{B}}$ of $\boldsymbol{x}$ relative to the given basis $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_n\}$.

3. $\boldsymbol{b}_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}, \boldsymbol{b}_2 = \begin{bmatrix} 2 \\ -5 \end{bmatrix}, \boldsymbol{x} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

4. $\boldsymbol{b}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \boldsymbol{b}_2 = \begin{bmatrix} 5 \\ -6 \end{bmatrix}, \boldsymbol{x} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$

5. $\boldsymbol{b}_1 = \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix}, \boldsymbol{b}_2 = \begin{bmatrix} -3 \\ 4 \\ 9 \end{bmatrix}, \boldsymbol{b}_3 = \begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix}, \boldsymbol{x} = \begin{bmatrix} 8 \\ -9 \\ 6 \end{bmatrix}$

6. $\boldsymbol{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \boldsymbol{b}_2 = \begin{bmatrix} 2 \\ 1 \\ 8 \end{bmatrix}, \boldsymbol{b}_3 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \boldsymbol{x} = \begin{bmatrix} 3 \\ -5 \\ 4 \end{bmatrix}$

In exercises 7 and 8, find the change of coordinates matrix from $\boldsymbol{B}$ to standard basis in $\boldsymbol{R}^n$.

7. $\boldsymbol{B} = \left\{ \begin{bmatrix} 2 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \end{bmatrix} \right\}$

8. $\boldsymbol{B} = \left\{ \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -5 \end{bmatrix}, \begin{bmatrix} 8 \\ -2 \\ 7 \end{bmatrix} \right\}$

In exercises 9 and 10, use an inverse matrix to find $[\boldsymbol{x}]_{\boldsymbol{B}}$ for the given $\boldsymbol{x}$ and $\boldsymbol{B}$.

9. $\boldsymbol{B} = \left\{\begin{bmatrix}4\\5\end{bmatrix}, \begin{bmatrix}6\\7\end{bmatrix}\right\}, \boldsymbol{x} = \begin{bmatrix}2\\0\end{bmatrix}$

10. $\boldsymbol{B} = \left\{\begin{bmatrix}3\\-5\end{bmatrix}, \begin{bmatrix}-4\\6\end{bmatrix}\right\}, \boldsymbol{x} = \begin{bmatrix}2\\-6\end{bmatrix}$

11. The set $\boldsymbol{B} = \{\mathbf{1} + \boldsymbol{t}^2, \boldsymbol{t} + \boldsymbol{t}^2, \mathbf{1} + 2\boldsymbol{t} + \boldsymbol{t}^2\}$ is a basis for $\boldsymbol{P}_2$. Find the coordinate vector of $\boldsymbol{p}(\boldsymbol{t}) = \boldsymbol{1} + 4\boldsymbol{t} + 7\boldsymbol{t}^2$ relative to $\boldsymbol{B}$.

12. The vectors $\boldsymbol{v}_1 = \begin{bmatrix}1\\-3\end{bmatrix}, \boldsymbol{v}_2 = \begin{bmatrix}2\\-8\end{bmatrix}, \boldsymbol{v}_3 = \begin{bmatrix}-3\\7\end{bmatrix}$ span $\boldsymbol{R}^2$ but do not form a basis. Find two different ways to express $\begin{bmatrix}1\\1\end{bmatrix}$ as a linear combination of $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$.

13. Let $\boldsymbol{B} = \left\{\begin{bmatrix}1\\-4\end{bmatrix}, \begin{bmatrix}-2\\9\end{bmatrix}\right\}$. Since the coordinate mapping determined by $\boldsymbol{B}$ is a linear transformation from $\boldsymbol{R}^2$ into $\boldsymbol{R}^2$, this mapping must be implemented by some 2 x 2 matrix $\boldsymbol{A}$. Find it.

In exercises 14-16, use coordinate vectors to test the linear independence of the sets of polynomials.

14. $\mathbf{1} + \boldsymbol{t}^3, \mathbf{3} + \boldsymbol{t} - 2\boldsymbol{t}^2, -\boldsymbol{t} + 3\boldsymbol{t}^2 - \boldsymbol{t}^3$

15. $(\boldsymbol{t}-\mathbf{1})^2, \boldsymbol{t}^3 - \mathbf{2}, (\mathbf{t} - \mathbf{2})^3$

16. $\mathbf{3} + 7\boldsymbol{t}, \mathbf{5} + \boldsymbol{t} - 2\boldsymbol{t}^3, \boldsymbol{t} - 2\boldsymbol{t}^2, \mathbf{1} + 16\boldsymbol{t} - 6\boldsymbol{t}^2 + 2\boldsymbol{t}^3$

17. Let $\boldsymbol{H}$ = Span $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ and $\boldsymbol{B} = \{\boldsymbol{v}_1, \boldsymbol{v}_2\}$. Show that $\boldsymbol{x}$ is in $\boldsymbol{H}$ and find the $\boldsymbol{B}$-coordinate vector of $\boldsymbol{x}$, for $\boldsymbol{v}_1 = \begin{bmatrix}11\\-5\\10\\7\end{bmatrix}, \boldsymbol{v}_2 = \begin{bmatrix}14\\-8\\13\\10\end{bmatrix}, \boldsymbol{x} = \begin{bmatrix}19\\-13\\18\\15\end{bmatrix}$.

18. Let $\boldsymbol{H}$ = Span $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ and $\boldsymbol{B} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$. Show that $\boldsymbol{B}$ is a basis for $\boldsymbol{H}$ and $\boldsymbol{x}$ is in $\boldsymbol{H}$, and find the $\boldsymbol{B}$-coordinate vector of $\boldsymbol{x}$, for $\boldsymbol{v}_1 = \begin{bmatrix}-6\\4\\-9\\4\end{bmatrix}, \boldsymbol{v}_2 = \begin{bmatrix}8\\-3\\7\\-3\end{bmatrix}, \boldsymbol{v}_3 = \begin{bmatrix}-9\\5\\-8\\3\end{bmatrix}, \boldsymbol{x} = \begin{bmatrix}4\\7\\-8\\3\end{bmatrix}$.

# Lecture 24

## Dimension of a Vector Space

In this lecture, we will focus over the dimension of the vector spaces. The dimension of a vector space V is the cardinality or the number of vectors in the basis B of the given vector space. If the basis B has *n* (say) elements then this number *n* (called the dimension) is an intrinsic property of the space ***V.*** That is it does not depend on the particular choice of basis rather, all the bases of V will have the same cardinality. Thus, we can say that the dimension of a vector space is always unique. The discussion of dimension will give additional insight into properties of bases.
The first theorem generalizes a well-known result about the vector space $\boldsymbol{R^n}$.

**Note**

A vector space ***V*** with a basis ***B*** containing *n* vectors is isomorphic to $\boldsymbol{R^n}$ i.e., there exist a one-to-one linear transformation from V to $\boldsymbol{R^n}$.

**Theorem 1** If a vector space ***V*** has a basis $\boldsymbol{B = \{b_1, \ldots, b_n\}}$, then any set in ***V*** containing more than *n* vectors must be linearly dependent.

**Theorem 2** If a vector space ***V*** has a basis of *n* vectors, then every basis of ***V*** must consist of exactly *n* vectors.

**Finite and infinite dimensional vector spaces**

If the vector space ***V*** is spanned or generated by a finite set, then ***V*** is said to be **finite-dimensional**, and the **dimension** of ***V***, written as dim *V*, is the number of vectors in a basis for ***V***. If ***V*** is not spanned by a finite set, then ***V*** is said to be **infinite-dimensional**. That is, if we are unable to find a finite set that can generate the whole vector space, then such a vector space is called infinite dimensional.

**Note**

(1) The dimension of the zero vector space $\boldsymbol{\{0\}}$ is defined to be zero.
(2) Every finite dimensional vector space contains a basis.

**Example 1** The n dimensional set of real numbers $\mathbf{R^n}$, set of polynomials of order n $\mathbf{P_n}$, and set of matrices of order $m \times n$ $\mathbf{M_{mn}}$ are all finite- dimensional vector spaces. However, the vector spaces **F** $(-\infty, \infty)$, **C** $(-\infty, \infty)$, and $\mathbf{C^m}$ $(-\infty, \infty)$ are infinite-dimensional.

**Example 2**

(a) Any pair of non-parallel vectors ***a***, ***b*** in the xy-plane, which are necessarily linearly independent, can be regarded as a basis of the subspace $\boldsymbol{R^2}$. In particular the set of unit vectors $\boldsymbol{\{i, j\}}$ forms a basis for $\boldsymbol{R^2}$. Therefore, dim $\boldsymbol{(R^2) = 2}$.

Any set of three non coplanar vectors $\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$ in ordinary (physical) space, which will be necessarily linearly independent, spans the space $\boldsymbol{R^3}$. Therefore any set of such vectors forms a basis for $\boldsymbol{R^3}$. In particular the set of unit vectors $\{\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}\}$ forms a basis of $\boldsymbol{R^3}$. This basis is called standard basis for $\boldsymbol{R^3}$. Therefore dim $(\boldsymbol{R^3}) = 3$.

The set of vectors $\{\boldsymbol{e_1}, \boldsymbol{e_2}, \ldots, \boldsymbol{e_n}\}$ where

$$\boldsymbol{e_1} = (1, 0, 0, 0, \ldots, 0),$$
$$\boldsymbol{e_2} = (0, 1, 0, 0, \ldots, 0),$$
$$\boldsymbol{e_3} = (0, 0, 1, 0, \ldots, 0),$$
$$\ldots$$
$$\ldots$$
$$\ldots$$
$$\boldsymbol{e_n} = (0, 0, 0, 0, \ldots, 1)$$

is linearly independent.
Moreover, any vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$ in $\boldsymbol{R^n}$ can be expressed as a linear combination of these vectors as

$$\boldsymbol{x} = x_1\boldsymbol{e_1} + x_2\boldsymbol{e_2} + x_3\boldsymbol{e_3} + \ldots + x_n\boldsymbol{e_n}.$$

Hence, the set $\{\boldsymbol{e_1}, \boldsymbol{e_2}, \ldots, \boldsymbol{e_n}\}$ forms a basis for $\boldsymbol{R^n}$. It is called the standard basis of $\boldsymbol{R^n}$, therefore dim $(\boldsymbol{R^n}) = n$. Any other set of $n$ linearly independent vectors in $\boldsymbol{R^n}$ will form a non-standard basis.

(b) The set $\boldsymbol{B} = \{\boldsymbol{1}, \boldsymbol{x}, \boldsymbol{x^2}, \ldots, \boldsymbol{x^n}\}$ forms a basis for the vector space $\boldsymbol{P_n}$ of polynomials of degree $\leq n$. It is called the standard basis with dim $(\boldsymbol{P_n}) = n + 1$.

(c) The set of 2 x 2 matrices with real entries (elements) $\{\boldsymbol{u_1}, \boldsymbol{u_2}, \boldsymbol{u_3}, \boldsymbol{u_4}\}$ where

$$\boldsymbol{u_1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \boldsymbol{u_2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \boldsymbol{u_3} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \boldsymbol{u_4} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

is a linearly independent and every 2 x 2 matrix with real entries can be expressed as their linear combination. Therefore, they form a basis for the vector space $\boldsymbol{M_{2X2}}$. This basis is called the standard basis for $\boldsymbol{M_{2X2}}$ with dim $(\boldsymbol{M_{2X2}}) = 4$.

**Note**

**(1)** dim $(\boldsymbol{R^n}) = n$ { The standard basis has $n$ vectors}.

**(2)** dim $(\boldsymbol{P_n}) = n + 1$ { The standard basis has $n+1$ vectors}.

**(3)** dim $(\boldsymbol{M_{m\times n}}) = mn$ { The standard basis has $mn$ vectors.}

**Example 3** Let $\boldsymbol{W}$ be the subspace of the set of all (2 x 2) matrices defined by

$$\boldsymbol{W} = \{\boldsymbol{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} : 2a - b + 3c + d = 0\}.$$

Determine the dimension of $\boldsymbol{W}$.
**Solution** The algebraic specification for $\boldsymbol{W}$ can be rewritten as $d = -2a + b - 3c$.

Now $\boldsymbol{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Substituting the value of d, it becomes

$$A = \begin{bmatrix} a & b \\ c & -2a+b-3c \end{bmatrix}$$

This can be written as

$$A = \begin{bmatrix} a & 0 \\ 0 & -2a \end{bmatrix} + \begin{bmatrix} 0 & b \\ 0 & b \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ c & -3c \end{bmatrix}$$

$$= a\begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} + c\begin{bmatrix} 0 & 0 \\ 1 & -3 \end{bmatrix}$$

$$= a\boldsymbol{A_1} + b\boldsymbol{A_2} + c\boldsymbol{A_3}$$

where $\boldsymbol{A_1} = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$, $\boldsymbol{A^2} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$, and $\boldsymbol{A_3} = \begin{bmatrix} 0 & 0 \\ 1 & -3 \end{bmatrix}$

The matrix $\boldsymbol{A}$ is in $\boldsymbol{W}$ if and only if $\boldsymbol{A} = a\boldsymbol{A_1} + b\boldsymbol{A_2} + c\boldsymbol{A_3}$, so $\{\boldsymbol{A_1}, \boldsymbol{A_2}, \boldsymbol{A_3}\}$ is a spanning set for $\boldsymbol{W}$. Now, check if this set is a basis for W or not. We will see whether $\{\boldsymbol{A_1}, \boldsymbol{A_2}, \boldsymbol{A_3}\}$ is linearly independent or not. $\{\boldsymbol{A_1}, \boldsymbol{A_2}, \boldsymbol{A_3}\}$ is said to be linearly independent if

$aA_1 + bA_2 + cA_3 = 0 \Rightarrow a=b=c=0$ *i.e.*,

$$a\begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} + c\begin{bmatrix} 0 & 0 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a & 0 \\ 0 & -2a \end{bmatrix} + \begin{bmatrix} 0 & b \\ 0 & b \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ c & -3c \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

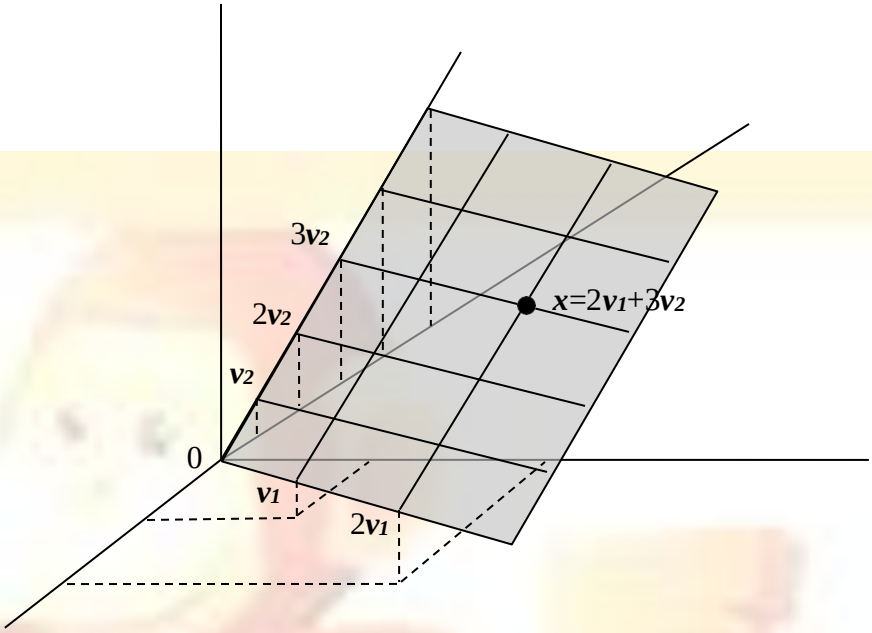
$$\begin{bmatrix} a & b \\ c & -2a+b-3c \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Equating the elements, we get

$a = 0, b = 0, c = 0$

This implies $\{\mathbf{A_1}, \mathbf{A_2}, \mathbf{A_3}\}$ is a linearly independent set that spans W. Hence, it's the basis of W with dim( W)= 3.

**Example 4** Let $\boldsymbol{H}$ = Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$, where $\boldsymbol{v_1} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$ and $\boldsymbol{v_2} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$. Then $\boldsymbol{H}$ is the plane studied in Example 10 of lecture 23**.** A basis for $\boldsymbol{H}$ is $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$, since $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ are not multiples and hence are linearly independent. Thus, dim $\boldsymbol{H}$ = *2*.




**A coordinate system on a plane $H$ in $R^3$**

**Example 5** Find the dimension of the subspace

$$H = \left\{ \begin{bmatrix} a - 3b + 6c \\ 5a + 4d \\ b - 2c - d \\ 5d \end{bmatrix} : a,b,c,d \in R \right\}$$

**Solution** The representative vector of H can be written as

$$\begin{bmatrix} a-3b+6c \\ 5a+4d \\ b-2c-d \\ 5d \end{bmatrix} = a\begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \end{bmatrix} + b\begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c\begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix} + d\begin{bmatrix} 0 \\ 4 \\ -1 \\ 5 \end{bmatrix}$$

Now, it is easy to see that $H$ is the set of all linear combinations of the vectors

$$v_1 = \begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 5 \end{bmatrix}$$

Clearly, $v_1 \neq 0, v_2$ is not a multiple of $v_1$, but $v_3$ is a multiple of $v_2$. By the Spanning Set Theorem, we may discard $v_3$ and still have a set that spans $H$. Finally; $v_4$ is not a linear

combination of $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$. So $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_4}\}$ is linearly independent and hence is a basis for $\boldsymbol{H}$. Thus dim $\boldsymbol{H} = 3$.

**Example 6** The subspaces of $\boldsymbol{R^3}$ can be classified by various dimensions as shown in Fig. 1.

***0*- dimensional subspaces**

The only 0-dimensional subspace of $\boldsymbol{R^3}$ is zero space.

***1*- dimensional subspaces**

1-dimensional subspaces include any subspace spanned by a single non-zero vector. Such subspaces are lines through the origin.

**2- dimensional subspaces**

Any subspace spanned by two linearly independent vectors. Such subspaces are planes through the origin.

**3- dimensional subspaces**

The only 3-dimensional subspace is $\boldsymbol{R^3}$ itself. Any three linearly independent vectors in $\boldsymbol{R^3}$ span all of $\boldsymbol{R^3}$, by the Invertible Matrix Theorem.

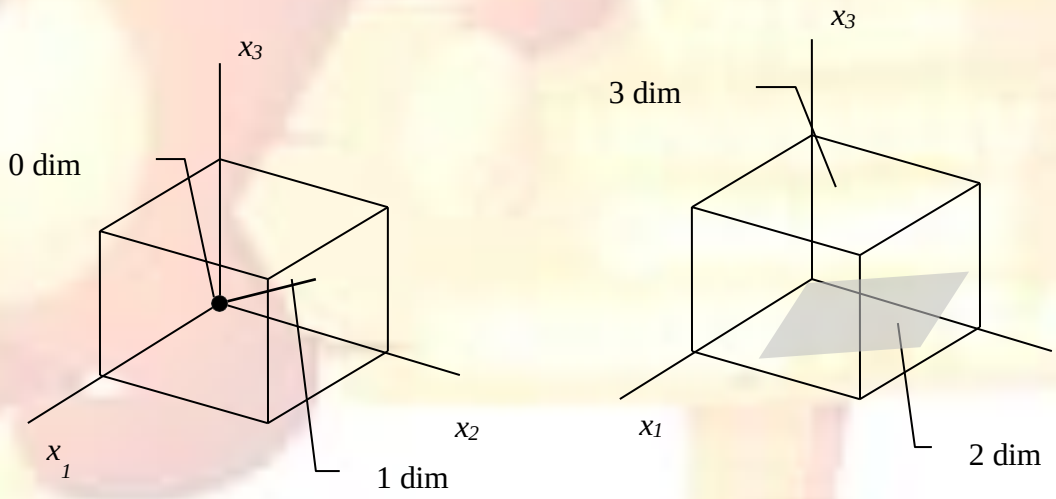



**Figure 1 – Sample subspaces of $\boldsymbol{R^3}$**

**Bases for Nul *A* and Col *A***

We already know how to find vectors that span the null space of a matrix $\boldsymbol{A}$. The discussion in Lecture 21 pointed out that our method always produces a linearly independent set. Thus the method produces a basis for Nul $\boldsymbol{A}$.

**Example 7** Find a basis for the null space of $\boldsymbol{A} = \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$.

**Solution** The null space of $\boldsymbol{A}$ is the solution space of homogeneous system

$$\begin{aligned} 2x_1 + 2x_2 - x_3 + x_5 &= 0 \\ -x_1 - x_2 + 2x_3 - 3x_4 + x_5 &= 0 \\ x_1 + x_2 - 2x_3 - x_5 &= 0 \\ x_3 + x_4 + x_5 &= 0 \end{aligned}$$

The most appropriate way to solve this system is to reduce its augmented matrix into reduced echelon form.

$$\begin{bmatrix} 2 & 2 & -1 & 0 & 1 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \\ 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} \quad R_4 \leftrightarrow R_2,\ R_3 \leftrightarrow R_1$$

$$\leftrightarrow \begin{bmatrix} 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & -1 & 0 & 1 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \end{bmatrix} \quad R_3 - 2R_1,\ R_3 - 3R_2$$

$$\leftrightarrow \begin{bmatrix} 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 3 & 0 & 3 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \end{bmatrix} \quad R_3 - 3R_2$$

$$\leftrightarrow \begin{bmatrix} 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \end{bmatrix} \quad -\frac{1}{3}R_3$$

$$\leftrightarrow \begin{bmatrix} 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \end{bmatrix} \quad R_4 + R_1$$

$$\begin{bmatrix} 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 \end{bmatrix} \quad R_4 + 3R_3$$

$$\begin{bmatrix} 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad R_1 + 2R_2$$

$$\begin{bmatrix} 1 & 1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad R_2 - R_3, R_1 - 2R_3$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Thus, the reduced row echelon form of the augmented matrix is

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

which corresponds to the system

$$\begin{aligned} 1x_1 + 1x_2 + \quad & 1x_5 = 0 \\ 1x_3 + \quad & 1x_5 = 0 \\ 1x_4 \quad & = 0 \\ 0 & = 0 \end{aligned}$$

No equation of this system has a form zero = nonzero. Therefore, the system is consistent. Since the number of unknowns is more than the number of equations, we will assign some arbitrary value to some variables. This will lead to infinite many solutions of the system.

$$x_1 = -1x_2 - 1x_5$$
$$x_2 = s$$
$$x_3 = -1x_5$$
$$x_4 = 0$$
$$x_5 = t$$

The general solution of the given system is

$$x_1 = -s - t \quad , \quad x_2 = s \quad , \quad x_3 = -t \quad , \quad x_4 = 0 \quad , \quad x_5 = t$$

Therefore, the solution vector can be written as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -s-t \\ s \\ -t \\ 0 \\ t \end{bmatrix} = \begin{bmatrix} -s \\ s \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -t \\ 0 \\ -t \\ 0 \\ t \end{bmatrix} = s\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

which shows that the vectors $\boldsymbol{v_1} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ and $\boldsymbol{v_2} = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$ span the solution space .Since they are also linearly independent,$\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is a basis for Nul $\boldsymbol{A}$.

The next two examples describe a simple algorithm for finding a basis for the column space.

**Example 8** Find a basis for Col $\boldsymbol{B}$, where $\boldsymbol{B} = [\boldsymbol{b}_1 \quad \boldsymbol{b}_2, \quad ..., \quad \boldsymbol{b}_5] = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

**Solution** Each non-pivot column of $\boldsymbol{B}$ is a linear combination of the pivot columns. In fact, $\boldsymbol{b_2} = 4\boldsymbol{b_1}$ and $\boldsymbol{b_4} = 2\boldsymbol{b_1} - \boldsymbol{b_3}$. By the Spanning Set Theorem, we may discard $\boldsymbol{b_2}$ and $\boldsymbol{b_4}$ and $\{\boldsymbol{b_1}, \boldsymbol{b_3}, \boldsymbol{b_5}\}$ will still span Col $\boldsymbol{B}$. Let

$$\boldsymbol{S} = \{\boldsymbol{b}_1 \ , \ \boldsymbol{b}_3 \ , \ \boldsymbol{b}_5\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Since $\boldsymbol{b_1} \neq \boldsymbol{0}$ and no vector in $\boldsymbol{S}$ is a linear combination of the vectors that precede it, $\boldsymbol{S}$ is linearly independent. Thus $\boldsymbol{S}$ is a basis for Col $\boldsymbol{B}$.

What about a matrix $\boldsymbol{A}$ that is not in reduced echelon form? Recall that any linear dependence relationship among the columns of $\boldsymbol{A}$ can be expressed in the form $\boldsymbol{Ax}$

$= \mathbf{0}$, where $\mathbf{x}$ is a column of weights. (If some columns are not involved in a particular dependence relation, then their weights are zero.) When $\mathbf{A}$ is row reduced to a matrix $\mathbf{B}$, the columns of $\mathbf{B}$ are often totally different from the columns of $\mathbf{A}$. However, the equations $\mathbf{Ax} = \mathbf{0}$ and $\mathbf{Bx} = \mathbf{0}$ have exactly the same set of solutions. That is, the columns of $\mathbf{A}$ have exactly the same linear dependence relationships as the columns of $\mathbf{B}$.

Elementary row operations on a matrix do not affect the linear dependence relations among the columns of the matrix.

**Example 9** It can be shown that the matrix

$$\mathbf{A} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_5] = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$$

is row equivalent to the matrix $\mathbf{B}$ in Example 8. Find a basis for Col $\mathbf{A}$.

**Solution** In Example 8, we have seen that $\mathbf{b}_2 = 4\mathbf{b}_1$ *and* $\mathbf{b}_4 = 2\mathbf{b}_1 - \mathbf{b}_3$ so we can expect that $\mathbf{a}_2 = 4\mathbf{a}_1$ *and* $\mathbf{a}_4 = 2\mathbf{a}_1 - \mathbf{a}_3$. This is indeed the case. Thus, we may discard $\mathbf{a}_2$ and $\mathbf{a}_4$ while selecting a minimal spanning set for Col $\mathbf{A}$. In fact, $\{\mathbf{a}_1, \mathbf{a}_3, \mathbf{a}_5\}$ must be linearly independent because any linear dependence relationship among $\mathbf{a}_1, \mathbf{a}_3, \mathbf{a}_5$ would imply a linear dependence relationship among $\mathbf{b}_1, \mathbf{b}_3, \mathbf{b}_5$. But we know that $\{\mathbf{b}_1, \mathbf{b}_3, \mathbf{b}_5\}$ is a linearly independent set. Thus $\{\mathbf{a}_1, \mathbf{a}_3, \mathbf{a}_5\}$ is a basis for Col $\mathbf{A}$. The columns we have used for this basis are the pivot columns of $\mathbf{A}$.

Examples 8 and 9 illustrate the following useful fact.

**Theorem 3** The pivot columns of a matrix $\mathbf{A}$ form a basis for Col $\mathbf{A}$.

**Proof** The general proof uses the arguments discussed above. Let $\mathbf{B}$ be the reduced echelon form of $\mathbf{A}$. The set of pivot columns of $\mathbf{B}$ is linearly independent, for no vector in the set is a linear combination of the vectors that precede it. Since $\mathbf{A}$ is row equivalent to $\mathbf{B}$, the pivot columns of $\mathbf{A}$ are linearly independent too, because any linear dependence relation among the columns of $\mathbf{A}$ corresponds to a linear dependence relation among the columns of $\mathbf{B}$. For this same reason, every non-pivot column of $\mathbf{A}$ is a linear combination of the pivot columns of $\mathbf{A}$. Thus the non-pivot columns of $\mathbf{A}$ may be discarded from the spanning set for Col $\mathbf{A}$, by the Spanning Set Theorem. This leaves the pivot columns of $\mathbf{A}$ as a basis for Col $\mathbf{A}$.

**Note** Be careful to use pivot columns of $\mathbf{A}$ itself for the basis of Col $\mathbf{A}$. The columns of an echelon form $\mathbf{B}$ are often not in the column space of $\mathbf{A}$. For instance, the columns of the $\mathbf{B}$ in Example 8 all have zeros in their last entries, so they cannot span the column space of the $\mathbf{A}$ in Example 9.

**Example 10** Let $\boldsymbol{v_1} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ and $\boldsymbol{v_2} = \begin{bmatrix} -2 \\ 7 \\ -9 \end{bmatrix}$. Determine if $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is a basis for $\boldsymbol{R^3}$. Is $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ a basis for $\boldsymbol{R^2}$?

**Solution** Let $A = [\boldsymbol{v_1} \quad \boldsymbol{v_2}]$. Row operations show that $A = \begin{bmatrix} 1 & -2 \\ -2 & 7 \\ 3 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 0 & 0 \end{bmatrix}$. Not every row of $A$ contains a pivot position. So the columns of $A$ do not span $\boldsymbol{R^3}$, by Theorem 4 in Lecture 6. Hence $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is not a basis for $\boldsymbol{R^3}$. Since $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ are not in $\boldsymbol{R^2}$, they cannot possibly be a basis for $\boldsymbol{R^2}$. However, since $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ are obviously linearly independent, they are a basis for a subspace of $\boldsymbol{R^3}$, namely, Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$.

**Example 11** Let $\boldsymbol{v_1} = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}, \boldsymbol{v_2} = \begin{bmatrix} 6 \\ 2 \\ -1 \end{bmatrix}, \boldsymbol{v_3} = \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}, \boldsymbol{v_4} = \begin{bmatrix} -4 \\ -8 \\ 9 \end{bmatrix}$. Find a basis for the subspace $\boldsymbol{W}$ spanned by $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}, \boldsymbol{v_4}\}$.

**Solution** Let $\boldsymbol{A}$ be the matrix whose column space is the space spanned by $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}, \boldsymbol{v_4}\}$,

$$\boldsymbol{A} = \begin{bmatrix} 1 & 6 & 2 & -4 \\ -3 & 2 & -2 & -8 \\ 4 & -1 & 3 & 9 \end{bmatrix}$$

Reduce the matrix $\boldsymbol{A}$ into its echelon form in order to find its pivot columns.

$$\boldsymbol{A} = \begin{bmatrix} 1 & 6 & 2 & -4 \\ -3 & 2 & -2 & -8 \\ 4 & -1 & 3 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 6 & 2 & -4 \\ 0 & 20 & 4 & -20 \\ 0 & -25 & -5 & 25 \end{bmatrix} \text{ by } R_2 + 3R_1,\ R_3 - 4R_1$$

$$\sim \begin{bmatrix} 1 & 6 & 2 & -4 \\ 0 & 5 & 1 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ by } \frac{1}{4}R_2,\ -\frac{1}{5}R_3,\ R_3 - R_2$$

The first two columns of $\boldsymbol{A}$ are the pivot columns and hence form a basis of Col $\boldsymbol{A} = \boldsymbol{W}$. Hence $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is a basis for $\boldsymbol{W}$.

Note that the reduced echelon form of $\boldsymbol{A}$ is not needed in order to locate the pivot columns.

**Procedure**

**Basis and Linear Combinations**

Given a set of vectors $S = \{v_1, v_2, \ldots, v_k\}$ in $R^n$, the following procedure produces a subset of these vectors that form a basis for span ($S$) and expresses those vectors of $S$ that are not in the basis as linear combinations of the basis vector.

Step1: Form the matrix $A$ having $v_1, v_2, \ldots, v_k$ as its column vectors.

Step2: Reduce the matrix $A$ to its reduced row echelon form $R$, and let $w_1, w_2, \ldots, w_k$ be the column vectors of $R$.

Step3: Identify the columns that contain the leading entries i.e., 1's in $R$. The corresponding column vectors of $A$ are the basis vectors for span ($S$).

Step4: Express each column vector of $R$ that does not contain a leading entry as a linear combination of preceding column vector that do contain leading entries (we will be able to do this by inspection). This yields a set of dependency equations involving the column vectors of $R$. The corresponding equations for the column vectors of $A$ express the vectors which are not in the basis as linear combinations of basis vectors.

**Example 12 Basis and Linear Combinations**

(a) Find a subset of the vectors $v_1 = (1, -2, 0, 3)$, $v_2 = (2, -4, 0, 6)$, $v_3 = (-1, 1, 2, 0)$ and $v_4 = (0, -1, 2, 3)$ that form a basis for the space spanned by these vectors.

(b) Express each vector not in the basis as a linear combination of the basis vectors.

**Solution** (a) We begin by constructing a matrix that has $v_1, v_2, v_3, v_4$ as its column vectors

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ -2 & -4 & 1 & -1 \\ 0 & 0 & 2 & 2 \\ 3 & 6 & 0 & 3 \end{bmatrix} \quad \text{(A)}$$

$$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$$

$$v_1 \quad v_2 \quad v_3 \quad v_4$$

Finding a basis for column space of this matrix can solve the first part of our problem.

Transforming Matrix to Reduced Row Echelon Form:

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ -2 & -4 & 1 & -1 \\ 0 & 0 & 2 & 2 \\ 3 & 6 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 3 & 3 \end{bmatrix} \quad 2R_1 + R_2$$

$$\quad -3R_1 + R_4$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 3 & 3 \end{bmatrix} \begin{matrix} \\ -1R_2 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ -2R_2 + R_3 \\ -3R_2 + R_4 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ R_2 + R_1 \\ \\ \end{matrix}$$

Labeling the column vectors of the resulting matrix as $w_1$, $w_2$, $w_3$ and $w_4$ yields

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{(B)}$$

$$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$$

$$w_1 \quad w_2 \quad w_3 \quad w_4$$

The leading entries occur in column 1 and 3 so $\{w_1, w_3\}$ is a basis for the column space of (B) and consequently $\{v_1, v_3\}$ is the basis for column space of (A).

(b) We shall start by expressing $w_2$ and $w_4$ as linear combinations of the basis vector $w_1$ and $w_3$. The simplest way of doing this is to express $w_2$ and $w_4$ in term of basis vectors with smaller subscripts. Thus we shall express $w_2$ as a linear combination of $w_1$, and we shall express $w_4$ as a linear combination of $w_1$ and $w_3$. By inspection of (B), these linear combinations are $w_2 = 2w_1$ and $w_4 = w_1 + w_3$. We call them the dependency equations. The corresponding relationship of (A) are $v_3 = 2v_1$ and $v_5 = v_1 + v_3$.

**Example 13 Basis and Linear Combinations**

(a) Find a subset of the vectors $v_1 = (1, -1, 5, 2)$, $v_2 = (-2, 3, 1, 0)$, $v_3 = (4, -5, 9, 4)$, $v_4 = (0, 4, 2, -3)$ and $v_5 = (-7, 18, 2, -8)$ that form a basis for the space spanned by these vectors.

(b) Express each vector not in the basis as a linear combination of the basis vectors

**Solution** (a) We begin by constructing a matrix that has $v_1, v_2, \ldots, v_5$ as its column vectors

$$\begin{bmatrix} 1 & -2 & 4 & 0 & -7 \\ -1 & 3 & -5 & 4 & 18 \\ 5 & 1 & 9 & 2 & 2 \\ 2 & 0 & 4 & -3 & -8 \end{bmatrix} \quad \text{(A)}$$

$$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$$

$$v_1 \quad v_2 \quad v_3 \quad v_4 \quad v_5$$

Finding a basis for column space of this matrix can solve the first part of our problem.

Transforming Matrix to Reduced Row Echelon Form:

$$\begin{bmatrix} 1 & -2 & 4 & 0 & -7 \\ -1 & 3 & -5 & 4 & 18 \\ 5 & 1 & 9 & 2 & 2 \\ 2 & 0 & 4 & -3 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 4 & 0 & -7 \\ 0 & 1 & -1 & 4 & 11 \\ 0 & 11 & -11 & 2 & 37 \\ 0 & 4 & -4 & -3 & 6 \end{bmatrix} \begin{matrix} \\ R_1 + R_2 \\ -5R_1 + R_3 \\ -2R_1 + R_4 \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 4 & 0 & -7 \\ 0 & 1 & -1 & 4 & 11 \\ 0 & 0 & 0 & -42 & -84 \\ 0 & 0 & 0 & -19 & -38 \end{bmatrix} \begin{matrix} \\ -11R_2 + R_3 \\ -4R_2 + R_4 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 4 & 0 & -7 \\ 0 & 1 & -1 & 4 & 11 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -19 & -38 \end{bmatrix} \begin{matrix} \\ \\ (-1/42)R_3 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 4 & 0 & -7 \\ 0 & 1 & -1 & 4 & 11 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ 19R_3 + R_4 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 4 & 0 & -7 \\ 0 & 1 & -1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ (-4)R_3 + R_2 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & -1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ 2R_2 + R_1 \\ \end{matrix}$$

Denoting the column vectors of the resulting matrix by $w_1$ , $w_2$ , $w_3$, $w_4$, and $w_5$ yields

$$\begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & -1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{(B)}$$

$$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$$

$$w_1 \quad w_2 \quad w_3 \quad w_4 \quad w_5$$

The leading entries occur in columns 1,2 and 4 so that $\{w_1, w_2, w_4\}$ is a basis for the column space of (B) and consequently $\{v_1, v_2, v_4\}$ is the basis for column space of (A).

(b) We shall start by expressing $w_3$ and $w_5$ as linear combinations of the basis vector $w_1$, $w_2$, $w_4$. The simplest way of doing this is to express $w_3$ and $w_5$ in term of basis vectors with smaller subscripts. Thus we shall express $w_3$ as a linear combination of $w_1$ and $w_2$, and we shall express $w_5$ as a linear combination of $w_1$, $w_2$, and $w_4$. By inspection of (B), these linear combination are $w_3 = 2w_1 - w_2$ and $w_5 = -w_1 + 3w_2 + 2w_4$.

The corresponding relationship of (A) are $v_3 = 2v_1 - v_2$ and $v_5 = -v_1 + 3v_2 + 2v_4$.

**Example 14 Basis and Linear Combinations**

(a) Find a subset of the vectors $v_1 = (1, -2, 0, 3)$, $v_2 = (2, -5, -3, 6)$, $v_3 = (0, 1, 3, 0)$, $v_4 = (2, -1, 4, -7)$ and $v_5 = (5, -8, 1, 2)$ that form a basis for the space spanned by these vectors.

(b) Express each vector not in the basis as a linear combination of the basis vectors.

**Solution** (a) We begin by constructing a matrix that has $v_1, v_2, \ldots, v_5$ as its column vectors

$$\begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & -8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & 6 & 0 & -7 & 2 \end{bmatrix} \quad \text{(A)}$$

$$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$$

$$v_1 \quad v_2 \quad v_3 \quad v_4 \quad v_5$$

Finding a basis for column space of this matrix can solve the first part of our problem. Reducing the matrix to reduced-row echelon form and denoting the column vectors of the resulting matrix by $w_1, w_2, w_3, w_4$, and $w_5$ yields

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{(B)}$$

$$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$$

$$w_1 \quad w_2 \quad w_3 \quad w_4 \quad w_5$$

The leading entries occur in columns 1, 2 and 4 so $\{w_1, w_2, w_4\}$ is a basis for the column space of (B) and consequently $\{v_1, v_2, v_4\}$ is the basis for column space of (A).

(b) Dependency equations are $\boldsymbol{w_3 = 2w_1 - w_2}$ and $\boldsymbol{w_5 = w_1 + w_2 + w_4}$
The corresponding relationship of (A) are $\boldsymbol{v_3 = 2v_1 - v_2}$ ***and*** $\boldsymbol{v_5 = v_1 + v_2 + v_4}$

**Subspaces of a Finite-Dimensional Space** The next theorem is a natural counterpart to the Spanning Set Theorem.

**Theorem 5** Let $\boldsymbol{H}$ be a subspace of a finite-dimensional vector space $\boldsymbol{V}$. Any linearly independent set in $\boldsymbol{H}$ can be expanded, if necessary, to a basis for $\boldsymbol{H}$. Also, $\boldsymbol{H}$ is finite-dimensional and $\dim \boldsymbol{H} \leq \dim \boldsymbol{V}$.

When the dimension of a vector space or subspace is known, the search for a basis is simplified by the next theorem. It says that if a set has the right number of elements, then one has only to show either that the set is linearly independent or that it spans the space. The theorem is of critical importance in numerous applied problems (involving differential equations or difference equations, for example) where linear independence is much easier to verify than spanning.

**Theorem 5 (The Basis Theorem)** Let $\boldsymbol{V}$ be a $p$-dimensional vector space, $p \geq 1$. Any linearly independent set of exactly $p$ elements in $\boldsymbol{V}$ is automatically a basis for $\boldsymbol{V}$. Any set of exactly $p$ elements that spans $\boldsymbol{V}$ is automatically a basis for $\boldsymbol{V}$.

**The Dimensions of Nul *A* and Col *A*** Since the pivot columns of a matrix $\boldsymbol{A}$ form a basis for Col $\boldsymbol{A}$, we know the dimension of Col $\boldsymbol{A}$ as soon as we know the pivot columns. The dimension of Nul $\boldsymbol{A}$ might seem to require more work, since finding a basis for Nul $\boldsymbol{A}$ usually takes more time than a basis for Col $\boldsymbol{A}$. Yet, there is a shortcut.

Let A be an $m \times n$ matrix, and suppose that the equation $\boldsymbol{Ax = 0}$ has $k$ free variables. From lecture 21, we know that the standard method of finding a spanning set for Nul $\boldsymbol{A}$ will produce exactly $k$ linearly independent vectors say, $\boldsymbol{u_1, \ldots, u_k}$, one for each free variable. So $\{\boldsymbol{u_1, \ldots, u_k}\}$ is a basis for Nul $\boldsymbol{A}$, and the number of free variables determines the size of the basis. Let us summarize these facts for future reference.

The dimension of Nul $\boldsymbol{A}$ is the number of free variables in the equation $\boldsymbol{Ax = 0}$, and the dimension of Col $\boldsymbol{A}$ is the number of pivot columns in $\boldsymbol{A}$.

**Example 15** Find the dimensions of the null space and column space of

$$\boldsymbol{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

**Solution** Row reduce the augmented matrix $[\boldsymbol{A} \quad \boldsymbol{0}]$ to echelon form and obtain

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Writing it in equations form, we get

$x_1 - 2x_2 + 2x_3 + 3x_4 - x_5 = 0$

$x_3 + 2x_4 - 2x_5 = 0$

Since the number of unknowns is more than the number of equations, we will introduce free variables here (say) $x_2$, $x_4$ and $x_5$. Hence the dimension of Nul ***A*** is *3*. Also dim Col ***A*** *is 2* because ***A*** has two pivot columns.

**Example 16** Decide whether each statement is true or false, and give a reason for each answer. Here ***V*** is a non-zero finite-dimensional vector space.

1. If dim ***V*** = *p* and if ***S*** is a linearly dependent subset of ***V***, then ***S*** contains more than *p* vectors.
2. If ***S*** spans ***V*** and if ***T*** is a subset of ***V*** that contains more vectors than ***S***, then ***T*** is linearly dependent.

**Solution**

1. False. Consider the set ***{0}***.
2. True. By the Spanning Set Theorem, ***S*** contains a basis for ***V***; call that basis ***S′*** . Then ***T*** will contain more vectors than ***S′*** . By Theorem 1, ***T*** is linearly dependent.

**Exercises**

For each subspace in exercises 1-6, (a) find a basis and (b) state the dimension.

1. $\left\{\begin{bmatrix} s-2t \\ s+t \\ 3t \end{bmatrix} : s,t \text{ in } \boldsymbol{R}\right\}$

2. $\left\{\begin{bmatrix} 2c \\ a-b \\ b-3c \\ a+2b \end{bmatrix} : a,b,c \text{ in } \boldsymbol{R}\right\}$

3. $\left\{\begin{bmatrix} a-4b-2c \\ 2a+5b-4c \\ -a+2c \\ -3a+7b+6c \end{bmatrix} : a,b,c \text{ in } \boldsymbol{R}\right\}$

4. $\left\{\begin{bmatrix} 3a+6b-c \\ 6a-2b-2c \\ -9a+5b+3c \\ -3a+b+c \end{bmatrix} : a,b,c \text{ in } \boldsymbol{R}\right\}$

5. {(a, b, c): a – 3b + c = 0, b – 2c = 0, 2b – c = 0}

6 {(a, b, c, d): a - 3b + c = 0}

7. Find the dimension of the subspace $\boldsymbol{H}$ of $\boldsymbol{R}^2$ spanned by

$$\begin{bmatrix} 2 \\ -5 \end{bmatrix}, \begin{bmatrix} -4 \\ 10 \end{bmatrix}, \begin{bmatrix} -3 \\ 6 \end{bmatrix}$$

8. Find the dimension of the subspace spanned by the given vectors.

$$\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} -7 \\ -3 \\ 1 \end{bmatrix}$$

Determine the dimensions of Nul $\boldsymbol{A}$ and Col $\boldsymbol{A}$ for the matrices shown in exercises 9 to 12.

9. $\boldsymbol{A} = \begin{bmatrix} 1 & -6 & 9 & 0 & -2 \\ 0 & 1 & 2 & -4 & 5 \\ 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

10. $\boldsymbol{A} = \begin{bmatrix} 1 & 3 & -4 & 2 & -1 & 6 \\ 0 & 0 & 1 & -3 & 7 & 0 \\ 0 & 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

11. $\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 9 & 5 \\ 0 & 0 & 1 & -4 \end{bmatrix}$

12. $\boldsymbol{A} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 4 & 7 \\ 0 & 0 & 5 \end{bmatrix}$

13. The first four Hermite polynomials are **1**, $2t$, $-2 + 4t^2$, and $-12t + 8t^3$. These polynomials arise naturally in the study of certain important differential equations in mathematical physics. Show that the first four Hermite polynomials form a basis of $P_3$.

14. Let $B$ be the basis of $P_3$ consisting of the Hermite polynomials in exercise 13, and let $p(t) = 7 - 12t - 8t^2 + 12t^3$. Find the coordinate vector of $p$ relative to $B$.

15. Extend the following vectors to a basis for $R^5$:

$$v_1 = \begin{bmatrix} -9 \\ -7 \\ 8 \\ -5 \\ 7 \end{bmatrix}, v_2 = \begin{bmatrix} 9 \\ 4 \\ 1 \\ 6 \\ -7 \end{bmatrix}, v_3 = \begin{bmatrix} 6 \\ 7 \\ -8 \\ 5 \\ -7 \end{bmatrix}$$

# Lecture 25

## Rank

With the help of vector space concepts, for a matrix several interesting and useful relationships in matrix rows and columns have been discussed.

For instance, imagine placing 2000 random numbers into a 40 x 50 matrix $\boldsymbol{A}$ and then determining both the maximum number of linearly independent columns in $\boldsymbol{A}$ and the maximum number of linearly independent columns in $\boldsymbol{A^T}$ (rows in $\boldsymbol{A}$). Remarkably, the two numbers are the same. Their common value is called the rank of the matrix. To explain why, we need to examine the subspace spanned by the subspace spanned by the rows of $\boldsymbol{A}$.

**The Row Space** If $\boldsymbol{A}$ is an $m \times n$ matrix, each row of $\boldsymbol{A}$ has $n$ entries and thus can be identified with a vector in $\boldsymbol{R^n}$. The set of all linear combinations of the row vectors is called the row space of $\boldsymbol{A}$ and is denoted by Row $\boldsymbol{A}$. Each row has $n$ entries, so Row $\boldsymbol{A}$ is a subspace of $\boldsymbol{R^n}$. Since the rows of $\boldsymbol{A}$ are identified with the columns of $\boldsymbol{A^T}$, we could also write Col $\boldsymbol{A^T}$ in place of Row $\boldsymbol{A}$.

**Example 1** Let $\boldsymbol{A} = \begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix}$ and

$\boldsymbol{r_1} = (-2,-5,8,0,-17)$
$\boldsymbol{r_2} = (1,3,-5,1,5)$
$\boldsymbol{r_3} = (3,11,-19,7,1)$
$\boldsymbol{r_4} = (1,7,-13,5,-3)$

The row space of $\boldsymbol{A}$ is the subspace of $\boldsymbol{R^5}$ spanned by $\{\boldsymbol{r_1}, \boldsymbol{r_2}, \boldsymbol{r_3}, \boldsymbol{r_4}\}$. That is, Row $\boldsymbol{A}$ = Span $\{\boldsymbol{r_1}, \boldsymbol{r_2}, \boldsymbol{r_3}, \boldsymbol{r_4}\}$. Naturally, we write row vectors horizontally; however, they could also be written as column vectors

**Example** **Let**

$\boldsymbol{A} = \begin{bmatrix} 2 & 1 & 0 \\ 3 & -1 & 4 \end{bmatrix}$ and

$\boldsymbol{r_1} = (2,1,0)$
$\boldsymbol{r_2} = (3,-1,4)$

That is Row **A**=Span $\{\boldsymbol{r_1}, \boldsymbol{r_2}\}$.

We could use the Spanning Set Theorem to shrink the spanning set to a basis.

Some times row operation on a matrix will not give us the required information but row reducing certainly worthwhile, as the next theorem shows

**Theorem 1** If two matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are row equivalent, then their row spaces are the same. If $\boldsymbol{B}$ is in echelon form, the nonzero rows of $\boldsymbol{B}$ form a basis for the row space of $\boldsymbol{A}$ as well as $\boldsymbol{B}$.

**Theorem 2** If $\boldsymbol{A}$ and $\boldsymbol{B}$ are row equivalent matrices, then

**(a)** A given set of column vectors of $\boldsymbol{A}$ is linearly independent if and only if the corresponding column vectors of $\boldsymbol{B}$ are linearly independent.
**(b)** A given set of column vector of $\boldsymbol{A}$ forms a basis for the column space of $\boldsymbol{A}$ if and only if the corresponding column vector of $\boldsymbol{B}$ forms a basis for the column space of $\boldsymbol{B}$.

**Example 2** **(Bases for Row and Column Spaces)**

Find the bases for the row and column spaces of $\boldsymbol{A} = \begin{bmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 2 & -6 & 9 & -1 & 8 & 2 \\ 2 & -6 & 9 & -1 & 9 & 7 \\ -1 & 3 & -4 & 2 & -5 & -4 \end{bmatrix}$.

**Solution** We can find a basis for the row space of $\boldsymbol{A}$ by finding a basis for the row space of any row-echelon form of $\boldsymbol{A}$.

Now
$$\begin{bmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 2 & -6 & 9 & -1 & 8 & 2 \\ 2 & -6 & 9 & -1 & 9 & 7 \\ -1 & 3 & -4 & 2 & -5 & -4 \end{bmatrix}$$
$$\begin{bmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 0 & 0 & 1 & 3 & -2 & -6 \\ 0 & 0 & 1 & 3 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ -2R_1 + R_2 \\ -2R_1 + R_3 \\ R_1 + R_4 \end{matrix}$$
$$\begin{bmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 0 & 0 & 1 & 3 & -2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ -1R_2 + R_3 \\ \end{matrix}$$

Row-echelon form of $\boldsymbol{A}$: $\boldsymbol{R} = \begin{bmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 0 & 0 & 1 & 3 & -2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Here Theorem 1 implies that that the non zero rows are the basis vectors of the matrix. So these bases vectors are

$$\boldsymbol{r}_1 = \begin{bmatrix} 1 & -3 & 4 & -2 & 5 & 4 \end{bmatrix}$$
$$\boldsymbol{r}_2 = \begin{bmatrix} 0 & 0 & 1 & 3 & -2 & -6 \end{bmatrix}$$
$$\boldsymbol{r}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 5 \end{bmatrix}$$

$\boldsymbol{A}$ and $\boldsymbol{R}$ may have different column spaces, we cannot find a basis for the column space of $\boldsymbol{A}$ directly from the column vectors of $\boldsymbol{R}$. however, it follows from the theorem (2b) if

we can find a set of column vectors of ***R*** that forms a basis for the column space of ***R***, then the corresponding column vectors of ***A*** will form a basis for the column space of ***A***.

The first, third, and fifth columns of ***R*** contains the leading 1's of the row vectors, so

$$\boldsymbol{c}_1' = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \boldsymbol{c}_3' = \begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \end{bmatrix} \qquad \boldsymbol{c}_5' = \begin{bmatrix} 5 \\ -2 \\ 1 \\ 0 \end{bmatrix}$$

form a basis for the column space of ***R***, thus the corresponding column vectors of ***A***

namely, $\boldsymbol{c}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ -1 \end{bmatrix} \qquad \boldsymbol{c}_3 = \begin{bmatrix} 4 \\ 9 \\ 9 \\ -4 \end{bmatrix} \qquad \boldsymbol{c}_5 = \begin{bmatrix} 5 \\ 8 \\ 9 \\ -5 \end{bmatrix}$

form a basis for the column space of ***A***.

**Example**

The matrix

$$R = \begin{bmatrix} 1 & -2 & 5 & 0 & 3 \\ 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

is in row-echelon form.

The vectors

$$\boldsymbol{r}_1 = \begin{bmatrix} 1 & -2 & 5 & 0 & 3 \end{bmatrix}$$

$$\boldsymbol{r}_2 = \begin{bmatrix} 0 & 1 & 3 & 0 & 0 \end{bmatrix}$$

$$\boldsymbol{r}_3 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

**form a basis for the row space of R, and the vectors**

$$c_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, c_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, c_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

**form a basis for the column space of R.**

**Example 3** **(Basis for a Vector Space using Row Operation)**

Find bases for the space spanned by the vectors

$v_1 = (1,-2,0,0,3)$ $v_2 = (2,-5,-3,-2,6)$

$v_3 = (0,5,15,10,0)$ $v_4 = (2,6,18,8,6)$

**Solution** The space spanned by these vectors is the row space of the matrix

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$$

Transforming Matrix to Row Echelon Form:

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 5 & 15 & 10 & 0 \\ 0 & 10 & 18 & 8 & 0 \end{bmatrix} \begin{matrix} \\ (-2)R_1 + R_2 \\ (-2)R_1 + R_4 \\ (-1)R_2 \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -12 & -12 & 0 \end{bmatrix} \begin{matrix} \\ (-5)R_2 + R_3 \\ (-10)R_2 + R_4 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & -12 & -12 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ R_{34} \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ (-1/12)R_3 \\ \\ \end{matrix}$$

Therefore, $$\boldsymbol{R} = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The non-zero row vectors in this matrix are

$$\boldsymbol{w_1} = (1,-2,0,0,3), \boldsymbol{w_2} = (0,1,3,2,0), \boldsymbol{w_3} = (0,0,1,1,0)$$

These vectors form a basis for the row space and consequently form a basis for the subspace of $\boldsymbol{R^5}$ spanned by $\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}$.

**Example 4** **(Basis for the Row Space of a Matrix)**

Find a basis for the row space of $\boldsymbol{A} = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$ consisting entirely of row vectors from $\boldsymbol{A}$.

**Solution** We find $\boldsymbol{A^T}$; then we will use the method of example (2) to find a basis for the column space of $\boldsymbol{A^T}$; and then we will transpose again to convert column vectors back to row vectors. Transposing $\boldsymbol{A}$ yields

$$\boldsymbol{A^T} = \begin{bmatrix} 1 & 2 & 0 & 2 \\ -2 & -5 & 5 & 6 \\ 0 & -3 & 15 & 18 \\ 0 & -2 & 10 & 8 \\ 3 & 6 & 0 & 6 \end{bmatrix}$$

Transforming Matrix to Row Echelon Form:

$$\begin{bmatrix} 1 & 2 & 0 & 2 \\ -2 & -5 & 5 & 6 \\ 0 & -3 & 15 & 18 \\ 0 & -2 & 10 & 8 \\ 3 & 6 & 0 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & -1 & 5 & 10 \\ 0 & -3 & 15 & 18 \\ 0 & -2 & 10 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ 2R_1 + R_2 \\ \\ (-3)R_1 + R_5 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 1 & -5 & -10 \\ 0 & -3 & 15 & 18 \\ 0 & -2 & 10 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (-1)R_2$$

$$\begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 1 & -5 & -10 \\ 0 & 0 & 0 & -12 \\ 0 & 0 & 0 & -12 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} (3)R_2 + R_3 \\ (2)R_2 + R_4 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 1 & -5 & -10 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -12 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (-1/12)R_3$$

$$\begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 1 & -5 & -10 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad 12\,R_3 + R_4$$

Now $$\boldsymbol{R} = \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 1 & -5 & -10 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The first, second and fourth columns contain the leading 1's, so the corresponding column vectors in $\boldsymbol{A^T}$ form a basis for the column space of $\boldsymbol{A^T}$; these are

$$\boldsymbol{c_1} = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \\ 3 \end{bmatrix}, \quad \boldsymbol{c_2} = \begin{bmatrix} 2 \\ -5 \\ -3 \\ -2 \\ 6 \end{bmatrix} \text{ and } \boldsymbol{c_4} = \begin{bmatrix} 2 \\ 6 \\ 18 \\ 8 \\ 6 \end{bmatrix}$$

Transposing again and adjusting the notation appropriately yields the basis vectors

$\boldsymbol{r_1} = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \end{bmatrix}, \boldsymbol{r_2} = \begin{bmatrix} 2 & -5 & -3 & -2 & 6 \end{bmatrix}$ *and* $\boldsymbol{r_4} = \begin{bmatrix} 2 & 6 & 18 & 8 & 6 \end{bmatrix}$

for the row space of $\boldsymbol{A}$.

The following example shows how one sequence of row operations on $A$ leads to bases for the three spaces: Row $A$, Col $A$, and Nul $A$.

**Example 5** Find bases for the row space, the column space and the null space of the matrix

$$A = \begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix}$$

**Solution** To find bases for the row space and the column space, row reduce $A$ to an echelon form: $A \sim B = \begin{bmatrix} 1 & 3 & -5 & 1 & 5 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & -4 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

By Theorem (1), the first three rows of $B$ form a basis for the row space of $A$ (as well as the row space of $B$). Thus Basis for Row $A$:

$$\{(1, 3, -5, 1, 5), (0, 1, -2, 2, -7), (0, 0, 0, -4, 20)\}$$

For the column space, observe from $B$ that the pivots are in columns *1*, *2* and *4*. Hence columns *1*, *2* and *4* of $A$ (not $B$) form a basis for Col $A$:

$$\text{Basis for Col } A: \left\{ \begin{bmatrix} -2 \\ 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 11 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 7 \\ 5 \end{bmatrix} \right\}$$

Any echelon form of $A$ provides (in its nonzero rows) a basis for Row $A$ and also identifies the pivot columns of $A$ for Col $A$. However, for Nul $A$, we need the reduced echelon form. Further row operations on $B$ yield

$$A \sim B \sim C = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The equation $Ax = 0$ is equivalent to $Cx = 0$, that is,

$$\begin{aligned} x_1 + \quad & \quad x_3 \quad + x_5 = 0 \\ x_2 - 2x_3 \quad & \quad + 3x_5 = 0 \\ x_4 \quad & - 5x_5 = 0 \end{aligned}$$

So $x_1 = -x_3 - x_5$, $x_2 = 2x_3 - 3x_5$, $x_4 = 5x_5$, with $x_3$ and $x_5$ free variables. The usual calculations (discussed in lecture 21) show that

$$\text{Basis for Nul } \boldsymbol{A}: \left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 1 \end{bmatrix} \right\}$$

Observe that, unlike the bases for Col $\boldsymbol{A}$, the bases for Row $\boldsymbol{A}$ and Nul $\boldsymbol{A}$ have no simple connection with the entries in $\boldsymbol{A}$ itself.

**Note**

1. Although the first three rows of $\boldsymbol{B}$ in Example (5) are linearly independent, it is wrong to conclude that the first three rows of $\boldsymbol{A}$ are linearly independent. (In fact, the third row of A is 2 times the first row plus 7 times the second row).
2. Row operations do not preserve the linear dependence relations among the rows of a matrix.

**Definition** The **rank** of $\boldsymbol{A}$ is the dimension of the column space of $\boldsymbol{A}$.
Since Row $\boldsymbol{A}$ is the same as Col $\boldsymbol{A}^T$, the dimension of the row space of $\boldsymbol{A}$ is the rank of $\boldsymbol{A}^T$. The dimension of the null space is sometimes called the **nullity** of $\boldsymbol{A}$.

**Theorem 3** **(The Rank Theorem)** The dimensions of the column space and the row space of an $m \times n$ matrix $\boldsymbol{A}$ are equal. This common dimension, the rank of $\boldsymbol{A}$, also equals the number of pivot positions in $\boldsymbol{A}$ and satisfies the equation

$$\text{rank } \boldsymbol{A} + \dim \text{Nul } \boldsymbol{A} = n$$

**Example 6**

(a) If $\boldsymbol{A}$ is a $7 \times 9$ matrix with a two – dimensional null space, what is the rank of $\boldsymbol{A}$?
(b). Could a $6 \times 9$ matrix have a two – dimensional null space?

**Solution**

(a) Since $\boldsymbol{A}$ has $9$ columns, $(\text{rank } \boldsymbol{A}) + 2 = 9$ and hence rank $\boldsymbol{A} = 7$.
(b) No, If a $6 \times 9$ matrix, call it $\boldsymbol{B}$, had a two – dimensional null space, it would have to have rank $7$, by the Rank Theorem. But the columns of $\boldsymbol{B}$ are vectors in $\boldsymbol{R}^6$ and so the dimension of Col $\boldsymbol{B}$ cannot exceed $6$; that is, rank $\boldsymbol{B}$ cannot exceed $6$.

The next example provides a nice way to visualize the subspaces we have been studying. Later on, we will learn that Row $\boldsymbol{A}$ and Nul $\boldsymbol{A}$ have only the zero vector in common and are actually "perpendicular" to each other. The same fact will apply to Row $\boldsymbol{A}^T$ (= Col $\boldsymbol{A}$) and Nul $\boldsymbol{A}^T$. So the figure in Example (7) creates a good mental image for the general case.

**Example 7** Let $A = \begin{bmatrix} 3 & 0 & -1 \\ 3 & 0 & -1 \\ 4 & 0 & 5 \end{bmatrix}$. It is readily checked that Nul $A$ is the $x_2$ – axis, Row $A$ is the $x_1x_3$ – plane, Col $A$ is the plane whose equation is $x_1 - x_2 = 0$ and Nul $A^T$ is the set of all multiples of $(1, -1, 0)$. Figure 1 shows Nul $A$ and Row $A$ in the domain of the linear transformation $x \to Ax$; the range of this mapping, Col $A$, is shown in a separate copy of $R^3$, along with Nul $A^T$.

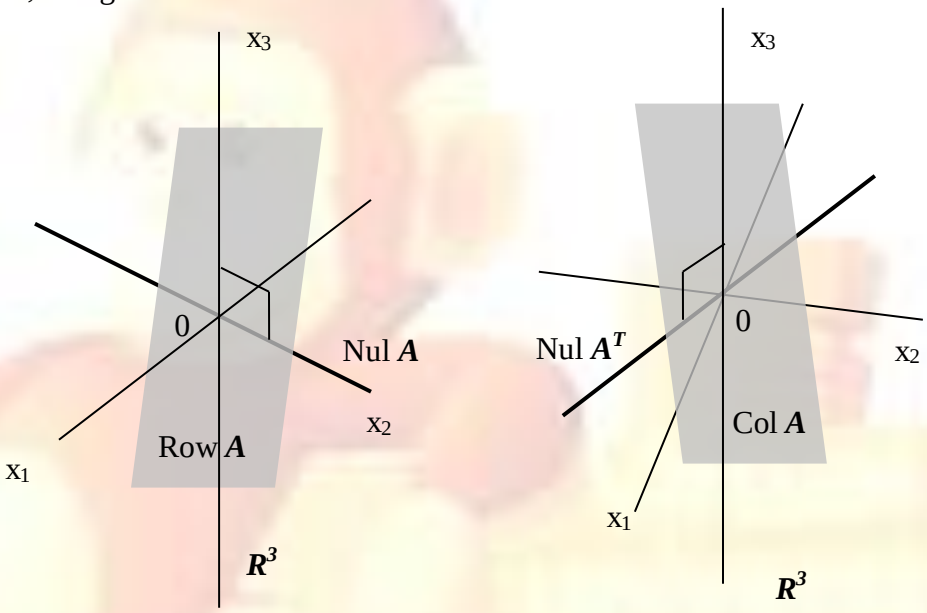



**Figure 1 – Subspaces associated with a matrix *A***

**Applications to Systems of Equations**

The Rank Theorem is a powerful tool for processing information about systems of linear equations. The next example simulates the way a real-life problem using linear equations might be stated, without explicit mention of linear algebra terms such as matrix, subspace and dimension.

**Example 8** A scientist has found two solutions to a homogeneous system of *40* equations in *42* variables. The two solutions are not multiples and all other solutions can be constructed by adding together appropriate multiples of these two solutions. Can the scientist be certain that an associated non-homogeneous system (with the same coefficients) has a solution?

**Solution** Yes. Let $A$ be the $40 \times 42$ coefficient matrix of the system. The given information implies that the two solutions are linearly independent and span Nul $A$. So dim Nul $A = 2$. By the Rank Theorem, dim Col $A = 42 - 2 = 40$. Since $R^{40}$ is the only subspace of $R^{40}$ whose dimension is *40*, Col $A$ must be all of $R^{40}$. This means that every non-homogeneous equation $Ax = b$ has a solution.

**Example 9** Find the rank and nullity of the matrix $A = \begin{bmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{bmatrix}$.

Verify that values obtained verify the dimension theorem.

**Solution**

$$\begin{bmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 0 & -4 & -5 & 3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{bmatrix} (-1)R_1$$

$$\begin{bmatrix} 1 & -2 & 0 & -4 & -5 & 3 \\ 0 & -1 & 2 & 12 & 16 & -5 \\ 0 & -1 & 2 & 12 & 16 & -5 \\ 0 & -1 & 2 & 12 & 16 & -5 \end{bmatrix} \begin{matrix} (-3)R_1 + R_2 \\ (-2)R_1 + R_3 \\ (-4)R_1 + R_4 \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & 0 & -4 & -5 & 3 \\ 0 & 1 & -2 & -12 & -16 & 5 \\ 0 & -1 & 2 & 12 & 16 & -5 \\ 0 & -1 & 2 & 12 & 16 & -5 \end{bmatrix} (-1)R_2$$

$$\begin{bmatrix} 1 & -2 & 0 & -4 & -5 & 3 \\ 0 & 1 & -2 & -12 & -16 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ R_2 + R_3 \\ R_2 + R_4 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & -4 & -28 & -37 & 13 \\ 0 & 1 & -2 & -12 & -16 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} 2R_2 + R_1$$

The reduced row-echelon form of $A$ is

$$\begin{bmatrix} 1 & 0 & -4 & -28 & -37 & 13 \\ 0 & 1 & -2 & -12 & -16 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (1)$$

The corresponding system of equations will be

$$x_1 - 4x_3 - 28x_4 - 37x_5 + 13x_6 = 0$$
$$x_2 - 2x_3 - 12x_4 - 16x_5 + 5x_6 = 0$$

or, on solving for the leading variables,

$$x_1 = 4x_3 - 28x_4 + 37x_5 - 13x_6 \qquad (2)$$
$$x_2 = 2x_3 + 12x_4 + 16x_5 - 5x_6$$

it follows that the general solution of the system is

$$x_1 = 4r + 28s + 37t - 13u$$
$$x_2 = 2r + 12s + 16t - 5u$$
$$x_3 = r$$
$$x_4 = s$$
$$x_5 = t$$
$$x_6 = u$$

or equivalently,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = r \begin{bmatrix} 4 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 28 \\ 12 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 37 \\ 16 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + u \begin{bmatrix} -13 \\ -5 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \qquad (3)$$

The four vectors on the right side of (3) form a basis for the solution space, so nullity $(\boldsymbol{A}) = 4$. The matrix $\boldsymbol{A} = \begin{bmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{bmatrix}$ has 6 columns,

so rank$(\boldsymbol{A})$ + nullity$(\boldsymbol{A}) = 2 + 4 = 6 = n$

**Example 10** Find the rank and nullity of the matrix; then verify that the values obtained satisfy the dimension theorem $A = \begin{bmatrix} 1 & -3 & 2 & 2 & 1 \\ 0 & 3 & 6 & 0 & -3 \\ 2 & -3 & -2 & 4 & 4 \\ 3 & -6 & 0 & 6 & 5 \\ -2 & 9 & 2 & -4 & -5 \end{bmatrix}$

**Solution** Transforming Matrix to the Reduced Row Echelon Form:

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 \\ 0 & 3 & 6 & 0 & -3 \\ 2 & -3 & -2 & 4 & 4 \\ 3 & -6 & 0 & 6 & 5 \\ -2 & 9 & 2 & -4 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 \\ 0 & 3 & 6 & 0 & -3 \\ 0 & 3 & -6 & 0 & 2 \\ 0 & 3 & -6 & 0 & 2 \\ 0 & 3 & 6 & 0 & -3 \end{bmatrix} \begin{matrix} \\ (-2)R_1 + R_3 \\ (-3)R_1 + R_4 \\ 2R_1 + R_5 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 \\ 0 & 1 & 2 & 0 & -1 \\ 0 & 3 & -6 & 0 & 2 \\ 0 & 3 & -6 & 0 & 2 \\ 0 & 3 & 6 & 0 & -3 \end{bmatrix} \begin{matrix} \\ \\ (1/3)R_2 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 \\ 0 & 1 & 2 & 0 & -1 \\ 0 & 0 & -12 & 0 & 5 \\ 0 & 0 & -12 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ (-3)R_2 + R_3 \\ (-3)R_2 + R_4 \\ (-3)R_2 + R_5 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 \\ 0 & 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/12 \\ 0 & 0 & -12 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ (-1/12)R_3 \\ \\ \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 2 & 2 & 1 \\ 0 & 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/12 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad 12R_3 + R_4$$

$$\begin{bmatrix} 1 & -3 & 0 & 2 & 11/6 \\ 0 & 1 & 0 & 0 & -1/6 \\ 0 & 0 & 1 & 0 & -5/12 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} (-2)R_3 + R_2 \\ (-2)R_3 + R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 2 & 4/3 \\ 0 & 1 & 0 & 0 & -1/6 \\ 0 & 0 & 1 & 0 & -5/12 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (3)R_2 + R_1 \qquad (1)$$

Since there are three nonzero rows (or equivalently, three leading 1's) the row space and column space are both three dimensional so rank $(\boldsymbol{A}) = 3$.

To find the nullity of $\boldsymbol{A}$, we find the dimension of the solution space of the linear system $\boldsymbol{Ax} = \boldsymbol{0}$. The system can be solved by reducing the augmented matrix to reduced row echelon form. The resulting matrix will be identical to (1), except with an additional last column of zeros, and the corresponding system of equations will be

$$x_1 + 0x_2 + 0x_3 + 2x_4 + \frac{4}{3}x_5 = 0$$

$$0x_1 + x_2 + 0x_3 + 0x_4 - \frac{1}{6}x_5 = 0$$

$$0x_1 + 0x_2 + x_3 + 0x_4 - \frac{5}{12}x_5 = 0$$

The system has infinitely many solutions:

$x_1 = -2\,x_4 + (-4/3)\,x_5$ $\qquad$ $x_2 = (1/6)\,x_5$

$x_3 = (5/12)\,x_5$ $\qquad$ $x_4 = s$

$x_5 = t$

The solution can be written in the vector form:

$\boldsymbol{c}_4 = (-2, 0, 0, 1, 0)$ $\qquad$ $\boldsymbol{c}_5 = (-4/3, 1/6, 5/12, 0, 1)$

Therefore the **null space** has a basis formed by the set

*{(-2, 0, 0, 1, 0), (-4/3, 1/6, 5/12,0,1)}*

The nullity of the matrix is 2. Now Rank $(A)$ + nullity $(A)$ = *3 + 2 =5 = n*

**Theorem 4** If $A$ is an *m* x *n*, matrix, then
**(a)** rank ($A$) = the number of leading variables in the solution of $Ax = 0$
**(b)** nullity ($A$) = the number of parameters in the general solution of $Ax = 0$

**Example 11** Find the number of parameters in the solution set of $Ax = 0$ if $A$ is a $5\times7$ matrix of rank *3*.
**Solution** nullity $(A)$ = *n* – rank $(A)$ = *7-3 =4*
Thus, there are four parameters.

**Example** Find the number of parameters in the solution set of $Ax = 0$ if $A$ is a $4\times4$ matrix of rank *0*.
**Solution** nullity $(A)$ = *n* – rank $(A)$ = *4-0 =4*
Thus, there are four parameters.

**Theorem 5** If $A$ is any matrix, then rank $(A)$ = rank $(A^T)$

**Four fundamental matrix spaces**

If we consider a matrix $A$ and its transpose $A^T$ together, then there are six vectors spaces of interest:

| | |
|---|---|
| Row space of $A$ | row space of $A^T$ |
| Column space of $A$ | column space of $A^T$ |
| Null space of $A$ | null space of $A^T$ |

However, transposing a matrix converts row vectors into column vectors and column vectors into row vectors, so that, except for a difference in notation, the row space of $A^T$ is the same as the column space of $A$ and the column space of $A^T$ is the same as row space of of $A$.
This leaves four vector spaces of interest:

| | |
|---|---|
| Row space of $A$ | column space of $A$ |
| Null space of $A$ | null space of $A^T$ |

These are known as the **fundamental matrix spaces** associated with $A$, if $A$ is an *m* x *n* matrix, then the row space of $A$ and null space of $A$ are subspaces of $R^n$ and the column space of $A$ and the null space of $A^T$ are subspaces of $R^m$.

Suppose now that $A$ is an *m* x *n* matrix of rank *r*, it follows from theorem (5) that $A^T$ is an *n* x *m* matrix of rank *r* . Applying theorem (3) on $A$ and $A^T$ yields

Nullity $(A)$=*n-r*, nullity $(A^T)$=*m-r*

From which we deduce the following table relating the dimensions of the four fundamental spaces of an *m* x *n* matrix $A$ of rank *r*.

| **Fundamental space** | **Dimension** |
|---|---|
| Row space of $\boldsymbol{A}$ | $r$ |
| Column space of $\boldsymbol{A}$ | $r$ |
| Null space of $\boldsymbol{A}$ | $n-r$ |
| Null space of $\boldsymbol{A^T}$ | $m-r$ |

**Example 12** If $\boldsymbol{A}$ is a $7$ x $4$ matrix, then the rank of $\boldsymbol{A}$ is at most $4$ and, consequently, the seven row vectors must be linearly dependent. If $\boldsymbol{A}$ is a $4$ x $7$ matrix, then again the rank of $\boldsymbol{A}$ is at most $4$ and, consequently, the seven column vectors must be linearly dependent.

**Rank and the Invertible Matrix Theorem** The various vector space concepts associated with a matrix provide several more statements for the Invertible Matrix Theorem. We list only the new statements here, but we reference them so they follow the statements in the original Invertible Matrix Theorem in lecture 13.

**Theorem 6 The Invertible Matrix Theorem (Continued)**

Let $\boldsymbol{A}$ be an $n$ x $n$ matrix. Then the following statements are each equivalent to the statement that $\boldsymbol{A}$ is an invertible matrix.

m. The columns of $\boldsymbol{A}$ form a basis of $\boldsymbol{R^n}$.
n. Col $\boldsymbol{A} = \boldsymbol{R^n}$.
o. dim Col $\boldsymbol{A} = n$
p. rank $\boldsymbol{A} = n$
q. Nul $\boldsymbol{A} = \{\boldsymbol{0}\}$
r. dim Nul $\boldsymbol{A} = 0$

**Proof** Statement (m) is logically equivalent to statements (e) and (h) regarding linear independence and spanning. The other statements above are linked into the theorem by the following chain of almost trivial implications:

$$(g) \Rightarrow (n) \Rightarrow (o) \Rightarrow (p) \Rightarrow (r) \Rightarrow (q) \Rightarrow (d)$$

Only the implication (p) $\Rightarrow$ (r) bears comment. It follows from the Rank Theorem because $\boldsymbol{A}$ is $n \times n$ . Statements (d) and (g) are already known to be equivalent, so the chain is a circle of implications.

We have refrained from adding to the Invertible Matrix Theorem obvious statements about the row space of $\boldsymbol{A}$, because the row space is the column space of $\boldsymbol{A^T}$. Recall from (1) of the Invertible Matrix Theorem that $\boldsymbol{A}$ is invertible if and only if $\boldsymbol{A^T}$ is invertible. Hence every statement in the Invertible Matrix Theorem can also be stated for $\boldsymbol{A^T}$.

**Numerical Note**

Many algorithms discussed in these lectures are useful for understanding concepts and making simple computations by hand. However, the algorithms are often unsuitable for large-scale problems in real life.

Rank determination is a good example. It would seem easy to reduce a matrix to echelon form and count the pivots. But unless exact arithmetic is performed on a matrix whose entries are specified exactly, row operations can change the apparent rank of a matrix.

For instance, if the value of $x$ in the matrix $\begin{bmatrix} 5 & 7 \\ 5 & x \end{bmatrix}$ is not stored exactly as *7* in a computer, then the rank may be *1* or *2*, depending on whether the computer treats $x - 7$ as zero.

In practical applications, the effective rank of a matrix $\boldsymbol{A}$ is often determined from the singular value decomposition of $\boldsymbol{A}$.

**Example 13** The matrices below are row equivalent

$$\boldsymbol{A} = \begin{bmatrix} 2 & -1 & 1 & -6 & 8 \\ 1 & -2 & -4 & 3 & -2 \\ -7 & 8 & 10 & 3 & -10 \\ 4 & -5 & -7 & 0 & 4 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 1 & -2 & -4 & 3 & -2 \\ 0 & 3 & 9 & -12 & 12 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1. Find rank $\boldsymbol{A}$ and dim Nul $\boldsymbol{A}$.
2. Find bases for Col $\boldsymbol{A}$ and Row $\boldsymbol{A}$.
3. What is the next step to perform if one wants to find a basis for Nul $\boldsymbol{A}$?
4. How many pivot columns are in a row echelon form of $\boldsymbol{A}^T$?

**Solution**

1. $\boldsymbol{A}$ has two pivot columns, so rank $\boldsymbol{A} = 2$. Since $\boldsymbol{A}$ has *5* columns altogether, dim Nul $\boldsymbol{A} = 5 - 2 = 3$.
2. The pivot columns of $\boldsymbol{A}$ are the first two columns. So a basis for Col $\boldsymbol{A}$ is

$$\{\boldsymbol{a}_1, \boldsymbol{a}_2\} = \left\{ \begin{bmatrix} 2 \\ 1 \\ -7 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 8 \\ -5 \end{bmatrix} \right\}$$

   The nonzero rows of $\boldsymbol{B}$ form a basis for Row $\boldsymbol{A}$, namely *{(1, –2, –4, 3, –2), (0, 3, 9, –12, 12)}*. In this particular example, it happens that any two rows of $\boldsymbol{A}$ form a basis for the row space, because the row space is two-dimensional and none of the rows of $\boldsymbol{A}$ is a multiple of another row. In general, the nonzero rows of an echelon form of $\boldsymbol{A}$ should be used as a basis for Row $\boldsymbol{A}$, not the rows of $\boldsymbol{A}$ itself.
3. For Nul $\boldsymbol{A}$, the next step is to perform row operations on $\boldsymbol{B}$ to obtain the reduced echelon form of $\boldsymbol{A}$.
4. Rank $\boldsymbol{A}^T$ = rank $\boldsymbol{A}$, by the Rank Theorem, because Col $\boldsymbol{A}^T$ = Row $\boldsymbol{A}$. So $\boldsymbol{A}^T$ has two pivot positions.

**Exercises**

In exercises 1 to 4, assume that the matrix $\boldsymbol{A}$ is row equivalent to $\boldsymbol{B}$. Without calculations, list rank $\boldsymbol{A}$ and dim Nul $\boldsymbol{A}$. Then find bases for Col $\boldsymbol{A}$, Row $\boldsymbol{A}$, and Nul $\boldsymbol{A}$.

1. $\boldsymbol{A} = \begin{bmatrix} 1 & -4 & 9 & -7 \\ -1 & 2 & -4 & 1 \\ 5 & -6 & 10 & 7 \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} 1 & 0 & -1 & 5 \\ 0 & -2 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

2. $\boldsymbol{A} = \begin{bmatrix} 1 & -3 & 4 & -1 & 9 \\ -2 & 6 & -6 & -1 & -10 \\ -3 & 9 & -6 & -6 & -3 \\ 3 & -9 & 4 & 9 & 0 \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} 1 & -3 & 0 & 5 & -7 \\ 0 & 0 & 2 & -3 & 8 \\ 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

3. $\boldsymbol{A} = \begin{bmatrix} 2 & -3 & 6 & 2 & 5 \\ -2 & 3 & -3 & -3 & -4 \\ 4 & -6 & 9 & 5 & 9 \\ -2 & 3 & 3 & -4 & 1 \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} 2 & -3 & 6 & 2 & 5 \\ 0 & 0 & 3 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

4. $\boldsymbol{A} = \begin{bmatrix} 1 & 1 & -3 & 7 & 9 & -9 \\ 1 & 2 & -4 & 10 & 13 & -12 \\ 1 & -1 & -1 & 1 & 1 & -3 \\ 1 & -3 & 1 & -5 & -7 & 3 \\ 1 & -2 & 0 & 0 & -5 & -4 \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} 1 & 1 & -3 & 7 & 9 & -9 \\ 0 & 1 & -1 & 3 & 4 & -3 \\ 0 & 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

5. If a 3 x 8 matrix $\boldsymbol{A}$ has rank 3, find dim Nul $\boldsymbol{A}$, dim Row $\boldsymbol{A}$, and rank $\boldsymbol{A^T}$.

6. If a 6 x 3 matrix $\boldsymbol{A}$ has rank 3, find dim Nul $\boldsymbol{A}$, dim Row $\boldsymbol{A}$, and rank $\boldsymbol{A^T}$.

7. Suppose that a 4 x 7 matrix $\boldsymbol{A}$ has four pivot columns. Is Col $\boldsymbol{A} = \boldsymbol{R^4}$? Is Nul $\boldsymbol{A} = \boldsymbol{R^3}$? Explain your answers.

8. Suppose that a 5 x 6 matrix $\boldsymbol{A}$ has four pivot columns. What is dim Nul $\boldsymbol{A}$? Is Col $\boldsymbol{A} = \boldsymbol{R^4}$? Why or why not?

9. If the null space of a 5 x 6 matrix $\boldsymbol{A}$ is 4-dimensional, what is the dimension of the column space of $\boldsymbol{A}$?

10. If the null space of a 7 x 6 matrix $\boldsymbol{A}$ is 5-dimensional, what is the dimension of the column space of $\boldsymbol{A}$?

11. If the null space of an 8 x 5 matrix $\boldsymbol{A}$ is 2-dimensional, what is the dimension of the row space of $\boldsymbol{A}$?

12. If the null space of a 5 x 6 matrix $\boldsymbol{A}$ is 4-dimensional, what is the dimension of the row space of $\boldsymbol{A}$?

13. If $\boldsymbol{A}$ is a 7 x 5 matrix, what is the largest possible rank of $\boldsymbol{A}$? If $\boldsymbol{A}$ is a 5 x 7 matrix, what is the largest possible rank of $\boldsymbol{A}$? Explain your answers.

14. If ***A*** is a 4 x 3 matrix, what is the largest possible dimension of the row space of ***A***? If ***A*** is a 3 x 4 matrix, what is the largest possible dimension of the row space of ***A***? Explain.

15. If ***A*** is a 6 x 8 matrix, what is the smallest possible dimension of Nul ***A***?

16. If ***A*** is a 6 x 4 matrix, what is the smallest possible dimension of Nul ***A***?



# Lecture 26

## Change of Basis

When a basis $\boldsymbol{B}$ is chosen for an $n$-dimensional vector space $\boldsymbol{V}$, the associated coordinate mapping onto $\boldsymbol{R^n}$ provides a coordinate system for $\boldsymbol{V}$. Each $\boldsymbol{x}$ in $\boldsymbol{V}$ is identified uniquely by its $\boldsymbol{B}$-coordinate vector $[\boldsymbol{x}]_{\boldsymbol{B}}$.

In some applications, a problem is described initially using a basis $\boldsymbol{B}$, but the problem's solution is aided by changing $\boldsymbol{B}$ to a new basis $\boldsymbol{C}$. Each vector is assigned a new $\boldsymbol{C}$-coordinate vector. In this section, we study how $[\boldsymbol{x}]_{\boldsymbol{C}}$ and $[\boldsymbol{x}]_{\boldsymbol{B}}$ are related for each $\boldsymbol{x}$ in $\boldsymbol{V}$.

To visualize the problem, consider the two coordinate systems in Fig. 1.

In Fig. 1(a), $\boldsymbol{x} = 3\boldsymbol{b}_1 + \boldsymbol{b}_2$, while in Fig. 1 (b), the same $\boldsymbol{x}$ is shown as $\boldsymbol{x} = 6\boldsymbol{c}_1 + 4\boldsymbol{c}_2$. That is

$$[\boldsymbol{x}]_{\boldsymbol{B}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \text{and} \quad [\boldsymbol{x}]_{\boldsymbol{C}} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

Example 1 shows how to find the connection between the two coordinate vectors. provided we know how $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ are formed from $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$.

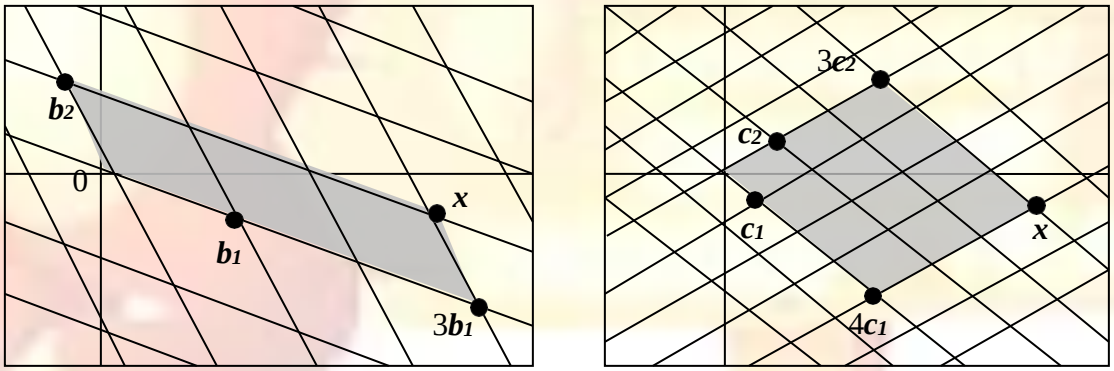



**Figure 1** – Two coordinate systems for the same vector space

**Example 1** Consider two bases $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2\}$ and $\boldsymbol{C} = \{\boldsymbol{c}_1, \boldsymbol{c}_2\}$ for a vector space $\boldsymbol{V}$, such that

$$\boldsymbol{b}_1 = 4\boldsymbol{c}_1 + \boldsymbol{c}_2 \text{ and } \boldsymbol{b}_2 = -6\boldsymbol{c}_1 + \boldsymbol{c}_2 \quad (1)$$

Suppose that $\boldsymbol{x} = 3\boldsymbol{b}_1 + \boldsymbol{b}_2$ (2)

suppose that $[\boldsymbol{x}]_{\boldsymbol{B}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Find $[\boldsymbol{x}]_{\boldsymbol{C}}$.

**Solution** Since the coordinate mapping is a linear transformation, we apply (2)

$$[\boldsymbol{x}]_C = [3\boldsymbol{b}_1 + \boldsymbol{b}_2]_C$$
$$= 3[\boldsymbol{b}_1]_C + [\boldsymbol{b}_2]_C$$

as a matrix equation,

$$[\boldsymbol{x}]_C = \begin{bmatrix} [\boldsymbol{b}_1]_C & [\boldsymbol{b}_2]_C \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad (3)$$

From (1), $[\boldsymbol{b}_1]_C = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $[\boldsymbol{b}_2]_C = \begin{bmatrix} -6 \\ 1 \end{bmatrix}$

from (3) $[\boldsymbol{x}]_C = \begin{bmatrix} 4 & -6 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$

The $C$ – coordinates of $\boldsymbol{x}$ match those of the $\boldsymbol{x}$ in Fig. 1.
The argument used to derive formula (3) is easily generalized to yield the following result.

**Theorem** Let $\boldsymbol{B} = \{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n\}$ and $\boldsymbol{C} = \{\boldsymbol{c}_1, \ldots, \boldsymbol{c}_n\}$ be bases of a vector space $\boldsymbol{V}$. Then there exists an $n$ x $n$ matrix $\underset{C \leftarrow B}{\boldsymbol{P}}$ such that

$$[\boldsymbol{x}]_C = \underset{C \leftarrow B}{\boldsymbol{P}} [\boldsymbol{x}]_B \quad (4)$$

The columns of $\underset{C \leftarrow B}{\boldsymbol{P}}$ are the $\boldsymbol{C}$-coordinate vectors of the vectors in the basis $\boldsymbol{B}$. So

$$\underset{C \leftarrow B}{\boldsymbol{P}} = \begin{bmatrix} [\boldsymbol{b}_1]_C & [\boldsymbol{b}_2]_C & \cdots & [\boldsymbol{b}_n]_C \end{bmatrix}$$

The matrix $\underset{C \leftarrow B}{\boldsymbol{P}}$ in above theorem is called the **change-of-coordinates matrix from $\boldsymbol{B}$ to $\boldsymbol{C}$**. Multiplication by $\underset{C \leftarrow B}{\boldsymbol{P}}$ converts $\boldsymbol{B}$-coordinates into $\boldsymbol{C}$-coordinates. Figure 2 illustrates the change-of-coordinates equation (4).
The columns of $\underset{C \leftarrow B}{\boldsymbol{P}}$ are linearly independent because they are the coordinate vectors of the linearly independent set $\boldsymbol{B}$. Thus $\underset{C \leftarrow B}{\boldsymbol{P}}$ is invertible. Left-multiplying both sides of (4) by $\left(\underset{C \leftarrow B}{\boldsymbol{P}}\right)^{-1}$, we obtain

$$\left(\underset{C \leftarrow B}{\boldsymbol{P}}\right)^{-1} [\boldsymbol{x}]_C = [\boldsymbol{x}]_B$$

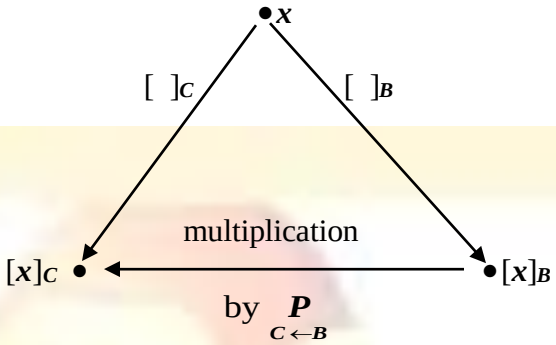



**Figure 2** Two coordinate system for $V$.

Thus $\left(\underset{C\leftarrow B}{P}\right)^{-1}$ is the matrix that converts $C$-coordinates into $B$-coordinates. That is,

$$\left(\underset{C\leftarrow B}{P}\right)^{-1} = \underset{B\leftarrow C}{P} \qquad (5)$$

**Change of Basis in $R^n$** If $B = \{b_1, \ldots, b_n\}$ and $E$ is the standard basis $\{e_1, \ldots, e_n\}$ in $R^n$, then $[b_1]_E = b_1$, and likewise for the other vectors in $B$. In this case, $\underset{E\leftarrow B}{P}$ is the same as the change-of-coordinates matrix $P_B$ introduced in Lecture 23, namely,

$$P_B = \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix}$$

To change coordinates between two nonstandard bases in $R^n$, we need above Theorem. The theorem shows that to solve the change-of-basis problem, we need the coordinate vector of the old basis relative to the new basis.

**Example 2** Let $D = \{d_1, d_2, d_3\}$ and $F = \{f_1, f_2, f_3\}$ be basis for vector space $V$, and suppose that $f_1 = 2d_1 - d_2 + d_3$, $f_2 = 3d_2 + d_3$, $f_3 = -3d_1 + 2d_3$.

(a) Find the change-of-coordinates matrix from $F$ to $D$.

(b) Find $[x]_D$ for $x = f_1 - 2f_2 + 2f_3$.

**Solution** (a) $$\underset{D\leftarrow F}{P} = \begin{bmatrix} [f_1]_D & [f_2]_D & [f_3]_D \end{bmatrix}$$

But $$[f_1]_D = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, [f_2]_D = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}, [f_3]_D = \begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix}$$

$$\therefore \quad \underset{D\leftarrow F}{P} = \begin{bmatrix} 2 & 0 & -3 \\ -1 & 3 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

(b) $$[x]_D = [f_1 - 2f_2 + 2f_3]_D = [f_1]_D - 2[f_2]_D + 2[f_3]_D$$

$$= \left[ [\boldsymbol{f}]_{1\,D} \quad [\boldsymbol{f}]_{2\,D} \quad [\boldsymbol{f}]_{3\,D} \right] \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & -3 \\ -1 & 3 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ -7 \\ 3 \end{bmatrix}$$

**Example 3** Let $\boldsymbol{b}_1 = \begin{bmatrix} -9 \\ 1 \end{bmatrix}$, $\boldsymbol{b}_2 = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$, $\boldsymbol{c}_1 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$, $\boldsymbol{c}_2 = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$ and consider the bases for $\boldsymbol{R}^2$ given by $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2\}$ and $\boldsymbol{C} = \{\boldsymbol{c}_1, \boldsymbol{c}_2\}$. Find the change of coordinates matrix from $\boldsymbol{B}$ to $\boldsymbol{C}$.

**Solution** The matrix $\boldsymbol{P}_{\boldsymbol{C} \leftarrow \boldsymbol{B}}$ involves the $\boldsymbol{C}$ – coordinate vectors of $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$.

Let $[\boldsymbol{b}_1]_C = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $[\boldsymbol{b}_2]_C = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$. Then, by definition,

$$[\boldsymbol{c}_1 \quad \boldsymbol{c}_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \boldsymbol{b}_1 \quad \text{and} \quad [\boldsymbol{c}_1 \quad \boldsymbol{c}_2] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \boldsymbol{b}_2$$

To solve both systems simultaneously augment the coefficient matrix with $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ and row reduce:

$$[\boldsymbol{c}_1 \quad \boldsymbol{c}_2 \quad \boldsymbol{b}_1 \quad \boldsymbol{b}_2] = \left[\begin{array}{cc:cc} 1 & 3 & -9 & -5 \\ -4 & -5 & 1 & -1 \end{array}\right]$$

$$\sim \left[\begin{array}{cc:cc} 1 & 3 & -9 & -5 \\ 0 & 7 & -35 & -21 \end{array}\right] R_2 + 4R_1$$

$$\sim \left[\begin{array}{cc:cc} 1 & 3 & -9 & -5 \\ 0 & 1 & -5 & -3 \end{array}\right] \frac{1}{7} R_2$$

$$\sim \left[\begin{array}{cc:cc} 1 & 0 & 6 & 4 \\ 0 & 1 & -5 & -3 \end{array}\right] R_1 - 3R_2 \qquad (6)$$

$$[\boldsymbol{b}_1]_C = \begin{bmatrix} 6 \\ -5 \end{bmatrix} \quad \text{and} \quad [\boldsymbol{b}_2]_C = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

therefore

$$\boldsymbol{P}_{\boldsymbol{C} \leftarrow \boldsymbol{B}} = \left[ [\boldsymbol{b}_1]_C \; [\boldsymbol{b}_2]_C \right] = \begin{bmatrix} 6 & 4 \\ -5 & -3 \end{bmatrix}$$

**Example 4** Let $\boldsymbol{b_1} = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$, $\boldsymbol{b_2} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $\boldsymbol{c_1} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $\boldsymbol{c_2} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$ and consider the bases for $\boldsymbol{R^2}$ given by $\boldsymbol{B} = \{\boldsymbol{b_1}, \boldsymbol{b_2}\}$ and $\boldsymbol{C} = \{\boldsymbol{c_1}, \boldsymbol{c_2}\}$.

(a) Find the change – of – coordinates matrix from $\boldsymbol{C}$ to $\boldsymbol{B}$.

(b) Find the change – of – coordinates matrix from $\boldsymbol{B}$ to $\boldsymbol{C}$.

**Solution**

(a) Notice that $\underset{B \leftarrow C}{\boldsymbol{P}}$ is needed rather than $\underset{C \leftarrow B}{\boldsymbol{P}}$ and compute

$$[\boldsymbol{b_1} \quad \boldsymbol{b_2} \vdots \boldsymbol{c_1} \quad \boldsymbol{c_2}] = \left[\begin{array}{cc:cc} 7 & 2 & 4 & 5 \\ -2 & -1 & 1 & 2 \end{array}\right]$$

$$\left[\begin{array}{cc:cc} 1 & \frac{2}{7} & \frac{4}{7} & \frac{5}{7} \\ -2 & -1 & 1 & 2 \end{array}\right] \frac{1}{7} R_1$$

$$\left[\begin{array}{cc:cc} 1 & \frac{2}{7} & \frac{4}{7} & \frac{5}{7} \\ 0 & -\frac{3}{7} & \frac{15}{7} & \frac{24}{7} \end{array}\right] 2R_1 + R_2$$

$$\left[\begin{array}{cc:cc} 1 & \frac{2}{7} & \frac{4}{7} & \frac{5}{7} \\ 0 & 1 & -5 & -8 \end{array}\right] -\frac{7}{3} R_2$$

$$\left[\begin{array}{cc:cc} 1 & 0 & 2 & 3 \\ 0 & 1 & -5 & -8 \end{array}\right] -\frac{2}{7} R_2 + R_1$$

so $\underset{B \leftarrow C}{\boldsymbol{P}} = \begin{bmatrix} 2 & 3 \\ -5 & -8 \end{bmatrix}$

(b) By part (a) and property (5) above (with $\boldsymbol{B}$ and $\boldsymbol{C}$ interchanged),

$$\underset{C \leftarrow B}{\boldsymbol{P}} = (\underset{B \leftarrow C}{\boldsymbol{P}})^{-1} = \frac{1}{-1}\begin{bmatrix} -8 & -3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 3 \\ -5 & -2 \end{bmatrix}.$$

**Example 5** Let $\boldsymbol{b_1} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$, $\boldsymbol{b_2} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$, $\boldsymbol{c_1} = \begin{bmatrix} -7 \\ 9 \end{bmatrix}$, $\boldsymbol{c_2} = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$ and consider the bases for $\boldsymbol{R^2}$ given by $\boldsymbol{B} = \{\boldsymbol{b_1}, \boldsymbol{b_2}\}$ and $\boldsymbol{C} = \{\boldsymbol{c_1}, \boldsymbol{c_2}\}$.

(a) Find the change – of – coordinates matrix from $\boldsymbol{C}$ to $\boldsymbol{B}$.

(b) Find the change – of – coordinates matrix from $\boldsymbol{B}$ to $\boldsymbol{C}$.

**Solution**

(a) Notice that $\underset{B \leftarrow C}{\boldsymbol{P}}$ is needed rather than $\underset{C \leftarrow B}{\boldsymbol{P}}$ and compute

$$[\boldsymbol{b_1} \quad \boldsymbol{b_2} \vdots \boldsymbol{c_1} \quad \boldsymbol{c_2}] = \left[\begin{array}{cc:cc} 1 & -2 & -7 & -5 \\ -3 & 4 & 9 & 7 \end{array}\right] \sim \left[\begin{array}{cc:cc} 1 & 0 & 5 & 3 \\ 0 & 1 & 6 & 4 \end{array}\right]$$

so $\underset{B\leftarrow C}{\boldsymbol{P}} = \begin{bmatrix} 5 & 3 \\ 6 & 4 \end{bmatrix}$

(b) By part (a) and property (5) above (with $\boldsymbol{B}$ and $\boldsymbol{C}$ interchanged),

$$\underset{C\leftarrow B}{\boldsymbol{P}} = (\underset{B\leftarrow C}{\boldsymbol{P}})^{-1} = \frac{1}{2}\begin{bmatrix} 4 & -3 \\ -6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & -3/2 \\ -3 & 5/2 \end{bmatrix}.$$

**Example 6**

1. Let $\boldsymbol{F} = \{\boldsymbol{f}_1, \boldsymbol{f}_2\}$ and $\boldsymbol{G} = \{\boldsymbol{g}_1, \boldsymbol{g}_2\}$ be bases for a vector space $\boldsymbol{V}$, and let $\boldsymbol{P}$ be a matrix whose columns are $[\boldsymbol{f}_1]_G$. Which of the following equations is satisfied by $\boldsymbol{P}$ for all $\boldsymbol{v}$ in $\boldsymbol{V}$?

(i) $[\boldsymbol{v}]_F = \boldsymbol{P}[\boldsymbol{v}]_G$ (ii) $[\boldsymbol{v}]_G = \boldsymbol{P}[\boldsymbol{v}]_F$

2. Let $\boldsymbol{B}$ and $\boldsymbol{C}$ be as in Example 1. Use the results of that example to find the change-of-coordinates matrix from $\boldsymbol{C}$ to $\boldsymbol{B}$.

**Solution**

1. Since the columns of $\boldsymbol{P}$ are $\boldsymbol{G}$-coordinate vectors, a vector of the form $\boldsymbol{Px}$ must be a $\boldsymbol{G}$-coordinate vector. Thus $\boldsymbol{P}$ satisfies equation (ii).
2. The coordinate vectors found in Example 1 show that

$$\underset{C\leftarrow B}{\boldsymbol{P}} = \begin{bmatrix} [\boldsymbol{b}_1]_C & [\boldsymbol{b}_2]_C \end{bmatrix} = \begin{bmatrix} 4 & -6 \\ 1 & 1 \end{bmatrix}$$

Hence $$\underset{B\leftarrow C}{\boldsymbol{P}} = \left(\underset{C\leftarrow B}{\boldsymbol{P}}\right)^{-1} = \frac{1}{10}\begin{bmatrix} 1 & 6 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 0.1 & 0.6 \\ -0.1 & 0.4 \end{bmatrix}$$

## Exercises

1. Let $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2\}$ and $\boldsymbol{C} = \{\boldsymbol{c}_1, \boldsymbol{c}_2\}$ be bases for a vector space $\boldsymbol{V}$, and suppose that $\boldsymbol{b}_1 = 6\,\boldsymbol{c}_1 - 2\,\boldsymbol{c}_2$ and $\boldsymbol{b}_2 = 9\,\boldsymbol{c}_1 - 4\,\boldsymbol{c}_2$.
(a) Find the change-of-coordinates matrix from $\boldsymbol{B}$ to $\boldsymbol{C}$.
(b) Find $[\boldsymbol{x}]_C$ for $\boldsymbol{x} = -3\,\boldsymbol{b}_1 + 2\,\boldsymbol{b}_2$. Use part (a).

2. Let $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2\}$ and $\boldsymbol{C} = \{\boldsymbol{c}_1, \boldsymbol{c}_2\}$ be bases for a vector space $\boldsymbol{V}$, and suppose that $\boldsymbol{b}_1 = -\boldsymbol{c}_1 + 4\,\boldsymbol{c}_2$ and $\boldsymbol{b}_2 = 5\,\boldsymbol{c}_1 - 3\,\boldsymbol{c}_2$.
(a) Find the change-of-coordinates matrix from $\boldsymbol{B}$ to $\boldsymbol{C}$.
(b) Find $[\boldsymbol{x}]_C$ for $\boldsymbol{x} = 5\,\boldsymbol{b}_1 + 3\,\boldsymbol{b}_2$.

3. Let $\boldsymbol{U} = \{\boldsymbol{u}_1, \boldsymbol{u}_2\}$ and $\boldsymbol{W} = \{\boldsymbol{w}_1, \boldsymbol{w}_2\}$ be bases for $\boldsymbol{V}$, and let $\boldsymbol{P}$ be a matrix whose columns are $[\boldsymbol{u}_1]_W$ and $[\boldsymbol{u}_2]_W$ . Which of the following equations is satisfied by $\boldsymbol{P}$ for all $\boldsymbol{x}$ in $\boldsymbol{V}$?

(i) $[\boldsymbol{x}]_U = \boldsymbol{P}[\boldsymbol{x}]_W$ (ii) $[\boldsymbol{x}]_W = \boldsymbol{P}[\boldsymbol{x}]_U$

4. Let $\boldsymbol{A} = \{\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3\}$and $\boldsymbol{D} = \{\boldsymbol{d}_1, \boldsymbol{d}_2, \boldsymbol{d}_3\}$ be bases for $\boldsymbol{V}$, and let $\boldsymbol{P} = \begin{bmatrix} [\boldsymbol{d}_1]_A & [\boldsymbol{d}_2]_A & [\boldsymbol{d}_3]_A \end{bmatrix}$. Which of the following equations is satisfied by $\boldsymbol{P}$ for all $\boldsymbol{x}$ in $\boldsymbol{V}$?

(i) $[\boldsymbol{x}]_A = \boldsymbol{P}[\boldsymbol{x}]_D$ (ii) $[\boldsymbol{x}]_D = \boldsymbol{P}[\boldsymbol{x}]_A$

5. Let $\boldsymbol{A} = \{\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3\}$ and $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3\}$ be bases for a vector space $\boldsymbol{V}$, and suppose that $\boldsymbol{a}_1 = 4\boldsymbol{b}_1 - \boldsymbol{b}_2$, $\boldsymbol{a}_2 = -\boldsymbol{b}_1 + \boldsymbol{b}_2 + \boldsymbol{b}_3$, and $\boldsymbol{a}_3 = \boldsymbol{b}_2 - 2\,\boldsymbol{b}_3$.
(a) Find the change-of-coordinates matrix from $\boldsymbol{A}$ to $\boldsymbol{B}$.
(b) Find $[\boldsymbol{x}]_{\boldsymbol{B}}$ for $\boldsymbol{x} = 3\,\boldsymbol{a}_1 + 4\,\boldsymbol{a}_2 + \boldsymbol{a}_3$.

In exercises 6 to 9, let $\boldsymbol{B} = \{\boldsymbol{b}_1, \boldsymbol{b}_2\}$ and $\boldsymbol{C} = \{\boldsymbol{c}_1, \boldsymbol{c}_2\}$ be bases for $\boldsymbol{R}^2$. In each exercise, find the change-of-coordinates matrix from $\boldsymbol{B}$ to $\boldsymbol{C}$, and change-of-coordinates matrix from $\boldsymbol{C}$ to $\boldsymbol{B}$.

6. $\boldsymbol{b}_1 = \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \boldsymbol{b}_2 = \begin{bmatrix} -3 \\ -1 \end{bmatrix}, \boldsymbol{c}_1 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}, \boldsymbol{c}_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$

7 $\boldsymbol{b}_1 = \begin{bmatrix} -1 \\ 8 \end{bmatrix}, \boldsymbol{b}_2 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}, \boldsymbol{c}_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \boldsymbol{c}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

8. $\boldsymbol{b}_1 = \begin{bmatrix} -6 \\ -1 \end{bmatrix}, \boldsymbol{b}_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \boldsymbol{c}_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \boldsymbol{c}_2 = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$

9. $\boldsymbol{b}_1 = \begin{bmatrix} 7 \\ -2 \end{bmatrix}, \boldsymbol{b}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \boldsymbol{c}_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \boldsymbol{c}_2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$

10. In $\boldsymbol{P}_2$, find the change-of-coordinate matrix from the basis $\boldsymbol{B} = \{\mathbf{1} - 2\,\boldsymbol{t} + \boldsymbol{t}^2, \mathbf{3} - 5\,\boldsymbol{t} + 4\,\boldsymbol{t}^2, 2\,\boldsymbol{t} + 3\,\boldsymbol{t}^2\}$ to the standard basis $\boldsymbol{C} = \{\mathbf{1}, \boldsymbol{t}, \boldsymbol{t}^2\}$. Then find the $\boldsymbol{B}$-coordinate vector for $-\mathbf{1} + 2\,\boldsymbol{t}$.

11. In $\boldsymbol{P}_2$, find the change-of-coordinates matrix from the basis $\boldsymbol{B} = \{\mathbf{1} - 3\,\boldsymbol{t}^2, \mathbf{2} + \boldsymbol{t} - 5\,\boldsymbol{t}^2, \mathbf{1} + 2\,\boldsymbol{t}\}$ to the standard basis. Then write $\boldsymbol{t}^2$ as a linear combination of the polynomials in $\boldsymbol{B}$.

12. Let $\boldsymbol{P} = \begin{bmatrix} 1 & 2 & -1 \\ -3 & -5 & 0 \\ 4 & 6 & 1 \end{bmatrix}, \boldsymbol{v}_1 = \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix}, \boldsymbol{v}_2 = \begin{bmatrix} -8 \\ 5 \\ 2 \end{bmatrix}, \boldsymbol{v}_3 = \begin{bmatrix} -7 \\ 2 \\ 6 \end{bmatrix}$

(a). Find a basis $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$ for $\boldsymbol{R}^3$ such that $\boldsymbol{P}$ is the change-of-coordinates matrix from $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$ to the basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$.
(b). Find a basis $\{\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3\}$ for $\boldsymbol{R}^3$ such that $\boldsymbol{P}$ is the change-of-coordinates matrix from $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ to $\{\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3\}$.

# Lecture 27

## Applications to Difference Equations

Recently Discrete or Digital data has been used preferably rather then continuous data in scientific and engineering problems. Difference equation is considered more reliable tool to analyze such type of data even if we are using a differential equation to analyze continuous process, a numerical solution is often produced from a related difference equation.
In this lecture we will study some fundamental properties of linear difference equation that are considered best tool in Linear Algebra.

**Discrete-Time Signals** Let ***S*** is the space of discrete-time signals. A signal in ***S*** is a function defined only on the integers and is visualized as a sequence of numbers, say, $\{y_k\}$.
Digital signals obviously arise in electrical and control systems engineering, but discrete-data sequences are also generated in biology, physics, economics, demography and many other areas, wherever a process is measured, or sampled, at discrete time intervals.
When a process beings at a specific time, it is sometimes convenient to write a signal as a sequence of the form $(y_0, y_1, y_2, \ldots)$. The term $y_k$ for k<0 either are assumed to be zero or are simply omitted.
Discrete time signals are functions defined on integers that are sequences or it takes only a discrete set of values.
For Example a Radio Station broadcast's weather report once in a day so the sampling period is discrete, but sampling may be uniform or constant accordingly.

**Example 1** The crystal clear sounds from a compact disc player are produced from music that has been sampled at the rate of 44,100 times per second. At each measurement, the amplitude of the music signal is recorded as a number, say, $y_k$. The original music is composed of many different sounds of varying frequencies, yet the sequence $\{y_k\}$ contains enough information to reproduce all the frequencies in the sound up to about 20,000 cycles per second, higher than the human ear can sense.

**Linear Independence in the Space *S* of Signals** To simplify notation, we consider a set of only three signals in ***S***, say, $\{u_k\}$,$\{v_k\}$and $\{w_k\}$. They are linearly independent precisely when the equation

$$c_1 u_k + c_2 v_k + c_3 w_k = 0 \quad \text{for all } k \qquad (1)$$

implies that $c_1 = c_2 = c_3 = 0$. The phrase "for all $k$" means for all integers positive, negative and zero. One could also consider signals that start with $k = 0$, for example, in a case "for all $k$" would mean for all integers $k \geq 0$.
Suppose $c_1, c_2, c_3$ satisfy (1). Then the equation (1) holds for any three consecutive values of $k$, say, $k$, $k + 1$ and $k + 2$. Thus (1) implies that

$$c_1 u_{k+1} + c_2 v_{k+1} + c_3 w_{k+1} = 0 \quad \text{for all } k$$

and

$$c_1 u_{k+2} + c_2 v_{k+2} + c_3 w_{k+2} = 0 \quad \text{for all } k$$

Hence $c_1$, $c_2$, $c_3$ satisfy

$$\begin{bmatrix} u_k & v_k & w_k \\ u_{k+1} & v_{k+1} & w_{k+1} \\ u_{k+2} & v_{k+2} & w_{k+2} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ for all } k \qquad (2)$$

The coefficient matrix in this system is called the **Casorati matrix** of the signals, and the determinant of the matrix is called the **Casoratian** of $\{u_k\}$,$\{v_k\}$, and $\{w_k\}$. If for at least one value of $k$ the Casorati matrix is invertible, then (2) will imply that $c_1 = c_2 = c_3 = 0$, which will prove that the three signals are linearly independent.

**Example 2** Verify that $1^k$, $(-2)^k$ and $3^k$ are linearly independent signals.

**Solution** The Casorati matrix is $\begin{bmatrix} 1^k & (-2)^k & 3^k \\ 1^{k+1} & (-2)^{k+1} & 3^{k+1} \\ 1^{k+2} & (-2)^{k+2} & 3^{k+2} \end{bmatrix}$

Row operations can show fairly easily that this matrix is always invertible. However; it is faster to substitute a value for $k$ – say, $k = 0$ – and row reduce the numerical matrix:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & 3 \\ 1 & 4 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & 2 \\ 1 & 4 & 9 \end{bmatrix} R_2 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & 2 \\ 0 & 3 & 8 \end{bmatrix} R_3 - R_1 \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & 2 \\ 0 & 0 & 10 \end{bmatrix} R_3 + R_2$$

The Casorati matrix is invertible for $k = 0$. So $1^k$, $(-2)^k$ and $3^k$ are linearly independent.

If a Casorati matrix is not invertible, the associated signals being tested may or may not be linearly dependent. However, it can be shown that if the signals are all solutions of the same homogeneous difference equation (described below), then either the Casorati matrix is invertible for all k and the signals are linearly independent, or else for all k the Casorati matrix is not invertible and the signals are linearly dependent.

**Activity**
Verify that $2^k$, $(-1)^k$ and $0^k$ are linearly independent signals.

**Linear Difference Equations** Given scalars $a_0, \ldots, a_n$, with $a_0$ and $a_n$ nonzero, and given a signal $\{z_k\}$, the equation

$$a_0 y_{k+n} + a_1 y_{k+n-1} + \ldots + a_{n-1} y_{k+1} + a_n y_k = z_k \quad \text{for all } k \qquad (3)$$

is called a **linear difference equation** (or **linear recurrence relation**) of order $n$. For simplicity, $a_0$ is often taken equal to 1. If $\{z_k\}$ is the zero sequence, the equation is **homogeneous;** otherwise, the equation is **non-homogeneous**.

In simple words, an equation which expresses a value of a sequence as a function of the other terms in the sequence is called a difference equation.
In particular an equation which expresses the value $a_n$ of a sequence $\{a_n\}$ as a function of the term $a_{n-1}$ is called a first order difference equation.

**Example 3** In digital signal processing, a difference equation such as (3) above describes a linear filter and $a_0, \ldots, a_n$ are called the **filter coefficients**. If $\{y_k\}$ is treated as the input and $\{z_k\}$ the output, then the solutions of the associated homogeneous equation are the signals that are filtered out and transformed into the zero signal. Let us feed two different signals into the filter

$$0.35y_{k+2} + 0.5y_{k+1} + 0.35y_k = z_k$$

Here 0.35 is an abbreviation for $\sqrt{2}/4$. The first signal is created by sampling the continuous signal $y = \cos(\pi t/4)$ at integer values of $t$, as in Fig. 3 (a). The discrete signals is $\{y_k\} = \{\ldots \cos(0), \cos(\pi/4), \cos(2\pi/4), \cos(3\pi/4) \ldots\}$

For simplicity, write $\pm 0.7$ in place of $\pm\sqrt{2}/2$, so that

$$\{y_k\} = \{ \ldots, 1, 0.7, 0, -0.7, -1, -0.7, 0, 0.7, 1, 0.7, 0, \ldots.\}$$

$k = 0$

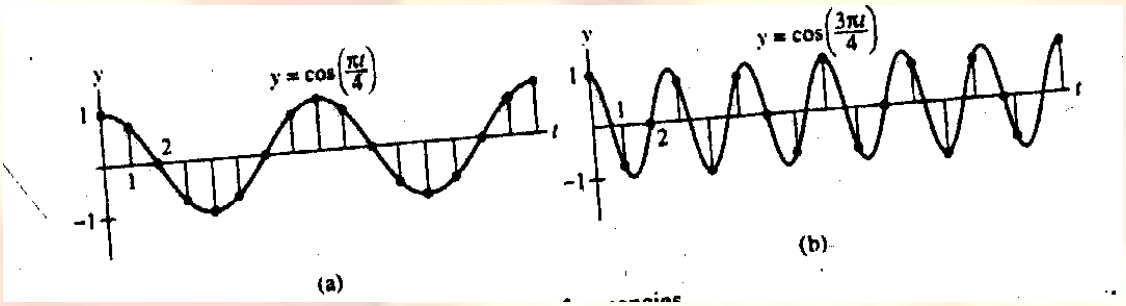



**Figure** Discrete signals with different frequencies

The following table shows a calculation of the output sequence $\{z_k\}$, where 0.35(0.7) is an abbreviation for $(\sqrt{2}/4).(\sqrt{2}/2) = 0.25$. The output is $\{y_k\}$, shifted by one term.

**Table** Computing the output of a filter

| $K$ | $y_k$ | $y_{k+1}$ | $y_{k+2}$ | $0.35y_k+0.5y_{k+1}+0.35y_{k+2} = z_k$ |
|---|---|---|---|---|
| 0 | **1** | **0.7** | **0** | $0.35(1)+0.5(0.7)+0.35(0)=0.7$ |
| 1 | **0.7** | **0** | **-0.7** | **0** |
| 2 | **0** | **-0.7** | **-1** | **-0.7** |
| 3 | **-0.7** | **-1** | **-0.7** | **-1** |
| 4 | **-1** | **-0.7** | **0** | **-0.7** |
| 5 | **-0.7** | **0** | **0.7** | **0** |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |

A different input signal is produced from the higher frequency signal $y = \cos(3\pi t/4)$. Sampling at the same rate as before produces a new input sequence:

$$\{w_k\} = \{\ldots, 1, -0.7, 0, 0.7, -1, 0.7, 0, -0.7, 1, -0.7, 0, \ldots\}$$
$$\uparrow$$
$$k = 0$$

When $\{w_k\}$ is fed into the filter, the output is the zero sequence. The filter, called a **low pass filter,** lets $\{y_k\}$ pass through, but stops the higher frequency $\{w_k\}$.
In many applications, a sequence$\{z_k\}$ is specified for the right side of a difference equation (3) and a $\{y_k\}$ that satisfies (3) is called a solution of the equation. The next example shows how to find solution for a homogeneous equation.

**Example 4** Solutions of a homogeneous difference equation often have the form $y_k = r^k$ for some $r$. Find some solutions of the equation

$$y_{k+3} - 2y_{k+2} - 5y_{k+1} + 6y_k = 0 \text{ for all } k \qquad (4)$$

**Solution** Substitute $r^k$ for $y_k$ in the equation and factor the left side:

$$r^{k+3} - 2r^{k+2} - 5r^{k+1} + 6r^k = 0 \qquad (5)$$

*By synthetic division*

$$r^k (r^3 - 2r^2 - 5r + 6) = 0$$

| | 1 | -2 | -5 | 6 |
|---|---|---|---|---|
| 1 | 0 | 1 | -1 | -6 |
| | 1 | -1 | -6 | 0 |

*So we get*

$$r^k [(x-1)(x^2 - x - 6)] = 0$$
$$r^k [(x-1)(x^2 - 3x + 2x - 6)] = 0$$
$$r^k [(x-1)\{x(x-3) + 2(x-3)\}] = 0$$
$$r^k (r-1)(r+2)(r-3) = 0 \qquad (6)$$

since (5) is equivalent to (6), $r^k$ satisfies the difference equation (4) if and only if $r$ satisfies (6). Thus $1^k$, $(-2)^k$ and $3^k$ are all solutions of (4). For instance, to verify that $3^k$ is a solution of (4), compute

$$3^{k+3} - 2.3^{k+2} - 5.3^{k+1} + 6.3^k = 3^k(27 - 18 - 15 + 6) = 0 \text{ for all k}$$

In general, a nonzero signal $r^k$ satisfies the homogeneous difference equation

$$y_{k+n} + a_1 y_{k+n-1} + \ldots + a_{n-1} y_{k+1} + a_n y_k = 0 \quad \text{for all } k$$

if and only if $r$ is a root of the **auxiliary equation**

$$r^n + a_1 r^{n-1} + \ldots + a_{n-1} r + a_n .1 = 0$$

We will not consider the case when $r$ is a repeated root of the auxiliary equation. When the auxiliary equation has a complex root, the difference equation has solutions of the form $s^k \cos kw$ and $s^k \sin kw$, for constants $s$ and $w$. This happened in Example 3.

**Solution Sets of Linear Difference Equations**

Given $a_1, \dots, a_n$, consider the mapping $T: S \to S$ that transforms a signal $\{y_k\}$into a signal $\{w_k\}$ given by

$$w_k = y_{k+n} + a_1 y_{k+n-1} + \dots + a_{n-1} y_{k+1} + a_n y_k$$

It is readily checked that $T$ is a linear transformation. This implies that the solution set of the homogeneous equation

$$y_{k+n} + a_1 y_{k+n-1} + \dots + a_{n-1} y_{k+1} + a_n y_k = 0 \quad \text{for all } k$$

is the kernel of $T$ (kernel is the set of signals that $T$ maps into the zero signal) and hence the solution set is a subspace of $S$. Any linear combination of solutions is again a solution.

**Theorem** If $a_n \neq 0$ and if $\{z_k\}$ is given, the equation

$$y_{k+n} + a_1 y_{k+n-1} + \dots + a_{n-1} y_{k+1} + a_n y_k = z_k \quad \text{for all } k \qquad (7)$$

has a unique solution whenever $y_0, \dots, y_{n-1}$ are specified.

**Proof** If $y_0, \dots, y_{n-1}$ are specified, use (7) to define

$$y_n = z_0 - [a_1 y_{n-1} + \dots + a_n y_0]$$

And now that $y_1, \dots, y_n$ are specified, use (7) to define $y_{n+1}$. In general, use the recursion relation

$$y_{n+k} = z_k - [a_1 y_{k+n-1} + \dots + a_n y_k] \qquad (8)$$

to define $y_{n+k}$ for $k \geq 0$. To define $y_k$ for $k<0$, use the recursion relation

$$y_k = \frac{1}{a_n} z_k - \frac{1}{a_n}[y_{k+n} + a_1 y_{k+n-1} + \dots + a_{n-1} y_{k+1}] \qquad (9)$$

This produces a signal that satisfies (7). Conversely, any signal that satisfies (7) for all $k$ certainly satisfies (8) and (9) so the solution of (7) is unique.

**Theorem** The set $H$ of all solutions of the $n$th-order homogeneous linear difference equation

$$y_{k+n} + a_1 y_{k+n-1} + \dots + a_{n-1} y_{k+1} + a_n y_k = 0 \quad \text{for all } k \qquad (10)$$

is an $n$-dimensional vector space.

**Proof** We explained earlier why $H$ is a subspace of $S$. For $\{y_k\}$in $H$, let $F\{y_k\}$be the vector in $R^n$ given by $(y_0, y_1, \dots, y_{n-1})$. It is readily verified that $F: H \to R^n$ is a linear transformation. Given any vector $(y_0, y_1, \dots, y_{n-1})$ in $R^n$, the previous theorem says that there is a unique signal $\{y_k\}$ in $H$ such that $F\{y_k\}=(y_0, y_1, \dots, y_{n-1})$. It means that $F$ is a one-to-one linear transformation of $H$ onto $R^n$; that is, $F$ is an isomorphism. Thus dim $H$ = dim $R^n = n$.

**Example 5** Find a basis for the set of all solutions to the difference equation

$$y_{k+3} - 2y_{k+2} - 5y_{k+1} + 6y_k = 0 \quad \text{for all } k$$

**Solution**

Generally it is difficult to identify directly that a set of signals spans the solution space, but this problem is being resolved by two above key theorems, from the last theorem the solution space is exactly three – dimensional and the Basis Theorem describes that a

linearly independent set of $n$ vectors in an $n$ – dimensional space is automatically a basis. So $1^k$, $(-2)^k$ and $3^k$ form a basis for the solution space.

The standard way to describe the "general solution" of (10) is to exhibit a basis for the subspace of all solutions. Such a basis is usually called a **fundamental set of solutions** of (10). In practice, if we can find $n$ linearly independent signals that satisfy (10), they will automatically span the $n$-dimensional solution space, as we saw in above example.

**Non-homogeneous Equations** The general solution of the non-homogeneous difference equation

$$y_{k+n} + a_1 y_{k+n-1} + \ldots + a_{n-1} y_{k+1} + a_n y_k = z_k \quad \text{for all } k \qquad (11)$$

can be written as one particular solution of (11) plus an arbitrary linear combination of a fundamental set of solutions of the corresponding homogeneous equation (10). This fact is analogous to the result in lecture 7 about how the solution sets of $\boldsymbol{Ax} = \boldsymbol{b}$ and $\boldsymbol{Ax} = \boldsymbol{0}$ are parallel. Both results have the same explanation: The mapping $\boldsymbol{x} \to \boldsymbol{Ax}$ is linear, and the mapping that transforms the signal $\{y_k\}$ into the signal $\{z_k\}$ in (11) is linear.

**Example 6** Verify that the signal $y_k = k^2$ satisfies the difference equation

$$y_{k+2} - 4y_{k+1} + 3y_k = -4k \quad \text{for all } k \qquad (12)$$

Then find a description of all solutions of this equation.

**Solution** Substitute $k^2$ for $y_k$ in the left side of (12):

$$(k+2)^2 - 4(k+1)^2 + 3k^2 = (k^2 + 4k + 4) - 4(k^2 + 2k + 1) + 3k^2 = -4k$$

So $k^2$ is indeed a solution of (12). The next step is to solve the homogeneous equation

$$y_{k+2} - 4y_{k+1} + 3y_k = 0 \qquad (13)$$

The auxiliary equation is

$$r^2 - 4r + 3 = r^2 - 3r - r + 3 = r(r-3) - 1(r-3) = (r-1)(r-3) = 0$$

The roots are $r = 1, 3$. So two solutions of the homogeneous difference equation are $\boldsymbol{1^k}$ and $\boldsymbol{3^k}$. They are obviously not multiples of each other, so they are linearly independent signals. (The Casorati test could have been used, too.) By the last Theorem, the solution space is two – dimensional, so $\boldsymbol{3^k}$ and $\boldsymbol{1^k}$ form a basis for the set of solutions of (13). Translating that set by a particular solution of the non-homogeneous equation (12), we obtain the general solution of (12):

$$\boldsymbol{k^2} + c_1 \boldsymbol{1^k} + c_2 \boldsymbol{3^k}, \quad \text{or} \quad \boldsymbol{k^2} + \boldsymbol{c_1} + c_2 \boldsymbol{3^k}$$

**Reduction to Systems of First-Order Equations** A modern way to study a homogeneous nth-order linear difference equation is to replace it by an equivalent system of first order difference equations, written in the form $\boldsymbol{x_{k+1}} = \boldsymbol{Ax_k}$ for k = 0, 1, 2, … Where the vectors $\boldsymbol{x_k}$ are in $\boldsymbol{R^n}$ and $\boldsymbol{A}$ is an $n$ x $n$ matrix.

**Example 7** Write the following difference equation as a first order system:

$$y_{k+3} - 2y_{k+2} - 5y_{k+1} + 6y_k = 0 \text{ for all } k$$

**Solution** For each $k$, set $\boldsymbol{x}_k = \begin{bmatrix} y_k \\ y_{k+1} \\ y_{k+2} \end{bmatrix}$

The difference equation says that $y_{k+3} = 2y_{k+2} + 5y_{k+1} - 6y_k$, so

$$x_{k+1} = \begin{bmatrix} y_{k+1} \\ y_{k+2} \\ y_{k+3} \end{bmatrix} = \begin{bmatrix} 0 + y_{k+1} + 0 \\ 0 + 0 + y_{k+2} \\ -6y_k + 5y_{k+1} + 2y_{k+2} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & 5 & 2 \end{bmatrix} \begin{bmatrix} y_k \\ y_{k+1} \\ y_{k+2} \end{bmatrix}$$

That is, $x_{k+1} = Ax_k$ for all $k$, where $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & 5 & 2 \end{bmatrix}$

In general, the equation $y_{k+n} + a_1 y_{k+n-1} + \ldots + a_{n-1} y_{k+1} + a_n y_k = 0$ for all $k$ can be written as $x_{k+1} = Ax_k$ for all $k$, where

$$x_k = \begin{bmatrix} y_k \\ y_{k+1} \\ \vdots \\ y_{k+n-1} \end{bmatrix}, A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix}$$

**Example 8** It can be shown that the signals $2^k$, $3^k \, Sin \frac{k\pi}{2}$, and $3^k \, Cos \frac{k\pi}{2}$ are solutions of $y_{k+3} - 2y_{k+2} + 9y_{k+1} - 18y_k = 0$. Show that these signals form a basis for the set of all solutions of the difference equation.

**Solution** Examine the Casorati matrix:

$$C(k) = \begin{bmatrix} 2^k & 3^k \, Sin \frac{k\pi}{2} & 3^k \, Cos \frac{k\pi}{2} \\ 2^{k+1} & 3^{k+1} \, Sin \frac{(k+1)\pi}{2} & 3^{k+1} \, Cos \frac{(k+1)\pi}{2} \\ 2^{k+2} & 3^{k+2} \, Sin \frac{(k+2\pi)}{2} & 3^{k+2} \, Cos \frac{(k+2\pi)}{2} \end{bmatrix}$$

Set $k = 0$ and row reduce the matrix to verify that it has three pivot positions and hence is invertible:

$$C(0) = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 0 \\ 4 & 0 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 3 & -2 \\ 4 & 0 & -9 \end{bmatrix} R_2 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 3 & -2 \\ 0 & 0 & -13 \end{bmatrix} R_3 - 4R_1$$

The Casorati matrix is invertible at $k = 0$, so signals are linearly independent. Since there are three signals, and the solution space $\boldsymbol{H}$ of the difference equation has three-dimensions (Theorem 2), the signals form a basis for $\boldsymbol{H}$, by the Basis Theorem.

## Exercises

Verify that the signals in exercises 1 and 2 are solution of the accompanying difference equation.

1. $\mathbf{2^k}$, $\mathbf{(-4)^k}$; $y_{k+2} + 2\,y_{k+1} - 8\,y_k = 0$

2. $\mathbf{3^k}$, $\mathbf{(-3)^k}$; $y_{k+2} - 9\,y_k = 0$

Show that the signals in exercises 3 to 6 form a basis for the solution set of accompanying difference equation.

3. $\mathbf{2^k}$, $\mathbf{(-4)^k}$; $y_{k+2} + 2\,y_{k+1} - 8\,y_k = 0$

4. $\mathbf{3^k}$, $\mathbf{(-3)^k}$; $y_{k+2} - 9\,y_k = 0$

5. $\mathbf{(-3)^k}$, $\boldsymbol{k}\mathbf{(-3)^k}$; $y_{k+2} + 6\,y_{k+1} + 9\,y_k = 0$

6. $\mathbf{5^k}\text{Cos}\dfrac{k\pi}{2}$, $\mathbf{5^k}\text{Sin}\dfrac{k\pi}{2}$; $y_{k+2} + 25\,y_k = 0$

In exercises 7 to 10, assume that the signals listed are solutions of the given difference equation. Determine if the signals form a basis for the solution space of the equation.

7. $\mathbf{1^k}$, $\mathbf{3^k}\text{Cos}\dfrac{k\pi}{2}$, $\mathbf{3^k}\text{Sin}\dfrac{k\pi}{2}$; $y_{k+3} - y_{k+2} + 9\,y_{k+1} - 9\,y_k = 0$

8. $\mathbf{(-1)^k}$, $\boldsymbol{k}\mathbf{(-1)^k}$, $\mathbf{5^k}$; $y_{k+3} - 3\,y_{k+2} - 9\,y_{k+1} - 5\,y_k = 0$

9. $\mathbf{(-1)^k}$, $\mathbf{3^k}$; $y_{k+3} + y_{k+2} - 9\,y_{k+1} - 9\,y_k = 0$

10. $\mathbf{1^k}$, $\mathbf{(-1)^k}$; $y_{k+4} - 2\,y_{k+2} + y_k = 0$

In exercises 11 and 12, find a basis for the solution space of the difference equation.

11. $y_{k+2} - 7\,y_{k+1} + 12\,y_k = 0$

12. $16y_{k+2} + 8\,y_{k+1} - 3\,y_k = 0$

In exercises 13 and 14, show that the given signal is a solution of the difference equation. Then find the general solution of that difference equation.

13. $y_k = \text{k}^2$; $y_{k+2} + 3\,y_{k+1} - 4\,y_k = 10\,\text{k} + 7$

14. $y_k = 2 - 2\,\text{k}$; $y_{k+2} - (9/2)\,y_{k+1} + 2\,y_k = 3\,\text{k} + 2$

Write the difference equations in exercises 15 and 16 as first order systems, $\boldsymbol{x}_{k+1} = \boldsymbol{A}\,\boldsymbol{x}_k$, for all $k$.

15. $y_{k+4} - 6\,y_{k+3} + 8\,y_{k+2} + 6\,y_{k+1} - 9\,y_k = 0$

16. $y_{k+3} - (3/4)y_{k+2} + (1/16)y_k = 0$

# Lecture 28

# Eigenvalues and Eigenvectors

In this lecture we will discuss linear equations of the form $Ax = x$ and, more generally, equations of the form $Ax = \lambda x$, where $\lambda$ is a scalar. Such equations arise in a wide variety of important applications and will be a recurring theme in the rest of this course.

**Fixed Points**

A fixed point of an $n \times n$ matrix $\boldsymbol{A}$ is a vector $\boldsymbol{x}$ in $\boldsymbol{R^n}$ such that $\boldsymbol{Ax} = \boldsymbol{x}$. Every square matrix $\boldsymbol{A}$ has at least one fixed point, namely $\boldsymbol{x} = \boldsymbol{0}$. We call this the trivial fixed point of $\boldsymbol{A}$.

The general procedure for finding the fixed points of a matrix $\boldsymbol{A}$ is to rewrite the equation $\boldsymbol{Ax} = \boldsymbol{x}$ as $\boldsymbol{Ax} = \boldsymbol{Ix}$ or, alternatively, as

$$(\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x} = \boldsymbol{0} \qquad (1)$$

Since this can be viewed as a homogeneous linear system of $n$ equations in $n$ unknowns with coefficient matrix $\boldsymbol{I} - \boldsymbol{A}$, we see that the set of fixed points of an $n \times n$ matrix is a subspace of $\boldsymbol{R^n}$ that can be obtained by solving (1).

The following theorem will be useful for ascertaining the nontrivial fixed points of a matrix.

**Theorem 1**

If $\boldsymbol{A}$ is an $n$ x $n$ matrix, then the following statements are equivalent.

(a) $\boldsymbol{A}$ has nontrivial fixed points.

(b) $\boldsymbol{I} - \boldsymbol{A}$ is singular.

(c) $\det(\boldsymbol{I} - \boldsymbol{A}) = 0$.

**Example 1**

In each part, determine whether the matrix has nontrivial fixed points; and, if so, graph the subspace of fixed points in an $xy$-coordinate system.

(*a*) $\boldsymbol{A} = \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix}$ (*b*) $\boldsymbol{A} = \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix}$

**Solution**

(a) The matrix has only the trivial fixed point since.

$$(I - A) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 3 & 6 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} -2 & -6 \\ -1 & -1 \end{pmatrix}$$

$$\det(I - A) = \det\begin{pmatrix} -2 & -6 \\ -1 & -1 \end{pmatrix} = (-1)(-2) - (-1)(-6) = -4 \neq 0$$

(b) The matrix has nontrivial fixed points since

$$(I - A) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix}$$

$$\det(I - A) = \det \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix} = 0$$

The fixed points $\boldsymbol{x} = (x, y)$ are the solutions of the linear system $(\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x} = \boldsymbol{0}$, which we can express in component form as

$$\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

A general solution of this system is

$$x = 2t, y = t \qquad (2)$$

which are parametric equations of the line $y = \frac{1}{2}x$. It follows from the corresponding vector form of this line that the fixed points are

$$\boldsymbol{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2t \\ t \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad (3)$$

As a check, $\boldsymbol{Ax} = \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2t \\ t \end{bmatrix} = \begin{bmatrix} 2t \\ t \end{bmatrix} = \boldsymbol{x}$

so every vector of form (3) is a fixed point of $\boldsymbol{A}$.

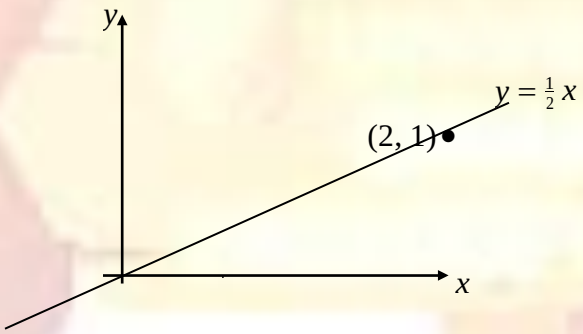



**Figure 1**

**Eigenvalues and Eigenvectors**

In a fixed point problem one looks for nonzero vectors that satisfy the equation $\boldsymbol{Ax} = \boldsymbol{x}$. One might also consider whether there are nonzero vectors that satisfy such equations as

$$\boldsymbol{Ax} = 2\boldsymbol{x}, \boldsymbol{Ax} = -3\boldsymbol{x}, \boldsymbol{Ax} = \sqrt{2}\,\boldsymbol{x}$$

or, more generally, equations of the form $\boldsymbol{Ax} = \lambda \boldsymbol{x}$ in which $\lambda$ is a scalar.

**Definition** If $\boldsymbol{A}$ is an $n$ x $n$ matrix, then a scalar $\lambda$ is called an **eigenvalue** of $\boldsymbol{A}$ if there is a nonzero vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \lambda \boldsymbol{x}$. If $\lambda$ is an eigenvalue of $\boldsymbol{A}$, then every nonzero vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \lambda \boldsymbol{x}$ is called an **eigenvector** of $\boldsymbol{A}$ corresponding to $\lambda$.

**Example 2**

Let $\boldsymbol{A}=\begin{bmatrix}1 & 6\\ 5 & 2\end{bmatrix}$, $\boldsymbol{u}=\begin{bmatrix}6\\ -5\end{bmatrix}$ and $\boldsymbol{v}=\begin{bmatrix}3\\ -2\end{bmatrix}$. Are $\boldsymbol{u}$ and $\boldsymbol{v}$ eigenvectors of $\boldsymbol{A}$?

**Solution**

$$\boldsymbol{Au}=\begin{bmatrix}1 & 6\\ 5 & 2\end{bmatrix}\begin{bmatrix}6\\ -5\end{bmatrix}=\begin{bmatrix}-24\\ 20\end{bmatrix}=-4\begin{bmatrix}6\\ -5\end{bmatrix}=-4\boldsymbol{u}$$

$$\boldsymbol{Av}=\begin{bmatrix}1 & 6\\ 5 & 2\end{bmatrix}\begin{bmatrix}3\\ -2\end{bmatrix}=\begin{bmatrix}-9\\ 11\end{bmatrix}\neq\lambda\begin{bmatrix}3\\ -2\end{bmatrix}$$

Thus $\boldsymbol{u}$ is an eigenvector corresponding to an eigenvalue – 4, but $\boldsymbol{v}$ is not an eigenvector of $\boldsymbol{A}$, because $\boldsymbol{Av}$ is not a multiple of $\boldsymbol{v}$.

**Example 3**

Show that 7 is an eigenvalue of $\boldsymbol{A}=\begin{bmatrix}1 & 6\\ 5 & 2\end{bmatrix}$, find the corresponding eigenvectors.

**Solution**

The scalar 7 is an eigenvalue of $\boldsymbol{A}$ if and only if the equation

$$\boldsymbol{Ax}=7\boldsymbol{x} \qquad \text{(A)}$$

has a nontrivial solution. But (A) is equivalent to $\boldsymbol{Ax}-7\boldsymbol{x}=\boldsymbol{0}$, or

$$(\boldsymbol{A}-7\boldsymbol{I})\,\boldsymbol{x}=\boldsymbol{0} \qquad \text{(B)}$$

To solve this homogeneous equation, form the matrix

$$\boldsymbol{A}-7\boldsymbol{I}=\begin{bmatrix}1 & 6\\ 5 & 2\end{bmatrix}-\begin{bmatrix}7 & 0\\ 0 & 7\end{bmatrix}=\begin{bmatrix}-6 & 6\\ 5 & -5\end{bmatrix}$$

The columns of $\boldsymbol{A}-7\boldsymbol{I}$ are obviously linearly dependent, so (B) has nontrivial solutions. Thus 7 is an eigenvalue of $\boldsymbol{A}$. To find the corresponding eigenvectors, use row operations:

$$\begin{bmatrix}-6 & 6 & 0\\ 5 & -5 & 0\end{bmatrix}\sim\begin{bmatrix}1 & -1 & 0\\ 5 & -5 & 0\end{bmatrix}(-1R_1-R_2)$$

$$\sim\begin{bmatrix}1 & -1 & 0\\ 0 & 0 & 0\end{bmatrix}(R_2-5R_1)$$

The general solution has the form $x_2\begin{bmatrix}1\\ 1\end{bmatrix}$. Each vector of this form with $x_2\neq 0$ is an eigenvector corresponding to $\lambda=7$.

The equivalence of equations (A) and (B) obviously holds for any $\lambda$ in place of $\lambda=7$. Thus $\lambda$ is an eigenvalue of $\boldsymbol{A}$ if and only if the equation

$$(\boldsymbol{A}-\lambda\boldsymbol{I})\boldsymbol{x}=\boldsymbol{0} \qquad \text{(C)}$$

has a nontrivial solution.

**Eigen space**

The set of all solutions of $(A - \lambda I)x = 0$ is just the null space of the matrix $A - \lambda I$. So this set is a subspace of $R^n$ and is called the eigenspace of $A$ corresponding to $\lambda$. The eigenspace consists of the zero vector and all the eigenvectors corresponding to $\lambda$.

Example 3 shows that for matrix $A$ in Example 2, the eigenspace corresponding to $\lambda = 7$ consists of all multiples of (1, 1), which is the line through (1, 1) and the origin. From Example 2, one can check that the eigenspace corresponding to $\lambda = -4$ is the line through (6, -5). These eigenspaces are shown in Fig. 1, along with eigenvectors (1, 1) and (3/2, -5/4) and the geometric action of the transformation $x \to Ax$ on each eigenspace.

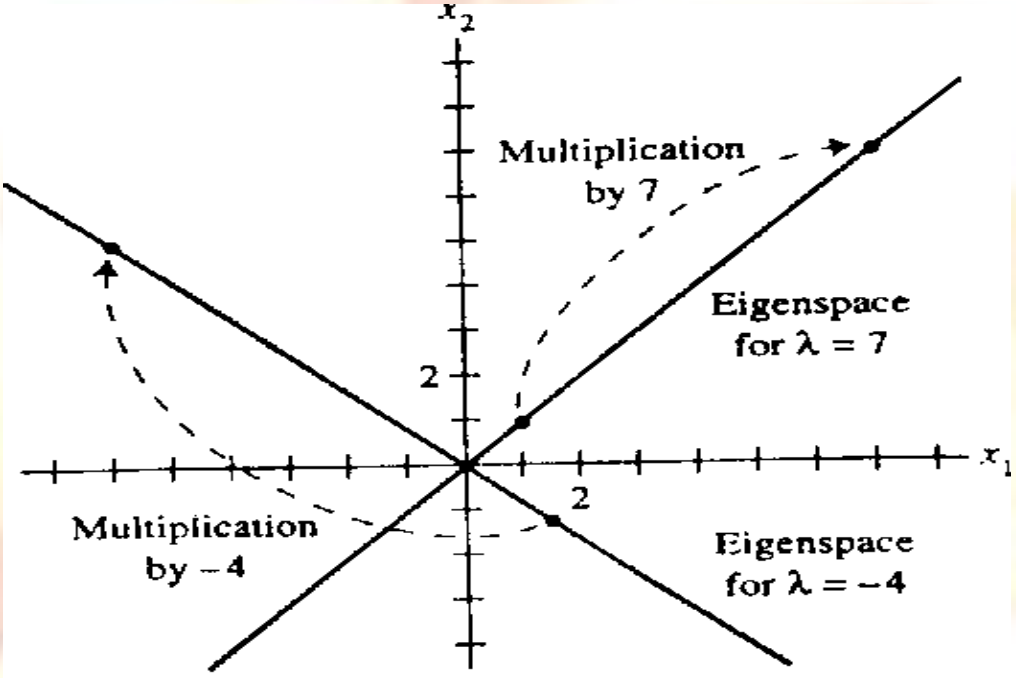



**FIGURE 2** Eigenspaces for $\lambda = -4$ and $\lambda = 7$.

**Example 4** Let $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$.

Find a basis for the corresponding eigenspace where eigen value of matrix is 2.

**Solution** Form $\boldsymbol{A}-2\boldsymbol{I}=\begin{bmatrix}4 & -1 & 6\\ 2 & 1 & 6\\ 2 & -1 & 8\end{bmatrix}-\begin{bmatrix}2 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 2\end{bmatrix}=\begin{bmatrix}2 & -1 & 6\\ 2 & -1 & 6\\ 2 & -1 & 6\end{bmatrix}$ and row reduce the augmented matrix for $(\boldsymbol{A}-2\boldsymbol{I})\,\boldsymbol{x}=\boldsymbol{0}$:

$$\begin{bmatrix}2 & -1 & 6 & 0\\ 2 & -1 & 6 & 0\\ 2 & -1 & 6 & 0\end{bmatrix}$$

$$\sim\begin{bmatrix}2 & -1 & 6 & 0\\ 0 & 0 & 0 & 0\\ 2 & -1 & 6 & 0\end{bmatrix}R_2-R_1$$

$$\sim\begin{bmatrix}2 & -1 & 6 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}R_3-R_1$$

At this point we are confident that 2 is indeed an eigenvalue of $\boldsymbol{A}$ because the equation $(\boldsymbol{A}-2\boldsymbol{I})\,\boldsymbol{x}=\boldsymbol{0}$ has free variables. The general solution is

$$2x_1-x_2+6x_3=0.....(a)$$

*Let* $x_2=t,\ x_3=s$ *then*

$$2x_1=t-6s$$

$$x_1=\left(\tfrac{1}{2}\right)t-3s$$

*then*

$$\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}t/2-3s\\ t\\ s\end{bmatrix}=\begin{bmatrix}t/2\\ t\\ 0\end{bmatrix}+\begin{bmatrix}-3s\\ 0\\ s\end{bmatrix}=t\begin{bmatrix}1/2\\ 1\\ 0\end{bmatrix}+s\begin{bmatrix}-3\\ 0\\ 1\end{bmatrix}$$

By back substitution the general solution is

$$\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=x_2\begin{bmatrix}1/2\\ 1\\ 0\end{bmatrix}+x_3\begin{bmatrix}-3\\ 0\\ 1\end{bmatrix},\ x_2 \text{ and } x_3 \text{ free}$$

The eigenspace, shown in Fig. 2, is a two – dimensional subspace of $\boldsymbol{R}^3$. A basis is

$$\left\{\begin{bmatrix}1\\ 2\\ 0\end{bmatrix},\begin{bmatrix}-3\\ 0\\ 1\end{bmatrix}\right\}\text{is a basis.}$$

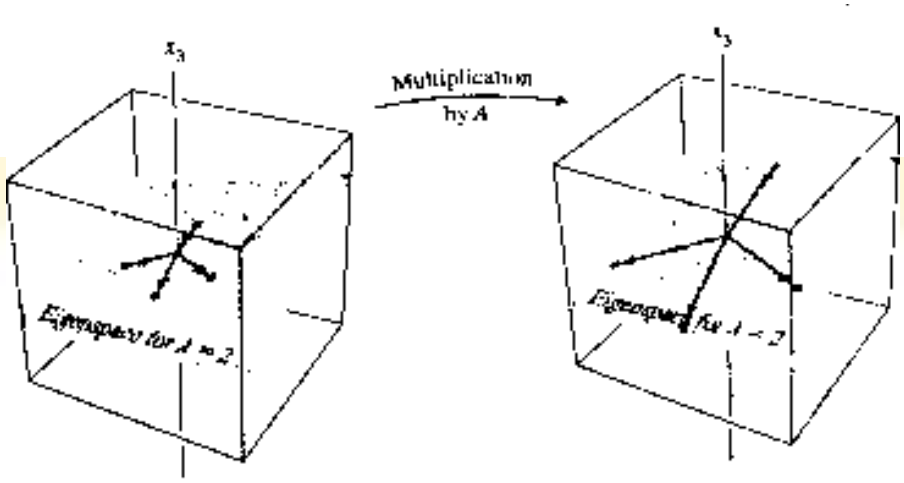



FIGURE 3 A acts as a dilation on the eigenspace.

The most direct way of finding the eigenvalues of an $n \times n$ matrix $\boldsymbol{A}$ is to rewrite the equation $Ax = \lambda x$ as $Ax = \lambda Ix$ , or equivalently, as

$$(\lambda I - A)x = 0 \qquad (4)$$

and then try to determine those values of $\lambda$ , if any, for which this system has nontrivial solutions. Since (4) have nontrivial solutions if and only if the coefficient matrix $\lambda \boldsymbol{I} - \boldsymbol{A}$ is singular, we see that the eigenvalues of $\boldsymbol{A}$ are the solutions of the equation

$$\det(\lambda I - A) = 0 \qquad (5)$$

Equation (5) is known as characteristic equation. Also, if $\lambda$ is an eigenvalue of $A$, then equation (4) has a nonzero solution space, which we call the eigenspace of $A$ corresponding to $\lambda$ . It is the nonzero vectors in the eigenspace of A corresponding to $\lambda$ that are the eigenvectors of A corresponding to $\lambda$ .

The above discussion is summarized by the following theorem.

**Theorem** If $\boldsymbol{A}$ is an $n \times n$ matrix and $\lambda$ is a scalar, then the following statements are equivalent.

(i) $\lambda$ is an eigenvalue of $\boldsymbol{A}$.

(ii) $\lambda$ is a solution of the equation $\det(\lambda I - A) = 0$ .

(iii) The linear system $(\lambda I - A)x = 0$ has nontrivial solutions.

**Eigenvalues of Triangular Matrices** If $\boldsymbol{A}$ is an $n \times n$ triangular matrix with diagonal entries $a_{11}$, $a_{22}$, …, $a_{nn}$, then $\lambda I - A$ is a triangular matrix with diagonal entries $\lambda - a_{11}, \lambda - a_{22}, \square, \lambda - a_{nn}$ . Thus, the characteristic polynomial of $\boldsymbol{A}$ is

$$\det(\lambda I - A) = (\lambda - a_{11})(\lambda - a_{22}) \cdots (\lambda - a_{nn})$$

which implies that the eigenvalues of $\boldsymbol{A}$ are

$$\lambda_1 = a_{11},\ \lambda_2 = a_{22}, \square,\ \lambda_n = a_{nn}$$

Thus, we have the following theorem.

**Theorem** If $A$ is a triangular matrix (upper triangular, lower triangular, or diagonal) then the eigenvalues of $A$ are the entries on the main diagonal of $A$.

**Example 5** (Eigenvalues of Triangular Matrices)

By inspection, the characteristic polynomial of the matrix $\boldsymbol{A} = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ -1 & -\frac{2}{3} & 0 & 0 \\ 7 & \frac{5}{8} & 6 & 0 \\ \frac{4}{9} & -4 & 3 & 6 \end{bmatrix}$ is

$p(\lambda) = (\lambda - \frac{1}{2})(\lambda + \frac{2}{3})(\lambda - 6)^2$. So the distinct eigenvalues of $\boldsymbol{A}$ are $\lambda = \frac{1}{2}$, $\lambda = -\frac{2}{3}$, and $\lambda = 6$.

**Eigenvalues of Powers of a Matrix** Once the eigenvalues and eigenvectors of a matrix $\boldsymbol{A}$ are found, it is a simple matter to find the eigenvalues and eigenvectors of any positive integer power of $\boldsymbol{A}$. For example, if $\lambda$ is an eigenvalue of $\boldsymbol{A}$ and $\boldsymbol{x}$ is a corresponding eigenvector, then $A^2x = A(Ax) = A(\lambda x) = \lambda(Ax) = \lambda(\lambda x) = \lambda^2 x$, which shows that $\lambda^2$ is an eigenvalue of $\boldsymbol{A^2}$ and $\boldsymbol{x}$ is a corresponding eigenvector. In general we have the following result.

**Theorem** If $\lambda$ is an eigenvalue of a matrix A and x is a corresponding eigenvector, and if k is any positive integer, then $\lambda^k$ is an eigenvalue of $A^k$ and x is a corresponding eigenvector.
Some problems that use this theorem are given in the exercises.

**A Unifying Theorem** Since $\lambda$ is an eigenvalue of a square matrix $\boldsymbol{A}$ if and only if there is a nonzero vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \lambda \boldsymbol{x}$, it follows that $\lambda = 0$ is an eigenvalue of $\boldsymbol{A}$ if and only if there is a nonzero vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \boldsymbol{0}$. However, this is true if and only if $\det(\boldsymbol{A}) = 0$, so we list the following

**Theorem** If $\boldsymbol{A}$ is an $n \times n$ matrix, then the following statements are equivalent.
(a) The reduced row echelon form of $\boldsymbol{A}$ is $\boldsymbol{I_n}$.
(b) $\boldsymbol{A}$ is expressible as a product of elementary matrices.
(c) $\boldsymbol{A}$ is invertible.
(d) $\boldsymbol{Ax} = \boldsymbol{0}$ has only the trivial solution.
(e) $\boldsymbol{Ax} = \boldsymbol{b}$ is consistent for every vector $\boldsymbol{b}$ in $\boldsymbol{R^n}$.
(f) $\boldsymbol{Ax} = \boldsymbol{b}$ has exactly one solution for every vector $\boldsymbol{b}$ in $\boldsymbol{R}^n$
(g) The column vectors of $\boldsymbol{A}$ are linearly independent.
(h) The row vectors of $\boldsymbol{A}$ are linearly independent.
(i) $\det(\boldsymbol{A}) \neq 0$.
(j) $\lambda = 0$ is not an eigenvalue of $\boldsymbol{A}$.

**Example 6**

(1) Is 5 an eigenvalue of $A=\begin{bmatrix}6 & -3 & 1\\ 3 & 0 & 5\\ 2 & 2 & 6\end{bmatrix}$?

(2) If $\boldsymbol{x}$ is an eigenvector for $\boldsymbol{A}$ corresponding to $\lambda$, what is $\boldsymbol{A^3x}$?

**Solution**

(1) The number 5 is an eigenvalue of $\boldsymbol{A}$ if and only if the equation $(\boldsymbol{A}-\lambda\boldsymbol{I})\,\boldsymbol{x}=\boldsymbol{0}$ has a nontrivial solution. Form

$$\boldsymbol{A}-5\boldsymbol{I}=\begin{bmatrix}6 & -3 & 1\\ 3 & 0 & 5\\ 2 & 2 & 6\end{bmatrix}-\begin{bmatrix}5 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & 5\end{bmatrix}=\begin{bmatrix}1 & -3 & 1\\ 3 & -5 & 5\\ 2 & 2 & 1\end{bmatrix}$$

and row reduce the augmented matrix:

$$\begin{bmatrix}1 & -3 & 1 & 0\\ 3 & -5 & 5 & 0\\ 2 & 2 & 1 & 0\end{bmatrix}$$

$$\sim\begin{bmatrix}1 & -3 & 1 & 0\\ 0 & 4 & 2 & 0\\ 2 & 2 & 1 & 0\end{bmatrix}\begin{matrix}\\ \\ R_2-3R_1\end{matrix}$$

$$\sim\begin{bmatrix}1 & -3 & 1 & 0\\ 0 & 4 & 2 & 0\\ 0 & 8 & -1 & 0\end{bmatrix}\begin{matrix}\\ \\ R_3-2R_1\end{matrix}$$

$$\sim\begin{bmatrix}1 & -3 & 1 & 0\\ 0 & 4 & 2 & 0\\ 0 & 0 & -5 & 0\end{bmatrix}\begin{matrix}\\ \\ R_3-2R_2\end{matrix}$$

At this point it is clear that the homogeneous system has no free variables. Thus $\boldsymbol{A}-5\boldsymbol{I}$ is an invertible matrix, which means that 5 is not an eigenvalue of $\boldsymbol{A}$.

(2). If $\boldsymbol{x}$ is an eigenvector for $\boldsymbol{A}$ corresponding to $\lambda$, then $\boldsymbol{Ax}=\lambda\boldsymbol{x}$ and so

$$\boldsymbol{A^2x}=\boldsymbol{A}(\lambda\boldsymbol{x})=\lambda\boldsymbol{Ax}=\lambda^2\boldsymbol{x}$$

Again $\boldsymbol{A^3x}=\boldsymbol{A}(\boldsymbol{A^2x})=\boldsymbol{A}(\lambda^2\boldsymbol{x})=\lambda^2\boldsymbol{Ax}=\lambda^3\boldsymbol{x}$. The general pattern, $\boldsymbol{A^kx}=\lambda^k\boldsymbol{x}$, is proved by induction.

**Exercises**

1. Is $\lambda=2$ an eigenvalue of $\begin{bmatrix}3 & 2\\ 3 & 8\end{bmatrix}$?

2. Is $\begin{bmatrix} -1+\sqrt{2} \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$? If so, find the eigenvalue.

3. Is $\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$? If so, find the eigenvalue.

4. Is $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 3 & 6 & 7 \\ 3 & 3 & 7 \\ 5 & 6 & 5 \end{bmatrix}$? If so, find the eigenvalue.

5. Is $\lambda = 4$ an eigenvalue of $\begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 1 \\ -3 & 4 & 5 \end{bmatrix}$? If so, find one corresponding eigenvector.

6. Is $\lambda = 3$ an eigenvalue of $\begin{bmatrix} 1 & 2 & 2 \\ 3 & -2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$? If so, find one corresponding eigenvector.

In exercises 7 to 12, find a basis for the eigenspace corresponding to each listed eigenvalue.

7. $\boldsymbol{A} = \begin{bmatrix} 4 & -2 \\ -3 & 9 \end{bmatrix}, \lambda = 10$

8. $\boldsymbol{A} = \begin{bmatrix} 7 & 4 \\ -3 & -1 \end{bmatrix}, \lambda = 15$

9. $\boldsymbol{A} = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 1 \\ -2 & 0 & 1 \end{bmatrix}, \lambda = 0, 1, 2, 3$

10. $\boldsymbol{A} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & -3 & 0 \\ 4 & -13 & 1 \end{bmatrix}, \lambda = -2$

11. $\boldsymbol{A} = \begin{bmatrix} 4 & 2 & 3 \\ -1 & 1 & -3 \\ 2 & 4 & 9 \end{bmatrix}, \lambda = -3$

12. $\boldsymbol{A} = \begin{bmatrix} 3 & 0 & 2 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}, \lambda = 4$

Find the eigenvalues of the matrices in Exercises 13 and 14.

13. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 5 \\ 0 & 0 & -1 \end{bmatrix}$

14. $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & -3 \end{bmatrix}$

15. For $\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$, find one eigenvalue, with no calculation. Justify your answer.

16. Without calculation, find one eigenvalue and two linearly independent vectors of $\boldsymbol{A} = \begin{bmatrix} 5 & 5 & 5 \\ 5 & 5 & 5 \\ 5 & 5 & 5 \end{bmatrix}$. Justify your answer.

# Lecture 29

## The Characteristic Equation

The Characteristic equation contains useful information about the eigenvalues of a square matrix **A.** It is defined as

$$\det(A-\lambda I)=0,$$

Where λ is the eigenvalue and I is the identity matrix. We will solve the Characteristic equation (also called the characteristic polynomial) to work out the eigenvalues of the given square matrix **A**.

**Example 1** Find the eigenvalues of $\boldsymbol{A}=\begin{bmatrix}2 & 3\\ 3 & -6\end{bmatrix}$.

**Solution** In order to find the eigenvalues of the given matrix, we must solve the matrix equation

$$(\boldsymbol{A}-\lambda\boldsymbol{I})\boldsymbol{x}=0$$

for the scalar $\lambda$ such that it has a nontrivial solution (since the matrix is non singular). By the Invertible Matrix Theorem, this problem is equivalent to finding all $\lambda$ such that the matrix $\boldsymbol{A}-\lambda\boldsymbol{I}$ is not invertible, where

$$\boldsymbol{A}-\lambda\boldsymbol{I}=\begin{bmatrix}2 & 3\\ 3 & -6\end{bmatrix}-\begin{bmatrix}\lambda & 0\\ 0 & \lambda\end{bmatrix}=\begin{bmatrix}2-\lambda & 3\\ 3 & -6-\lambda\end{bmatrix}.$$

By definition, this matrix $\boldsymbol{A}-\lambda\boldsymbol{I}$ fails to be invertible precisely when its determinant is zero. Thus, the eigenvalues of $\boldsymbol{A}$ are the solutions of the equation

$$\det(\boldsymbol{A}-\lambda\boldsymbol{I})=\det\begin{bmatrix}2-\lambda & 3\\ 3 & -6-\lambda\end{bmatrix}=0.$$

Recall that $\det\begin{bmatrix}a & b\\ c & d\end{bmatrix}=ad-bc$

So
$$\begin{aligned}\det(\boldsymbol{A}-\lambda\boldsymbol{I}) &= (2-\lambda)(-6-\lambda)-(3)(3)\\ &= -12+6\lambda-2\lambda+\lambda^2-9\\ &= \lambda^2+4\lambda-21\\ \lambda^2+4\lambda-21 &= 0,\\ (\lambda-3)(\lambda+7) &= 0,\end{aligned}$$

so the eigenvalues of $\boldsymbol{A}$ are 3 and –7.

**Example 2** Compute det $\boldsymbol{A}$ for $A=\begin{bmatrix}1 & 5 & 0\\ 2 & 4 & -1\\ 0 & -2 & 0\end{bmatrix}$

**Solution**

Firstly, we will reduce the given matrix in echelon form by applying elementary row operations

$$by\ R_2 - 2R_1$$

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix},$$

$$by\ R_2 \leftrightarrow R_3$$

$$\square \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & -6 & -1 \end{bmatrix},$$

$$by\ R_3 - 3R_2$$

$$\square \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix},$$

which is an upper triangular matrix. Therefore,

$$\det A = (1)(-2)(-1)$$
$$= 2.$$

**Theorem 1 Properties of Determinants**

Let $A$ and $B$ be two matrices of order $n$ then

(a) $A$ is invertible if and only if $\det A \neq 0$.
(b) $\det AB = (\det A)(\det B)$.
(c) $\det A^T = \det A$.
(d) If $A$ is triangular, then $\det A$ is the product of the entries on the main diagonal of $A$.
(e) A row replacement operation on A does not change the determinant.
(f) A row interchange changes the sign of the determinant.
(g) A row scaling also scales the determinant by the same scalar factor.

**Note** These Properties will be helpful in using the characteristic equation to find eigenvalues of a matrix **A**.

**Example 3** (a) Find the eigenvalues and corresponding eigenvectors of the matrix

$$\boldsymbol{A} = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$$

(b) Graph the eigenspaces of $\boldsymbol{A}$ in an xy-coordinate system.

**Solution** (a) The eigenvalues will be worked out by solving the characteristic equation of $\boldsymbol{A}$. Since

$$\lambda \boldsymbol{I} - \boldsymbol{A} = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} \lambda - 1 & -3 \\ -4 & \lambda - 2 \end{bmatrix}.$$

The characteristic equation $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$ becomes

$$\begin{vmatrix} \lambda - 1 & -3 \\ -4 & \lambda - 2 \end{vmatrix} = 0.$$

Expanding and simplifying the determinant, it yields

$$\lambda^2 - 3\lambda - 10 = 0,$$

or

$$(\lambda + 2)(\lambda - 5) = 0. \qquad (1)$$

Thus, the eigenvalues of $\boldsymbol{A}$ are $\lambda = -2$ and $\lambda = 5$.

Now, to work out the eigenspaces corresponding to these eigenvalues, we will solve the system

$$\begin{bmatrix} \lambda - 1 & -3 \\ -4 & \lambda - 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad (2)$$

for $\lambda = -2$ and $\lambda = 5$. Here are the computations for the two cases.

**(i) Case λ = -2**

In this case Eq. (2) becomes

$$\begin{bmatrix} -3 & -3 \\ -4 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which can be written as

$$-3x - 3y = 0,$$

$$-4x - 4y = 0 \Rightarrow x = -y.$$

In parametric form,

$$x = -t, y = t \ . \qquad (3)$$

Thus, the eigenvectors corresponding to $\lambda = -2$ are the nonzero vectors of the form

$$\boldsymbol{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}. \qquad (4)$$

It can be verified as

$$\begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} -t \\ t \end{bmatrix} = \begin{bmatrix} 2t \\ -2t \end{bmatrix} = -2 \begin{bmatrix} -t \\ t \end{bmatrix} = -2\boldsymbol{x}$$

Thus,

$$Ax = \lambda x$$

**(ii) Case λ = 5**

In this case Eq. (2) becomes

$$\begin{bmatrix} 4 & -3 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which can be written as

$$4x - 3y = 0$$

$$-4x + 3y = 0 \Rightarrow x = \frac{3}{4}y.$$

In parametric form,

$$x = \frac{3}{4}t, \quad y = t. \tag{5}$$

Thus, the eigenvectors corresponding to $\lambda = 5$ are the nonzero vectors of the form

$$\boldsymbol{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{3}{4}t \\ t \end{bmatrix} = t\begin{bmatrix} \frac{3}{4} \\ 1 \end{bmatrix}. \tag{6}$$

It can be verified as

$$\boldsymbol{Ax} = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}\begin{bmatrix} \frac{3}{4}t \\ t \end{bmatrix} = \begin{bmatrix} \frac{15}{4}t \\ 5t \end{bmatrix} = 5\begin{bmatrix} \frac{3}{4}t \\ t \end{bmatrix} = 5\boldsymbol{x}.$$

**(b)** The eigenspaces corresponding to $\lambda = -2$ and $\lambda = 5$ can be sketched from the parametric equations (3) and (5) as shown in figure 1(a).

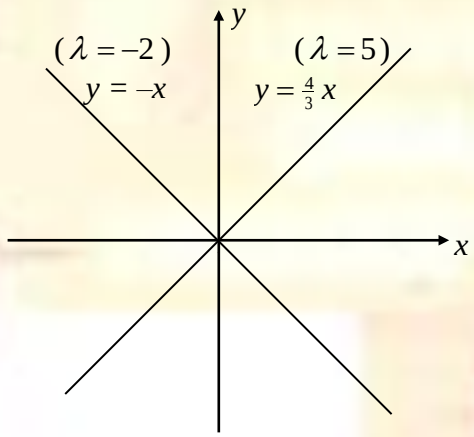



**Figure 1(a)**

It can also be drawn using the vector equations (4) and (6) as shown in Figure 1(b). When an eigenvector $\boldsymbol{x}$ in the eigenspace for $\lambda = 5$ is multiplied by $\boldsymbol{A}$, the resulting vector has the same direction as $\boldsymbol{x}$ but the length is increased by a factor of 5 and when an eigenvector $\boldsymbol{x}$ in the eigenspace for $\lambda = -2$ is multiplied by $\boldsymbol{A}$, the resulting vector is oppositely directed to $\boldsymbol{x}$ and the length is increased by a factor of 2. In both cases, multiplying an eigenvector by $\boldsymbol{A}$ produces a vector in the same eigenspace.

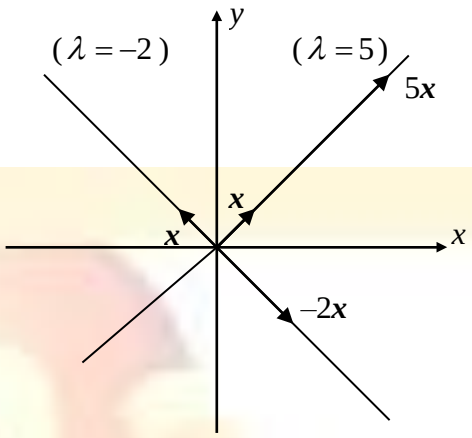



**Figure 1(b)**

**Eigenvalues of an $n \times n$ matrix**

Eigen values of an $n \times n$ matrix can be found in the similar fashion. However, for the higher values of n, it is more convenient to work them out using various available mathematical software. Here is an example for a $3 \times 3$ matrix.

**Example 4** Find the eigen values of the matrix $\boldsymbol{A} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -4 & -17 & 8 \end{bmatrix}$

**Solution**

$$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \det\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -4 & -17 & 8 \end{bmatrix}$$

$$= \begin{vmatrix} \lambda & 1 & 0 \\ 0 & \lambda & -1 \\ 4 & 17 & \lambda - 8 \end{vmatrix} \qquad (7)$$

$$= \lambda^3 - 8\lambda^2 + 17\lambda - 4,$$

which yields the characteristic equation

$$\lambda^3 - 8\lambda^2 + 17\lambda - 4 = 0 \qquad (8)$$

To solve this equation, firstly, we will look for integer solutions. This can be done by using the fact that if a polynomial equation has integer coefficients, then its integer solutions, if any, must be divisors of the constant term of the given polynomial. Thus, the only possible integer solutions of Eq.(8) are the divisors of –4, namely $\pm 1, \pm 2,$ and $\pm 4$. Substituting these values successively into Eq. (8) yields that $\lambda$ =4 is an integer solution. This implies that $\lambda - 4$ is a factor of Eq.(7), Thus, dividing the polynomial by $\lambda - 4$ and rewriting Eq.(8), we get

$$(\lambda - 4)(\lambda^2 - 4\lambda + 1) = 0.$$

Now, the remaining solutions of the characteristic equation satisfy the quadratic equation

$$\lambda^2 - 4\lambda + 1 = 0.$$

Solving the above equation by the quadratic formula, we get the eigenvalues of $\boldsymbol{A}$ as

$$\lambda = 4, \qquad \lambda = 2+\sqrt{3}, \qquad \lambda = 2-\sqrt{3}$$

**Example 5** Find the characteristic equation of $A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

**Solution** Clearly, the given matrix is an upper triangular matrix. Forming $\boldsymbol{A} - \lambda \boldsymbol{I}$, we get

$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \det \begin{bmatrix} 5-\lambda & -2 & 6 & -1 \\ 0 & 3-\lambda & -8 & 0 \\ 0 & 0 & 5-\lambda & 4 \\ 0 & 0 & 0 & 1-\lambda \end{bmatrix}$$

Now using the fact that determinant of a triangular matrix is equal to product of its diagonal elements, the characteristic equation becomes

$$(5-\lambda)^2(3-\lambda)(1-\lambda) = 0.$$

Expanding the product, we can also write it as

$$\lambda^4 - 14\lambda^3 + 68\lambda^2 - 130\lambda + 75 = 0.$$

Here, the eigenvalue 5 is said to have multiplicity 2 because $(\lambda - 5)$ occurs two times as a factor of the characteristic polynomial. In general, the (**algebraic**) multiplicity of an eigenvalue $\lambda$ is its multiplicity as a root of the characteristic equation.

**Note**

From the above mentioned examples, it can be easily observed that if A is an $n \times n$ matrix, then det (A – $\lambda$ I) is a polynomial of degree n called the characteristic polynomial of A.

**Example 6** The characteristic polynomial of a $6 \times 6$ matrix is $\lambda^6 - 4\lambda^5 - 12\lambda^4$. Find the eigenvalues and their multiplicities.

**Solution**

In order to find the eigenvalues, we will factorize the polynomial as

$$\lambda^6 - 4\lambda^5 - 12\lambda^4$$
$$= \lambda^4(\lambda^2 - 4\lambda - 12)$$
$$= \lambda^4(\lambda - 6)(\lambda + 2)$$

The eigenvalues are 0 (multiplicity 4), 6 (multiplicity 1) and – 2 (multiplicity 1).We could also list the eigenvalues in Example 6 as 0, 0, 0, 0, 6 and –2, so that the eigenvalues are repeated according to their multiplicities

**Activity**

Work out the eigenvalues and eigenvectors for the following square matrix.

$$A = \begin{bmatrix} 5 & 8 & 16 \\ 4 & 1 & 8 \\ -4 & -4 & -11 \end{bmatrix}.$$

**Similarity**

Let $A$ and $B$ be two $n$ x $n$ matrices, $A$ is said to be similar to $B$ if there exist an invertible matrix $P$ such that

$$P^{-1}AP = B,$$

or equivalently,

$$A = PBP^{-1}.$$

Replacing $Q$ by $P^{-1}$, we have

$$Q^{-1}BQ = A.$$

So $B$ is also similar to $A$. Thus, we can say that $A$ and $B$ are similar.

**Similarity transformation**

The act of changing $A$ into $P^{-1}AP$ is called a similarity transformation.

The following theorem illustrates use of the characteristic polynomial and it provides the foundation for several iterative methods that approximate eigenvalues.

**Theorem 2**

If $n$ x $n$ matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

**Proof** If $\boldsymbol{B} = \boldsymbol{P^{-1}AP}$, then

$$B - \lambda I = P^{-1}AP - \lambda I = P^{-1}AP - \lambda P^{-1}P = P^{-1}(AP - \lambda P) = P^{-1}(A - \lambda I)P$$

Using the multiplicative property (b) of Theorem 1, we compute

$$\det(B - \lambda I) = \det\left[P^{-1}(A - \lambda I)P\right]$$

$$= \det(P^{-1}).\det(A - \lambda I).\det(P) \qquad \text{(A)}$$

Since

$$\det(\boldsymbol{P^{-1}}).\ \det(\boldsymbol{P}) = \det(\boldsymbol{P^{-1}P})$$
$$= \det \boldsymbol{I}$$
$$= 1,$$

Eq. (A) implies that

$$\det(B - \lambda I) = \det(A - \lambda I).$$

Hence, both the matrices have the same characteristic polynomials and therefore, same eigenvalues.

**Note** It must be clear that Similarity and row equivalence are two different concepts. ( If $\boldsymbol{A}$ is row equivalent to $\boldsymbol{B}$, then $\boldsymbol{B} = \boldsymbol{EA}$ for some invertible matrix $\boldsymbol{E}$.) Row operations on a matrix usually change its eigenvalues.

**Application to Dynamical Systems**

Dynamical system is the one which evolves with the passage of time. Eigenvalues and eigenvectors play a vital role in the evaluation of a dynamical system. Let's consider an example of a dynamical system.

**Example 7** Let $A = \begin{bmatrix} .95 & .03 \\ .05 & .97 \end{bmatrix}$. Analyze the long term behavior of the dynamical system defined by $\boldsymbol{x}_{k+1} = A\boldsymbol{x}_k$ $(k = 0, 1, 2, \ldots)$, with $\boldsymbol{x}_0 = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$.

**Solution** The first step is to find the eigenvalues of $A$ and a basis for each eigenspace. The characteristic equation for $A$ is

$$0 = \det(A - \lambda I)$$

$$0 = \det \begin{bmatrix} 0.95 - \lambda & 0.03 \\ 0.05 & 0.97 - \lambda \end{bmatrix} = (.95 - \lambda)(.97 - \lambda) - (.03)(.05)$$

$$= \lambda^2 - 1.92\lambda + .92$$

By the quadratic formula

$$\lambda = \frac{1.92 \pm \sqrt{(1.92)^2 - 4(.92)}}{2} = \frac{1.92 \pm \sqrt{.0064}}{2} = \frac{1.92 \pm .08}{2} = 1 \text{ or } .92$$

Firstly, the eigenvectors will be found as given below.

$$Ax = \lambda x,$$

$$(Ax - \lambda x) = 0,$$

$$(A - \lambda I)x = 0.$$

For $\lambda = 1$

$$\left[ \begin{pmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0,$$

$$\begin{pmatrix} -0.05 & 0.03 \\ 0.05 & -0.03 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0,$$

which can be written as

$$-0.05x_1 + 0.03x_2 = 0$$

$$0.05x_1 - 0.03x_2 = 0 \Rightarrow x_1 = \frac{0.03}{0.05} x_2 \text{ or } x_1 = \frac{3}{5} x_2.$$

In parametric form, it becomes

$$x_1 = \frac{3}{5} t \text{ and } x_2 = t.$$

For $\lambda = 0.92$

$$\left[\begin{pmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{pmatrix} - \begin{pmatrix} 0.92 & 0 \\ 0 & 0.92 \end{pmatrix}\right]\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0,$$

$$\begin{pmatrix} 0.03 & 0.03 \\ 0.05 & 0.05 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0.$$

It can be written as

$$0.03x_1 + 0.03x_2 = 0$$

$$0.05x_1 + 0.05x_2 = 0 \Rightarrow x_1 = -x_2$$

In parametric form, it becomes

$$x_1 = t \text{ and } x_2 = -t$$

Thus, the eigenvectors corresponding to $\lambda = 1$ and $\lambda = .92$ are multiples of $\boldsymbol{v}_1 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ and $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ respectively.

The next step is to write the given $\boldsymbol{x}_0$ in terms of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$. This can be done because $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ is obviously a basis for $\boldsymbol{R}^2$. So there exists weights $c_1$ and $c_2$ such that

$$\boldsymbol{x}_0 = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 = [\boldsymbol{v}_1 \quad \boldsymbol{v}_2]\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \qquad (1)$$

In fact, $$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = [\boldsymbol{v}_1 \quad \boldsymbol{v}_2]^{-1}\boldsymbol{x}_0 = \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}^{-1}\begin{bmatrix} .60 \\ .40 \end{bmatrix}$$

Here,

$$\begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}^{-1} = \frac{1}{\begin{vmatrix} 3 & 1 \\ 5 & -1 \end{vmatrix}} Adj\begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix} = -\frac{1}{8}\begin{bmatrix} -1 & -1 \\ -5 & 3 \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = -\frac{1}{8}\begin{bmatrix} -1 & -1 \\ -5 & 3 \end{bmatrix}\begin{bmatrix} .60 \\ .40 \end{bmatrix} = \begin{bmatrix} .125 \\ .225 \end{bmatrix} \qquad (2)$$

Because $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in Eq.(1) are eigenvectors of $\boldsymbol{A}$, with $\boldsymbol{Av}_1 = \boldsymbol{v}_1$ and $\boldsymbol{Av}_2 = (.92)\,\boldsymbol{v}_2$, $\boldsymbol{x}_k$ can be computed as

$$\boldsymbol{x}_1 = \boldsymbol{Ax}_0 = c_1\boldsymbol{Av}_1 + c_2\boldsymbol{Av}_2 \quad \text{(Using linearity of } \boldsymbol{x} \to \boldsymbol{Ax}\text{)}$$

$$= c_1\boldsymbol{v}_1 + c_2(.92)\boldsymbol{v}_2 \quad (\boldsymbol{v}_1 \text{ and } \boldsymbol{v}_2 \text{ are eigenvectors.})$$

$$\boldsymbol{x}_2 = \boldsymbol{Ax}_1 = c_1\boldsymbol{Av}_1 + c_2(.92)\boldsymbol{Av}_2 = c_1\boldsymbol{v}_1 + c_2(.92)^2\boldsymbol{v}_2.$$

Continuing in the same way, we get the general equation as

$$\boldsymbol{x}_k = c_1\boldsymbol{v}_1 + c_2(.92)^k\boldsymbol{v}_2 \qquad (k = 0, 1, 2, \ldots).$$

Using $c_1$ and $c_2$ from Eq.(2),

$$\boldsymbol{x}_k = .125\begin{bmatrix} 3 \\ 5 \end{bmatrix} + .225(.92)^k\begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad (k = 0,1,2,\ldots) \qquad (3)$$

This explicit formula for $\boldsymbol{x}_k$ gives the solution of the difference equation $\boldsymbol{x}_{k+1} = \boldsymbol{Ax}_k$.

As $k \to \infty$, $(.92)^k$ tends to zero and $\boldsymbol{x}_k$ tends to $\begin{bmatrix} .375 \\ .625 \end{bmatrix} = .125\boldsymbol{v}_1$.

**Example 8** Find the characteristic equation and eigenvalues of $\boldsymbol{A} = \begin{bmatrix} 1 & -4 \\ 4 & 2 \end{bmatrix}$.

**Solution** The characteristic equation is

$$0 = \det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \det \begin{bmatrix} 1-\lambda & -4 \\ 4 & 2-\lambda \end{bmatrix}$$

$$= (1-\lambda)(2-\lambda) - (-4)(4),$$

$$= \lambda^2 - 3\lambda + 18,$$

which is a quadratic equation whose roots are given as

$$\lambda = \frac{3 \pm \sqrt{(-3)^2 - 4(18)}}{2}$$

$$= \frac{3 \pm \sqrt{-63}}{2}$$

Thus, we see that the characteristic equation has no real roots, so $\boldsymbol{A}$ has no real eigenvalues. $\boldsymbol{A}$ is acting on the real vector space $\boldsymbol{R}^2$ and there is no non-zero vector $\boldsymbol{v}$ in $\boldsymbol{R}^2$ such that $\boldsymbol{Av} = \lambda \boldsymbol{v}$ for some scalar $\lambda$.

## Exercises

Find the characteristic polynomial and the eigenvalues of matrices in exercises 1 to 12.

1. $\begin{bmatrix} 3 & -2 \\ & -1 \\ 1 & \end{bmatrix}$

2. $\begin{bmatrix} 5 & -3 \\ -4 & 3 \end{bmatrix}$

3. $\begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$

4. $\begin{bmatrix} 3 & -4 \\ 4 & 8 \end{bmatrix}$

5. $\begin{bmatrix} 5 & 3 \\ -4 & 4 \end{bmatrix}$

6. $\begin{bmatrix} 7 & -2 \\ 2 & 3 \end{bmatrix}$

7. $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & -1 \\ 0 & 6 & 0 \end{bmatrix}$

8. $\begin{bmatrix} 0 & 3 & 1 \\ 3 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$

9. $\begin{bmatrix} 4 & 0 & 0 \\ 5 & 3 & 2 \\ -2 & 0 & 2 \end{bmatrix}$

10. $\begin{bmatrix} -1 & 0 & 1 \\ -3 & 4 & 1 \\ 0 & 0 & 2 \end{bmatrix}$

11. $\begin{bmatrix} 6 & -2 & 0 \\ -2 & 9 & 0 \\ 5 & 8 & 3 \end{bmatrix}$

12. $\begin{bmatrix} 5 & -2 & 3 \\ 0 & 1 & 0 \\ 6 & 7 & -2 \end{bmatrix}$

For the matrices in exercises 13 to 15, list the eigenvalues, repeated according to their multiplicities.

13. $\begin{bmatrix} 4 & -7 & 0 & 2 \\ 0 & 3 & -4 & 6 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

14. $\begin{bmatrix} 5 & 0 & 0 & 0 \\ 8 & -4 & 0 & 0 \\ 0 & 7 & 1 & 0 \\ 1 & -5 & 2 & 1 \end{bmatrix}$

15. $\begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ -5 & 1 & 0 & 0 & 0 \\ 3 & 8 & 0 & 0 & 0 \\ 0 & -7 & 2 & 1 & 0 \\ -4 & 1 & 9 & -2 & 3 \end{bmatrix}$

16. It can be shown that the algebraic multiplicity of an eigenvalue $\lambda$ is always greater than or equal to the dimension of the eigenspace corresponding to $\lambda$. Find $h$ in the matrix $\boldsymbol{A}$ below such that the eigenspace for $\lambda=5$ is two-dimensional:

$$\boldsymbol{A} = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & h & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

# Lecture 30

## Diagonalization

Diagonalization is a process of transforming a vector A to the form $\boldsymbol{A = PDP^{-1}}$ for some invertible matrix **P** and a diagonal matrix **D**. In this lecture, the factorization enables us to compute $\boldsymbol{A^k}$ quickly for large values of $k$ which is a fundamental idea in several applications of linear algebra. Later, the factorization will be used to analyze (and decouple) dynamical systems.

The "*D*" in the factorization stands for diagonal. Powers of such a *D* are trivial to compute.

**Example 1** If $\boldsymbol{D} = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$, then $\boldsymbol{D^2} = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix}$ and

$$\boldsymbol{D}^3 = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix} = \begin{bmatrix} 5^3 & 0 \\ 0 & 3^3 \end{bmatrix}$$

In general, $\boldsymbol{D^k} = \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix}$ for $k \geq 1$

The next example shows that if $\boldsymbol{A = PDP^{-1}}$ for some invertible $\boldsymbol{P}$ and diagonal $\boldsymbol{D}$, then it is quite easy to compute $\boldsymbol{A^k}$.

**Example 2** Let $\boldsymbol{A} = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Find a formula for $\boldsymbol{A^k}$, given that $\boldsymbol{A = PDP^{-1}}$, where

$$\boldsymbol{P} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \text{ and } \boldsymbol{D} = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

**Solution** The standard formula for the inverse of a $2 \times 2$ matrix yields

$$\boldsymbol{P^{-1}} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

By associative property of matrix multiplication,

$$\boldsymbol{A^2} = (\boldsymbol{PDP^{-1}})(\boldsymbol{PDP^{-1}}) = \boldsymbol{PD}\underbrace{(\boldsymbol{P^{-1}P})}_{1}\boldsymbol{DP^{-1}} = PDIDP^{-1} = \boldsymbol{PDDP^{-1}}$$

where I is the identity matrix.

$$= \boldsymbol{PD^2P^{-1}} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

Again, $\boldsymbol{A^3} = (\boldsymbol{PDP^{-1}})\boldsymbol{A^2} = (\boldsymbol{PD}\underbrace{\boldsymbol{P^{-1}}}_{1})\boldsymbol{PD^2P^{-1}} = \boldsymbol{PDD^2P^{-1}} = \boldsymbol{PD^3P^{-1}}$

Thus, in general, for $k \geq 1$, $\boldsymbol{A^k} = \boldsymbol{PD^kP^{-1}} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$,

$$= \begin{bmatrix} 5^k & 3^k \\ -5^k & -2.3^k \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix},$$

$$= \begin{bmatrix} 2.5^k - 3^k & 5^k - 3^k \\ 2.3^k - 2.5^k & 2.3^k - 5^k \end{bmatrix}.$$

**Activity**

Work out $C^4$, given that $C = PDP^{-1}$ where

$$P = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

**Remarks**

A square matrix ***A*** is said to be **diagonalizable** if ***A*** is similar to a diagonal matrix, that is, if $\boldsymbol{A} = \boldsymbol{PDP^{-1}}$ for some invertible matrix ***P*** and some diagonal matrix ***D***. The next theorem gives a characterization of diagonalizable matrices and tells how to construct a suitable factorization.

**Theorem 1 The Diagonalization Theorem**

An $n$ x $n$ matrix $A$ is diagonalizable if and only if $A$ has $n$ linearly independent eigenvectors.

In fact, $\boldsymbol{A} = \boldsymbol{PDP^{-1}}$, with ***D*** a diagonal matrix, if and only if the columns of ***P*** are $n$ linearly independent eigenvectors of ***A***. In this case, the diagonal entries of ***D*** are eigenvalues of ***A*** that correspond, respectively, to the eigenvectors in ***P***.

In other words, ***A*** is diagonalizable if and only if there are enough eigenvectors to form a basis of $\boldsymbol{R^n}$. We call such a basis an eigenvector basis.

**Proof** First, observe that if ***P*** is any $n \times n$ matrix with columns $\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}$ and if ***D*** is any diagonal matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ then

$$\boldsymbol{AP} = \boldsymbol{A}\begin{bmatrix} \boldsymbol{v_1} & \boldsymbol{v_2} & \ldots & \boldsymbol{v_n} \end{bmatrix} = \begin{bmatrix} \boldsymbol{Av_1} & \boldsymbol{Av_2} & \ldots & \boldsymbol{Av_n} \end{bmatrix}, \tag{1}$$

while

$$\boldsymbol{PD} = \boldsymbol{P}\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} = \begin{bmatrix} \lambda_1 \boldsymbol{v}_1 & \lambda_2 \boldsymbol{v}_2 & \cdots & \lambda_n \boldsymbol{v}_n \end{bmatrix} \tag{2}$$

Suppose now that ***A*** is diagonalizable and $\boldsymbol{A} = \boldsymbol{PDP^{-1}}$. Then right-multiplying this relation by ***P***, we have $\boldsymbol{AP} = \boldsymbol{PD}$. In this case, (1) and (2) imply that

$$\begin{bmatrix} \boldsymbol{Av_1} & \boldsymbol{Av_2} & \cdots & \boldsymbol{Av_n} \end{bmatrix} = \begin{bmatrix} \lambda_1 \boldsymbol{v_1} & \lambda_2 \boldsymbol{v_2} & \cdots & \lambda_n \boldsymbol{v_n} \end{bmatrix} \tag{3}$$

Equating columns, we find that

$$\boldsymbol{Av_1} = \lambda_1 \boldsymbol{v_1}, \boldsymbol{Av_2} = \lambda_2 \boldsymbol{v_2}, \cdots, \boldsymbol{Av_n} = \lambda_n \boldsymbol{v_n} \tag{4}$$

Since ***P*** is invertible, its columns $\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}$ must be linearly independent. Also, since these columns are nonzero, Eq.(4) shows that $\lambda_1, \ldots, \lambda_n$ are eigenvalues and $\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}$ are

corresponding eigenvectors. This argument proves the "only if" parts of the first and second statements along with the third statement, of the theorem.

Finally, given any $n$ eigenvectors $\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}$ use them to construct the columns of $\boldsymbol{P}$ and use corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$ to construct $\boldsymbol{D}$. By Eqs. (1) – (3). $\boldsymbol{AP} = \boldsymbol{PD}$. This is true without any condition on the eigenvectors. If, in fact, the eigenvectors are linearly independent, then $\boldsymbol{P}$ is invertible (by the Invertible Matrix Theorem), and $\boldsymbol{AP} = \boldsymbol{PD}$ implies that $\boldsymbol{A} = \boldsymbol{PDP^{-1}}$.

**Diagonalizing Matrices**

**Example 3** Diagonalize the following matrix, if possible $\boldsymbol{A} = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$

**Solution** To diagonalize the given matrix, we need to find an invertible matrix **P** and a diagonal matrix **D** such that $\boldsymbol{A} = \boldsymbol{PDP^{-1}}$ which can be done in following four steps.

**Step 1** Find the eigenvalues of $\boldsymbol{A}$.

The characteristic equation becomes

$$0 = \det(A - \lambda I) = -\lambda^3 - 3\lambda^2 + 4$$
$$= -(\lambda - 1)(\lambda + 2)^2$$

The eigenvalues are $\lambda = 1$ and $\lambda = -2$ (multiplicity 2)

**Step 2** Find three linearly independent eigenvectors of $\boldsymbol{A}$. Since $\boldsymbol{A}$ is a $3 \times 3$ matrix and we have obtained three eigen values, we need three eigen vectors. This is the critical step. If it fails, then above Theorem says that $\boldsymbol{A}$ cannot be diagonalized. Now we will produce basis for these eigen values.

**Basis vector for** $\lambda = 1$:

$(A - \lambda I)x = 0$

$$\begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

After applying few row operations on the matrix $(A - \lambda I)$, we get

$$\begin{bmatrix} 0 & 1 & 1 \\ 3 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

which can be written as

$x_2 + x_3 = 0$

$3x_1 + 3x_2 = 0$

In parametric form, it becomes

$x_1 = t,\ x_2 = -t,\ x_3 = t$

Thus, the basis vector for $\lambda = 1$ is $v_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

**Basis vector for** $\lambda = -2$

$$(A - \lambda I)x = 0$$

$$\begin{bmatrix} 3 & 3 & 3 \\ -3 & -3 & -3 \\ 3 & 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

which can be written as

$$3x_1 + 3x_2 + 3x_3 = 0$$
$$-3x_1 - 3x_2 - 3x_3 = 0$$
$$3x_1 + 3x_2 + 3x_3 = 0$$

In parametric form, it becomes

$x_1 = -s - t, \quad x_2 = s, \quad x_3 = t$

Now,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -s - t \\ s \\ t \end{bmatrix} = \begin{bmatrix} -s \\ s \\ 0 \end{bmatrix} + \begin{bmatrix} -t \\ 0 \\ t \end{bmatrix},$$

$$= s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix},$$

$$= x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Thus, the basis for $\lambda = -2$ is $v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $v_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

We can check that $\{v_1, v_2, v_3\}$ is a linearly independent set.

**Step 3** Check that $\{v_1, v_2, v_3\}$ is a linearly independent set.

Construct $P$ from the vectors in step 2. The order of the vectors is not important. Using the order chosen in step 2, form $P = [v_1 \quad v_2 \quad v_3] = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

**Step 4** Form $\boldsymbol{D}$ from the corresponding eigen values. For this purpose, the order of the eigen values must match the order chosen for the columns of $\boldsymbol{P}$. Use the eigen value $\lambda = -2$ twice, once for each of the eigenvectors corresponding to $\lambda = -2$:

$$\boldsymbol{D} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Now, we need to check do $\boldsymbol{P}$ and $\boldsymbol{D}$ really work. To avoid computing $\boldsymbol{P}^{-1}$, simply verify that $\boldsymbol{AP} = \boldsymbol{PD}$. This is equivalent to $\boldsymbol{A} = \boldsymbol{PDP}^{-1}$ when $\boldsymbol{P}$ is invertible.We compute

$$\boldsymbol{AP} = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

$$\boldsymbol{PD} = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

**Example 4** Diagonalize the following matrix, if possible.

$$\boldsymbol{A} = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

**Solution** The characteristic equation of $\boldsymbol{A}$ turns out to be exactly the same as that in example 3 i.e.,

$$0 = \det(\boldsymbol{A} - \lambda \boldsymbol{I})$$

$$= \begin{vmatrix} 2-\lambda & 4 & 3 \\ -4 & -6-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{vmatrix}$$

$$= -\lambda^3 - 3\lambda^2 + 4$$

$$= -(\lambda - 1)(\lambda + 2)^2$$

The eigen values are $\lambda = 1$ and $\lambda = -2$ (multiplicity 2). However, when we look for eigen vectors, we find that each eigen space is only one – dimensional.

Basis for $\lambda = 1$: $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Basis for $\lambda = -2$: $\boldsymbol{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

There are no other eigen values and every eigen vector of $\boldsymbol{A}$ is a multiple of either $\boldsymbol{v_1}$ or $\boldsymbol{v_2}$. Hence it is impossible to form a basis of $\boldsymbol{R^3}$ using eigenvectors of $\boldsymbol{A}$. By above Theorem, $\boldsymbol{A}$ is not diagonalizable.

**Theorem 2** An *n* x *n* matrix with *n* distinct eigenvalues is diagonalizable.

The condition in Theorem 2 is sufficient but not necessary i.e., it is not necessary for an *n* x *n* matrix to have *n* distinct eigen values in order to be diagonalizable. Example 3 serves as a counter example of this case where the 3 x 3 matrix is diagonalizable even though it has only two distinct eigen values.

**Example 5** Determine if the following matrix is diagonalizable.

$$\boldsymbol{A} = \begin{bmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{bmatrix}$$

**Solution** In the light of Theorem 2, the answer is quite obvious. Since the matrix is triangular, its eigen values are obviously 5, 0, and –2.Since A is a 3 x 3 matrix with three distinct eigen values, $\boldsymbol{A}$ is diagonalizable.

**Matrices Whose Eigenvalues Are Not Distinct**

If an n x n matrix $\boldsymbol{A}$ has *n* distinct eigen values, with corresponding eigen vectors $\boldsymbol{v_1}$ ,..., $\boldsymbol{v_n}$ and if $\boldsymbol{P} = [\boldsymbol{v_1} \ldots \boldsymbol{v_n}]$ , then $\boldsymbol{P}$ is automatically invertible because its columns are linearly independent , by Theorem 2 of lecture 28. When $\boldsymbol{A}$ is diagonalizable but has fewer than *n* distinct eigen values, it is still possible to build $\boldsymbol{P}$ in a way that makes $\boldsymbol{P}$ automatically invertible, as shown in the next theorem.

**Theorem 3** Let $\boldsymbol{A}$ be an *n* x *n* matrix whose distinct eigen values are $\lambda_1, \ldots, \lambda_p$ .

a. For $1 \le k \le p$, the dimension of the eigen space for $\lambda_k$ is less than or equal to the multiplicity of the eigen value $\lambda_k$
b. The matrix $\boldsymbol{A}$ is diagonalizable if and only if the sum of the dimensions of the distinct eigen spaces is equal to *n*, and this happens if and only if the dimension of the eigen space for each of $\lambda_k$ equals the multiplicity of $\lambda_k$ .
c. If $\boldsymbol{A}$ is diagonalizable and $\boldsymbol{B_k}$ is basis for the eigen space corresponding to $\lambda_k$ for each k, then the total collection of vectors in the sets $\boldsymbol{B_1}$, ..., $\boldsymbol{B_p}$ form an eigenvector basis for $\boldsymbol{R^n}$ .

**Example 6** Diagonalize the following matrix, if possible.

$$\boldsymbol{A}=\begin{bmatrix}5 & 0 & 0 & 0\\ 0 & 5 & 0 & 0\\ 1 & 4 & -3 & 0\\ -1 & -2 & 0 & -3\end{bmatrix}$$

**Solution** Since $\boldsymbol{A}$ is triangular matrix, the eigenvalues are 5 and –3, each with multiplicity 2. Using the method of lecture 28, we find a basis for each eigen space.

$$\text{Basis for } \lambda=5:\boldsymbol{v}_1=\begin{bmatrix}-8\\ 4\\ 1\\ 0\end{bmatrix} \text{ and } \boldsymbol{v}_2=\begin{bmatrix}-16\\ 4\\ 0\\ 1\end{bmatrix}$$

$$\text{Basis for } \lambda=-3:\boldsymbol{v}_3=\begin{bmatrix}0\\ 0\\ 1\\ 0\end{bmatrix} \text{ and } \boldsymbol{v}_4=\begin{bmatrix}0\\ 0\\ 0\\ 1\end{bmatrix}$$

The set $\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_4\}$ is linearly independent, by Theorem 3. So the matrix $\boldsymbol{P}=[\boldsymbol{v}_1\ldots\boldsymbol{v}_4]$ is invertible, and $\boldsymbol{A}=\boldsymbol{PDP}^{-1}$, where

$$\boldsymbol{P}=\begin{bmatrix}-8 & -16 & 0 & 0\\ 4 & 4 & 0 & 0\\ 1 & 0 & 1 & 0\\ 0 & 1 & 0 & 1\end{bmatrix} \text{ and } \boldsymbol{D}=\begin{bmatrix}5 & 0 & 0 & 0\\ 0 & 5 & 0 & 0\\ 0 & 0 & -3 & 0\\ 0 & 0 & 0 & -3\end{bmatrix}$$

**Example 7**

(1) Compute $\boldsymbol{A}^8$ where $\boldsymbol{A}=\begin{bmatrix}4 & -3\\ 2 & -1\end{bmatrix}$

(2) Let $\boldsymbol{A}=\begin{bmatrix}-3 & 12\\ -2 & 7\end{bmatrix}, \boldsymbol{v}_1=\begin{bmatrix}3\\ 1\end{bmatrix}, \text{and } \boldsymbol{v}_2=\begin{bmatrix}2\\ 1\end{bmatrix}$. Suppose you are told that $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are eigenvectors of $\boldsymbol{A}$. Use this information to diagonalize $\boldsymbol{A}$.

(3) Let $\boldsymbol{A}$ be a 4 x 4 matrix with eigenvalues 5, 3, and -2, and suppose that you know the eigenspace for $\lambda=3$ is two-dimensional. Do you have enough information to determine if $\boldsymbol{A}$ is diagonalizable?

**Solution**

Here, $\det(\boldsymbol{A}-\lambda\boldsymbol{I})=\lambda^2-3\lambda+2=(\lambda-2)(\lambda-1)$.

The eigen values are 2 and 1, and corresponding eigenvectors are $\boldsymbol{v}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, and $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Next, form

$$\boldsymbol{P} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}, \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \text{ and } \boldsymbol{P}^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

Since $\boldsymbol{A} = \boldsymbol{PDP}^{-1}$, $\boldsymbol{A}^8 = \boldsymbol{PD}^8\boldsymbol{P} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2^8 & 0 \\ 0 & 1^8 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$

$$= \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 256 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 766 & -765 \\ 510 & -509 \end{bmatrix}$$

(2) Here, $\boldsymbol{Av}_1 = \begin{bmatrix} -3 & 12 \\ -2 & 7 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 1.\boldsymbol{v}_1$, and

$$\boldsymbol{Av}_2 = \begin{bmatrix} -3 & 12 \\ -2 & 7 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} = 3.\boldsymbol{v}_2$$

Clearly, $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are eigenvectors for the eigenvalues 1 and 3 , respectively. Thus

$\boldsymbol{A} = \boldsymbol{PDP}^{-1}$, where $\boldsymbol{P} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$ and $\boldsymbol{D} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

(3) Yes $\boldsymbol{A}$ is diagonalizable. There is a basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ for the eigen space corresponding to $\lambda=3$. Moreover, there will be at least one eigenvector for $\lambda=5$ and one for $\lambda=-2$ say $\boldsymbol{v}_3$ and $\boldsymbol{v}_4$. Then $\{\boldsymbol{v}_1, \ldots., \boldsymbol{v}_4\}$ is linearly independent and $\boldsymbol{A}$ is diagonalizable , by Theorem 3. There can be no additional eigen vectors that are linearly independent from $\boldsymbol{v}_1$ to $\boldsymbol{v}_4$ because the vectors are all in $\boldsymbol{R}^4$ .Hence the eigenspaces for $\lambda=5$ and $\lambda=-2$ are both one–dimensional.

**Exercise**

In exercises 1 and 2, let $A = PDP^{-1}$ and compute $A^4$.

1. $P = \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix}, D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

2. $P = \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$

In exercises 3 and 4, use the factorization $A = PDP^{-1}$ to compute $A^k$, where $k$ represents an arbitrary positive integer.

3. $\begin{bmatrix} a & 0 \\ 3(a-b) & b \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$

4. $\begin{bmatrix} -2 & 12 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 4 \\ 1 & -3 \end{bmatrix}$

In exercises 5 and 6, the matrix $A$ is factored in the form $PDP^{-1}$. Use the Diagonalization Theorem to find the eigenvalues of $A$ and a basis for each eigenspace.

5. $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/4 & 1/2 & -3/4 \\ 1/4 & -1/2 & 1/4 \end{bmatrix}$

6. $\begin{bmatrix} 4 & 0 & -2 \\ 2 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} -2 & 0 & -1 \\ 0 & 1 & 2 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 4 \\ -1 & 0 & -2 \end{bmatrix}$

Diagonalize the matrices in exercises 7 to 18, if possible.

7. $\begin{bmatrix} 3 & -1 \\ 1 & 5 \end{bmatrix}$

8. $\begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$

9. $\begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$

10. $\begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$

11. $\begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{bmatrix}$

12. $\begin{bmatrix} 4 & 0 & -2 \\ 2 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix}$

13. $\begin{bmatrix} 7 & 4 & 16 \\ 2 & 5 & 8 \\ -2 & -2 & -5 \end{bmatrix}$

14. $\begin{bmatrix} 0 & -4 & -6 \\ -1 & 0 & -3 \\ 1 & 2 & 5 \end{bmatrix}$

15. $\begin{bmatrix} 4 & 0 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

16. $\begin{bmatrix} -7 & -16 & 4 \\ 6 & 13 & -2 \\ 12 & 16 & 1 \end{bmatrix}$

17. $\begin{bmatrix} 5 & -3 & 0 & 9 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

18. $\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$

# Lecture 31

# Eigenvectors and Linear Transformations

The goal of this lecture is to investigate the relationship between eigenvectors and linear transformations. Previously, we have learned how to find the eigenvalues and eigenvectors. We shall see that the matrix factorization $\boldsymbol{A = PDP^{-1}}$ is also a type of linear transformations for some invertible matrix $\boldsymbol{P}$ and some diagonal matrix $\boldsymbol{D}$.

## The Matrix of a Linear Transformation

Let $\boldsymbol{V}$ be an $n$-dimensional vector space, $\boldsymbol{W}$ an $m$-dimensional vector space, and $\boldsymbol{T}$ any linear transformation from $\boldsymbol{V}$ to $\boldsymbol{W}$. To associate a matrix with $\boldsymbol{T}$, we choose (ordered) bases $\boldsymbol{B}$ and $\boldsymbol{C}$ for $\boldsymbol{V}$ and $\boldsymbol{W}$, respectively.
Given any $\boldsymbol{x}$ in $\boldsymbol{V}$, the coordinate vector $[\boldsymbol{x}]_B$ is in $\boldsymbol{R^n}$ and the coordinate vector of its image, $[\boldsymbol{T}(\boldsymbol{x})]_C$, is in $\boldsymbol{R^m}$, as shown in Fig. 1.

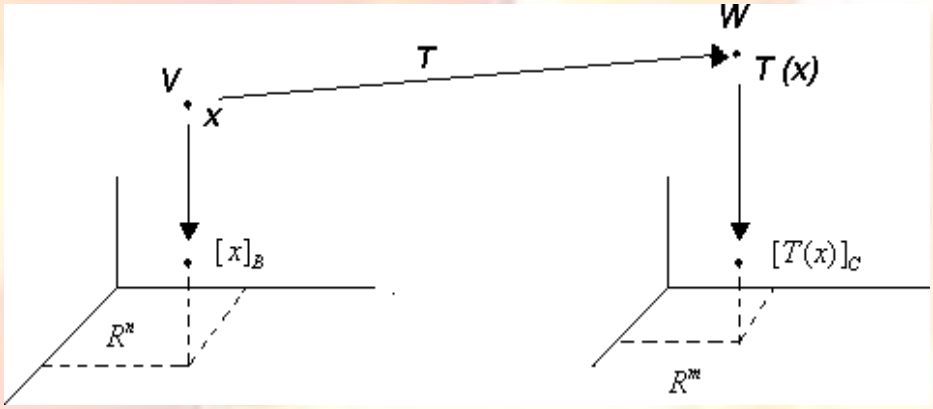



**Figure 1** A linear transformation from $\boldsymbol{V}$ to $\boldsymbol{W}$

Let $\boldsymbol{B}=\{\boldsymbol{b_1}, \ldots, \boldsymbol{b_n}\}$ be the basis for $\boldsymbol{V}$. If $\boldsymbol{x} = r_1\boldsymbol{b_1} + \cdots + r_n\boldsymbol{b_n}$,

then $[\boldsymbol{x}]_B = \begin{bmatrix} r_1 \\ \vdots \\ r_n \end{bmatrix}$

and T is a linear transformation

$$\boldsymbol{T}(\boldsymbol{x}) = \boldsymbol{T}(r_1\boldsymbol{b_1} + \cdots + r_n\boldsymbol{b_n}) = r_1\boldsymbol{T}(\boldsymbol{b_1}) + \cdots + r_n\boldsymbol{T}(\boldsymbol{b_n}) \qquad (1)$$

Using the basis $\boldsymbol{C}$ in $\boldsymbol{W}$, we can rewrite (1) in terms of $\boldsymbol{C}$-coordinate vectors:

$$[\boldsymbol{T}(\boldsymbol{x})]_C = r_1[\boldsymbol{T}(\boldsymbol{b_1})]_C + \cdots + r_n[\boldsymbol{T}(\boldsymbol{b_n})]_C \quad (2)$$

Since $\boldsymbol{C}$-coordinate vectors are in $\boldsymbol{R^m}$, the vector equation (2) can be written as a matrix equation, namely

$$[\boldsymbol{T}(\boldsymbol{x})]_C = \boldsymbol{M}[\boldsymbol{x}]_B \quad (3)$$

where $$\boldsymbol{M} = \left[[\boldsymbol{T}(\boldsymbol{b_1})]_C \quad [\boldsymbol{T}(\boldsymbol{b_2})]_C \quad \cdots \quad [\boldsymbol{T}(\boldsymbol{b_n})]_C\right] \quad (4)$$

The matrix $\boldsymbol{M}$ is a matrix representation of $\boldsymbol{T}$, called the **matrix for $\boldsymbol{T}$ relative to the bases $\boldsymbol{B}$ and $\boldsymbol{C}$**.

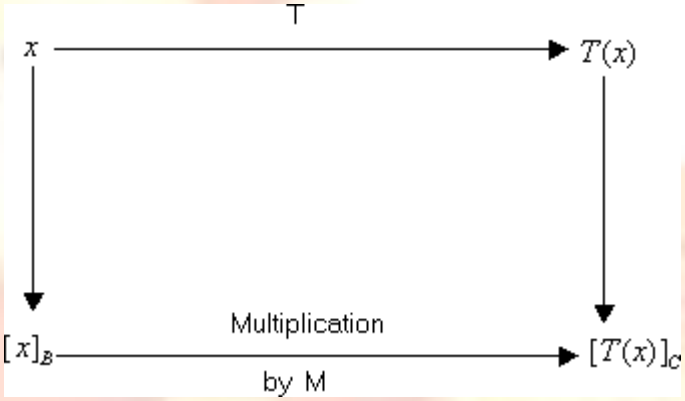



**Figure 2**

**Example 1**

Suppose that $\boldsymbol{B} = \{\boldsymbol{b_1}, \boldsymbol{b_2}\}$ is a basis for $\boldsymbol{V}$ and $\boldsymbol{C} = \{\boldsymbol{c_1}, \boldsymbol{c_2}, \boldsymbol{c_3}\}$ is a basis for $\boldsymbol{W}$. Let $\boldsymbol{T}: \boldsymbol{V} \to \boldsymbol{W}$ be a linear transformation with the property that

$$\boldsymbol{T}(\boldsymbol{b_1}) = 3\boldsymbol{c_1} - 2\boldsymbol{c_2} + 5\boldsymbol{c_3} \text{ and } \boldsymbol{T}(\boldsymbol{b_2}) = 4\boldsymbol{c_1} + 7\boldsymbol{c_2} - \boldsymbol{c_3}$$

Find the matrix $\boldsymbol{M}$ for $\boldsymbol{T}$ relative to $\boldsymbol{B}$ and $\boldsymbol{C}$.

**Solution**

Since $\boldsymbol{M} = \left[[\boldsymbol{T}(\boldsymbol{b_1})]_C \quad [\boldsymbol{T}(\boldsymbol{b_2})]_C \quad \cdots \quad [\boldsymbol{T}(\boldsymbol{b_n})]_C\right]$ and here

$$[\boldsymbol{T}(\boldsymbol{b}_1)]_C = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} \text{ and } [\boldsymbol{T}(\boldsymbol{b}_2)]_C = \begin{bmatrix} 4 \\ 7 \\ -1 \end{bmatrix}$$

Hence, $$\boldsymbol{M} = \begin{bmatrix} 3 & 4 \\ -2 & 7 \\ 5 & -1 \end{bmatrix}$$

**Linear Transformations from V into V**

In general when $W$ is the same as $V$ and the basis $C$ is the same as $B$, the matrix $M$ in (4) is called the **matrix for $T$ relative to $B$**, or simply the **$B$-matrix** for $T$, is denoted by $[T]_B$. See Fig. 3.

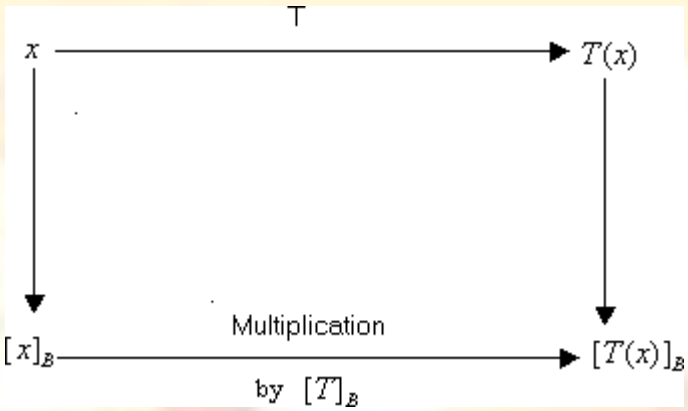



**Figure 3**

The $B$-matrix of $T: V \to V$ satisfies

$$[T(x)]_B = [T]_B [x]_B, \quad \text{for all } x \text{ in } V \qquad (5)$$

**Example 2**

The mapping $T: P_2 \to P_2$ defined by $T(a_0 + a_1 t + a_2 t^2) = a_1 + 2a_2 t$ is a linear transformation.

(a) Find the $B$-matrix for $T$, when $B$ is the basis $\{1, t, t^2\}$.
(b) Verify that $[T(p)]_B = [T]_B [p]_B$ for each $p$ in $P_2$.

**Solution**

**(a)** We have to find the $B$-matrix of $T: V \to V$ satisfies

$$[T(x)]_B = [T]_B [x]_B, \quad \text{for all } x \text{ in } V$$

Since $T(a_0 + a_1 t + a_2 t^2) = a_1 + 2a_2 t$ therefore

$$T(1) = 0,\ T(t) = 1,\ T(t^2) = 2t$$

$$[T(1)]_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, [T(t)]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, [T(t^2)]_B = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

$$[T]_B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

**(b)** For a general $p(t) = a_0 + a_1 t + a_2 t^2$, we have

$$[T(p)]_B = [a_1 + 2a_2 t]_B = \begin{bmatrix} a_1 \\ 2a_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = [T]_B [p]_B$$

See figure 4.

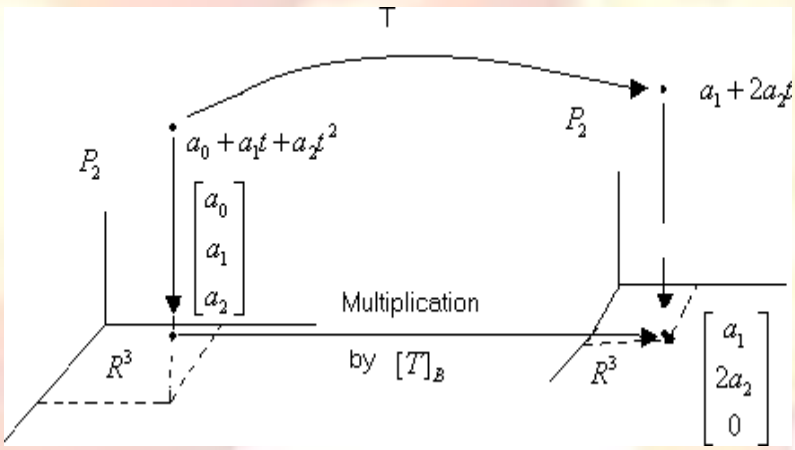



**Figure 4** Matrix representation of a linear transformation

## Linear Transformations on $R^n$

In an applied problem involving $R^n$, a linear transformation $T$ usually appears first as a matrix transformation, $x \to Ax$ . If $A$ is diagonalizable then there is a basis $B$ for $R^n$ consisting of eigenvectors of $A$. Theorem below shows that in this case, the $B$-matrix of $T$ is diagonal. Diagonalizing $A$ amounts to finding a diagonal matrix representation of $x \to Ax$ .

**Theorem: Diagonal Matrix Representation**

Suppose $A = PDP^{-1}$, where $D$ is diagonal $n$ x $n$ matrix. If $B$ is the basis for $R^n$ formed from the columns of $P$, then $D$ is the $B$-matrix of the transformation $x \to Ax$ .

**Proof** Denote the columns of $P$ by $b_1, \ldots, b_n$, so that $B = \{b_1, \ldots, b_n\}$ and $P = [b_1 \quad \ldots \quad b_n]$. In this case, $P$ is the change-of-coordinates matrix $P_B$ discussed in lecture 23, where $P[x]_B = x$ and $[x]_B = P^{-1}x$

If $T(x) = Ax$ for $x$ in $R^n$, then

$$[T]_B = \left[[T(b_1)]_B \quad \cdots \quad [T(b_n)]_B\right] \qquad \text{Definition of } [T]_B$$

$$= \left[[Ab_1]_B \quad \cdots \quad [Ab_n]_B\right] \qquad \text{Since } T(x) = Ax$$

$$= \left[P^{-1}Ab_1 \quad \cdots \quad P^{-1}Ab_n\right] \qquad \text{Change of Coordinates}$$

$$= P^{-1}A\left[b_1 \quad \cdots \quad b_n\right] \qquad \text{Matrix multiplication}$$

$$= P^{-1}AP \qquad (6)$$

Since $A = PDP^{-1}$, we have $[T]_B = P^{-1}AP = D$.

**Example 3**

Define $T : R^2 \to R^2$ by $T(x) = Ax$, where $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Find a basis $B$ for $R^2$ with the property that the $B$-matrix of $T$ is a diagonal matrix.

**Solution**

Since the eigenvalues of matrix A are 5 and 3 and the eigenvectors corresponding to eigenvalue 5 is $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and the eigenvectors corresponding to eigenvalue 3 is $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

Therefore $A = PDP^{-1}$, where $P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$

By the above theorem $D$ matrix is the $B$ matrix of T when $B = \left\{\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix}\right\}$.

**Similarity of Matrix Representations**

We know that $A$ is similar to $C$ if there is an invertible matrix $P$ such that $A = PCP^{-1}$, therefore if $A$ is similar to $C$ , then $C$ is the $B$-matrix of the transformation $x \to Ax$ **when the basis** $B$ **is formed from the columns of** $P$ by the theorem above.(Since in the proof , the information that $D$ is a diagonal matrix was not used).

The factorization $A = PCP^{-1}$ is shown in Fig. 5.

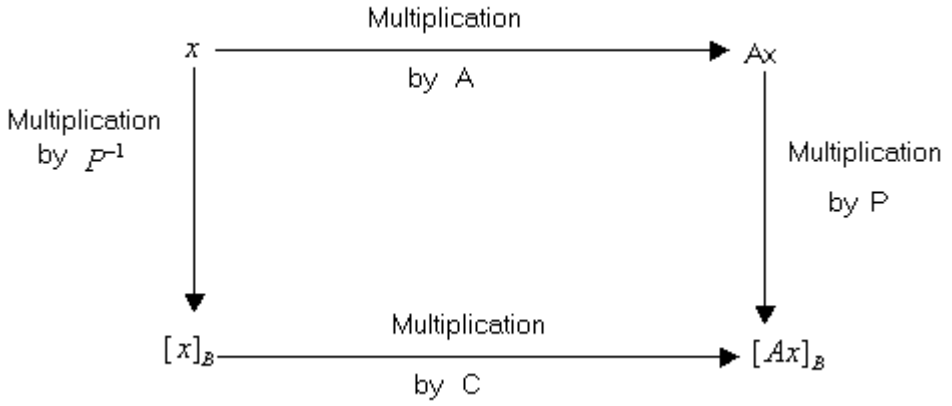



**Figure 5** Similarity of two matrix representations: $A = PCP^{-1}$

Conversely, if $T : R^n \to R^n$ is defined by $T(x) = Ax$, and if $B$ is any basis for $R^n$, then the $B$-matrix of $T$ is similar to $A$. In fact, the calculations in (6) show that if $P$ is the matrix whose columns come from the vectors in $B$, then $[T]_B = P^{-1}AP$. Thus, the set of all matrices similar to a matrix $A$ coincides with the set of all matrix representations of the transformation $x \to Ax$.

An efficient way to compute a $B$-matrix $P^{-1}AP$ is to compute $AP$ and then to row reduce the augmented matrix $[P \quad AP]$ to $[I \quad P^{-1}AP]$. A separate computation of $P^{-1}$ is unnecessary.

**Example 4**

Find $T(a_0 + a_1t + a_2t^2)$, if $T$ is the linear transformation from $P_2$ to $P_2$ whose matrix relative to $B = \{1, t, t^2\}$ is $[T]_B = \begin{bmatrix} 3 & 4 & 0 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix}$

**Solution**

Let $p(t) = a_0 + a_1t + a_2t^2$ and compute

$$[T(p)]_B = [T]_B [p]_B = \begin{bmatrix} 3 & 4 & 0 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 3a_0 + 4a_1 \\ 5a_1 - a_2 \\ a_0 - 2a_1 + 7a_2 \end{bmatrix}$$

So $T(p) = (3a_0+4a_1)1 + (5a_1-a_2)t + (a_0-2a_1+7a_2)t^2$ .

**Example 5**

Let $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$ be $n$ x $n$ matrices. Verify that
(a) $\boldsymbol{A}$ is similar to $\boldsymbol{A}$.
(b) If $\boldsymbol{A}$ is similar to $\boldsymbol{B}$ and $\boldsymbol{B}$ is similar to $\boldsymbol{C}$, then $\boldsymbol{A}$ is similar to $\boldsymbol{C}$.

**Solution**

(a) $\boldsymbol{A} = (\boldsymbol{I})^{-1}\boldsymbol{AI}$, so $\boldsymbol{A}$ is similar to $\boldsymbol{A}$.
(b) By hypothesis, there exist invertible matrices $\boldsymbol{P}$ and $\boldsymbol{Q}$ with the property that $\boldsymbol{B} = \boldsymbol{P}^{-1}\boldsymbol{AP}$ and $\boldsymbol{C} = \boldsymbol{Q}^{-1}\boldsymbol{BQ}$. Substituting, and using facts about the inverse of a product we have

$$\boldsymbol{C} = \boldsymbol{Q}^{-1}\boldsymbol{BQ} = \boldsymbol{Q}^{-1}(\boldsymbol{P}^{-1}\boldsymbol{AP})\boldsymbol{Q} = (\boldsymbol{PQ})^{-1}\boldsymbol{A}(\boldsymbol{PQ})$$

This equation has the proper form to show that A is similar to C.

# Lecture 32

## Eigenvalues and Eigenvectors

**Definition**

A complex scalar λ satisfies

$$\det(A-\lambda I) = 0$$

if and only if there is a nonzero vector x in $C^n$ such that

$$Ax = \lambda x.$$

We call λ a (complex) eigenvalue and x a (complex) eigenvector corresponding to λ.

**Example 1**

If $\boldsymbol{A} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, then the linear transformation $\boldsymbol{x} \to \boldsymbol{Ax}$ on $\boldsymbol{R^2}$ rotates the plane counterclockwise through a quarter-turn.

**Solution**

The action of $\boldsymbol{A}$ is periodic since after four quarter-turns, a vector is back where it started. Obviously, no nonzero vector is mapped into a multiple of itself, so $\boldsymbol{A}$ has no eigenvectors in $\boldsymbol{R^2}$ and hence no real eigenvalues. In fact, the characteristic equation of $\boldsymbol{A}$ is $\lambda^2 + 1 = 0 \Rightarrow \lambda^2 = -1 \Rightarrow \lambda = \pm i$

The only roots are complex: $\lambda = i$ and $\lambda = -i$. However, if we permit $\boldsymbol{A}$ to act on $\boldsymbol{C^2}$ then

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} i \\ 1 \end{bmatrix} = i\begin{bmatrix} 1 \\ -i \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} -i \\ 1 \end{bmatrix} = -i\begin{bmatrix} 1 \\ i \end{bmatrix}$$

Thus $i$ and $-i$ are the eigenvalues, with $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ i \end{bmatrix}$ as corresponding eigenvectors.

**Example 2**

Let $\boldsymbol{A} = \begin{bmatrix} 0.5 & -0.6 \\ 0.75 & 1.1 \end{bmatrix}$. Find the eigenvalues of A, and find a basis for each eigen-space.

**Solution**

The characteristic equation of $\boldsymbol{A}$ is

$$0 = \det\begin{bmatrix} 0.5-\lambda & -0.6 \\ 0.75 & 1.1-\lambda \end{bmatrix} = (0.5-\lambda)(1.1-\lambda) - (-0.6)(0.75)$$

$$= \lambda^2 - 1.6\lambda + 1$$

From the quadratic formula, $\lambda = \frac{1}{2}[1.6 \pm \sqrt{(-1.6)^2 - 4}] = 0.8 \pm 0.6i$. For the eigenvalue $\lambda = 0.8 - 0.6i$, we study

$$\boldsymbol{A} - (0.8 - 0.6i) = \begin{bmatrix} 0.5 & -0.6 \\ 0.75 & 1.1 \end{bmatrix} - \begin{bmatrix} 0.8 - 0.6i & 0 \\ 0 & 0.8 - 0.6i \end{bmatrix} = \begin{bmatrix} -0.3 + 0.6i & -0.6 \\ 0.75 & 0.3 + 0.6i \end{bmatrix} \quad (1)$$

Since 0.8 – 0.6$i$ is an eigenvalue, we know that the system

$$\begin{aligned} (-0.3 + 0.6i)x_1 - \quad & 0.6x_2 = 0 \\ 0.75x_1 + (0.3 + 0.6i)x_2 & = 0 \end{aligned} \quad (2)$$

has a nontrivial solution (with $x_1$ and $x_2$ possibly complex numbers).

$$\begin{aligned} & 0.75x_1 = (-0.3 - 0.6i)x_2 \\ \Rightarrow \quad & x_1 = (-0.4 - 0.8i)x_2 \end{aligned}$$

Taking $x_2 = 5$ to eliminate the decimals, we have $x_1 = -2 - 4i$. A basis for the eigenspace corresponding to $\lambda = 0.8 - 0.6i$ is $\boldsymbol{v}_1 = \begin{bmatrix} -2 - 4i \\ 5 \end{bmatrix}$

Analogous calculations for $\lambda = 0.8 + 0.6i$ produce the eigenvector $\boldsymbol{v}_2 = \begin{bmatrix} -2 + 4i \\ 5 \end{bmatrix}$

**Check**

Compute

$$\boldsymbol{A}\boldsymbol{v}_2 = \begin{bmatrix} 0.5 & -0.6 \\ 0.75 & 1.1 \end{bmatrix} \begin{bmatrix} -2 + 4i \\ 5 \end{bmatrix} = \begin{bmatrix} -4 + 2i \\ 4 + 3i \end{bmatrix} = (0.8 + 0.6i)\boldsymbol{v}_2$$

**Example 3**

One way to see how multiplication by the $\boldsymbol{A}$ in Example 2 affects points is to plot an arbitrary initial point-say, $\mathbf{x}_0 = (2, 0)$ and then to plot successive images of this point under repeated multiplications by $\boldsymbol{A}$. That is, plot

$$\boldsymbol{x}_1 = \boldsymbol{A}\boldsymbol{x}_0 = \begin{bmatrix} 0.5 & -0.6 \\ .75 & 1.1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1.0 \\ 1.5 \end{bmatrix}$$

$$\boldsymbol{x}_2 = \boldsymbol{A}\boldsymbol{x}_1 = \begin{bmatrix} 0.5 & -0.6 \\ 0.75 & 1.1 \end{bmatrix} \begin{bmatrix} 1.0 \\ 1.5 \end{bmatrix} = \begin{bmatrix} -0.4 \\ 2.4 \end{bmatrix}$$

$\boldsymbol{x}_3 = \boldsymbol{A}\boldsymbol{x}_2$, □

Figure 1 shows $\mathbf{x}_0, \ldots, \mathbf{x}_8$ as heavy dots. The smaller dots are the locations of $\boldsymbol{x}_9, \ldots, \boldsymbol{x}_{100}$. The sequence lies along an elliptical orbit.

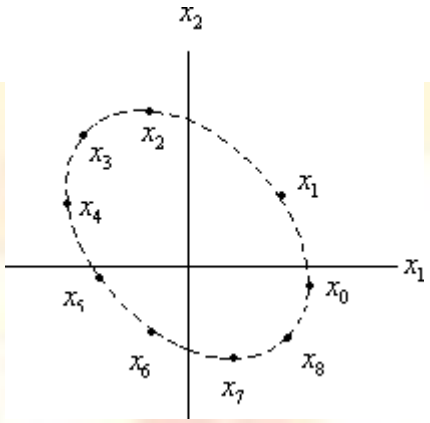



**Figure 1** Iterates of a point $x_0$ under the action of a matrix with a complex eigenvalue

Of course Figure 1 does not explain why the rotation occurs. The secret to the rotation is hidden in the real and imaginary parts of a complex eigenvector.

**Real and Imaginary Parts of Vectors**

The complex conjugate of a complex vector x in $C^n$ is the vector $\bar{x}$ in $C^n$ whose entries are the complex conjugates of the entries in x. The real and imaginary parts of a complex vector x are the vectors **Re** $x$ and **Im** $x$ formed from the real and imaginary parts of the entries of $x$.

**Example 4**

$$\text{If } \boldsymbol{x} = \begin{bmatrix} 3-i \\ i \\ 2+5i \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} + i \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}, \text{ then}$$

$$\text{Re}\,\boldsymbol{x} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}, \text{Im}\,\boldsymbol{x} = \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}, \text{and} \quad \bar{\boldsymbol{x}} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} - i \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 3+i \\ -i \\ 2-5i \end{bmatrix}$$

If $\boldsymbol{B}$ is an $m \times n$ matrix with possibly complex entries, then $\bar{\boldsymbol{B}}$ denotes the matrix whose entries are the complex conjugates of the entries in $\boldsymbol{B}$. Properties of conjugates for complex numbers carry over to complex matrix algebra:

$$\overline{r\boldsymbol{x}} = \bar{r}\,\bar{\boldsymbol{x}}, \quad \overline{B\boldsymbol{x}} = \bar{B}\,\bar{\boldsymbol{x}}, \text{and} \quad \overline{rB} = \bar{r}\,\bar{B}$$

**Eigenvalues and Eigenvectors of a Real Matrix that Acts on $C^n$**

Let $A$ be $n \times n$ matrix whose entries are real. Then $\overline{A\boldsymbol{x}} = \overline{A}\,\overline{\boldsymbol{x}} = A\overline{\boldsymbol{x}}$. If $\lambda$ is an eigenvalue of $A$ with $x$ a corresponding eigenvector in $C^n$, then $A\overline{\boldsymbol{x}} = \overline{A\boldsymbol{x}} = \overline{\lambda \boldsymbol{x}} = \overline{\lambda}\,\overline{\boldsymbol{x}}$ Hence $\overline{\lambda}$ is also an eigenvalue of $\boldsymbol{A}$, with $\overline{x}$ a corresponding eigenvector. This shows that when $\boldsymbol{A}$ is real, its complex eigenvalues occur in conjugate pairs. (Here and elsewhere, we use the term complex eigenvalue to refer to an eigenvalue $\lambda = a + bi$ with $b \neq 0$ .)

**Example 5**

The eigenvalues of the real matrix in example 2 are complex conjugates namely, 0.8 – 0.6*i* and 0.8 + 0.6*i*. The corresponding eigenvectors found in example 2 are also conjugates: $\boldsymbol{v}_1 = \begin{bmatrix} -2-4i \\ 5 \end{bmatrix}$ and $\boldsymbol{v}_2 = \begin{bmatrix} -2+4i \\ 5 \end{bmatrix} = \bar{\boldsymbol{v}}_1$

**Example 6**

If $\boldsymbol{C} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, where a and b are real and not both zero, then the eigenvalues of $\boldsymbol{C}$ are $\lambda = a \pm bi$. Also, if $r = |\lambda| = \sqrt{a^2 + b^2}$ , then

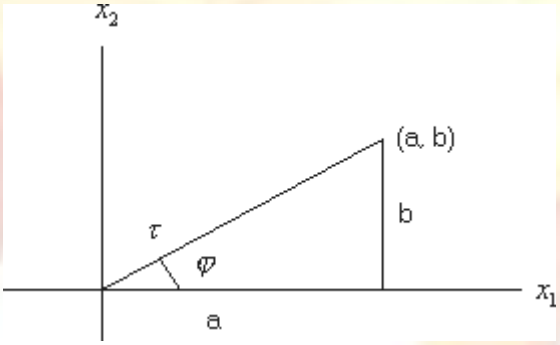



**Figure 2**

$$\boldsymbol{C} = r\begin{bmatrix} a/r & -b/r \\ b/r & a/r \end{bmatrix} = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix}\begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$$

where $\phi$ is the angle between the positive x-axis and the ray from (0, 0) through (a, b). See Fig. 2. The angle $\phi$ is called the argument of $\lambda = a + bi$.

Thus, the transformation $\boldsymbol{x} \to \boldsymbol{Cx}$ may be viewed as the composition of a rotation through the angle $\phi$ and a scaling by $|\lambda|$.

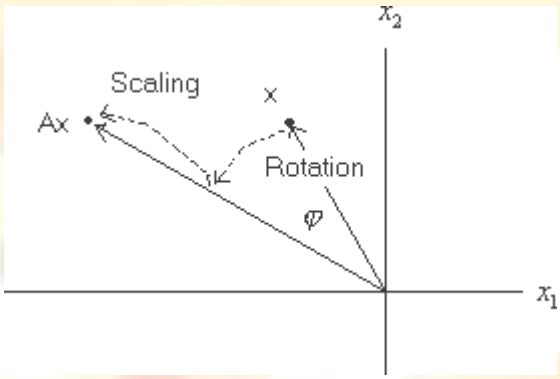



**Figure 3** A rotation followed by a scaling

Finally, we are ready to uncover the rotation that is hidden within a real matrix having a complex eigenvalue.

**Example 7**

Let $\boldsymbol{A} = \begin{bmatrix} 0.5 & -0.6 \\ 0.75 & 1.1 \end{bmatrix}, \lambda = 0.8 - 0.6i,$ and $\boldsymbol{v}_1 = \begin{bmatrix} -2-4i \\ 5 \end{bmatrix},$ as in example 2. Also, let $\boldsymbol{P}$ be the $2 \times 2$ real matrix $\boldsymbol{P} = \begin{bmatrix} \text{Re}\,\boldsymbol{v}_1 & \text{Im}\,\boldsymbol{v}_1 \end{bmatrix} = \begin{bmatrix} -2 & -4 \\ 5 & 0 \end{bmatrix}$

and let $\boldsymbol{C} = \boldsymbol{P}^{-1}\boldsymbol{AP} = \frac{1}{20}\begin{bmatrix} 0 & 4 \\ -5 & -2 \end{bmatrix}\begin{bmatrix} 0.5 & -0.6 \\ 0.75 & 1.1 \end{bmatrix}\begin{bmatrix} -2 & -4 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$

By Example 6, $\boldsymbol{C}$ is a pure rotation because $|\lambda|^2 = (.8)^2 + (.6)^2 = 1.$

From $\boldsymbol{C} = \boldsymbol{P}^{-1}\boldsymbol{AP}$, we obtain

$$\boldsymbol{A} = \boldsymbol{PCP}^{-1} = \boldsymbol{P}\begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}\boldsymbol{P}^{-1}$$

Here is the rotation "inside" *A*! The matrix $\boldsymbol{P}$ provides a change of variable, say, $\boldsymbol{x} = \boldsymbol{Pu}$. The action of $\boldsymbol{A}$ amounts to a change of variable from $\boldsymbol{x}$ to $\boldsymbol{u}$, followed by a rotation, and then return to the original variable. See Figure 4.

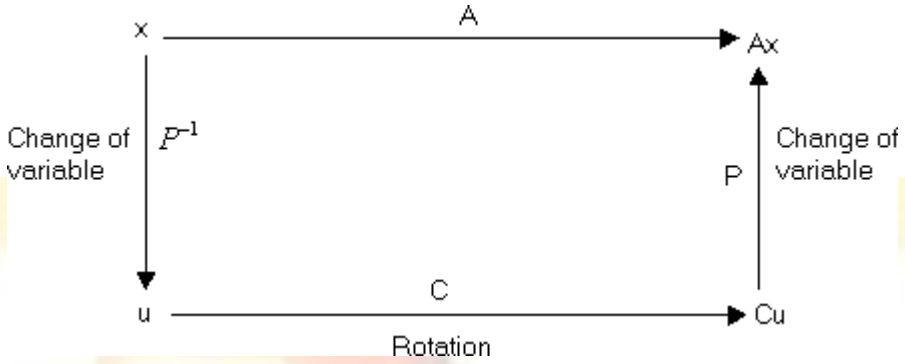



**Figure 4** Rotation due to a complex eigenvalue

**Theorem**

Let A be a real 2 x 2 matrix with complex Eigen values $\lambda = a - b_i \quad (b \neq 0)$ and associated eigenvectors $v$ in $\square^2$, then

$$A = PCP^{-1}, \quad C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix},$$

$$P = \begin{bmatrix} \mathrm{Re}\, v & \mathrm{Im}\, v \end{bmatrix}$$

# Lecture 33

## Discrete Dynamical Systems

For a dynamical system described by the difference equation $x_{k+1} = Ax_k$, eigenvalues and eigenvectors provide the key to understand the long term behavior and evolution of the system. Here the vector $x_k$ gives information about the system as time (which is denoted by k) passes. Discrete dynamical system has a great application in many scientific fields. For example, modern state space design method of standard undergraduate courses in control system and steady state response of a control system relies heavily on dynamical system.

In this section, we will suppose that a matrix $A$ is diagonalizable, with $n$ linearly independent eigenvectors, $v_1, \ldots, v_n$, and corresponding eigenvalues $\lambda_1, \lambda_2, \ldots \lambda_n$.

For convenience, assume that the eigenvectors are arranged so that

$$|\lambda_1| \geq |\lambda_2| \geq \ldots \geq |\lambda_n|$$

Since $\{ v_1, \ldots, v_n \}$ is a basis for $R^n$, any initial vector $x_0$ can be written uniquely as

$$x_0 = c_1 v_1 + c_2 v_2 + \ldots + c_n v_n$$

Since the $v$i are eigenvectors,

$$\begin{aligned} x_1 &= Ax_0 \\ &= c_1 Av_1 + c_2 Av_2 + \ldots + c_n Av_n \\ &= c_1 \lambda_1 v_1 + c_2 \lambda_2 v_2 + \ldots + c_n \lambda_n v_n \end{aligned}$$

In general,

$$x_k = c_1(\lambda_1)^k v_1 + \ldots + c_n(\lambda_n)^k v_n \qquad (k = 0,1,2\ldots)$$

We will discuss in the following examples what can happen to $x_k$ if $k \to \infty$.

**Example 1**

Denote the owl and wood rat populations at time $k$ by $\boldsymbol{x}_k = \begin{bmatrix} O_k \\ R_k \end{bmatrix}$, where $k$ is the time in months, $O_k$ is the number of owls in the region studied, and $R_k$ is the number of rats (measured in thousands). Suppose that

$$\begin{aligned} O_{k+1} &= (.5)O_k + (.4)R_k \\ R_{k+1} &= -p \cdot O_k + (1.1)R_k \end{aligned}$$

where $p$ is a positive parameter to be specified. The $(.5)O_k$ in the first equation says that with no wood rats for food only half of the owls will survive each month, while the $(1.1)R_k$ in the second equation says that with no owls as predators, the rat population will grow by 10% per month. If rats are plentiful, then $(.4)R_k$ will tend to make the owl population rise, while the negative term $-p.O_k$ measures the deaths of rats due to

predation by owls. (In fact, 1000*p* is the average number of rats eaten by one owl in one month.) Determine the evolution of this system when the predation parameter *p* is 0.104.

**Solution**

The coefficient matrix ***A*** for the given equations is

$$A = \begin{bmatrix} 0.5 & 0.4 \\ -p & 1.1 \end{bmatrix}$$

When $p = .104$,
The matrix becomes

$$A = \begin{bmatrix} 0.5 & 0.4 \\ -0.104 & 1.1 \end{bmatrix}$$

For eigenvalues of the coefficient matrix put
$\det(A - \lambda I) = 0$

The eigenvectors corresponding to the eigenvalues $\lambda_1 = 1.02$ and $\lambda_2 = .58$ are $\boldsymbol{v_1} = \begin{bmatrix} 10 \\ 13 \end{bmatrix}, \boldsymbol{v_2} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$.

An initial $\boldsymbol{x_0}$ can be written as $\boldsymbol{x_0} = c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2}$ then, for $k \geq 0$,

$$\boldsymbol{x}_k = c_1(1.02)^k \boldsymbol{v}_1 + c_2(.58)^k \boldsymbol{v}_2 = c_1(1.02)^k \begin{bmatrix} 10 \\ 13 \end{bmatrix} + c_2(.58)^k \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

As $k \to \infty, (.58)^k$ rapidly approaches zero. Assume $c_1 \neq 0$. Then, for all sufficiently large *k*, $\boldsymbol{x_k}$ is approximately the same as $c_1(1.02)^k\boldsymbol{v_1}$, and we write

$$\boldsymbol{x}_k \approx c_1(1.02)^k \begin{bmatrix} 10 \\ 13 \end{bmatrix} \quad \ldots\ldots\ldots\ldots\ldots\ldots(1)$$

The approximation in (1) improves as *k* increases, and so for large *k*,

$$\boldsymbol{x}_{k+1} \approx c_1(1.02)^{k+1} \begin{bmatrix} 10 \\ 13 \end{bmatrix} = (1.02)c_1(1.02)^k \begin{bmatrix} 10 \\ 13 \end{bmatrix} \approx 1.02\boldsymbol{x}_k$$

This approximation says that eventually both $x_k$ entries i.e owls and rats grow by a factor of almost 1.02 each month. Also equation (1) shows that $x_k$ is approximately a multiple of (10,13).So we can say that the entries in $x_k$ are almost in the same ratio as 10 to 13.It means that for every 10 owls there are about 13 thousand rats.

**Trajectory of a dynamical system**

In a 2×2 matrix, geometric description of a system's evolution can enhance the algebraic calculations. We can view that what happens to an initial point $x_0$ in $R^2$, when it is transformed repeatedly by the mapping x→Ax. The graph of $x_0, x_1 \ldots$is called a trajectory of the dynamical system.

**Example 2**

Plot several trajectories of the dynamical system $\boldsymbol{x}_{k+1} = \boldsymbol{A}\boldsymbol{x}_k$, when

$$\boldsymbol{A} = \begin{bmatrix} .80 & 0 \\ 0 & .64 \end{bmatrix}$$

**Solution**

In a diagonal matrix the eigen values are the diagonal entries and as $A$ is a diagonal matrix with diagonal enteries 0.8 and 0.64,Therefore the eigenvalues of $\boldsymbol{A}$ are .8 and .64, with eigenvectors $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\boldsymbol{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

If $\boldsymbol{x}_0 = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2$ , then $\boldsymbol{x}_k = c_1(.8)^k \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2(.64)^k \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Of course, $\boldsymbol{x}_k$ tends to 0 because $(.8)^k$ and $(.64)^k$ both approach to 0 as $k \to \infty$. But the way $\boldsymbol{x}_k$ goes towards 0 is interesting. Figure 1 shows the first few terms of several trajectories that begin at points on the boundary of the box with corners at $(\pm 3, \pm 3)$. The points on a trajectory are connected by a thin curve, to make the trajectory easier to see.

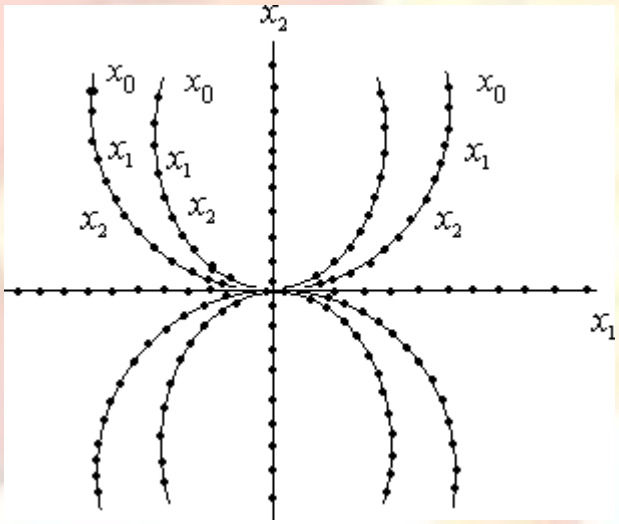



**Figure 1** The origin as an attractor

In this example, the origin is called an **attractor** of the dynamical system because all trajectories tend towards **O**. This occurs whenever both eigenvalues are less than 1 in magnitude. The direction of greatest attraction is along the line through **O** and the eigenvector $\boldsymbol{v}_2$ as it has smaller eigenvalue.

**Example 3**

Plot several typical solution of the equation $x_{k+1} = Ax_k$ , when

$$A = \begin{bmatrix} 1.44 & 0 \\ 0 & 1.2 \end{bmatrix}$$

**Solution**

In a diagonal matrix the eigen values are the diagonal entries and as $A$ is a diagonal matrix with diagonal entries 1.44 and 1.2,Therefore the eigenvalues of $\boldsymbol{A}$ are 1.44 and 1.2, with eigenvectors $\boldsymbol{v_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\boldsymbol{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

As

$$\begin{aligned} x_0 &= c_1 v_1 + c_2 v_2 \\ &= c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} c_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ c_2 \end{bmatrix} \\ &= \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}, \quad \textit{then} \\ x_k &= c_1 (1.44)^k \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 (1.2)^k \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}$$

Both terms grow in size as $k \to \infty$ ,but the first term grows faster because it has larger eigenvalue. Figure 2 shows several trajectories that begin at points quite close to 0.

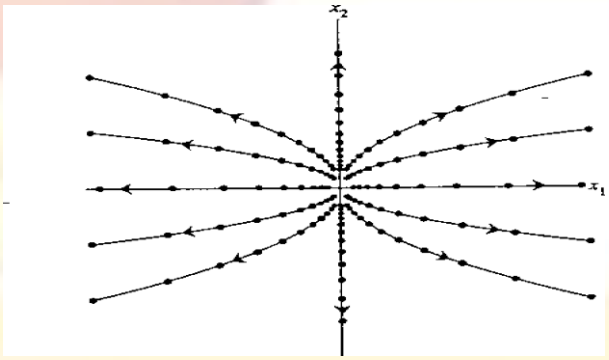



**Figure 2** The origin as a repellor

In this example, the origin is called a **repellor** of the dynamical system because all trajectories tend away from **O**. This occurs whenever both eigenvalues are greater than 1

in magnitude. The direction of the greatest repulsion is the line through 0 and that eigen vector which has a larger eigenvalue.

**Remark**

From the above two examples it may be noticed that when eigenvalues are less than 1 in magnitude, the origin behaves as an attractor and when the eigenvalues are greater than 1 in magnitude, the origin behaves as a repellor.

**Example 4**

Plot several typical solutions of the equation

$y_{k+1} = Dy_k$ ,where

$$D = \begin{bmatrix} 2.0 & 0 \\ 0 & 0.5 \end{bmatrix}$$

Show that a solution $\{y_k\}$ is unbounded if its initial point is not on the $x_2$-axis.

**Solution**

Mistakes

The eigenvalues of D are 2.0 and 0.5, with eigenvectors $\boldsymbol{v_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\boldsymbol{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

*If*

$y_0 = c_1\lambda_1 v_1 + c_2\lambda_2 v_2$

$y_0 = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$, *then*

$$y_k = c_1(2.0)^k \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2(0.5)^k \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

If $y_0$ is on the $x_2$-axis,then $c_1$ =0 and $y_k \to 0$ as $k \to \infty$ .But if $y_0$ is not on the $x_2$-axis, then the first term in the sum for $y_k$ becomes arbitrarily large, and so $\{y_k\}$ is unbounded. Figure 3 shows ten trajectories that begin near or on the axis.

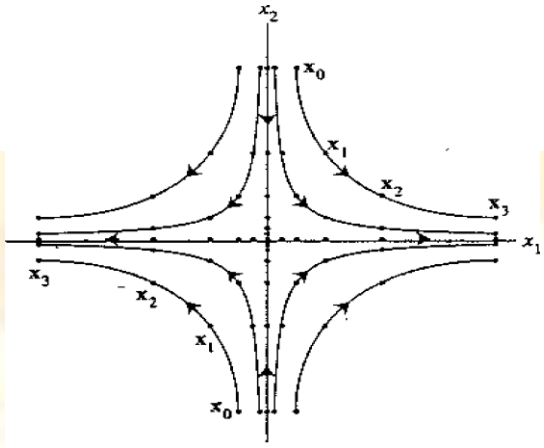



FIGURE 3 The origin as a saddle point.

In this example 'O' is called the saddle point because one eigenvalue is greater than '1' in magnitude and one is less than '1' in magnitude. The origin attracts solution from some directions and repels them in other directions.

## Change of Variable

The preceding three examples involved diagonal matrices. To handle the nondiagonal case, we return for a moment to the $n \times n$ case in which eigenvectors of $\boldsymbol{A}$ form a basis $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ for $\boldsymbol{R}^n$. Let $\boldsymbol{P} = [\boldsymbol{v}_1 \quad \cdots \quad \boldsymbol{v}_n]$, and let $\boldsymbol{D}$ be the diagonal matrix with the corresponding eigenvalues on the diagonal. Given a sequence $\{\boldsymbol{x}_k\}$ satisfying $\boldsymbol{x}_{k+1} = \boldsymbol{A}\boldsymbol{x}_k$, define a new sequence $\{\boldsymbol{y}_k\}$ by $\boldsymbol{y}_k = \boldsymbol{P}^{-1}\boldsymbol{x}_k$, or equivalently, $\boldsymbol{x}_k = \boldsymbol{P}\boldsymbol{y}_k$.

Substituting these relations into the equation $\boldsymbol{x}_{k+1} = \boldsymbol{A}\boldsymbol{x}_k$, and using the fact that $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$, we find that $\boldsymbol{P}\boldsymbol{y}_{k+1} = \boldsymbol{A}\boldsymbol{P}\boldsymbol{y}_k = (\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1})\boldsymbol{P}\boldsymbol{y}_k = \boldsymbol{P}\boldsymbol{D}\boldsymbol{y}_k$

Left-multiplying both sides by $\boldsymbol{P}^{-1}$, we obtain $\boldsymbol{y}_{k+1} = \boldsymbol{D}\boldsymbol{y}_k$

If we write $\boldsymbol{y}_k$ as $\boldsymbol{y}(k)$ and denote the entries in $\boldsymbol{y}(k)$ by $\boldsymbol{y}_1(k), \ldots, \boldsymbol{y}_n(k)$, then

$$\begin{bmatrix} y_1(k+1) \\ y_2(k+1) \\ \vdots \\ y_n(k+1) \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix} \begin{bmatrix} y_1(k) \\ y_2(k) \\ \vdots \\ y_n(k) \end{bmatrix}$$

The change of variable from $\boldsymbol{x}_k$ to $\boldsymbol{y}_k$ has decoupled the system of difference equations. The evolution of $\boldsymbol{y}_1(k)$, for example, is unaffected by what happens to $\boldsymbol{y}_2(k), \ldots, \boldsymbol{y}_n(k)$, because $y_1(k+1) = \lambda_1 . y_1(k)$ for each $k$.

**Example 5**

Show that the origin is a saddle point for solutions of $\boldsymbol{x_{k+1}} = \boldsymbol{A}\boldsymbol{x_k}$, where

$$\boldsymbol{A} = \begin{bmatrix} 1.25 & -.75 \\ -.75 & 1.25 \end{bmatrix}$$

find the directions of greatest attraction and greatest repulsion.

**Solution**

For eigenvalues of the given matrix put

$\det(A - \lambda I) = 0$

we find that $\boldsymbol{A}$ has eigenvalues 2 and .5, with corresponding eigenvectors $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, respectively. Since $|2| > 1$ and $|.5| < 1$, the origin is a saddle point of the dynamical system. If $\boldsymbol{x_0} = c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2}$, then

$$\boldsymbol{x}_k = c_1(2)^k \boldsymbol{v}_1 + c_2(.5)^k \boldsymbol{v}_2$$

In figure 4, the direction of the greatest repulsion is the line through $v_1$, the eigenvector whose eigenvalue is greater than 1.

The direction of the greatest attraction is determined by the eigenvector $v_2$ whose eigenvalue is less than 1.

A number of trajectories are shown in figure 4.When this graph is viewed in terms of eigenvector axes, the diagram looks same as figure 4.

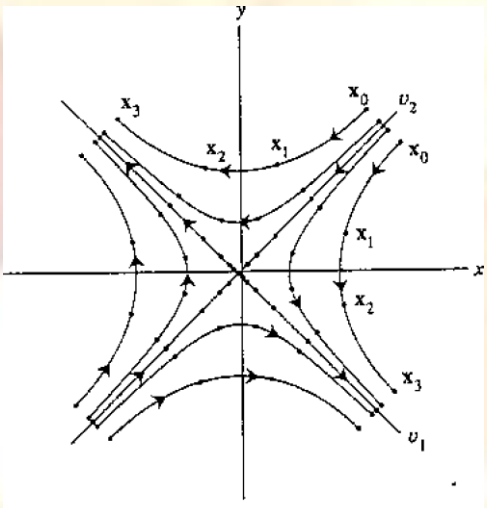



FIGURE 4 The origin as a saddle point.

**Complex Eigenvalues**

Since the characteristic equation of n× n matrix involves a polynomial of degree n, the equation always has exactly n roots, in which complex roots are possibly included. If we have complex eigenvalues then we get complex eigen vectors which may be divided into two parts: which is real part of the vector and imaginary part of the vector.

**Note**

If a matrix has two complex eigenvalues in which absolute value of one eigenvalue is greater than 1,then origin behaves as a repellor and iterates of $x_0$ will spiral outwards around the origin .If the absolute value of an eigenvalue is less than 1,then the origin behaves as an attractor and iterates of $x_0$ will spiral inward towards the origin.

**Example 6**

Let

$A = \begin{bmatrix} 0.8 & 0.5 \\ -0.1 & 1.0 \end{bmatrix}$ *haseigenvalue* $0.9 \pm 0.2i$ *with eigenvectors* $\begin{bmatrix} 1 \pm 2i \\ 1 \end{bmatrix}$.

*Find the trajectories of the system* $x_{k+1} = Ax_k$ *with initial vectors* $\begin{bmatrix} 0 \\ 2.5 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$ *and* $\begin{bmatrix} 0 \\ -2.5 \end{bmatrix}$.

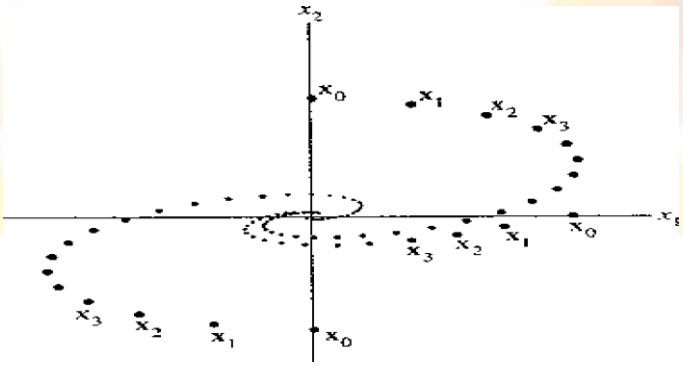
**Solution**

Given a matrix

$$A = \begin{bmatrix} 0.8 & 0.5 \\ -0.1 & 1.0 \end{bmatrix}$$

The eigenvalues of the matrix are $0.9 \pm 0.2i$ and corresponding eigenvectors are $\begin{bmatrix} 1 \pm 2i \\ 1 \end{bmatrix}$

also the initial vectors are given as $\begin{bmatrix} 0 \\ 2.5 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ *and* $\begin{bmatrix} 0 \\ -2.5 \end{bmatrix}$.



**Example 7**

Suppose the search survival rate of a young bird is 50%,and the stage matrix A is

$$A=\begin{bmatrix}0 & 0 & 0.33\\ 0.3 & 0 & 0\\ 0 & 0.71 & 0.94\end{bmatrix}$$

What does the stage matrix model predict about this bird?

**Solution**

Given a matrix $A=\begin{bmatrix}0 & 0 & 0.33\\ 0.3 & 0 & 0\\ 0 & 0.71 & 0.94\end{bmatrix}$

for eigenvalues of A we will put det(A- $\lambda$ I)=0

$$\det(A-\lambda I)=\begin{vmatrix}-\lambda & 0 & 0.33\\ 0.3 & -\lambda & 0\\ 0 & 0.71 & 0.94-\lambda\end{vmatrix}$$

$$=-\lambda(-0.94+\lambda^2)+0.33(0.3\times 0.71)$$

$$=0.94\lambda-\lambda^3+0.07029$$

*for eigenvalues put* $\det(A-\lambda I)=0$

$$0.94\lambda-\lambda^3+0.07029=0$$

$$\lambda^3-0.94\lambda-0.07029=0$$

it gives 3 values of $\lambda$ which are approximately 1. 01,-0.03+0.26i,-0.03-0.26i

eigen vector for first value of $\lambda$ will be $v_1=(10,3,31)$ and for the next two values of $\lambda$ we willget complex eigen vectors denoted by $v_2$ and $v_3$.

Now

$$x_k=c_1(1.01)^k v_1+c_2(-0.03+0.26i)^k v_2+c_3(-0.03-0.26i)^k v_3$$

As $k\to\infty$ ,the second two vectors tends to 0.So $x_k$ becomes more and more like the (real) vector $c_1(1.01)^k v_1$ .Thus the long term growth rate of the owl population will be 1.01,and the population will grow slowly. The eigenvector describes the eventual distribution of the owls by life stages: For every 31 adults, there will be about 10 juveniles and three subadults.

**Exercise**

Suppose that a $3\times 3$ matrix A has eigenvalues 2,2/5,3/5 and corresponding eigenvectors

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} \quad v_2 = \begin{bmatrix} -2 \\ 1 \\ -5 \end{bmatrix} \quad v_3 = \begin{bmatrix} -3 \\ 3 \\ 7 \end{bmatrix}$$

$$also \quad x_0 = \begin{bmatrix} -2 \\ -5 \\ -3 \end{bmatrix}$$

Find a general solution of the equation $x_{k+1} = Ax_k$ also find what happens to $x_k$ when $k \to \infty$.

# Lecture 34

# Applications to Differential Equations

## Differential equation

A differential equation is any equation which contains derivatives, either ordinary derivatives or partial derivatives. Differential equations play an extremely important and useful role in applied mathematics, engineering, and physics. Moreover, a lot of mathematical and numerical machinery has been developed for the solution of differential equations.

## System of linear differential equations

A system of linear differential equations can be expressed as:

$$\begin{aligned} x_1' &= a_{11}x_1 + \ldots + a_{1n}x_n \\ x_2' &= a_{21}x_1 + \ldots + a_{2n}x_n \\ &\vdots \\ x_n' &= a_{n1}x_1 + \ldots + a_{nn}x_n \end{aligned}$$

where $x_i(t)$ is a function of time, i = 1, . . . n, and the matrix of constant coefficient is $A = \lceil \lfloor a_{ij} \rfloor \rceil$

In many applied problems, several quantities are continuously varying in time, and they are related by a system of differential equations:

$$\begin{aligned} x_1' &= a_{11}x_1 + \ldots + a_{1n}x_n \\ x_2' &= a_{21}x_1 + \ldots + a_{2n}x_n \\ &\vdots \\ x_n' &= a_{n1}x_1 + \ldots + a_{nn}x_n \end{aligned}$$

Here $x_1, x_2, \ldots, x_n$ are differentiable functions of t ,with derivatives $x_1', x_2', \ldots, x_n'$ ,and the $a_{ij}$ are constants. Write the system as a matrix differential equation

$$x' = Ax$$

Where

$$x(t) = \begin{bmatrix} x_1(t) \\ . \\ . \\ . \\ x_n(t) \end{bmatrix}, x'(t) = \begin{bmatrix} x_1'(t) \\ . \\ . \\ . \\ x_n'(t) \end{bmatrix}, and\ A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ . & & . \\ . & & . \\ . & & . \\ a_{n1} & & a_{nn} \end{bmatrix}$$

**Superposition of solutions**

If $u$ and $v$ are two solutions of $x' = Ax$, then $cu + dv$ is also a solution.

Proof:

If $u$ and $v$ are two solutions of $x' = Ax$ then $u' = Au$ and $v' = Av$ .Therefore,

$$\begin{aligned} (cu + dv)' &= cu' + dv' \\ &= cAu + dAv \\ &= A(cu + dv) \end{aligned}$$

Linear combination of any set of solutions is also a solution. This property is called **superposition** of solutions.

**Fundamental set of Solutions**

If A is an n× n matrix, then there are n linearly independent functions in a fundamental set, and each solution of $x' = Ax$ is a unique linear combination of these 'n' functions. So we can say that fundamental set of solutions is a basis for the set of all solutions of $x' = Ax$ ,and the solution set is n-dimensional vector space of functions.

**Initial value problem**

If a vector $x_0$ is specified, then the initial value problem is to construct the unique function $x$ such that $x' = Ax\ and\ x(0) = x_0$

**Example 1**

For the system $x' = Ax$ .What will be the solution when A is a diagonal matrix and is given by

$$A = \begin{pmatrix} 3 & 0 \\ 0 & -5 \end{pmatrix}$$

**Solution**

The solution of the given system can be obtained by elementary calculus. Consider

$$\begin{bmatrix} x_1'(t) \\ x_2'(t) \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & -5 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

*from here wehave*

$x_1'(t) = 3x_1(t)$ *and* $x_2'(t) = -5x_2(t)$

*first of all we will find solutionof* $x_1'(t) = 3x_1(t)$ *by usin g calculus.*

$$x_1'(t) = 3x_1(t)$$

$$\frac{dx_1}{dt} = 3x_1$$

$$\frac{dx_1}{x_1} = 3dt$$

*Integrating both sides*

$$\int \frac{dx_1}{x_1} = 3\int dt$$

$$\ln x_1 = 3t + \ln c_1$$

$$\ln x_1 - \ln c_1 = 3t$$

$$\ln\left(\frac{x_1}{c_1}\right) = 3t$$

*taking anti*log *on both sides*

$$\frac{x_1}{c_1} = e^{3t}$$

$$x_1(t) = c_1 e^{3t}$$

Similarly Solution of

$x_2'(t) = -5x_2(t)$

*will be*

$x_2(t) = c_2 e^{-5t}$

And it can be written in the form

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} c_1 e^{3t} \\ c_2 e^{-5t} \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{-5t}$$

The given system is said to be decoupled because each derivative of the function depends only on the function itself, not on some combination or coupling of $x_1(t)$ *and* $x_2(t)$.

**Observation**

From the above example we observe that a solution of the general equation $x' = Ax$ might be a linear combination of functions of the form

$$x(t) = ve^{\lambda t}$$

Where $\lambda$ is some scalar and v is some fixed non-zero vector.

$$x(t) = ve^{\lambda t} \quad \cdots\cdots(1)$$

*differentiating w.r.t 't'*

$$x'(t) = \lambda ve^{\lambda t}$$

*Multiplying both sides of* (1) *by A*

$$Ax(t) = Ave^{\lambda t}$$

*Since* $e^{\lambda t}$ *is never zero,* $x'(t)$ *will equal to* $Ax(t)$ *iff* $\lambda v = Av$.

**Eigenfunctions**

Each pair of eigenvalue and its corresponding eigenvector provides a solution of the equation $x' = Ax$ which is called **eigenfunctions** of the differential equation.

**Example 2**

The circuit in figure 1 can be described by

$$\begin{bmatrix} v_1'(t) \\ v_2'(t) \end{bmatrix} = \begin{bmatrix} -(R_1+R_2)/C_1 & R_2/C_1 \\ R_2/C_2 & -R_2/C_2 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}$$

Where $v_1(t)$ and $v_2(t)$ are the voltages across the two capacitors at time *t*.

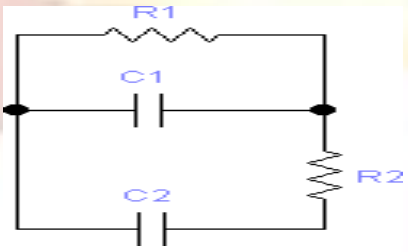



Figure 1

Suppose that resistor $R_1$ is 2 ohms, $R_2$ is 1 ohms, capacitor $C_1$ is 2 farads, and capacitor $C_2$ is 1 farad and suppose that there is an initial charge of 5 volts on capacitor $C_1$ and 4 volts on capacitor $C_2$. Find formulas for $v_1(t)$ and $v_2(t)$ that describe how the voltages change over time.

**Solution**

For the data given, we can form A as

$$A=\begin{bmatrix} -(R_1+R_2)/C_1 & R_2/C_1 \\ R_2/C_2 & -R_2/C_2 \end{bmatrix}$$

$R_1=2$

$R_2=1$

$C_1=2$

$C_2=1$ *so*

$-(R_1+R_2)/C_1 = -(2+1)/2 \Rightarrow -3/2 \Rightarrow -1.5$

$R_2/C_1 = 1/2 \Rightarrow 0.5$

$R_2/C_2 = 1/1 \Rightarrow 1$

$-R_2/C_2 = -1/1 \Rightarrow -1$

$\boldsymbol{A}=\begin{bmatrix} -1.5 & .5 \\ 1 & -1 \end{bmatrix}, \boldsymbol{x}=\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$, and $\boldsymbol{x}_0=\begin{bmatrix} 5 \\ 4 \end{bmatrix}$. The vector $\boldsymbol{x}_0$ lists the initial values of $\boldsymbol{x}$. From $\boldsymbol{A}$, we obtain eigenvalues $\lambda_1=-.5$ and $\lambda_2=-2$, by using det (A-λI)=0 with corresponding eigenvectors $\boldsymbol{v_1}=\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\boldsymbol{v_2}=\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

The eigenfunctions $\boldsymbol{x_1}(t)=\boldsymbol{v_1}e^{\lambda_1 t}$ and $\boldsymbol{x_2}(t)=\boldsymbol{v_2}e^{\lambda_2 t}$ both satisfy $\boldsymbol{x}'=\boldsymbol{A}\boldsymbol{x}$, and so does any linear combination of $\boldsymbol{x_1}$ and $\boldsymbol{x_2}$. Set

$$\boldsymbol{x}(t)=c_1\boldsymbol{v}_1e^{\lambda_1 t}+c_2\boldsymbol{v}_2e^{\lambda_2 t}$$
$$=c_1\begin{bmatrix} 1 \\ 2 \end{bmatrix}e^{-.5t}+c_2\begin{bmatrix} -1 \\ 1 \end{bmatrix}e^{-2t} \quad \cdots\cdots\cdots\cdots (1)$$

and note that $\boldsymbol{x}(0) = c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2}$. Since $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ are obviously linearly independent and hence span $\boldsymbol{R^2}$, $c_1$ and $c_2$ can be found to make $\boldsymbol{x}(0)$ equal to $\boldsymbol{x_0}$.

We can find $c_1$ *and* $c_2$ as

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}=\begin{bmatrix} v_1 & v_2 & x_0 \end{bmatrix}$$
$$=\begin{bmatrix} 1 & -1 & 5 \\ 2 & 1 & 4 \end{bmatrix} \quad \textit{row reduction of this matrix gives}$$
$$=\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \end{bmatrix}$$

*hence* $c_1=3$ *and* $c_2=-2$

Put the value of $c_1$ *and* $c_2$ in (1)

$$\boldsymbol{x}(t)=3\begin{bmatrix} 1 \\ 2 \end{bmatrix}e^{-.5t}-2\begin{bmatrix} -1 \\ 1 \end{bmatrix}e^{-2t}$$

Thus the desired solution of the differential equation $\boldsymbol{x}'=\boldsymbol{A}\boldsymbol{x}$ is

$$x(t) = \begin{bmatrix} 3e^{-0.5t} \\ 6e^{-0.5t} \end{bmatrix} - \begin{bmatrix} -2e^{-2t} \\ 2e^{-2t} \end{bmatrix}$$

or $$\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} = \begin{bmatrix} 3e^{-0.5t} + 2e^{-2t} \\ 6e^{-0.5t} - 2e^{-2t} \end{bmatrix}$$

As both eigenvalues are negative so origin in this case behaves as an attractor and the direction of greatest attraction will be along the more negative eigenvalue $\lambda = -2$.

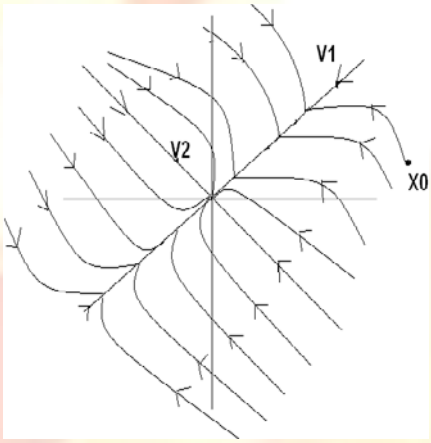



Figure 2 The Origin as an attractor

**Example 3**

Suppose a particle is moving in a planar force field and its position vector $x$ satisfies $x' = Ax$ and $x(0) = x_0$, where

$$\boldsymbol{A} = \begin{bmatrix} 4 & -5 \\ -2 & 1 \end{bmatrix}, \ \boldsymbol{x^0} = \begin{bmatrix} 2.9 \\ 2.6 \end{bmatrix}$$

Solve this initial value problem for $t \geq 0$, and sketch the trajectory of the particle.

**Solution**

The eigenvalues of the given matrix can be found by using det(A-λI)=0 which are turned out to be $\lambda_1 = 6$ and $\lambda_2 = -1$ with corresponding eigen vectors $v_1 = \begin{bmatrix} -5 \\ 2 \end{bmatrix}$ *and* $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

For any constants $c_1$ *and* $c_2$ ,the function

$$x(t) = c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_2 t}$$
$$= c_1 \begin{bmatrix} -5 \\ 2 \end{bmatrix} e^{6t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t}$$

We have to find $c_1$ *and* $c_2$ to satisfy x(0)= $x_0$ ,so

$$x_0 = c_1\begin{bmatrix}-5\\2\end{bmatrix} + c_2\begin{bmatrix}1\\1\end{bmatrix}$$

$$\begin{bmatrix}2.9\\2.6\end{bmatrix} = \begin{bmatrix}-5c_1\\2c_1\end{bmatrix} + \begin{bmatrix}c_2\\c_2\end{bmatrix}$$

$$= \begin{bmatrix}-5c_1 + c_2\\2c_1 + c_2\end{bmatrix}$$

$$2.9 = -5c_1 + c_2 \quad \text{.....(1)}$$

$$2.6 = 2c_1 + c_2 \quad \text{.....(2)}$$

*Subtracting* 1 *from* 2 *we get*

$c_1 = -\frac{3}{70}$ *and substituting this value of* $c_1$ *in* (1) *we get* $c_2 = \frac{188}{70}$

$$x(t) = -\frac{3}{70}\begin{bmatrix}-5\\2\end{bmatrix}e^{6t} + \frac{188}{70}\begin{bmatrix}1\\1\end{bmatrix}e^{-t}$$

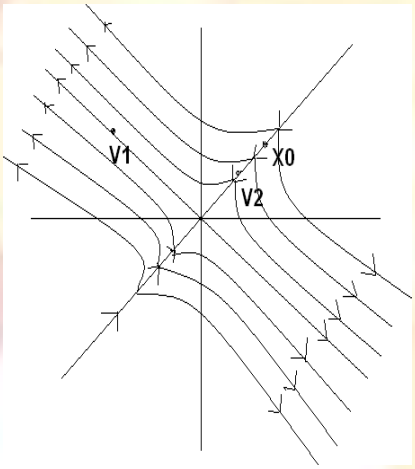



Figure 3 The Origin as a saddle point

As the matrix A has both positive and negative eigenvalues so in this case origin behaves as a saddle point. The direction of greatest repulsion is the line through $v_1$ *and* 0 corresponding to the positive eigenvalue. The direction of greatest attraction is the line through $v_2$ *and* 0 , corresponding to the negative eigenvalue.

## Decoupling a Dynamical System

For any dynamical system described by $\boldsymbol{x}' = \boldsymbol{A}\boldsymbol{x}$ when $\boldsymbol{A}$ is diagonalizable, the fundamental set of solutions can be obtained by the methods that we have discussed in examples 2 and 3.

Suppose that the eigenfunctions for $\boldsymbol{A}$ are

$$\boldsymbol{v}_1 e^{\lambda_1 t}, \cdots, \boldsymbol{v}_n e^{\lambda_n t}$$

with $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ linearly independent eigenvectors. Let $\boldsymbol{P} = [\boldsymbol{v}_1 \ \ldots \ \boldsymbol{v}_n]$ and let $\boldsymbol{D}$ be the diagonal matrix with entries $\lambda_1, \ldots, \lambda_n$ so that $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$. Now make a change of variable, defining a new function $\boldsymbol{y}$ by

$$\boldsymbol{y}(t) = \boldsymbol{P}^{-1}\boldsymbol{x}(t), \text{ or equivalently, } \boldsymbol{x}(t) = \boldsymbol{P}\boldsymbol{y}(t)$$

Substituting these relations into $\boldsymbol{x}' = \boldsymbol{A}\boldsymbol{x}$ gives

$$Py'(t) = (PDP^{-1})Py(t)$$
$$Py'(t) = PDy(t)$$
$$\textit{premultiplying both sides by } P^{-1}$$
$$P^{-1}Py'(t) = P^{-1}PDy(t)$$
$$y'(t) = Dy(t) \ \textit{ or}$$
$$y' = Dy$$

$$\begin{bmatrix} y_1'(t) \\ y_2'(t) \\ \vdots \\ y_n'(t) \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix} \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{bmatrix}$$

The change of variable from $\boldsymbol{x}$ to $\boldsymbol{y}$ has decoupled the system of differential equations because the derivative of each scalar function $\boldsymbol{y}_k$ depends only on $\boldsymbol{y}_k$. Since $\boldsymbol{y}_1' = \lambda_1 \boldsymbol{y}_1$, we have $\boldsymbol{y}_1(t) = c_1 e^{\lambda_1 t}$, with similar formulas for $\boldsymbol{y}_2, \ldots, \boldsymbol{y}_n$. Thus

$$\boldsymbol{y}(t) = \begin{bmatrix} c_1 e^{\lambda_1 t} \\ \vdots \\ c_n e^{\lambda_n t} \end{bmatrix}, \quad \text{where} \quad \begin{bmatrix} c_1 \\ \vdots \\ c_2 \end{bmatrix} = \boldsymbol{y}(0) = \boldsymbol{P}^{-1}\boldsymbol{x}(0) = \boldsymbol{P}^{-1}\boldsymbol{x}_0$$

To obtain the general solution $\boldsymbol{x}$ of the original system, compute

$$\boldsymbol{x}(t) = \boldsymbol{P}\boldsymbol{y}(t) = \begin{bmatrix} \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_n \end{bmatrix} \boldsymbol{y}(t) = c_1 \boldsymbol{v}_1 e^{\lambda_1 t} + \cdots + c_n \boldsymbol{v}_n e^{\lambda_n t}$$

This is the eigenfunction expansion as constructed in example 2.

## Complex Eigenvalues

As we know that for a real matrix A , complex eigenvalues and associated eigen vectors come in conjugate pairs i.e if $\lambda$ is an eigenvalue of A then $\bar{\lambda}$ will be the 2nd

eigenvalue. Similarly if one eigen vector is $v$ then the other will be $\bar{v}$. So there will be two solutions of $x' = Ax$ and they are

$x_1(t) = ve^{\lambda t}, x_2(t) = \bar{v}e^{\bar{\lambda} t}$

Here $x_2(t) = \overline{x_1(t)}$

$$\mathrm{Re}(ve^{\lambda t}) = \frac{1}{2}\left[x_1(t) + \overline{x_1(t)}\right]$$

$$\mathrm{Im}(ve^{\lambda t}) = \frac{1}{2i}\left[x_1(t) - \overline{x_1(t)}\right]$$

From Calculus

$$e^x = 1 + x + \frac{x^2}{2!} + \ldots + \frac{x^n}{n!} + \ldots$$

$$e^{\lambda t} = 1 + \lambda t + \frac{(\lambda t)^2}{2!} + \ldots + \frac{(\lambda t)^n}{n!} + \ldots$$

*If we write* $\lambda = a + ib$

$$e^{(a+ib)t} = e^{at}.e^{ibt} = e^{at}(\cos bt + i \sin bt)$$

*So*

$$ve^{\lambda t} = (\mathrm{Re}\, v + i\, \mathrm{Im}\, v).e^{at}(\cos bt + i \sin bt)$$

$$ve^{\lambda t} = \mathrm{Re}\, v.e^{at} \cos bt + \mathrm{Re}\, v.e^{at} i \sin bt + i\, \mathrm{Im}\, v.e^{at} \cos bt - \mathrm{Im}\, v.e^{at} \sin bt$$

$$ve^{\lambda t} = e^{at}(\mathrm{Re}\, v \cos bt - \mathrm{Im}\, v \sin bt) + ie^{at}(\mathrm{Re}\, v \sin bt + \mathrm{Im}\, v \cos bt)$$

*So two real solutions of* $x' = Ax$ *are*

$$y_1(t) = \mathrm{Re}\, x_1(t) = \left[(\mathrm{Re}\, v)\cos bt - (\mathrm{Im}\, v)\sin bt\right]e^{at}$$

$$y_2(t) = \mathrm{Im}\, x_1(t) = \left[(\mathrm{Re}\, v)\sin bt + (\mathrm{Im}\, v)\cos bt\right]e^{at}$$

**Example 4**

The circuit in figure 4 can be described by the equation

$$\begin{bmatrix} i_L' \\ v_c' \end{bmatrix} = \begin{bmatrix} -R_2/L & -1/L \\ 1/C & -1/(RC) \end{bmatrix} \begin{bmatrix} i_L \\ v_c \end{bmatrix}$$

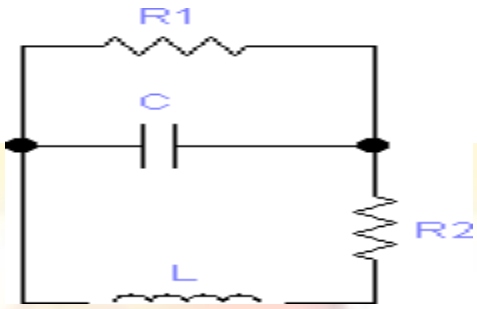



Figure 4

Where $i_L$ is the current passing through the inductor $L$ and $v_c$ is the voltage drop across the capacitor $C$. Suppose $R_1$ is 5 ohms, $R_2$ is .8 ohm, $C$ is .1 farad, and $L$ is .4 henry. Find formulas for $i_L$ and $v_c$, if the initial current through the inductor is 3 amperes and the initial voltage across the capacitor is 3 volts.

**Solution**

From the data given, We can form A as

$$A = \begin{bmatrix} -R_2/L & -1/L \\ 1/C & -1/(R_1 C) \end{bmatrix}$$

$$R_1 = 5$$

$$R_2 = 0.8$$

$$C = 0.1$$

$$L = 0.4$$

$$-R_2/L = -0.8/0.4 \Rightarrow -8/10 \times 10/4 \Rightarrow -2$$

$$-1/L = -1/0.4 \Rightarrow -10/4 \Rightarrow -2.5$$

$$1/C = 1/0.1 \Rightarrow 10$$

$$-1/(R_1 C) = -1/5 \times 0.1 \Rightarrow -1/0.5 \Rightarrow -2$$

$$\boldsymbol{A} = \begin{bmatrix} -2 & -2.5 \\ 10 & -2 \end{bmatrix}$$

$$\det(A-\lambda I)=\begin{bmatrix}-2-\lambda & -2.5\\ 10 & -2-\lambda\end{bmatrix}$$

$$=(-2-\lambda)^2+25$$

$$=4+\lambda^2+4\lambda+25$$

$$=\lambda^2+4\lambda+29$$

*put* $\det(A-\lambda I)=0$

$$\lambda^2+4\lambda+29=0$$

the eigenvalues of A is $-2\pm 5i$ with corresponding eigenvectors $\boldsymbol{v}_1=\begin{bmatrix} i\\ 2\end{bmatrix}$ *and* $\begin{bmatrix}-i\\ 2\end{bmatrix}$ The complex solutions of $\boldsymbol{x}'=\boldsymbol{A}\boldsymbol{x}$ are complex linear combinations of

$$\boldsymbol{x}_1(t)=\begin{bmatrix} i\\ 2\end{bmatrix}e^{(-2+5i)t} \quad \text{and} \quad \boldsymbol{x}_2(t)=\begin{bmatrix}-i\\ 2\end{bmatrix}e^{(-2-5i)t}$$

$$=\begin{bmatrix} i\\ 2\end{bmatrix}e^{-2t}.e^{5it}$$

$$\boldsymbol{x}_1(t)=\begin{bmatrix} i\\ 2\end{bmatrix}e^{-2t}(\cos 5t+i\sin 5t)$$

By taking real and imaginary parts of $\boldsymbol{x}_1$, we obtain real solutions:

$$\boldsymbol{y}_1(t)=\begin{bmatrix}-\sin 5t\\ 2\cos 5t\end{bmatrix}e^{-2t},\ \boldsymbol{y}_2(t)=\begin{bmatrix}\cos 5t\\ 2\sin 5t\end{bmatrix}e^{-2t}$$

Since $\boldsymbol{y}_1$ and $\boldsymbol{y}_2$ are linearly independent functions, they form a basis for the two-dimensional real vector space of solutions $\boldsymbol{x}'=\boldsymbol{A}\boldsymbol{x}$. Thus, the general solution is

$$\boldsymbol{x}(t)=c_1\begin{bmatrix}-\sin 5t\\ 2\cos 5t\end{bmatrix}e^{-2t}+c^2\begin{bmatrix}\cos 5t\\ 2\sin 5t\end{bmatrix}e^{-2t}$$

To satisfy $\boldsymbol{x}(0)=x_0=\begin{bmatrix}3\\ 3\end{bmatrix}$ we need s$c_1$ and $c_2$

$$c_1\begin{bmatrix}0\\ 2\end{bmatrix}+c_2\begin{bmatrix}1\\ 0\end{bmatrix}=\begin{bmatrix}3\\ 3\end{bmatrix},$$

$$\begin{bmatrix}0\\ 2c_1\end{bmatrix}+\begin{bmatrix}c_2\\ 0\end{bmatrix}=\begin{bmatrix}3\\ 3\end{bmatrix}$$

$$0+c_2=3$$

Thus $2c_1+0=3$

$$c_2=3$$

$$c_1=\frac{3}{2}=1.5$$

$$\boldsymbol{x}(t) = c_1 \begin{bmatrix} -\sin 5t \\ 2\cos 5t \end{bmatrix} e^{-2t} + c^2 \begin{bmatrix} \cos 5t \\ 2\sin 5t \end{bmatrix} e^{-2t}$$

$$= 1.5 \begin{bmatrix} -\sin 5t \\ 2\cos 5t \end{bmatrix} e^{-2t} + 3 \begin{bmatrix} \cos 5t \\ 2\sin 5t \end{bmatrix} e^{-2t}$$

$$= \begin{bmatrix} -1.5\sin 5te^{-2t} \\ 3\cos 5te^{-2t} \end{bmatrix} + \begin{bmatrix} 3\cos 5te^{-2t} \\ 6\sin 5te^{-2t} \end{bmatrix}$$

$$= \begin{bmatrix} -1.5\sin 5te^{-2t} + 3\cos 5te^{-2t} \\ 3\cos 5te^{-2t} + 6\sin 5te^{-2t} \end{bmatrix}$$

$$\begin{bmatrix} i_L(t) \\ v_C(t) \end{bmatrix} = \begin{bmatrix} -1.5\sin 5t + 3\cos 5t \\ 3\cos 5t + 6\sin 5t \end{bmatrix} e^{-2t}$$

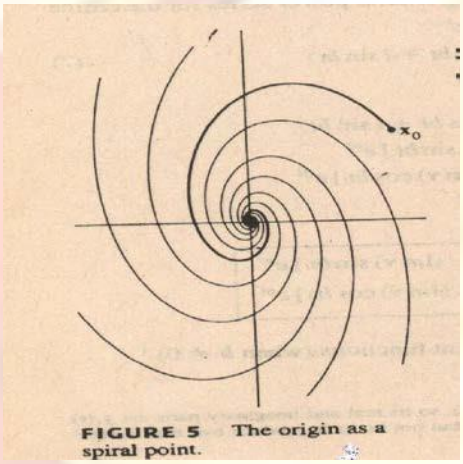



FIGURE 5 The origin as a spiral point.

**Figure 5** The origin as a spiral point

In this figure, due to the complex eigenvalues a rotation is caused by the sine and cosine function and hence the origin is called a spiral point. The trajectories spiral inwards as in this example the real part of the eigenvalue is negative.

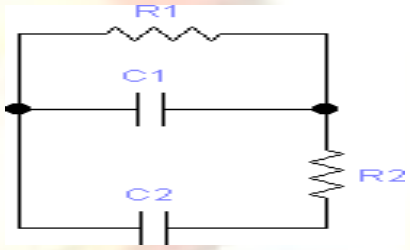
Similarly, it is important to note that when A has a complex eigenvalue with positive real part, the trajectories spiral outwards .And if the real part of the eigenvalue is zero, then the trajectories form ellipses around the origin.

**Exercise 1**

The circuit in given figure can be described by

$$\begin{bmatrix} v_1'(t) \\ v_2'(t) \end{bmatrix} = \begin{bmatrix} -(1/R_1 + 1/R_2)/C_1 & 1/R_2C_1 \\ 1/R_2C_2 & -1/R_2C_2 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}$$

Where $v_1(t)$ and $v_2(t)$ are the voltages across the two capacitors at time *t*.




Suppose that resistor $R_1$ is 1 ohms, $R_2$ is 2 ohms, capacitor $C_1$ is 1 farad, and capacitor $C_2$ is 0.5 farad, and suppose that there is an initial charge of 7 volts on capacitor $C_1$ and 6 volts on capacitor $C_2$. Find formulas for $v_1(t)$ and $v_2(t)$ that describe how the voltages change over time.

**Exercise 2**

Find formulas for the current $i_L$ and the voltage $V_C$ for the circuit in Example 4, assuming that $R_1 = 10\,ohms, R_2 = 0.2\,ohm, C = 0.4\ farad, L = 0.2\,henry$
the initial current is 0 amp, and the initial voltage is 16 volts.

# Lecture 35
# Iterative Estimates for Eigenvalues

## Eigenvalues and eigenvectors

Let $A$ be an $n \times n$ square matrix, $\lambda$ is any scalar value. If there exists a non-zero vector $X$, such that it satisfies the equation $AX = \lambda X$ then $\lambda$ is called the eigenvalue and $X$ is called the corresponding eigenvector.

i.e. if

$$AX = \lambda X$$
$$AX - \lambda X = 0$$
$$AX - \lambda IX = 0$$
$$(A - \lambda I)X = 0$$

To find the eigenvalues we have to solve the equation

$$|A - \lambda I| = 0$$

**Example 1**

Find the eigenvalues of $A = \begin{bmatrix} 1 & 0 \\ 1 & -2 \end{bmatrix}$.

**Solution**

$$A = \begin{bmatrix} 1 & 0 \\ 1 & -2 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 0 \\ 1 & -2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 1-\lambda & 0 \\ 1 & -2-\lambda \end{bmatrix}$$

$$|A - \lambda I| = (1-\lambda)(-2-\lambda)$$

Solving $|A - \lambda I| = 0$, we get

$$(1-\lambda)(-2-\lambda) = 0$$

So, either $(1-\lambda) = 0$ or $(-2-\lambda) = 0$

$so, \lambda = 1 \quad \lambda = -2$ are the eigenvalues of A.

## Power Method

Let a matrix A is diagonalizable, with n linearly independent eigenvectors, $v_1, v_2, \ldots v_n$ and corresponding eigenvalues $\lambda_1, \lambda_2, \ldots \lambda_n$ such that

$$|\lambda_1| > |\lambda_2| \geq \ldots \geq |\lambda_n|$$

Since $\{v_1, v_2, \ldots v_n\}$ is a basis for $R^n$, any vector $x$ can be written as

$$x = c_1 v_1 + c_2 v_2 + \ldots + c_n v_n$$

Multiplying both sides by A, we get

$$\begin{aligned} Ax &= A(c_1v_1 + c_2v_2 + \ldots + c_nv_n) \\ &= A(c_1v_1) + A(c_2v_2) + \ldots A(c_nv_n) \\ &= c_1(Av_1) + c_2(Av_2) + \ldots c_n(Av_n) \\ &= c_1(\lambda_1 v_1) + c_2(\lambda_2 v_2) + \ldots c_n(\lambda_n v_n) \end{aligned}$$

Again multiplying by A and simplifying as above, we get

$$A^2x = c_1(\lambda_1^2 v_1) + c_2(\lambda_2^2 v_2) + \ldots c_n(\lambda_n^2 v_n)$$

Continuing this process we get

$$A^k x = c_1(\lambda_1^k v_1) + c_2(\lambda_2^k v_2) + \ldots c_n(\lambda_n^k v_n)$$

$$A^k x = c_1(\lambda_1)^k v_1 + c_2(\lambda_2)^k v_2 + \ldots c_n(\lambda_n)^k v_n \quad (k = 1, 2, \ldots)$$

$$\left(\frac{1}{\lambda_1}\right)^k A^k x = c_1v_1 + c_2\left(\frac{\lambda_1}{\lambda_1}\right)^k v_2 + \ldots c_n\left(\frac{\lambda_n}{\lambda_1}\right)^k v_n \quad (k = 1, 2, \ldots)$$

$$(\lambda_1)^{-k} A^k x \to c_1v_1$$

$as\ k \to \infty$ Since $|\lambda_1| > |\lambda_2| \geq \ldots \geq |\lambda_n|$

Thus for large k, a scalar multiple of $A^k x$ determines almost the same direction as the eigenvector $c_1v_1$. Since, positive scalar multiples do not change the direction of the vector, $A^k x$ itself points almost in the same direction as $v_1$ or $-v_1$, provided $c_1 \neq 0$.

**Procedure for finding the Eigenvalues and Eigenvectors**
**Power Method**

To compute the largest eigenvalue and the corresponding eigenvector of the system

$$A(X) = \lambda(X)$$

Where A is a real, symmetric or un-symmetric matrix, the power method is widely used in practice.

**Procedure**

**Step 1**: Choose the initial vector such that the largest element is unity.

**Step 2**: The normalized vector $v^{(0)}$ is pre-multiplied by the matrix A.

**Step 3**: The resultant vector is again normalized.

**Step 4**: This process of iteration is continued and the new normalized vector is repeatedly pre-multiplied by the matrix A until the required accuracy is obtained.
At this point, the result looks like

$$u^{(k)} = [A]v^{(k-1)} = q_k v^{(k)}$$

Here, $q_k$ is the desired largest eigenvalue and $v^{(k)}$ is the corresponding eigenvector.

**Example 2**

Let $A = \begin{pmatrix} 1.8 & .8 \\ 0.2 & 1.2 \end{pmatrix}$, $v_1 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$ and $X = \begin{pmatrix} -0.5 \\ 1 \end{pmatrix}$. Then A has Eigenvalues 2 and 1, and the eigenspace for $\lambda_1 = 2$ is the line through 0 and $v_1$. For k=0, …, 8, compute $A^k x$ and construct the line through 0 and $A^k x$. What happens as k increases?

**Solution**

The first three calculations are

$$\boldsymbol{Ax} = \begin{bmatrix} 1.8 & 0.8 \\ 0.2 & 1.2 \end{bmatrix}\begin{bmatrix} -0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.1 \\ 1.1 \end{bmatrix}$$

$$\boldsymbol{A^2x} = \boldsymbol{A(Ax)} = \begin{bmatrix} 1.8 & 0.8 \\ 0.2 & 1.2 \end{bmatrix}\begin{bmatrix} -0.1 \\ 1.1 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 1.3 \end{bmatrix}$$

$$\boldsymbol{A^3x} = \boldsymbol{A(A^2x)} = \begin{bmatrix} 1.8 & 0.8 \\ 0.2 & 1.2 \end{bmatrix}\begin{bmatrix} 0.7 \\ 1.3 \end{bmatrix} = \begin{bmatrix} 2.3 \\ 1.7 \end{bmatrix}$$

Analogous calculations complete Table 1.

**Table 1** Iterates of a Vector

| $k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| $\boldsymbol{A^kx}$ | $\begin{bmatrix} -.5 \\ 1 \end{bmatrix}$ | $\begin{bmatrix} -0.1 \\ 1.1 \end{bmatrix}$ | $\begin{bmatrix} 0.7 \\ 1.3 \end{bmatrix}$ | $\begin{bmatrix} 2.3 \\ 1.7 \end{bmatrix}$ | $\begin{bmatrix} 5.5 \\ 2.5 \end{bmatrix}$ | $\begin{bmatrix} 11.9 \\ 4.1 \end{bmatrix}$ | $\begin{bmatrix} 24.7 \\ 7.3 \end{bmatrix}$ | $\begin{bmatrix} 50.3 \\ 13.7 \end{bmatrix}$ | $\begin{bmatrix} 101.5 \\ 26.5 \end{bmatrix}$ |

The vectors $x, Ax, A^2x, ..., A^kx$ are shown in Fig. 1. The other vectors are growing too long to display. However, line segments are drawn showing the directions of those vectors. In fact, the directions of the vectors are what we really want to see, not the vectors themselves. The lines seem to be approaching the line representing the eigenspaces spanned by $\boldsymbol{v_1}$. More precisely, the angle between the line (subspace) determined by $A^kx$ and the line (eigenspaces) determined by $v_1$ goes to zero as $k \to \infty$.

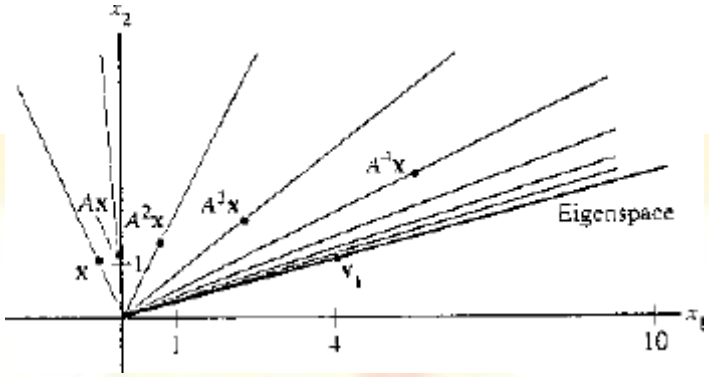



**Figure 1** Directions determined by $\boldsymbol{x}, \boldsymbol{Ax}, \boldsymbol{A^2x}, \ldots, \boldsymbol{A^7x}$

The vectors $(\lambda_1)^{-k} \boldsymbol{A^k x}$ in (3) are scaled to make them converge to $c_1 v_1$, provided $c_1 \neq 0$. We cannot scale $A^k x$ in this way because we do not know $\lambda_1$ but we can scale each $A^k x$ to make its largest entry 1. It turns out that the resulting sequence $\{x_k\}$ will converge to a multiple of $v_1$ whose largest entry is 1. Figure 2 shows the scaled sequence for Example 1. The eigenvalue $\lambda_1$ can be estimated from the sequence $\{x_k\}$ too. When $x_k$ is close to an eigenvector for $\lambda_1$ the vector $Ax_k$ is close to $\lambda_1 \boldsymbol{x_k}$ with each entry in $Ax_k$ approximately $\lambda_1$ times the corresponding entry in $x_k$. Because the largest entry in $x_k$ is 1 the largest entry in $Ax_k$ is close to $\lambda_1$.

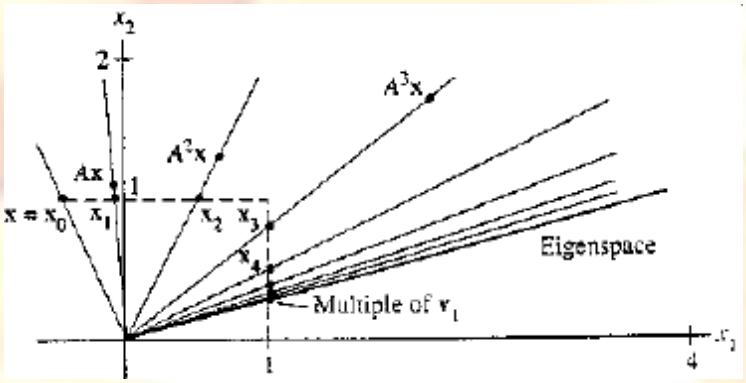



**Figure 2** Scaled multiples of $x, Ax, A^2x, \ldots, A^7x$.

**Steps for finding the eigenvalue and the eigenvector**

1. Select an initial vector $x_0$ whose largest entry is 1.
2. For k=0,1,….
   a. Compute $Ax_k$
   b. Let $\mu_k$ be an entry in $Ax_k$ whose absolute value is as large as possible.
   c. Compute $x_{k+1} = \left(\frac{1}{\mu_k}\right) Ax_k$

3. For almost all choices of $x_0$, the sequence $\{\mu_k\}$ approaches the dominant eigenvalue, and the sequence $\{x_k\}$ approaches a dominant eigenvector.

**Example 3**

Apply the power method to $A = \begin{pmatrix} 6 & 5 \\ 1 & 2 \end{pmatrix}$ with $x_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Stop when $k = 5$, and estimate the dominant eigenvalue and a corresponding eigenvector of A.

**Solution**

To begin, we calculate $Ax_0$ and identify the largest entry $\mu_0$ in $Ax_0$.

$$\boldsymbol{Ax_0} = \begin{bmatrix} 6 & 5 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix},\ \mu_0 = 5$$

Scale $Ax_0$ by $1/\mu_0$ to get $x_1$, compute $Ax_1$ and identify the largest entry in $Ax_1$

$$\boldsymbol{x_1} = \frac{1}{\mu_0}\boldsymbol{Ax_0} = \frac{1}{5}\begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ .4 \end{bmatrix}$$

$$\boldsymbol{Ax_1} = \begin{bmatrix} 6 & 5 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ .4 \end{bmatrix} = \begin{bmatrix} 8 \\ 1.8 \end{bmatrix},\ \mu_1 = 8$$

Scale $Ax_1$ by $1/\mu_1$ to get $\boldsymbol{x_2}$, compute $Ax_2$, and identify the largest entry in $Ax_2$.

$$\boldsymbol{x_2} = \frac{1}{\mu_1}\boldsymbol{Ax_1} = \frac{1}{8}\begin{bmatrix} 8 \\ 1.8 \end{bmatrix} = \begin{bmatrix} 1 \\ .225 \end{bmatrix}$$

$$\boldsymbol{Ax_2} = \begin{bmatrix} 6 & 5 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ .225 \end{bmatrix} = \begin{bmatrix} 7.125 \\ 1.450 \end{bmatrix},\ \mu_2 = 7.125$$

Scale $Ax_2$ by $1/\mu_2$ to get $x_3$, and so on. The results of MATLAB calculations for the first five iterations are arranged in the Table 2.

The evidence from the Table 2 strongly suggests that $\{x_k\}$ approaches (1, .2) and $\{\mu_k\}$ approaches 7. If so, then (1, .2) is an eigenvector and 7 is the dominant eigenvalue. This is easily verified by computing

$$\boldsymbol{A}\begin{bmatrix} 1 \\ .2 \end{bmatrix} = \begin{bmatrix} 6 & 5 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ .2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1.4 \end{bmatrix} = 7\begin{bmatrix} 1 \\ .2 \end{bmatrix}$$

**TABLE 2** The Power Method for Example 2

| $k$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $x_k$ | $\begin{bmatrix}0\\1\end{bmatrix}$ | $\begin{bmatrix}1\\.4\end{bmatrix}$ | $\begin{bmatrix}1\\.225\end{bmatrix}$ | $\begin{bmatrix}1\\.2035\end{bmatrix}$ | $\begin{bmatrix}1\\.2005\end{bmatrix}$ | $\begin{bmatrix}1\\.20007\end{bmatrix}$ |
| $Ax_k$ | $\begin{bmatrix}5\\2\end{bmatrix}$ | $\begin{bmatrix}8\\1.8\end{bmatrix}$ | $\begin{bmatrix}7.125\\1.450\end{bmatrix}$ | $\begin{bmatrix}7.0175\\1.4070\end{bmatrix}$ | $\begin{bmatrix}7.0025\\1.4010\end{bmatrix}$ | $\begin{bmatrix}7.00036\\1.40014\end{bmatrix}$ |
| $\mu_k$ | 5 | 8 | 7.125 | 7.0175 | 7.0025 | 7.00036 |

**Example 4**

Find the first three iterations of the power method applied on the following matrices

$$\begin{bmatrix}1 & 1 & 0\\2 & 4 & 2\\0 & 1 & 2\end{bmatrix} \quad use\ x_0 = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

**Solution**

**1st Iteration**

$$u_1 = Ax_0 = \begin{bmatrix}1 & 1 & 0\\2 & 4 & 2\\0 & 1 & 2\end{bmatrix}\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}0+0+0\\0+0+2\\0+0+2\end{bmatrix} = \begin{bmatrix}0\\2\\2\end{bmatrix}$$

Now, we normalize the resultant vector to get

$$u_1 = \begin{bmatrix}0\\2\\2\end{bmatrix} = 2\begin{bmatrix}0\\1\\1\end{bmatrix} = q_1 x_1$$

**2nd Iteration**

$$u_2 = Ax_1 = \begin{bmatrix}1 & 1 & 0\\2 & 4 & 2\\0 & 1 & 2\end{bmatrix}\begin{bmatrix}0\\1\\1\end{bmatrix} = \begin{bmatrix}0+1+0\\0+4+2\\0+1+2\end{bmatrix} = \begin{bmatrix}1\\6\\3\end{bmatrix}$$

Now, we normalize the resultant vector to get

$$u_2 = \begin{bmatrix}1\\6\\3\end{bmatrix} = 6\begin{bmatrix}\frac{1}{6}\\1\\\frac{1}{2}\end{bmatrix} = q_2 x_2$$

**3rd Iteration**

$$u_3 = Ax_2 = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 4 & 2 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{6} \\ 1 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{6}+1+0 \\ \frac{1}{3}+4+1 \\ 0+1+1 \end{bmatrix} = \begin{bmatrix} \frac{7}{6} \\ \frac{16}{3} \\ 2 \end{bmatrix}$$

Now, we normalize the resultant vector to get

$$u_3 = \begin{bmatrix} \frac{7}{6} \\ \frac{16}{3} \\ 2 \end{bmatrix} = \frac{16}{3} \begin{bmatrix} \frac{7}{32} \\ 1 \\ \frac{3}{8} \end{bmatrix} = q_3 x_3$$

Hence the largest eigenvalue after 3 iterations is $\frac{16}{3}$.

The corresponding eigenvector is $\begin{bmatrix} \frac{7}{32} \\ 1 \\ \frac{3}{8} \end{bmatrix}$

**The inverse power method**

1. Select an initial estimate sufficiently close to $\lambda$.
2. Select an initial vector $x_0$ whose large entry is 1.
3. For k=0, 1, 2,…

   Solve $(A-\alpha I)y_k = x_k$.

   Let $\mu_k$ be an entry in $y_k$ whose absolute value is as large as possible.

   Compute $v_k = \alpha + (\frac{1}{\mu_k})$.

   Compute $x_{k+1} = (\frac{1}{\mu_k}) y_k$

4. For almost all the choice of $x_0$, the sequence $\{v_k\}$ approaches the eigenvalue $\lambda$ of A, and the sequence $\{x_k\}$ approaches a corresponding eigenvector.

**Example 5**

It is not uncommon in some applications to need to know the smallest eigenvalue of a matrix $A$ and to have at hand rough estimates of the eigenvalues. Suppose 21, 3.3, and 1.9 are estimates for the eigenvalues of the matrix $A$ below. Find the smallest eigenvalue, accurate to six decimal places.

$$\boldsymbol{A} = \begin{bmatrix} 10 & -8 & -4 \\ -8 & 13 & 4 \\ -4 & 5 & 4 \end{bmatrix}$$

**Solution**

The two smallest eigenvalues seem close together, so we use the inverse power method for $A-1.9I$. Results of a MATLAB calculation are shown in Table 3. Here $x_0$ was chosen arbitrarily, $y_k = (A-1.9I)^{-1}x_k$, $\mu_k$ is the largest entry in $y_k$, $\boldsymbol{v_k} = 1.9 + 1/\mu_k$, and $\boldsymbol{x_{k+1}} = (1/\mu_k)\boldsymbol{y_k}$.

**Table 3:** The Inverse Power Method

| $K$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $x_0$ | $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ | $\begin{bmatrix} 0.5736 \\ 0.0646 \\ 1 \end{bmatrix}$ | $\begin{bmatrix} 0.5054 \\ 0.0045 \\ 1 \end{bmatrix}$ | $\begin{bmatrix} 0.5004 \\ 0.0003 \\ 1 \end{bmatrix}$ | $\begin{bmatrix} 0.50003 \\ 0.00002 \\ 1 \end{bmatrix}$ |
| $y_k$ | $\begin{bmatrix} 4.45 \\ 0.50 \\ 7.76 \end{bmatrix}$ | $\begin{bmatrix} 5.0131 \\ 0.0442 \\ 9.9197 \end{bmatrix}$ | $\begin{bmatrix} 5.0012 \\ 0.0031 \\ 9.9949 \end{bmatrix}$ | $\begin{bmatrix} 5.0001 \\ 0.0002 \\ 9.9996 \end{bmatrix}$ | $\begin{bmatrix} 5.000006 \\ 0.000015 \\ 9.999975 \end{bmatrix}$ |
| $\mu_k$ | 7.76 | 9.9197 | 9.9949 | 9.9996 | 9.999975 |
| $v_k$ | 2.03 | 2.0008 | 2.00005 | 2.000004 | 2.0000002 |

Therefore, we can say that eigenvalue is 2 from the matrix

$$\boldsymbol{A} = \begin{bmatrix} 10 & -8 & -4 \\ -8 & 13 & 4 \\ -4 & 5 & 4 \end{bmatrix}$$

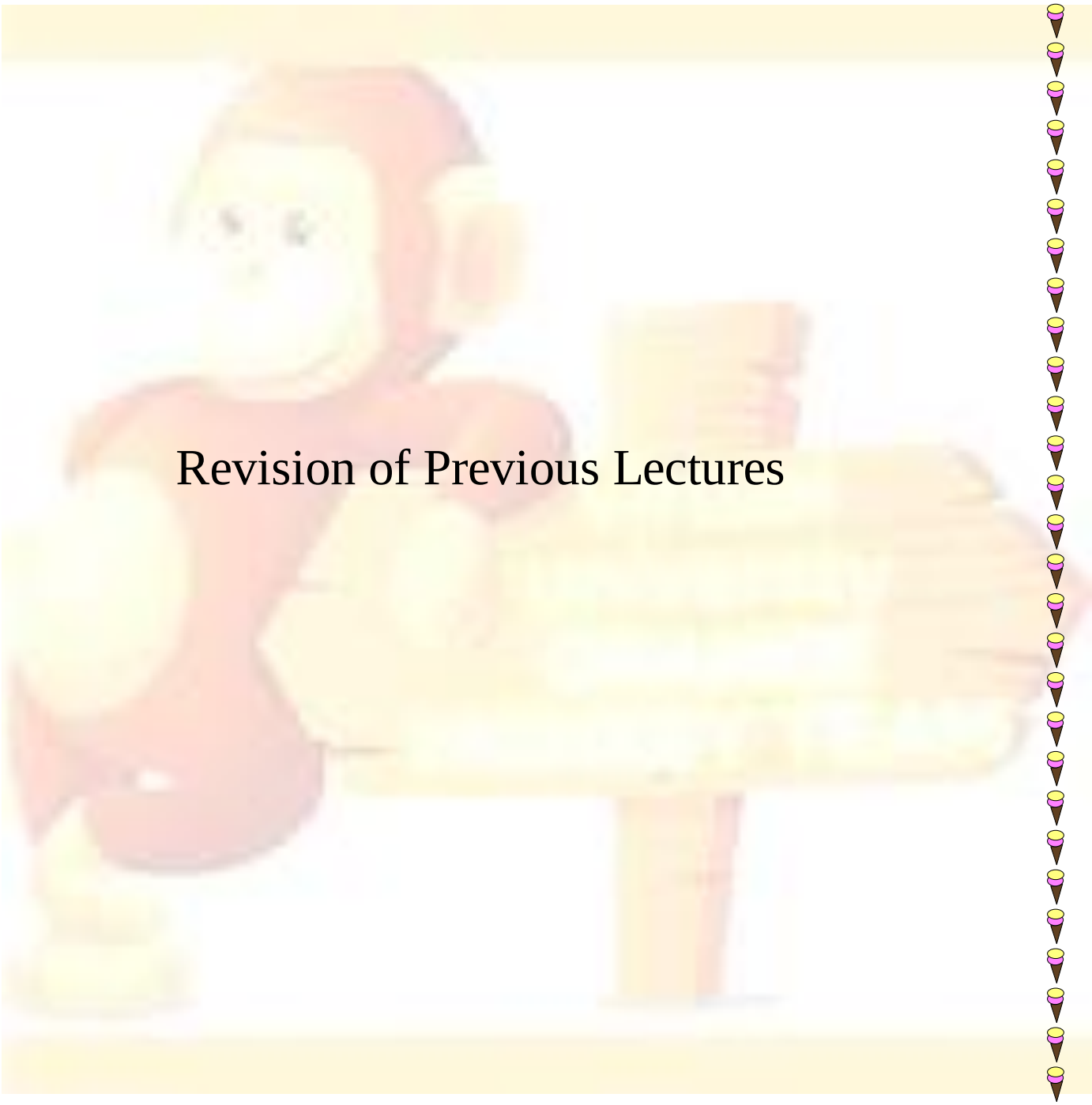
**Exercise**

How can you tell that if a given vector $X$ is a good approximation to an eigenvector of a matrix $A$: if it is, how would you estimate the corresponding eigenvalue? Experiment with

$$A = \begin{bmatrix} 5 & 8 & 4 \\ 8 & 3 & -1 \\ 4 & -1 & 2 \end{bmatrix} \quad \text{And} \quad X = \begin{bmatrix} 1.0 \\ -4.3 \\ 8.1 \end{bmatrix}$$

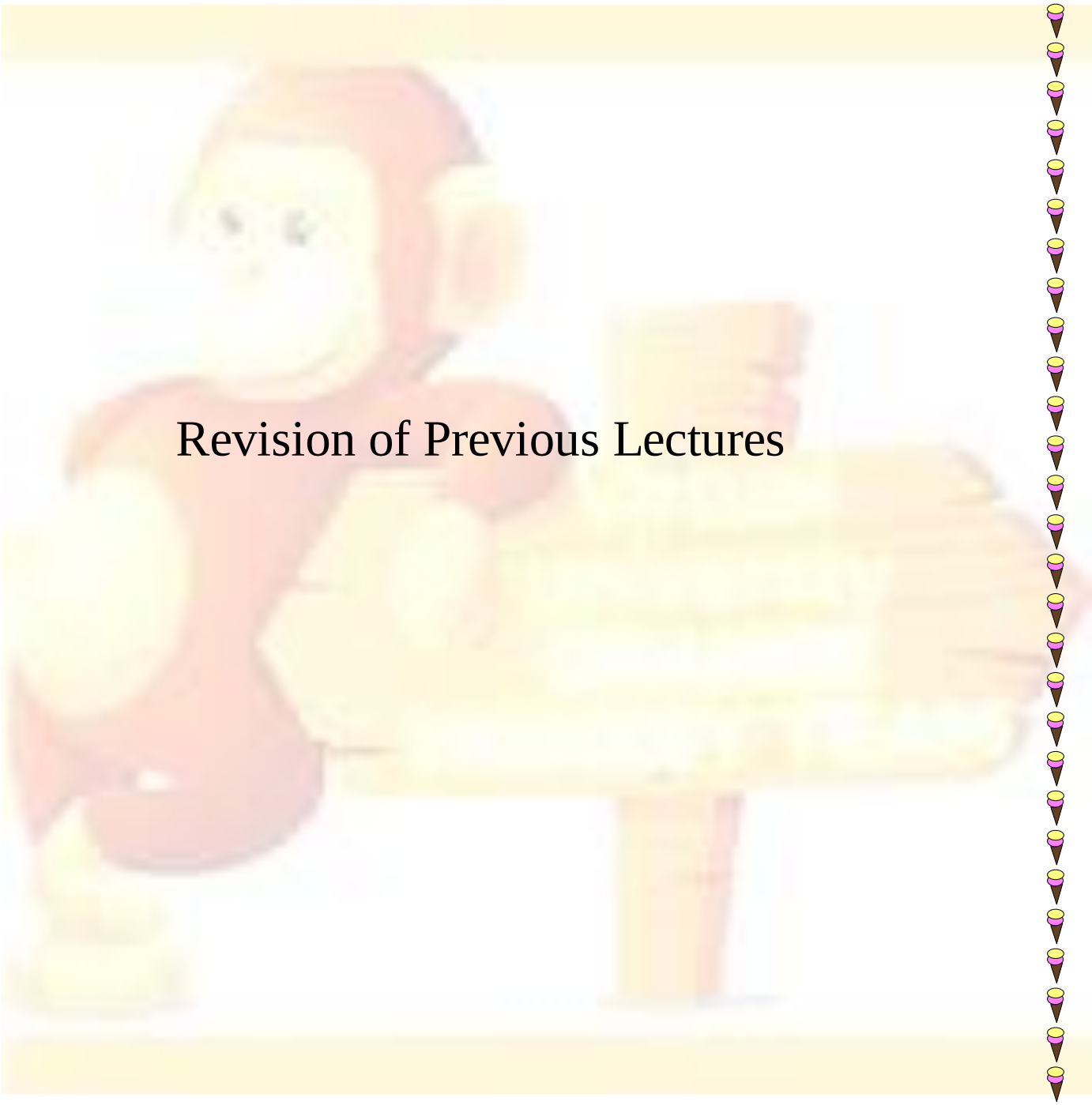
**Lecture 36**

**Revision**



**Lecture 37**

**Revision**



# Lecture 38

## Inner Product

If $u$ and $v$ are vectors in $R^n$, then we regard $u$ and $v$ as $n \times 1$ matrices. The transpose $u^t$ is a $1 \times n$ matrix, and the matrix product $u^t v$ is a $1 \times 1$ matrix which we write as a single real number (a scalar) without brackets.

The number $u^t v$ is called the inner product of $u$ and $v$. And often is written as $u.v$

This inner product is also referred to as a dot product.

If $u = \begin{bmatrix} u_1 \\ u_2 \\ . \\ . \\ . \\ u_n \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ v_2 \\ . \\ . \\ . \\ v_n \end{bmatrix}$

Then the inner product of $u$ and $v$ is

$$\begin{bmatrix} u_1 & u_2 & . & . & . & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ . \\ . \\ . \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \ldots + u_n v_n$$

**Example 1**

Compute $u.v$ and $v.u$ when $u = \begin{bmatrix} 2 \\ -5 \\ -1 \end{bmatrix}$ and $v = \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix}$.

**Solution**

$$u = \begin{bmatrix} 2 \\ -5 \\ -1 \end{bmatrix} \text{ and } v = \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix}$$

$$u^t = \begin{bmatrix} 2 & -5 & -1 \end{bmatrix}$$

$$u.v = u^t v = \begin{bmatrix} 2 & -5 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix} = 2(3) + (-5)(2) + (-1)(-3)$$

$$= 6 - 10 + 3 = -1$$

$$v^t = \begin{bmatrix} 3 & 2 & -3 \end{bmatrix}$$

$$v.u = v^t u = \begin{bmatrix} 3 & 2 & -3 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \\ -1 \end{bmatrix} = 3(2) + (2)(-5) + (-3)(-1)$$

$$= 6 - 10 + 3 = -1$$

**Theorem**

Let $u$, $v$ and $w$ be vectors in $R^n$, and let $c$ be a scalar. Then

a. $u.v = v.u$
b. $(u+v).w = u.w + v.w$
c. $(cu).v = c(u.v) = u.(cv)$
d. $u.u \geq 0$ and $u.u = 0$ if and only if $u = 0$

**Observation**

$$(c_1u_1 + c_2u_2 + .... c_pu_p).w$$
$$= c_1(u_1.w) + c_2(u_2.w) + ... c_p(u_p.w)$$

**Length or Norm**

The length or Norm of $v$ is the nonnegative scalar $\|v\|$ defined by

$$\|v\| = \sqrt{v.v} = \sqrt{v_1^2 + v_1^2 + ... + v_n^2}$$

$$\|v\|^2 = v.v$$

Note: For any scalar $c$, $\|cv\| = |c|\|v\|$

**Unit vector**

A vector whose length is 1 is called a unit vector. If we divide a non-zero vector $v$ by its length $\|v\|$, we obtain a unit vector $u$ as

$$u = \frac{v}{\|v\|}$$

The length of $u$ is $\|u\| = \frac{1}{\|v\|}\|v\| = 1$

**Definition**

The process of creating the unit vector $u$ from $v$ is sometimes called normalizing $v$, and we say that $u$ is in the same direction as $v$. In this case "$u$" is called the normalized vector.

**Example 2**

Let $v = (1, 2, 2, 0)$ in $R^4$. Find a unit vector $u$ in the same direction as $v$.

**Solution**

The length of $v$ is given by

$$\|v\| = \sqrt{v.v} = \sqrt{v_1^2 + v_2^2 + v_3^2 + v_4^2}$$

So,

$$\|v\| = \sqrt{1^2 + 2^2 + 2^2 + 0^2} = \sqrt{1+4+4+0} = \sqrt{9} = 3$$

The unit vector $u$ in the direction of $v$ is given as

$$u = \frac{1}{\|v\|} v = \frac{1}{3}\begin{bmatrix} 1 \\ 2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix}$$

To check that $\|u\| = 1$

$$\|u\| = \sqrt{u.u} = \sqrt{(\frac{1}{3})^2 + (\frac{-2}{3})^2 + (\frac{2}{3})^2 + (0)^2} = \sqrt{\frac{1}{9} + \frac{4}{9} + \frac{4}{9} + 0} = 1$$

**Example 3**

Let W be the subspace of $R^2$ spanned by $X = (\frac{2}{3}, 1)$ . Find a unit vector Z that is a basis for W.

**Solution**

***W*** consists of all multiples of ***x***, as in Fig. 2(a). Any nonzero vector in ***W*** is a basis for ***W***. To simplify the calculation, ***x*** is scaled to eliminate fractions. That is, multiply ***x*** by 3 to get

$$\boldsymbol{y} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Now compute $\|\boldsymbol{y}\|^2 = 2^2 + 3^2 = 13, \|\boldsymbol{y}\| = \sqrt{13},$ and normalize ***y*** to get

$$\boldsymbol{z} = \frac{1}{\sqrt{13}}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{13} \\ 3/\sqrt{13} \end{bmatrix}$$

See Fig. 2(b). Another unit vector is $(-2/\sqrt{13}, -3/\sqrt{13})$.

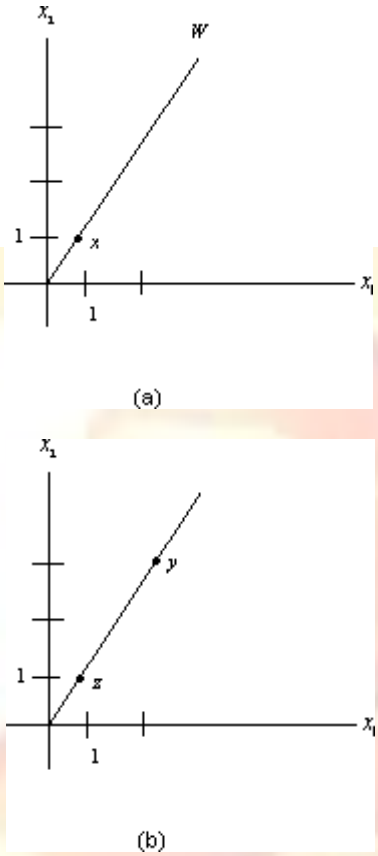



**Figure 2** Normalizing a vector to produce a unit vector.

**Definition**

For $u$ and $v$ vectors in $R^n$, the distance between $u$ and $v$, written as dist $(u, v)$, is the length of the vector $u - v$. That is

$$dist(u, v) = \|u - v\|$$

**Example 4**

Compute the distance between the vectors $u = (7, 1)$ and $v = (3, 2)$

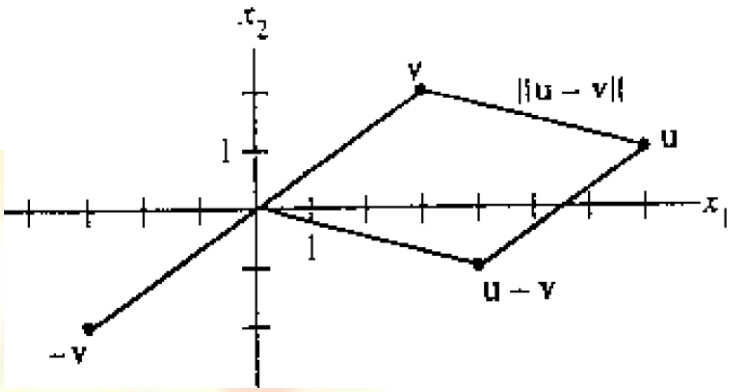
**Solution**

$$u - v = \begin{bmatrix} 7 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 7-3 \\ 1-2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$\text{dist}(u, v) = \|u - v\| = \sqrt{(4)^2 + (-1)^2} = \sqrt{16+1} = \sqrt{17}$$

**Law of Parallelogram of vectors**

The vectors, $u, v$ and $u - v$ are shown in the fig. below. When the vector $u - v$ is added to $v$, the result is $u$. Notice that the parallelogram in the fig. below shows that the distance from $u$ to $v$ is the same as the distance of $u - v$ to $o$.




**Example 5**

If $u=(u_1,u_2,u_3)$ and $v=(v_1,v_2,v_3)$, then

$$dist(u,v)=\|u-v\|=\sqrt{(u-v).(u-v)}$$

$$=\sqrt{(u_1-v_1)^2+(u_2-v_2)^2+(u_3-v_3)^2}$$

**Definition**

Two vectors in $u$ and $v$ in $R^n$ are orthogonal (to each other) if $u.v=0$

**Note**

The zero vector is orthogonal to every vector in $R^n$ because $0^t.v=0$ for all $v$ in $R^n$.

**The Pythagorean Theorem**

Two vectors $u$ and $v$ are orthogonal if and only if $\|u+v\|^2=\|u\|^2+\|v\|^2$

**Orthogonal Complements**

The set of all vectors z that are orthogonal to w in W is called the orthogonal complement of W and is denoted by $w^{\perp}$

**Example 6**

Let $\boldsymbol{W}$ be a plane through the origin in $\boldsymbol{R^3}$, and let $\boldsymbol{L}$ be the line through the origin and perpendicular to $\boldsymbol{W}$. If $\boldsymbol{z}$ and $\boldsymbol{w}$ are nonzero, $\boldsymbol{z}$ is on $\boldsymbol{L}$, and $\boldsymbol{w}$ is in $\boldsymbol{W}$, then the line segment from $\boldsymbol{0}$ to $\boldsymbol{z}$ is perpendicular to the line segment from $\boldsymbol{0}$ to $\boldsymbol{w}$; that is, $\boldsymbol{z}\cdot\boldsymbol{w}=0$. So each vector on $\boldsymbol{L}$ is orthogonal to every $\boldsymbol{w}$ in $\boldsymbol{W}$. In fact, $\boldsymbol{L}$ consists of all vectors that are orthogonal to the $\boldsymbol{w}$'s in $\boldsymbol{W}$, and $\boldsymbol{W}$ consists of all vectors orthogonal to the $\boldsymbol{z}$'s in $\boldsymbol{L}$. That is,

$$\boldsymbol{L}=\boldsymbol{W}^{\perp} \text{ and } \boldsymbol{W}=\boldsymbol{L}^{\perp}$$

**Remarks**

The following two facts about $W^{\perp}$, with $W$ a subspace of $R^n$, are needed later in the segment.

(1) A vector $x$ is in $W^{\perp}$ if and only if $x$ is orthogonal to every vector in a set that spans $W$.

(2) $W^{\perp}$ is a subspace of $R^n$.

**Theorem 3**

Let $A$ be $m$ x $n$ matrix. Then the orthogonal complement of the row space of $A$ is the null space of $A$, and the orthogonal complement of the column space of $A$ is the null space of $A^T$: (Row $A$)$^{\perp}$ = Nul $A$, (Col $A$)$^{\perp}$ = Nul $A^T$

**Proof**

The row-column rule for computing $Ax$ shows that if $x$ is in Nul $A$, then $x$ is orthogonal to each row of $A$ (with the rows treated as vectors in $R^n$). Since the rows of $A$ span the row space, $x$ is orthogonal to Row $A$. Conversely, if $x$ is orthogonal to Row $A$, then $x$ is certainly orthogonal to each row of $A$, and hence $Ax = 0$. This proves the first statement. The second statement follows from the first by replacing $A$ with $A^T$ and using the fact that Col $A$ = Row $A^T$.

**Angles in $R^2$ and $R^3$**

If $u$ and $v$ are nonzero vectors in either $R^2$ or $R^3$, then there is a nice connection between their inner product and the angle $\vartheta$ between the two line segments from the origin to the points identified with $u$ and $v$. The formula is

$$u \cdot v = \|u\|\|v\|\cos\vartheta \qquad (2)$$

To verify this formula for vectors in $R^2$, consider the triangle shown in Fig. 7, with sides of length $\|u\|, \|v\|$, and $\|u-v\|$. By the law of cosines,

$$\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos\vartheta$$

which can be rearranged to produce

$$\|u\|\|v\|\cos\vartheta = \frac{1}{2}\left[\|u\|^2 + \|v\|^2 - \|u-v\|^2\right]$$

$$= \frac{1}{2}\left[u_1^2 + u_2^2 + v_1^2 + v_2^2 - (u_1 - v_1)^2 - (u_2 - v_2)^2\right] = u_1 v_1 + u_2 v_2 = u \cdot v$$

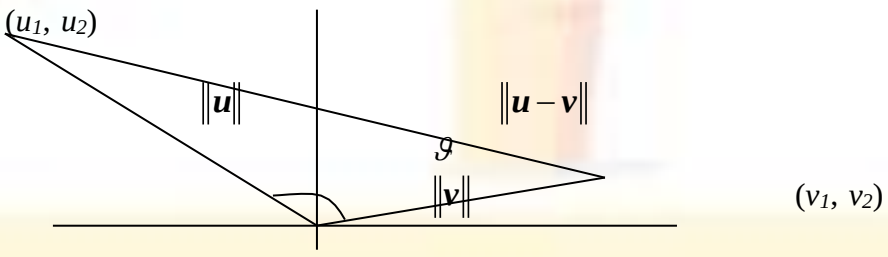



**Figure 7** The angle between two vectors.

**Example 7**

Find the angle between the vectors $u=(1,-1,2),\quad v=(2,1,0)$

**Solution**

$u.v=(1)(2)+(-1)(1)+(2)(0)=2-1+0=1$

And

$\|u\|=\sqrt{(1)^2+(-1)^2+(2)^2}=\sqrt{1+1+4}=\sqrt{6}$

$\|v\|=\sqrt{(2)^2+(1)^2+(0)^2}=\sqrt{4+1}=\sqrt{5}$

Angle between the two vectors is given by

$$\cos\vartheta=\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{\|\boldsymbol{u}\|\|\boldsymbol{v}\|}$$

Putting the values, we get

$$\cos\theta=\frac{1}{\sqrt{\boldsymbol{6}}\sqrt{5}}=\frac{1}{\sqrt{30}}$$

$$\cos\theta=\frac{1}{\sqrt{30}}$$

$$\theta=\cos^{-1}\frac{1}{\sqrt{30}}=79.48^{\circ}$$

**Exercises**

**Q.1**

Compute $u.v$ and $v.u$ when $u=\begin{bmatrix}1\\5\\3\end{bmatrix}$ *and* $v=\begin{bmatrix}-3\\1\\5\end{bmatrix}$

**Q.2**

Let $v=(2,1,0,3)$ in $R^4$. Find a unit vector $u$ in the direction opposite to that of $v$.

**Q.3**

Let W be the subspace of $R^3$ spanned by $X=(\frac{1}{2},\frac{3}{2},\frac{5}{2})$. Find a unit vector Z that is a basis for W.

**Q.4**

Compute the distance between the vectors $u$ = (1, 5, 7) and $v$ = (2, 3, 5).

**Q.5**

Find the angle between the vectors $u=(2,1,3),\quad v=(1,0,2)$.

# Lecture No.39
# Orthogonal and Orthonormal sets

**Objectives**

The objectives of the lecture are to learn about:

- Orthogonal Set.
- Orthogonal Basis.
- Unique representation of a vector as a linear combination of Basis vectors.
- Orthogonal Projection.
- Decomposition of a vector into sum of two vectors.
- Orthonormal Set.
- Orthonormal Basis.
- Some examples to verify the definitions and the statements of the theorems.

**Orthogonal Set**

Let $S=\{u_1,u_2,...,u_p\}$ be the set of non-zero vectors in $R^n$, is said to be an orthogonal set if all vectors in $S$ are mutually orthogonal. That is
$O \notin S$ and $u_i.u_j = o \; \forall i \neq j, i, j = 1,2,..., p.$

**Example**

Show that $S=\{u_1,u_2,u_3\}$ is an orthogonal set. Where

$$u_1=\begin{bmatrix}3\\1\\1\end{bmatrix}, u_2=\begin{bmatrix}-1\\2\\1\end{bmatrix} \text{ and } u_3=\begin{bmatrix}-1\\ \frac{-}{2}\\-2\\7\\ \frac{}{2}\end{bmatrix}.$$

**Solution**

To show that $S$ is orthogonal, we show that each vector in $S$ is orthogonal to other. That is
$u_i.u_j = o \; \forall i \neq j, i, j = 1,2,3.$
For $i=1, j=2$

$$u_1.u_2=\begin{bmatrix}3\\1\\1\end{bmatrix}.\begin{bmatrix}-1\\2\\1\end{bmatrix}$$

$$=-3+2+1=0$$

Which implies $u_1$ is orthogonal to $u_2$.

For $i=1, j=3$

$$u_1 . u_3 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} . \begin{bmatrix} -1 \\ -2 \\ \frac{7}{2} \end{bmatrix}$$

$$= \frac{-3}{2} - 2 + \frac{7}{2}$$

$$= 0.$$

Which implies $u_1$ is orthogonal to $u_3$.

For $i=2, j=3$

$$u_2 . u_3 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} . \begin{bmatrix} -1 \\ -2 \\ \frac{7}{2} \end{bmatrix}$$

$$= \frac{1}{2} - 4 + \frac{7}{2}$$

$$= 0.$$

Which implies $u_2$ is orthogonal to $u_3$.

Thus $S = \{u_1, u_2, u_3\}$ is an orthogonal set.

**Theorem**

Suppose that $S = \{u_1, u_2, ..., u_p\}$ is an orthogonal set of non-zero vectors in $R^n$ and $W = Span\{u_1, u_2, ..., u_p\}$. Then $S$ is linearly independent set and a basis for $W$.

**Proof**

Suppose

$0 = c_1 u_1 + c_2 u_2 + ... + c_p u_p$.

Where $c_1, c_2 ..., c_p$ are scalars.

$u_1 . 0 = u_1 . (c_1 u_1 + c_2 u_2 + ... + c_p u_p)$

$0 = u_1 . (c_1 u_1) + u_1 . (c_2 u_2) + ... + u_1 . (c_p u_p)$

$= c_1 (u_1 . u_1) + c_2 (u_1 . u_2) + ... + c_p (u_1 . u_p)$

$= c_1 (u_1 . u_1)$

Since $S$ is orthogonal set, so, $u_1.u_2 + ... + u_1.u_p = 0$ but $u_1.u_1 > 0.$

Therefore $c_1 = 0$. Similarly, it can be shown that $c_2 = c_3 = ... = c_p = 0$

Therefore by definition $S = \{u_1, u_2, ..., u_p\}$ is linearly independent set and by definition of basis is a basis for subspace $W$.

**Example**

If $S = \{u_1, u_2\}$ is an orthogonal set of non-zero vector in $R^2$ . Show that $S$ is linearly independent set. Where

$u_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $u_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$.

**Solution**

To show that $S = \{u_1, u_2\}$ is linearly independent set, we show that the following vector equation

$c_1u_1 + c_2u_2 = 0.$

has only the trivial solution. i.e. $c_1 = c_2 = 0.$

$$c_1u_1 + c_2u_2 = 0$$

$$c_1\begin{bmatrix} 3 \\ 1 \end{bmatrix} + c_2\begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3c_1 \\ c_1 \end{bmatrix} + \begin{bmatrix} -c_2 \\ 3c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$3c_1 - c_2 = 0$$

$$c_1 + 3c_2 = 0$$

Solve them simultaneously, gives

$c_1 = c_2 = 0.$

Therefore if $S$ is an orthogonal set then it is linearly independent.

**Orthogonal basis**

Let $S = \{u_1, u_2, ..., u_p\}$ be a basis for a subspace $W$ of $R^n$, is also an orthogonal basis if $S$ is an orthogonal set.

**Theorem**

If $S = \{u_1, u_2, ..., u_p\}$ is an orthogonal basis for a subspace $W$ of $R^n$. Then each $y$ in $W$ can be uniquely expressed as a linear combination of $u_1, u_2, ..., u_p$. That is

$y = c_1u_1 + c_2u_2 + ... + c_pu_p.$

Where

$$c_j = \frac{y.u_j}{u_j.u_j}$$

**Proof**

$$\begin{aligned} y.u_1 &= (c_1u_1 + c_2u_2 + \ldots + c_pu_p).u_1 \\ &= (c_1u_1).u_1 + (c_2u_2).u_1 + \ldots + (c_pu_p).u_1 \\ &= c_1(u_1.u_1) + c_2(u_1.u_2) + \ldots + c_p(u_1.u_p) \\ &= c_1(u_1.u_1). \end{aligned}$$

Since $S$ is orthogonal set, so, $u_1.u_2 + \ldots + u_1.u_p = 0$ but $u_1.u_1 > 0$.

Hence

$c_1 = \frac{y.u_1}{u_1.u_1}$ and similarly $c_2 = \frac{y.u_2}{u_2.u_2}, \ldots\ c_p = \frac{y.u_p}{u_p.u_p}$.

**Example**

The set $S = \{u_1, u_2, u_3\}$ as in first example is an orthogonal basis for $R^3$. Express $y$ as a linear combination of the vectors in $S$. Where

$y = [6 \quad 1 \quad -8]^T$

**Solution**

We want to write

$y = c_1u_1 + c_2u_2 + c_3u_3$

Where $c_1, c_2$ and $c_3$ are to be determined.

By the above theorem

$$c_1 = \frac{y.u_1}{u_1.u_1}$$

$$= \frac{\begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix} . \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} . \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}} = \frac{11}{11} = 1$$

$$c_2 = \frac{y.u_2}{u_2.u_2}$$

$$= \frac{\begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}}{\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}} = \frac{-12}{6} = -2$$

And

$$c_3 = \frac{y.u_3}{u_3.u_3}$$

$$= \frac{\begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix} \cdot \begin{bmatrix} -\frac{1}{2} \\ -2 \\ \frac{7}{2} \end{bmatrix}}{\begin{bmatrix} -\frac{1}{2} \\ -2 \\ \frac{7}{2} \end{bmatrix} \cdot \begin{bmatrix} -\frac{1}{2} \\ -2 \\ \frac{7}{2} \end{bmatrix}} = \frac{-33}{33/2} = -2$$

Hence

$y = u_1 - 2u_2 - 2u_3.$

**Example**

The set $S = \{u_1, u_2, u_3\}$ is an orthogonal basis for $R^3$. Write $y$ as a linear combination of the vectors in $S$. Where

$$y = \begin{bmatrix} 3 \\ 7 \\ 4 \end{bmatrix}, u_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \text{ and } u_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

**Solution**

We want to write

$y = c_1u_1 + c_2u_2 + c_3u_3$

Where $c_1, c_2$ and $c_3$ are to be determined.

By the above theorem

$$c_1 = \frac{y.u_1}{u_1.u_1} = \frac{\begin{bmatrix}3\\7\end{bmatrix}.\begin{bmatrix}1\\-1\end{bmatrix}}{\begin{bmatrix}1\\-1\end{bmatrix}.\begin{bmatrix}1\\-1\end{bmatrix}} = \frac{3-7+0}{1+1+0} = -2$$

$$c_2 = \frac{y.u_2}{u_2.u_2} = \frac{\begin{bmatrix}3\\7\end{bmatrix}.\begin{bmatrix}1\\1\end{bmatrix}}{\begin{bmatrix}1\\1\end{bmatrix}.\begin{bmatrix}1\\1\end{bmatrix}} = \frac{3+7+0}{1+1} = 5$$

And

$$c_3 = \frac{y.u_3}{u_3.u_3} = \frac{\begin{bmatrix}3\\7\end{bmatrix}.\begin{bmatrix}0\\0\end{bmatrix}}{\begin{bmatrix}0\\0\end{bmatrix}.\begin{bmatrix}0\\0\end{bmatrix}} = \frac{4}{1} = 4$$

Hence

$y = -2u_1 + 5u_2 + 4u_3.$

**Exercise**

The set $S = \{u_1, u_2, u_3\}$ is an orthogonal basis for $R^3$. Write $y$ as a linear combination of the vectors in $S$. where

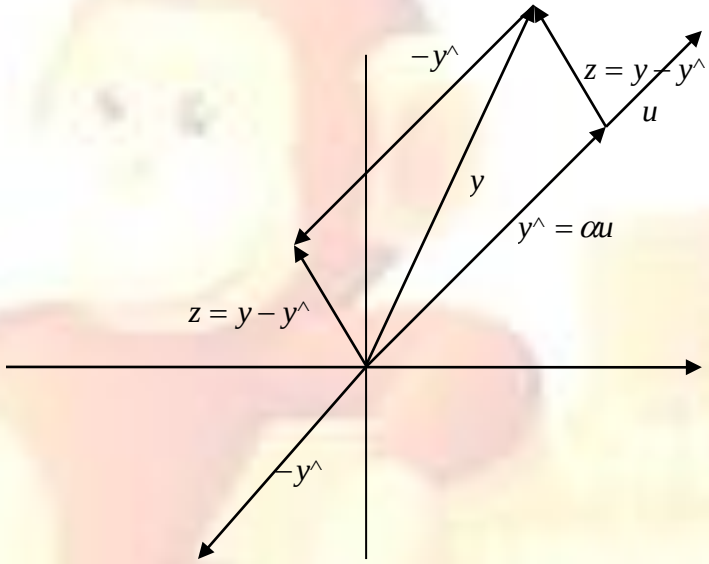
$$y = \begin{bmatrix}2\\1\\3\end{bmatrix}, u_1 = \begin{bmatrix}2\\-1\\0\end{bmatrix}, u_2 = \begin{bmatrix}1\\5\\2\\5\\-1\end{bmatrix} \text{ and } u_3 = \begin{bmatrix}8\\3\\16\\3\\8\\3\end{bmatrix}$$

**An Orthogonal Projection (Decomposition of a vector into the sum of two vectors**

Decomposition of a non- zero vector $y \in R^n$ into the sum of two vectors in such a way, one is multiple of $u \in R^n$ and the other orthogonal to $u$ **.** That is

$$y = y^{\wedge} + z$$

Where $y^{\wedge} = \alpha u$ for some scalar $\alpha$ and $z$ is orthogonal to $u$ .




In the above figure a vector $y$ is decomposed into two vectors $z = y - y^{\wedge}$ and $y^{\wedge} = \alpha u$.
Clearly it can be seen that $z = y - y^{\wedge}$ is orthogonal to $u$ and $y^{\wedge} = \alpha u$ is a multiple of $u$ .
Since $z = y - y^{\wedge}$ is orthogonal to u.
Therefore

$$z.u = 0$$
$$(y - y^{\wedge}).u = 0$$
$$(y - \alpha u).u = 0$$
$$y.u - \alpha(u.u) = 0$$
$$\Rightarrow \alpha = \frac{y.u}{u.u}$$

And

$$z = y - y^{\wedge}$$
$$= y - \frac{y.u}{u.u}u$$

Hence

$y^{\wedge} = \frac{y.u}{u.u}u$, which is an orthogonal projection of $y$ onto $u$.

And

$z = y - y^{\wedge}$

$= y - \frac{y.u}{u.u}u$ is a component of $y$

**Example**

Let $\boldsymbol{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ and $\boldsymbol{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$.

Find the orthogonal projection of **y** onto **u**. Then write **y** as a sum of two orthogonal vectors, one in span {u} and one orthogonal to **u**.

**Solution**

Compute

$$\boldsymbol{y} \cdot \boldsymbol{u} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 40$$

$$\boldsymbol{u} \cdot \boldsymbol{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 20$$

The orthogonal projection of $\boldsymbol{y}$ onto $\boldsymbol{u}$ is $\hat{y} = \frac{y \cdot u}{u \cdot u}u = \frac{40}{20}u = 2\begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$ and the component of $\boldsymbol{y}$ orthogonal to $\boldsymbol{u}$ is $y - \hat{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} - \begin{bmatrix} 8 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

The sum of these two vectors is $\boldsymbol{y}$. That is, $\begin{bmatrix} 7 \\ 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$y \qquad \hat{y} \quad (y - \hat{y})$

This decomposition of $\boldsymbol{y}$ is illustrated in Fig. 3. **Note**: If the calculations above are correct, then $\{\hat{y}, y - \hat{y}\}$ will be an orthogonal set. As a check, compute

$$\hat{y} \cdot (y - \hat{y}) = \begin{bmatrix} 8 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = -8 + 8 = 0$$

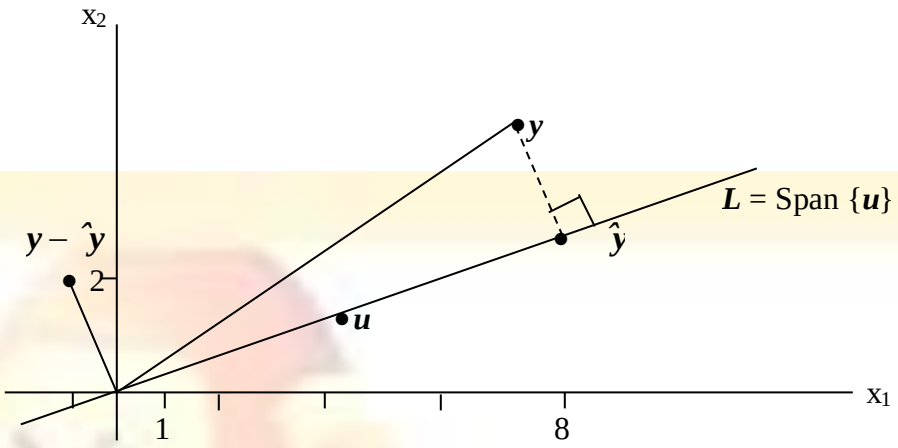
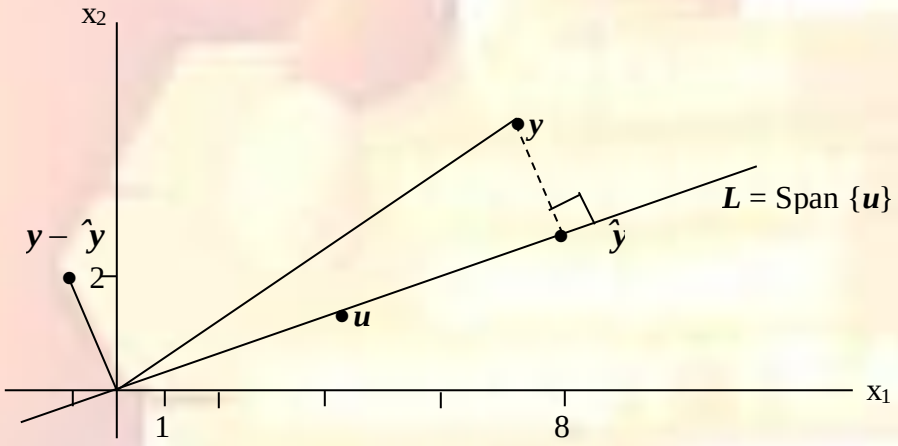



**Figure 3** The orthogonal projection of ***y*** on to a line through the origin.

**Example**

Find the distance in figure below from *y* to *L*.




**Solution**

The distance from y to L is the length of the perpendicular line segment from y to the orthogonal projection $\hat{y}$ .

The length equals the length of $y - \hat{y}$ .

This distance is

$$\|y - \hat{y}\| = \sqrt{(-1)^2 + 2^2} = \sqrt{5}$$

**Example**

Decompose $\boldsymbol{y} = (-3,-4)$ into two vectors $\hat{y}$ and $\boldsymbol{z}$, where $\hat{y}$ is a multiple of $\boldsymbol{u} = (-3, 1)$ and $\boldsymbol{z}$ is orthogonal to $\boldsymbol{u}$. Also prove that $\hat{y}.z = 0$

**Solution**

It is very much clear that $\hat{y}$ is an orthogonal projection of $\boldsymbol{y}$ onto $\boldsymbol{u}$ and it is calculated by applying the following formula

$$\hat{y} = \frac{y.u}{u.u}u$$

$$= \frac{\begin{bmatrix}-3\\-4\end{bmatrix}\cdot\begin{bmatrix}-3\\1\end{bmatrix}}{\begin{bmatrix}-3\\1\end{bmatrix}\cdot\begin{bmatrix}-3\\1\end{bmatrix}}\begin{bmatrix}-3\\1\end{bmatrix} = \frac{9-4}{9+1}\begin{bmatrix}-3\\1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}-3\\1\end{bmatrix} = \begin{bmatrix}-\frac{3}{2}\\\frac{1}{2}\end{bmatrix}$$

$$z = y - \hat{y} = \begin{bmatrix}-3\\-4\end{bmatrix} - \begin{bmatrix}-\frac{3}{2}\\\frac{1}{2}\end{bmatrix} = \begin{bmatrix}-3+3/2\\-4-1/2\end{bmatrix} = \begin{bmatrix}-\frac{3}{2}\\-\frac{9}{2}\end{bmatrix}$$

So,

$$\hat{y} = \begin{bmatrix}-\frac{3}{2}\\\frac{1}{2}\end{bmatrix} \text{ and } z = \begin{bmatrix}-\frac{3}{2}\\-\frac{9}{2}\end{bmatrix}$$

Now

$$y^{\wedge}.z = \begin{bmatrix} -\frac{3}{2} \\ 1 \\ \frac{}{2} \end{bmatrix} \cdot \begin{bmatrix} -\frac{3}{2} \\ _\frac{9}{2} \end{bmatrix}$$

$$= \frac{9}{4} - \frac{9}{4}$$

$$= 0$$

Therefore, $y^{\wedge}$ is orthogonal to ***z***

**Exercise**

Find the orthogonal projection of a vector ***y* = (-3, 2)** onto ***u* = (2, 1).** Also prove that $y = y^{\wedge} + z$, where $y^{\wedge}$ a multiple of ***u*** and **z** is is an orthogonal to ***u.***

**Orthonormal Set**

Let $S = \{u_1, u_2, ..., u_p\}$ be the set of non-zero vectors in $R^n$, is said to be an orthonormal set if $S$ is an orthogonal set of unit vectors.

**Example**

Show that $S = \{u_1, u_2, u_3\}$ is an orthonormal set. Where

$$u_1 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ 0 \\ \frac{-1}{\sqrt{5}} \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \text{ and } u_3 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ 0 \\ \frac{2}{\sqrt{5}} \end{bmatrix}.$$

**Solution**

To show that $S$ is an orthonormal set, we show that it is an orthogonal set of unit vectors.

It can be easily prove that $S$ is an orthogonal set because

$u_i . u_j = 0 \; \forall \; i \neq j, i, j = 1, 2, 3.$

Furthermore

$$u_1 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ 0 \\ \frac{-1}{\sqrt{5}} \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} u_3 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ 0 \\ \frac{2}{\sqrt{5}} \end{bmatrix}.$$

$$u_1.u_1 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ 0 \\ \frac{-1}{\sqrt{5}} \end{bmatrix}. \begin{bmatrix} \frac{2}{\sqrt{5}} \\ 0 \\ \frac{-1}{\sqrt{5}} \end{bmatrix}$$

$$= \frac{4}{5} + 0 + \frac{1}{5}$$

$$= 1$$

$$u_2.u_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}. \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

$$= 0 + 1 + 0$$

$$= 1$$

And

$$u_3.u_3 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ 0 \\ \frac{2}{\sqrt{5}} \end{bmatrix}. \begin{bmatrix} \frac{1}{\sqrt{5}} \\ 0 \\ \frac{2}{\sqrt{5}} \end{bmatrix}$$

$$= \frac{1}{5} + \frac{4}{5}$$

$$= 1$$

Hence

$S = \{u_1, u_2, u_3\}$ is an orthonormal set.

**Orthonormal basis**

Let $S = \{u_1, u_2, ..., u_p\}$ be a basis for a subspace $W$ of $R^n$, is also an orthonormal basis if $S$ is an orthonormal set.

**Example**

Show that $S=\{u_1,u_2,u_3\}$ is an orthonormal basis of $R^3$, where

$$u_1=\begin{bmatrix}\frac{3}{\sqrt{11}}\\ \frac{1}{\sqrt{11}}\\ \frac{1}{\sqrt{11}}\end{bmatrix},\ u_2=\begin{bmatrix}\frac{-1}{\sqrt{6}}\\ \frac{2}{\sqrt{6}}\\ \frac{1}{\sqrt{6}}\end{bmatrix} \text{ and } u_3=\begin{bmatrix}\frac{-1}{\sqrt{66}}\\ \frac{-4}{\sqrt{66}}\\ \frac{7}{\sqrt{66}}\end{bmatrix}.$$

**Solution**

To show that $S=\{u_1,u_2,u_3\}$ is an orthonormal basis, it is sufficient to show that it is an orthogonal set of unit vectors. That is

$u_i.u_j=0\ \forall i\neq j, i,j=1,2,3.$

And

$u_i.u_j=1\forall i=j, i,j=1,2,3.$

Clearly it can be seen that

$u_1.u_2=0,$

$u_1.u_3=0$

And

$u_2.u_3=0.$

Furthermore

$u_1.u_1=1,$

$u_2.u_2=1$

And

$u_3.u_3=1.$

Hence $S$ is an orthonormal basis of $R^3$.

**Theorem**

A $m\times n$ matrix U has orthonormal columns if and only if $U^tU=I$

**Proof**

Keep in mind that in an if and only if statement, one part depends on the other, so, each part is proved separately. That is, we consider one part and then prove the other part with the help of that assumed part.

Before proving both sides of the statements, we have to do some extra work which is necessary for the better understanding.

Let $u_1,u_2,...,u_m$ be the columns of $\boldsymbol{U}$. Then $\boldsymbol{U}$ can be written in matrix form as

$$U=\begin{bmatrix}u_1 & u_2 & u_3 & \dots & u_m\end{bmatrix}$$

Taking transpose, it becomes

$$U^t = \begin{bmatrix} u_1^t \\ u_2^t \\ u_3^t \\ . \\ . \\ . \\ u_m^t \end{bmatrix}$$

$$U^t U = \begin{bmatrix} u_1^t \\ u_2^t \\ u_3^t \\ . \\ . \\ u_m^t \end{bmatrix} \begin{bmatrix} u_1 & u_2 & u_3 & \dots & u_m \end{bmatrix} = \begin{bmatrix} u_1^t u_1 & u_1^t u_2 & u_1^t u_3 & \dots & u_1^t u_m \\ u_2^t u_1 & u_2^t u_2 & u_2^t u_3 & \dots & u_2^t u_m \\ u_3^t u_1 & u_3^t u_2 & u_3^t u_3 & \dots & u_3^t u_m \\ . & . & . & & . \\ . & . & . & & . \\ . & . & . & & . \\ u_m^t u_1 & u_m^t u_2 & u_m^t u_3 & \dots & u_m^t u_m \end{bmatrix}$$

*As* $u.v = v^t u$

*Therefore*

$$U^t U = \begin{bmatrix} u_1.u_1 & u_1.u_2 & u_1.u_3 & \dots & u_1.u_m \\ u_2.u_1 & u_2.u_2 & u_2.u_3 & \dots & u_2.u_m \\ u_3.u_1 & u_3.u_2 & u_3.u_3 & \dots & u_3.u_m \\ . & . & . & & . \\ . & . & . & & . \\ . & . & . & & . \\ u_m.u_1 & u_m.u_2 & u_m.u_3 & \dots & u_m.u_m \end{bmatrix}$$

Now, we come to prove the theorem.

First suppose that $U^t U = I$, and we prove that columns of U are orthonormal.

Since, we assume that

$$U^tU = \begin{bmatrix} u_1.u_1 & u_1.u_2 & u_1.u_3 & \dots & u_1.u_m \\ u_2.u_1 & u_2.u_2 & u_2.u_3 & \dots & u_2.u_m \\ u_3.u_1 & u_3.u_2 & u_3.u_3 & \dots & u_3.u_m \\ . & . & . & & . \\ . & . & . & & . \\ . & . & . & & . \\ u_m.u_1 & u_m.u_2 & u_m.u_3 & \dots & u_m.u_m \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ . & . & . & & . \\ . & . & . & & . \\ . & . & . & & . \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

Clearly, it can be seen that

$u_i.u_j = 0 \; for \; i \neq j \quad i, j = 1, 2, \dots m$

*and*

$u_i.u_j = 1 \; for \; i = j \quad i, j = 1, 2, \dots m$

Therefore, columns of U are orthonormal.

Next suppose that the columns of U are orthonormal and we will show that $U^tU = I$.

Since we assume that columns of U are orthonormal, so, we can write

$u_i.u_j = 0 \; for \; i \neq j \quad i, j = 1, 2, \dots m$

*and*

$u_i.u_j = 1 \; for \; i = j \quad i, j = 1, 2, \dots m$

Hence, $U^tU = \begin{bmatrix} u_1.u_1 & u_1.u_2 & u_1.u_3 & \dots & u_1.u_m \\ u_2.u_1 & u_2.u_2 & u_2.u_3 & \dots & u_2.u_m \\ u_3.u_1 & u_3.u_2 & u_3.u_3 & \dots & u_3.u_m \\ . & . & . & & . \\ . & . & . & & . \\ . & . & . & & . \\ u_m.u_1 & u_m.u_2 & u_m.u_3 & \dots & u_m.u_m \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ . & . & . & & . \\ . & . & . & & . \\ . & . & . & & . \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

That is

$U^t U = I.$

Which is our required result.

**Exercise**

Prove that the following matrices have orthonormal columns using above theorem.

(1) $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

(2) $\begin{bmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{bmatrix}$

(3) $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

**Solution (1)**

**Let**

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$U^t = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

*Then*

$$U^t U = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$U^t U = I$

Therefore, by the above theorem, U has orthonormal columns.

**(2) And (3)** are left for reader.

**Theorem**

Let U be an $m \times n$ matrix with orthonormal columns, and let x and y be in $R^n$. Then

a) $\|Ux\| = \|x\|$

b) $(Ux).(Uy) = x.y$

c) $(Ux).(Uy) = 0$ *iff* $x.y = 0$

**Example**

$$Let\ U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix} \quad and \quad X = \begin{bmatrix} \sqrt{2} \\ 3 \end{bmatrix}$$

Verify that $\|Ux\| = \|x\|$

**Solution**

Notice that U has orthonormal columns and

$$U^T U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 2/3 & -2/3 & 1/3 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$Ux = \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} \sqrt{2} \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$$

$$\|Ux\| = \sqrt{9+1+1} = \sqrt{11}$$

$$\|x\| = \sqrt{2+9} = \sqrt{11}$$

# Lecture No.40
# Orthogonal Decomposition

**Objectives**

The objectives of the lecture are to learn about:

- Orthogonal Decomposition Theorem.
- Best Approximation Theorem.

**Orthogonal Projection**

The orthogonal projection of a point in $R^2$ onto a line through the origin has an important analogue in $R^n$ .

That is given a vector $Y$ and a subspace W in $R^n$, there is a vector $\hat{y}$ in W such that

1) $\hat{y}$ is the unique vector in W for which $y - \hat{y}$ is orthogonal to W, and
2) $\hat{y}$ is the unique vector in W closest to $y$.




We observe that whenever a vector y is written as a linear combination of vectors $u_1, u_2, ..., u_n$ in a basis of $R^n$, the terms in the sum for y can be grouped into two parts so that y can be written as $y = z_1 + z_2$, where $z_1$ is a linear combination of some of the $u_i$'s, and $z_2$ is a linear combination of the rest of the $u_i$'s. This idea is particularly useful when $\{u_1, u_2, ..., u_n\}$ is an orthogonal basis.

**Example 1**

Let $\{u_1, u_2, ..., u_5\}$ be an orthogonal basis for $R^5$ and let $y = c_1 u_1 + c_2 u_2 + ... + c_5 u_5$.

Consider the subspace W= Span{$u_1$, $u_2$} and write y as the sum of a vector $z_1$ in W and a vector $z_2$ in $W^{\perp}$.

**Solution**

Write

$$y = \underbrace{c_1 u_1 + c_2 u_2}_{z_1} + \underbrace{c_3 u_3 + c_4 u_4 + c_5 u_5}_{z_2}$$

where $z_1 = c_1u_1 + c_2u_2$ is in Span of {$\boldsymbol{u_1}$, $\boldsymbol{u_2}$}, and $z_2 = c_3u_3 + c_4u_4 + c_5u_5$ is in Span of {$\boldsymbol{u_3}$, $\boldsymbol{u_4}$, $\boldsymbol{u_5}$}.

To show that $\boldsymbol{z_2}$ is in $\boldsymbol{W}^{\perp}$, it suffices to show that $\boldsymbol{z_2}$ is orthogonal to the vectors in the basis {$\boldsymbol{u_1}$, $\boldsymbol{u_2}$} for $\boldsymbol{W}$. Using properties of the inner product, compute

$$z_2 \cdot u_1 = (c_3u_3 + c_4u_4 + c_5u_5) \cdot u_1 = c_3u_3 \cdot u_1 + c_4u_4 \cdot u_1 + c_5u_5 \cdot u_1 = 0$$

since $\boldsymbol{u_1}$ is orthogonal to $\boldsymbol{u_3}$, $\boldsymbol{u_4}$, and $\boldsymbol{u_5}$, a similar calculation shows that $\boldsymbol{z_2} \cdot \boldsymbol{u_2} = 0$

Thus, $\boldsymbol{z_2}$ is in $\boldsymbol{W}^{\perp}$.

**Orthogonal decomposition theorem**

Let W be a subspace of $R^n$, then each y in $R^n$ can be written uniquely in the form

$$y = \hat{y} + z$$

Where $\hat{y} \in W$ and $z \in W^{\perp}$

Furthermore, if $\{u_1, u_2, ..., u_p\}$ is any orthogonal basis for $W$, then

$$\hat{y} = c_1 u_1 + c_2 u_2 + ... + c_n u_n, \text{ where } c_j = \frac{y.u_j}{u_j.u_j}$$

**Proof**

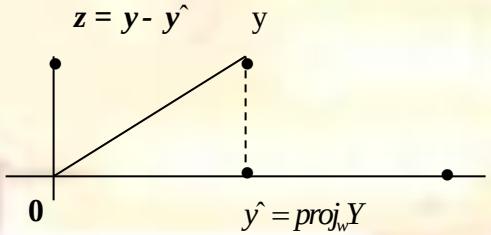



**Fig: Orthogonal projection of y on to W.**

Firstly, we show that $\hat{y} \in W$, $z \in W^{\perp}$. Then we will show that $y = \hat{y} + z$ can be represented in a unique way.

Suppose W is a subspace of $R^n$ and let $\{u_1, u_2, ..., u_p\}$ be an orthogonal basis for W.

As $\hat{y} = c_1 u_1 + c_2 u_2 ... c_p u_p$ where $c_j = \frac{y.u_j}{u_j.u_j}$

Since $\{u_1, u_2, ..., u_p\}$ is the basis for W and $y^\wedge$ is written as a linear combination of these basis vectors. Therefore, by definition of basis $y^\wedge \in W$.

Now, we will show that $z = y - y^\wedge \in W^\perp$. For this it is sufficient to show that $z \perp u_j$ for each $j = 1, 2, ... p$.

Let $u_1 \in W$ be an arbitrary vector.

$$\begin{aligned} z \,.\, u_1 &= (y - y^\wedge).u_1 \\ &= y.u_1 - y^\wedge.u_1 \\ &= y.u_1 - (c_1u_1 + c_2u_2 + ... + c_pu_p).u_1 \\ &= y.u_1 - c_1(u_1.u_1) - c_2(u_2.u_1) - ...c_p(u_p.u_1) \\ &= y.u_1 - c_1(u_1.u_1) \quad \text{where } u_j.u_1 = 0,\ j = 2, 3, ...p \\ &= y.u_1 - \frac{y.u_1}{u_1.u_1}(u_1.u_1) \\ &= y.u_1 - y.u_1 \\ &= 0 \end{aligned}$$

Therefore, $z \perp u_1$.

Since $u_1$ is an arbitrary vector, therefore $z \perp u_j$ for $j = 1, 2, ... p$.

Hence by definition of $W^\perp, z \in W^\perp$

Now, we must show that $y = y^\wedge + z$ is unique by contradiction.

Let $y = y^\wedge + z$ and $y = y_1^\wedge + z_1$, where $y^\wedge, y_1^\wedge \in W$ and $z, z_1 \in W^\perp$,

also $z \neq z_1$ *and* $y^\wedge \neq y_1^\wedge$. Since above representations for y are equal, that is

$$y^\wedge + z = y_1^\wedge + z_1$$

$$\Rightarrow \quad y^\wedge - y_1^\wedge = z_1 - z$$

Let

$$s = y^\wedge - y_1^\wedge$$

Then

$$s = z_1 - z$$

Since W is a subspace, therefore, by closure property

$$s = y^\wedge - y_1^\wedge \in W$$

Furthermore, $W^\perp$ is also a subspace, therefore by closure property

$$s = z_1 - z \in w^\perp$$

Since

$s \in W$ and $s \in W^\perp$. Therefore by definition $s \perp s$

That is **s. s**=0

Therefore

$$s = y^\wedge - y_1^\wedge = 0$$

$$\Rightarrow y^\wedge = y_1^\wedge$$

Also

$z_1 = z$

This shows that representation is unique.

**Example**

Let $u_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, and $y = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

Observe that $\{u_1, u_2\}$ is an orthogonal basis for W=span$\{u_1, u_2\}$, write y as the sum of a vector in W and a vector orthogonal to W.

**Solution**

Since $y^{\wedge} \in W$, therefore $y^{\wedge}$ can be written as following:

$$y^{\wedge} = c_1 u_1 + c_2 u_2$$

$$= \frac{y.u_1}{u_1.u_1} u_1 + \frac{y.u_2}{u_2.u_2} u_2$$

$$= \frac{9}{30}\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{3}{6}\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \frac{9}{30}\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{15}{30}\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} -2 \\ 10 \\ 1 \end{bmatrix}$$

$$y - y^{\wedge} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} = \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$

$$= 7/5 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

Above theorem ensures that $y - y^{\wedge}$ is in $W^{\perp}$.

You can also verify by $(y - y^{\wedge}).u_1 = 0$ and $(y - y^{\wedge}).u_2 = 0$.

The desired decomposition of y is

$$y=\begin{bmatrix}1\\2\\3\end{bmatrix}=\begin{bmatrix}-2/5\\2\\1/5\end{bmatrix}+\begin{bmatrix}7/5\\0\\14/5\end{bmatrix}$$

**Example**

Let W=span$\{u_1,u_2\}$, where $u_1=\begin{bmatrix}1\\-3\\2\end{bmatrix}$ and $u_2=\begin{bmatrix}4\\2\\1\end{bmatrix}$

Decompose $y=\begin{bmatrix}2\\-2\\5\end{bmatrix}$ into two vectors; one in $W$ and one in $W^{\perp}$. Also verify that these two vectors are orthogonal.

**Solution**

Let $y^{\wedge}\in W$ and $z=y-y^{\wedge}\in W^{\perp}$.

Since $y^{\wedge}\in W$, therefore $y^{\wedge}$ can be written as following:

$$y^{\wedge}=c_1u_1+c_2u_2$$

$$=\frac{y.u_1}{u_1.u_1}u_1+\frac{y.u_2}{u_2.u_2}u_2$$

$$=\frac{9}{7}\begin{bmatrix}1\\-3\\2\end{bmatrix}+\frac{3}{7}\begin{bmatrix}4\\2\\1\end{bmatrix}$$

$$y^{\wedge}=\begin{bmatrix}3\\-3\\3\end{bmatrix}$$

Now

$$z=y-y^{\wedge}=\begin{bmatrix}2\\-2\\5\end{bmatrix}-\begin{bmatrix}3\\-3\\3\end{bmatrix}$$

$$z=\begin{bmatrix}-1\\1\\2\end{bmatrix}$$

Now we show that $z\perp y^{\wedge}$, i.e. $z.y^{\wedge}=0$

$$z.y^{\wedge} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} . \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix}$$

$$= 0$$

Therefore $z \perp y^{\wedge}$.

**Exercise**

Let W=span$\{u_1, u_2\}$, where $u_1 = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$ and $u_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$

Write $y = \begin{bmatrix} 6 \\ -8 \\ 12 \end{bmatrix}$ as a sum of two vectors; one in $W$ and one in $W^{\perp}$. Also verify that these two vectors are orthogonal.

**Best Approximation Theorem**

Let W is a finite dimensional subspace of an inner product space V and y is any vector in V. The best approximation to y from W is then $\Pr oj_w^y$, i.e for every w (that is not $\Pr oj_w^y$) in W, we have

$\|y - \Pr oj_w^y\| < \|y - w\|$.

**Example**

Let W=span$\{u_1, u_2\}$, where $u_1 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$, $u_2 = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$ and $y = \begin{bmatrix} 2 \\ -2 \\ 5 \end{bmatrix}$. Then using above theorem, find the distance from y to W.

**Solution**

Using above theorem the distance from y to W is calculated using the following formula

$\|y - \Pr oj_w^y\| = \|y - y^{\wedge}\|$

Since, we have already calculated

$$y - y^{\wedge} = \begin{bmatrix} 2 \\ -2 \\ 5 \end{bmatrix} - \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

So $\|y - y^{\wedge}\| = \sqrt{6}$

**Example**

The distance from a point y in $R^n$ to a subspace W is defined as the distance from y to the nearest point in W.

Find the distance from $\boldsymbol{y}$ to $\boldsymbol{W} = \text{span}\{u_1, u_2\}$, where

$$y = \begin{bmatrix} -1 \\ -5 \\ 10 \end{bmatrix}, u_1 = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

By the Best Approximation Theorem, the distance from $\boldsymbol{y}$ to $\boldsymbol{W}$ is $\|y - \hat{y}\|$, where $\hat{y} = \text{proj}_W \boldsymbol{y}$. Since, $\{\boldsymbol{u_1}, \boldsymbol{u_2}\}$ is an orthogonal basis for $\boldsymbol{W}$, we have

$$\hat{y} = \frac{15}{30} u_1 + \frac{-21}{6} u_2 = \frac{1}{2}\begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix} - \frac{7}{2}\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -8 \\ 4 \end{bmatrix}$$

$$\boldsymbol{y} - \hat{\boldsymbol{y}} = \begin{bmatrix} -1 \\ -5 \\ 10 \end{bmatrix} - \begin{bmatrix} -1 \\ -8 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix}$$

$$\|y - \hat{y}\|^2 = 3^2 + 6^2 = 45$$

The distance from $\boldsymbol{y}$ to $\boldsymbol{W}$ is $\sqrt{45} = 3\sqrt{5}$.

**Theorem**

If $\{u_1, u_2, ..., u_p\}$ is an orthonormal basis for a subspace W of $R^n$, then

$$\text{Proj}_w y = (y.u_1)u_1 + (y.u_2)u_2 + ...(y.u_p)u_p$$

$$If\ U = [u_1\ u_2 \ldots\ldots u_p]$$

$$then\ \text{Proj}_w^y = UU^T y \quad \forall y in\ R^n$$

**Example**

**Let** $u_1 = \begin{bmatrix} -7 \\ 1 \\ 4 \end{bmatrix}, u_2 = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}, y = \begin{bmatrix} -9 \\ 1 \\ 6 \end{bmatrix}$

and $\boldsymbol{W} = \text{span}\{u_1, u_2\}$. Use the fact that $u_1$ and $u_2$ are orthogonal to compute $\text{Proj}_w y$.

**Solution**

$$\text{Proj}_w y = \frac{y.u_1}{u_1.u_1}u_1 + \frac{y.u_2}{u_2.u_2}u_2$$

$$= \frac{88}{66}u_1 + \frac{-2}{6}u_2$$

$$= \frac{4}{3}\begin{bmatrix}-7\\1\\4\end{bmatrix} - \frac{1}{3}\begin{bmatrix}-1\\1\\2\end{bmatrix} = \begin{bmatrix}-9\\1\\6\end{bmatrix} = y$$

In this case,y happens to be a linear combination of $u_1$ and $u_2$ . So y is in W. The closest point in W to y is y itself.

# Lecture # 41

## Orthogonal basis, Gram-Schmidt Process, Orthonormal basis

**Example 1**

Let $W$ = Span $\{x_1, x_2\}$, where

$$x_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} \text{ and } x_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Find the orthogonal basis $\{v_1, v_2\}$ for $W$.

**Solution**

Let P be a projection of $\boldsymbol{x_2}$ on to $\boldsymbol{x_1}$. The component of $\boldsymbol{x_2}$ orthogonal to $\boldsymbol{x_1}$ is $\boldsymbol{x_2} - \boldsymbol{P}$, which is in $\boldsymbol{W}$ as it is formed from $\boldsymbol{x_2}$ and a multiple of $\boldsymbol{x_1}$.

Let $\boldsymbol{v_1} = \boldsymbol{x_1}$ and compute

$$v_2 = x_2 - P = x_2 - \frac{x_2 . v_1}{v_1 . v_1} v_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{15}{45} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

Thus, $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is an orthogonal set of nonzero vectors in $\boldsymbol{W}$, dim $\boldsymbol{W}$ = 2 and $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$ is a basis of $\boldsymbol{W}$.

**Example 2**

For the given basis of a subspace $W$ = Span $\{x_1, x_2\}$,

$$x_1 = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix} \text{ and } x_2 = \begin{bmatrix} 5 \\ 6 \\ -7 \end{bmatrix}$$

Find the orthogonal basis $\{v_1, v_2\}$ for $W$.

**Solution**

Set $\boldsymbol{v_1} = \boldsymbol{x_1}$ and compute

$$v_2 = x_2 - \frac{x_2.v_1}{v_1.v_1} v_1 = \begin{bmatrix} 5 \\ 6 \\ -7 \end{bmatrix} - \frac{10}{20} \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 6 \\ -7 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix}$$

Thus, an orthogonal basis for W is $\left\{ \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix} \right\}$

**Theorem**

Given a basis $\{x_1, \ldots, x_p\}$ for a subspace $W$ of $R^n$. Define

$$v_1 = x_1 \quad v_2 = x_2 - \frac{x_2.v_1}{v_1.v_1} v_1$$

$$v_3 = x_3 - \frac{x_3.v_1}{v_1.v_1} v_1 - \frac{x_3.v_2}{v_2.v_2} v_2$$

$$v_p = x_p - \frac{x_p.v_1}{v_1.v_1} v_1 - \frac{x_p.v_2}{v_2.v_2} v_2 - \ldots - \frac{x_p.v_{p-1}}{v_{p-1}.v_{p-1}} v_{p-1}$$

Then $\{v_1, \ldots, v_p\}$ is an orthogonal basis for $W$.

In addition

$$\text{Span } \{v_1, \ldots, v_k\} = \text{Span } \{x_1, \ldots, x_k\} \text{ for } 1 \le k \le p$$

**Example 3**

The following vectors $\{x_1, x_2, x_3\}$ are linearly independent

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

Construct an orthogonal basis for $W$ by Gram-Schmidt Process.

**Solution**

To construct orthogonal basis we have to perform the following steps.

**Step 1** Let $v_1 = x_1$

**Step 2**

$$\text{Let } v_2 = x_2 - \frac{x_2 . v_1}{v_1 . v_1} v_1$$

Since $$v_2 = x_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{3}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

**Step 3**

$$v_3 = x_3 - \left[ \frac{x_3 . v_1}{v_1 . v_1} v_1 + \frac{x_3 . v_2}{v_2 . v_2} v_2 \right] = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \left[ \begin{bmatrix} 2 \\ 4 \\ 1 \\ 1 \end{bmatrix} + \frac{2}{12} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2/3 \\ 2/3 \\ 2/3 \end{bmatrix} \right]$$

$$v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 2/3 \\ 2/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

Thus, $\{v_1, v_2, v_3\}$ is an orthogonal set of nonzero vectors in ***W.***

**Example 4**

Find an orthogonal basis for the column space of the following matrix by Gram-Schmidt Process.

$$\begin{bmatrix} -1 & 6 & 6 \\ 3 & -8 & 3 \\ 1 & -2 & 6 \\ 1 & -4 & -3 \end{bmatrix}$$

**Solution**

Name the columns of the above matrix as $x_1$, $x_2$, $x_3$ and perform the Gram-Schmidt Process on these vectors.

$$x1 = \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}, x2 = \begin{bmatrix} 6 \\ -8 \\ -2 \\ -4 \end{bmatrix}, x3 = \begin{bmatrix} 6 \\ 3 \\ 6 \\ -3 \end{bmatrix}$$

Set $v_1 = x_1$

$$v_2 = x_2 - \frac{x_2.v_1}{v_1.v_1} v_1$$

$$= \begin{bmatrix} 6 \\ -8 \\ -2 \\ -4 \end{bmatrix} - (-3) \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$$v_3 = x_3 - \frac{x_3.v_1}{v_1.v_1} v_1 - \frac{x_3.v_2}{v_2.v_2} v_2$$

$$= \begin{bmatrix} 6 \\ 3 \\ 6 \\ -3 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix} - \frac{5}{2} \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix}$$

Thus, orthogonal basis is $\left\{ \begin{bmatrix} -1 \\ 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 3 \\ -1 \end{bmatrix} \right\}$

**Example 5**

Find the orthonormal basis of the subspace spanned by the following vectors.

$$x_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

**Solution**

Since from example # 1, we have

$$v_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \; v_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

*Orthonormal Basis*

$$u_1 = \frac{1}{\|v_1\|} v_1 = \frac{1}{\sqrt{45}} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix}, u_2 = \frac{1}{\|v_2\|} v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

**Example 6**

Find the orthonormal basis of the subspace spanned by the following vectors.

$$x_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix} \text{ and } x_2 = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$$

**Solution**

Firstly we find $v_1$ and $v_2$ by Gram-Schmidt Process as

$$v_2 = x_2 = x_2 - \frac{x_2 . v_1}{v_1 . v_1} v_1$$

Set v1 = $x_1$

$$v_2 = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} - \frac{15}{30} \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ -5/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 3 \end{bmatrix}$$

$$Now \|v_2\| = \frac{1}{\sqrt{54}} = \frac{1}{3\sqrt{6}} \; and \; since \|v_1\| = \sqrt{30}$$

*Thus, the orthonormal basis for W is*

$$\left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|} \right\} = \left\{ \begin{bmatrix} 2/\sqrt{30} \\ -5/\sqrt{30} \\ 1/\sqrt{30} \end{bmatrix}, \begin{bmatrix} 2/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} \right\}$$

**Theorem**

If $A$ is an $m$ x $n$ matrix with linearly independent columns, then $A$ can be factored as $A = QR$, where $Q$ is an $m$ x $n$ matrix whose columns form an orthonormal basis for Col $A$ and $R$ is an $n$ x $n$ upper triangular invertible matrix with positive entries on its diagonal.

**Example 7**

Find a $QR$ factorization of matrix $A = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}$

**Solution**

Firstly find the orthonormal basis by applying Gram Schmidt process on the columns of A. We get the following matrix Q.

$$Q = \begin{bmatrix} 1/\sqrt{5} & 1/2 & 1/2 \\ -1/\sqrt{5} & 0 & 0 \\ -1/\sqrt{5} & 1/2 & 1/2 \\ 1/\sqrt{5} & -1/2 & 1/2 \\ 1/\sqrt{5} & 1/2 & -1/2 \end{bmatrix}$$

$$Now\ R = Q^T A = \begin{bmatrix} \sqrt{5} & -\sqrt{5} & 4\sqrt{5} \\ 0 & 6 & -2 \\ 0 & 0 & 4 \end{bmatrix}$$

Verify that A=QR.

**Theorem**

*If* $\{u_1, ..., u_p\}$ *is an orthonormal basis for a subspace* $W$ *of* $R^n$, *then*

$proj_w y = (y \cdot u_1)u_1 + (y \cdot u_2)u_2 + \cdots + (y \cdot u_p)u_p$

*If* $U = [u_1 \ u_2 \ \ldots \ u_p]$,

*then* $proj_w y = UU^T y \ \ \forall\ y$ *in* $R^n$

**The Orthogonal Decomposition Theorem**

Let $W$ be a subspace of $R^n$ Then each $y$ in $R^n$ can be written uniquely in the form $y = \hat{y} + z$
where $W^{\perp}$ is in $W$ and $z$ is in

In fact, if $\{u_1, ..., u_p\}$ is any orthogonal basis of $W$, then

$$\hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \cdots + \frac{y \cdot u_p}{u_p \cdot u_p} u_p$$

and $z = y - \hat{y}$ . The vector $\hat{y}$ is called the orthogonal projection of $y$ onto $W$ and is often written as $proj_w\ y$.

**Best Approximation Theorem**

Let $W$ be a subspace of $R^n$, $y$ is any vector in $R^n$ and $\hat{y}$ the orthogonal projection of $y$ onto $W$. Then $\hat{y}$ is the closest point in $W$ to $y$, in the sense that

for all $v$ in $W$ distinct from $\hat{y}$ .

$$\|y - \hat{y}\| < \|y - v\|$$

The vector $\hat{y}$ in this theorem is called the best approximation to $y$ by elements of $W$.

**Exercise 1**

Let $W$ = Span $\{x1, x2\}$, where

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ and } x_2 = \begin{bmatrix} 1/3 \\ 1/3 \\ -2/3 \end{bmatrix}$$

Construct an orthonormal basis for $W$.

**Exercise 2**

Find an orthogonal basis for the column space of the following matrix by Gram-Schmidt Process.

$$\begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & -5 & -2 \\ 3 & -7 & 8 \end{bmatrix}$$

**Exercise 3**

Find a $QR$ factorization of

$$A = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}$$

# Lecture # 42
# Least Square Solution

## Best Approximation Theorem

Let $W$ be a subspace of $R^n$, $y$ *be* any vector in $R^n$ and $y\hat{}$ the orthogonal projection of $y$ onto $W$. Then $\hat{y}$ is the closest point in $W$ to $y$, in the sense that $\|y-y\hat{}\| < \|y-v\|$ for all $v$ in $W$ distinct from $y\hat{}$

The vector $y\hat{}$ in this theorem is called the best approximation to $y$ by elements of $W$.

## Least-squares solution

The most important aspect of the least-squares problem is that no matter what "$x$" we select, the vector $Ax$ will necessarily be in the column space Col $A$. So we seek an $x$ that makes $Ax$ the closest point in Col $A$ to $b$. Of course, if $b$ happens to be in Col $A$, then $b$ is $Ax$ for some $x$ and such an $x$ is a "least-squares solution."

## Solution of the General Least-Squares Problem

Given $\boldsymbol{A}$ and $\boldsymbol{b}$ as above, apply the Best Approximation Theorem stated above to the subspace Col $\boldsymbol{A}$. Let $b\hat{} = proj_{Col\,A}b$

Since $b\hat{}$ is in the column space of $\boldsymbol{A}$, the equation $Ax = b\hat{}$ is consistent, and there is an $x\hat{}$ in $\boldsymbol{R^n}$ such that

$$Ax\hat{} = b\hat{} \qquad (1)$$

Since $b\hat{}$ is the closest point in Col $\boldsymbol{A}$ to $\boldsymbol{b}$, a vector $x\hat{}$ is a least-squares solution of $\boldsymbol{Ax} = \boldsymbol{b}$ if and only if $\boldsymbol{x\hat{}}$ satisfies $Ax\hat{} = b\hat{}$. Such an $x\hat{}$ in $\boldsymbol{R^n}$ is a list of weights that will build $b\hat{}$ out of the columns of $\boldsymbol{A}$.

## Normal equations for $\boldsymbol{x\hat{}}$

Suppose that $\boldsymbol{x\hat{}}$ satisfies $Ax\hat{} = b\hat{}$. By the Orthogonal Decomposition Theorem the projection $b\hat{}$ has the property that $b - b\hat{}$ is orthogonal to Col $\boldsymbol{A}$, so $b - Ax\hat{}$ is orthogonal to each column of $\boldsymbol{A}$. If $\boldsymbol{a_j}$ is any column of $\boldsymbol{A}$, then $a_j \cdot (b - Ax\hat{}) = 0$, and $a_j^T(b - Ax\hat{}) = 0$.

Since each $\boldsymbol{a_i^T}$ is a row of $\boldsymbol{A^T}$,

$$A^T(b - Ax\hat{}) = 0 \qquad (2)$$

$$A^Tb - A^TAx\hat{} = 0$$

$$A^TAx\hat{} = A^Tb \qquad (3)$$

The matrix equation (3) represents a system of linear equations commonly referred to as the **normal equations for $\boldsymbol{\hat{x}}$.**

Since set of least-squares solutions is nonempty and any such $\hat{x}$ satisfies the normal equations. Conversely, suppose that $\hat{x}$ satisfies $A^T A\hat{x} = A^T b$. Then it satisfy that $b - A\hat{x}$ is orthogonal to the rows of $A^T$ and hence is orthogonal to the columns of $A$. Since the columns of $A$ span Col $A$, the vector $b - A\hat{x}$ is orthogonal to all of Col $A$. Hence the equation $b = A\hat{x} + (b - A\hat{x})$ is a decomposition of $b$ into the sum of a vector in Col $A$ and a vector orthogonal to Col $A$. By the uniqueness of the orthogonal decomposition, $A\hat{x}$ must be the orthogonal projection of $b$ onto Col $A$. That is, $A\hat{x} = \hat{b}$ and $\hat{x}$ is a least-squares solution.

**Definition**

If $A$ is $m$ x $n$ and $b$ is in $R^n$, a least-squares solution of $Ax = b$ is an I $\hat{x}$ in $R^n$ such that

$$\|b - A\hat{x}\| \le \|b - Ax\| \quad \forall\ x \in R^n$$

**Theorem**

The set of least-squares solutions of $Ax = b$ coincides with the nonempty set of solutions of the normal equations

$$A^T A\hat{x} = A^T b$$

**Example 1**

Find the least squares solution and its error from the following matrices,

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$$

**Solution**

Firstly we find

$$A^T A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \quad \textit{and}$$

$$A^T b = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

Then the equation $A^T A\hat{x} = A^T b$ becomes $\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$

Row operations can be used to solve this system, but since $A^TA$ is invertible and $2\times 2$, it is probably faster to compute $(A^TA)^{-1} = \frac{1}{84}\begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix}$

Therefore, $\hat{x} = (A^TA)^{-1}A^Tb = \frac{1}{84}\begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix}\begin{bmatrix} 19 \\ 11 \end{bmatrix} = \frac{1}{84}\begin{bmatrix} 84 \\ 168 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Now again as A= $\begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$

Then $\boldsymbol{A}\hat{\boldsymbol{x}} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 3 \end{bmatrix}$

Hence $\boldsymbol{b} - \boldsymbol{A}\hat{\boldsymbol{x}} = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ 8 \end{bmatrix}$

So $\|\boldsymbol{b} - \boldsymbol{A}\hat{\boldsymbol{x}}\| = \sqrt{(-2)^2 + (-4)^2 + 8^2} = \sqrt{84}$

The least-squares error is $\sqrt{84}$. For any $x$ in $R^2$, the distance between $b$ and the vector $Ax$ is at least $\sqrt{84}$.

**Example 2**

Find the general least-squares solution of $Ax = b$ in the form of a free variable with

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}, b = \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix}$$

**Solution**

Firstly we find, $A^TA = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 2 & 0 & 0 & 2 \end{bmatrix}$ and

$A^Tb = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ 2 \\ 6 \end{bmatrix}$

Then augmented matrix for $A^TA\hat{x} = A^Tb$ is

$$\begin{bmatrix} 6 & 2 & 2 & 2 & 4 \\ 2 & 2 & 0 & 0 & -4 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 0 & 2 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & -1 & -5 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The general solution is $x_1 = 3 - x_4, x_2 = -5 + x_4, x_3 = -2 + x_4,$ and $x_4$ is free.

So the general least-squares solution of $Ax = b$ has the form

$$\hat{x} = \begin{bmatrix} 3 \\ -5 \\ -2 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

**Theorem**

The matrix $A^TA$ is invertible iff the columns of $A$ are linearly independent. In this case, the equation $Ax = b$ has only one least-squares solution $\hat{x}$, and it is given by

$$\hat{x} = (A^TA)^{-1}A^Tb$$

**Example 3**

Find the least squares solution to the following system of equations.

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & -3 & 0 \\ 7 & 1 & 4 \\ 1 & 0 & 5 \end{bmatrix}, \begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix}, b = \begin{bmatrix} 0 \\ 1 \\ -2 \\ 4 \end{bmatrix}$$

**Solution**

As

$$A^T = \begin{bmatrix} 2 & 1 & 7 & 1 \\ 4 & -3 & 1 & 0 \\ 6 & 0 & 4 & 5 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 2 & 1 & 7 & 1 \\ 4 & -3 & 1 & 0 \\ 6 & 0 & 4 & 5 \end{bmatrix} \begin{bmatrix} 2 & 4 & 6 \\ 1 & -3 & 0 \\ 7 & 1 & 4 \\ 1 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 55 & 12 & 45 \\ 12 & 26 & 28 \\ 45 & 28 & 77 \end{bmatrix}$$

Now

$$A^T b = \begin{bmatrix} 2 & 1 & 7 & 1 \\ 4 & -3 & 1 & 0 \\ 6 & 0 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -2 \\ 4 \end{bmatrix} = \begin{bmatrix} -9 \\ -5 \\ 12 \end{bmatrix}$$

As $A^T A\hat{x} = A^T b$

$$\begin{bmatrix} 55 & 12 & 45 \\ 12 & 26 & 28 \\ 45 & 28 & 77 \end{bmatrix} \begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} = \begin{bmatrix} -9 \\ -5 \\ 12 \end{bmatrix}$$

$$\begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} = \begin{bmatrix} -.676 \\ -.776 \\ .834 \end{bmatrix}$$

**Example 4**

Compute the least square error for the solution of the following equation

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & -3 & 0 \\ 7 & 1 & 4 \\ 1 & 0 & 5 \end{bmatrix}, \begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix}, b = \begin{bmatrix} 0 \\ 1 \\ -2 \\ 4 \end{bmatrix}$$

.

**Solution**

$$Ax = \begin{bmatrix} 2 & 4 & 6 \\ 1 & -3 & 0 \\ 7 & 1 & 4 \\ 1 & 0 & 5 \end{bmatrix} \begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 \\ 1 & -3 & 0 \\ 7 & 1 & 4 \\ 1 & 0 & 5 \end{bmatrix} \begin{bmatrix} -.676 \\ -.776 \\ .834 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 0.548 \\ 1.652 \\ -2.172 \\ 3.494 \end{bmatrix}$$

As least square error

$$\|\bar{\in}\| = \|b - A\bar{x}\|$$

is as small as possible, or in other words is smaller than all other possible choices of x.

As

$$b - A\hat{x} = \begin{bmatrix} 0 \\ 1 \\ -2 \\ 4 \end{bmatrix} - \begin{bmatrix} 0.548 \\ 1.652 \\ -2.172 \\ 3.494 \end{bmatrix} = \begin{bmatrix} -0.548 \\ 0.652 \\ 0.172 \\ 1.494 \end{bmatrix}$$

$$\|\in\|^2 = \in_1^2 + \in_2^2 + \in_3^2 + \in_4^2$$

Thus, least square error is

$$\|\bar{\in}\| = \|b - A\bar{x}\| = \sqrt{(-0.548)^2 + (0.652)^2 + (0.172)^2 + (1.494)^2}$$

$$=0.3003+0.4251+.02958+2.23=2.987$$

**Theorem**

Given an *m* x *n* matrix *A* with linearly independent columns.Let *A* = *QR* be a *QR* factorization of *A* ,then for each *b* in *Rm*, the equation *Ax* = *b* has a unique least-squares solution, given by

$\hat{x} = R^{-1}Q^T b$

**Example 1**

Find the least square solution for $A=\begin{bmatrix}1 & 3 & 5\\ 1 & 1 & 0\\ 1 & 1 & 2\\ 1 & 3 & 3\end{bmatrix}, b=\begin{bmatrix}3\\ 5\\ 7\\ -3\end{bmatrix}$

**Solution**

First of all we find QR factorization of the given matrix A. For this we have to find out orthonormal basis for the column space of A by applying Gram-Schmidt Process, we get the matrix of orthonormal basis Q,

$$Q=\begin{bmatrix}1/2 & 1/2 & 1/2\\ 1/2 & -1/2 & -1/2\\ 1/2 & -1/2 & 1/2\\ 1/2 & 1/2 & -1/2\end{bmatrix} \text{ And}$$

$$R=Q^TA=\begin{bmatrix}1/2 & 1/2 & 1/2 & 1/2\\ 1/2 & -1/2 & -1/2 & 1/2\\ 1/2 & -1/2 & 1/2 & -1/2\end{bmatrix}\begin{bmatrix}1 & 3 & 5\\ 1 & 1 & 0\\ 1 & 1 & 2\\ 1 & 3 & 3\end{bmatrix}=\begin{bmatrix}2 & 4 & 5\\ 0 & 2 & 3\\ 0 & 0 & 2\end{bmatrix}$$

Then $$Q^Tb=\begin{bmatrix}1/2 & 1/2 & 1/2 & 1/2\\ 1/2 & -1/2 & -1/2 & 1/2\\ 1/2 & -1/2 & 1/2 & -1/2\end{bmatrix}\begin{bmatrix}3\\ 5\\ 7\\ -3\end{bmatrix}=\begin{bmatrix}6\\ -6\\ 4\end{bmatrix}$$

The least-squares solution $\hat{x}$ satisfies $R\hat{x}=Q^Tb$; that is, $\begin{bmatrix}2 & 4 & 5\\ 0 & 2 & 3\\ 0 & 0 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}6\\ -6\\ 4\end{bmatrix}$

This equation is solved easily and yields $\hat{x}=\begin{bmatrix}10\\ -6\\ 2\end{bmatrix}$.

**Example 2**

Find the least squares solution $R\hat{x}=Q^Tb$ to the given matrices,

$$A=\begin{bmatrix}3 & 1\\ 6 & 2\\ 0 & 2\end{bmatrix}, b=\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}$$

**Solution**

First of all we find QR factorization of the given matrix A. Thus, we have to make

Orthonormal basis by applying Gram Schmidt process on the columns of A,

Let $v_1 = x_1$

$$v_2 = x_2 - P_{v_1} x_2 = x_2 - \frac{x_2 . v_1}{v_1 . v_1} v_1 = \begin{bmatrix}1\\ 2\\ 2\end{bmatrix} - \frac{15}{45}\begin{bmatrix}3\\ 6\\ 0\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 2\end{bmatrix}$$

*Thus, the orthonormal basis are*

$$\left\{\frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}\right\} = \left\{\begin{bmatrix}1\\ 2\\ 0\end{bmatrix}, \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}\right\}$$

Thus

$$Q=\begin{bmatrix}1 & 0\\ 2 & 0\\ 0 & 1\end{bmatrix} \text{ and } Q^T=\begin{bmatrix}1 & 2 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$\text{Now } R = Q^T A = \begin{bmatrix}1 & 2 & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}3 & 1\\ 6 & 2\\ 0 & 2\end{bmatrix}$$

$$= \begin{bmatrix}15 & 5\\ 0 & 2\end{bmatrix} \quad \text{And } Q^T b = \begin{bmatrix}1 & 2 & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1\\ 2\\ 3\end{bmatrix} = \begin{bmatrix}5\\ 3\end{bmatrix}$$

Thus, least squares solution of $R\hat{x} = Q^T b$ is

$$\begin{bmatrix}15 & 5\\ 0 & 2\end{bmatrix}\hat{x} = \begin{bmatrix}5\\ 3\end{bmatrix}$$

$$\hat{x} = \begin{bmatrix}1\\ 1.8\end{bmatrix}$$

**Exercise 1**

Find a least-squares solution of $Ax = b$ for

$$\boldsymbol{A} = \begin{bmatrix} 1 & -6 \\ 1 & -2 \\ 1 & 1 \\ 1 & 7 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 6 \end{bmatrix}$$

**Exercise 2**

Find the least-squares solution and its error of $Ax = b$ for

$$\boldsymbol{A} = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$$

**Exercise 3**

Find the least squares solution $R\hat{x} = Q^T b$ to the given matrices,

$$A = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix}, b = \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix}$$

# Lecture # 43
# Inner Product Space

## Inner Product Space

In mathematics, an inner product space is a vector space with the additional structure called an inner product. This additional structure associates each pair of vectors in the space with a scalar quantity known as the inner product of the vectors. Inner products allow the rigorous introduction of intuitive geometrical notions such as the length of a vector or the angle between two vectors. They also provide the means of defining orthogonality between vectors (zero inner product). Inner product spaces generalize Euclidean spaces (in which the inner product is the dot product, also known as the scalar product) to vector spaces of any (possibly infinite) dimension, and are studied in functional analysis.

## Definition

An inner product on a vector space $V$ is a function that to each pair of vectors $u$ and $v$ associates a real number $\langle u,v\rangle$ and satisfies the following axioms,

For all $u, v, w$ in $V$ and all scalars $C$:

1) $\langle u,v\rangle = \langle v,u\rangle$
2) $\langle u+v,w\rangle = \langle u,w\rangle + \langle v,w\rangle$
3) $\langle cu,v\rangle = c\langle u,v\rangle$
4) $\langle u,u\rangle \geq 0$ *and* $\langle u,u\rangle = 0$ *iff* $u = 0$

A vector space with an inner product is called inner product space.

## Example 1

Fix any two positive numbers say 4 & 5 and for vectors $u=\langle u_1,u_2\rangle$ and $v=\langle v_1,v_2\rangle$ in $R^2$ set

$$\langle u,v\rangle = 4u_1v_1 + 5u_2v_2$$

Show that it defines an inner product.

## Solution

Certainly Axiom 1 is satisfied, because

$$\langle u,v\rangle = 4u_1v_1 + 5u_2v_2 = 4v_1u_1 + 5v_2u_2 = \langle \mathbf{v},u\rangle .$$

If $\mathbf{w} = (w_1, w_2)$, then

$$\langle u+v,w\rangle = 4(u_1+v_1)w_1 + 5(u_2+v_2)w_2$$

$$= 4u_1w_1 + 5u_2w_2 + 4v_1w_1 + 5v_2w_2 = \langle u,w\rangle + \langle v,w\rangle$$

This verifies Axiom 2.

For Axiom 3, we have $\langle c\mathbf{u},\mathbf{v}\rangle = 4(cu_1)v_1 + 5(cu_2)v_2 = c(4u_1v_1 + 5u_2v_2) = c\langle \mathbf{u},\mathbf{v}\rangle$

For Axiom 4, note that $\langle u,u\rangle = 4u_1^2 + 5u_2^2 \geq 0$, and $4u_1^2 + 5u_2^2 = 0$ only if $u_1 = u_2 = 0$, that is, if $\mathbf{u} = \mathbf{0}$. Also, $\langle 0,0\rangle = 0$.

So $\langle u,v\rangle = 4u_1v_1 + 5u_2v_2 = 4v_1u_1 + 5v_2u_2$ defines an inner product on $\boldsymbol{R^2}$.

**Example 2**

Let A be symmetric, positive definite $n\times n$ matrix and let u and v be vectors in $\Re^n$ .
Show that $\langle u,v\rangle = u^tAv$ defines and inner product.

**Solution**

We check that

$$\langle u,v\rangle = u^tAv = u.Av = Av.u$$

$$= A^tv.u = \left(v^tA\right)^t u = v^tAu = \langle v,u\rangle$$

Also

$$\langle u,v+w\rangle = u^tA(v+w) = u^tAv + u^tAw$$

$$= \langle u,v\rangle + \langle u,w\rangle$$

And

$$\langle cu,v\rangle = (cu)^tAv = c\left(u^tAv\right) = c\langle u,v\rangle$$

Finally since A is positive definite

$$\langle u,u\rangle = u^tAu > 0 \qquad for\ all\ u \neq 0$$

$$So \quad \langle u,u\rangle = u^tAu = 0 \qquad iff \quad u = 0$$

So $\langle u,v\rangle = u^tAv$ is an inner product space.

**Example 3**

Let $t_0, \ldots, t_n$ be distinct real numbers. For $p$ and $q$ in $\boldsymbol{P_n}$, define

$$\langle p,q\rangle = p(t_0)q(t_0) + p(t_1)q(t_1) + \cdots + p(t_n)q(t_n)$$

Show that it defines inner product.

**Solution**

Certainly Axiom 1 is satisfied, because

$$\langle p,q\rangle = p(t_0)q(t_0) + p(t_1)q(t_1) + \cdots + p(t_n)q(t_n)$$

$$= q(t_0)p(t_0) + q(t_1)p(t_1) + \cdots + q(t_n)p(t_n) = \langle q,p\rangle$$

If $\boldsymbol{r} = r(t_0) + r(t_1) + \cdots + r(t_n)$, then

$$\langle p+q,r\rangle = \left[p(t_0) + q(t_0)\right]r(t_0) + \left[p(t_1) + q(t_1)\right]r(t_1) + \cdots + \left[p(t_n) + q(t_n)\right]r(t_n)$$

$$= \left[p(t_0)r(t_0) + p(t_1)r(t_1) + \ldots + p(t_n)r(t_n)\right] + \left[q(t_0)r(t_0) + q(t_1)r(t_1) + \ldots + q(t_n)r(t_n)\right]$$

$$= \langle p,r\rangle + \langle q,r\rangle$$

This verifies Axiom 2.

For Axiom 3, we have

$$\langle cp,q\rangle = \left[cp(t_0)\right]q(t_0) + \left[cp(t_1)\right]q(t_1) + \cdots + \left[cp(t_n)\right]q(t_n)$$

$$= c\left[p(t_0)q(t_0) + p(t_1)q(t_1) + \cdots + p(t_n)q(t_n)\right] = c\langle p,q\rangle$$

For Axiom 4, note that

$$\langle p,p\rangle = [p(t_0)]^2 + [p(t_1)]^2 + \cdots + [p(t_n)]^2 \geq 0$$

Also, $\langle \mathbf{0}, \mathbf{0} \rangle$=0. (We still use a boldface zero for the zero polynomial, the zero vector in $\boldsymbol{P_n}$.) If $\langle p, p \rangle$=0, then $\boldsymbol{p}$ must vanish at $n$ + 1 points: $t_0$, …, $t_n$. This is possible only if $\boldsymbol{p}$ is the zero polynomial, because the degree of $\boldsymbol{p}$ is less than $n$ + 1. Thus $\langle p, q \rangle = p(t_0)q(t_0) + p(t_1)q(t_1) + \cdots + p(t_n)q(t_n)$ defines an inner product on $\boldsymbol{P_n}$.

**Example 4**

Compute $\langle p, q \rangle$ where p(t)= 4+t q(t) = 5-4t$^2$

Refer to $P_2$ with the inner product given by evaluation at -1, 0 and 1 in example 2.

**Solution**

$$P(-1) = 3,\ P(0) = 4,\ P(1) = 5$$
$$q(-1) = 1,\ q(0) = 5,\ q(1) = 1$$

$$\langle p, q \rangle = P(-1)q(-1) + P(0)q(0) + P(1)q(1)$$
$$= (3)(1) + (4)(5) + (5)(1)$$
$$= 3 + 20 + 5$$

**Example 5**

Compute the orthogonal projection of q onto the subspace spanned by p, for p and q in the above example.

**Solution**

The orthogonal projection of q onto the subspace spanned by p

$$P(-1) = 3,\ P(0) = 4,\ P(1) = 5$$
$$q(-1) = 1,\ q(0) = 5,\ q(1) = 1$$

$$q.p = 28 \qquad p.p = 50$$

$$\hat{q} = \frac{q.p}{p.p}p = \frac{28}{50}(4 + t)$$
$$= \frac{56}{25} + \frac{14}{25}t$$

**Example 6**

Let $V$ be $P_2$ , with the inner product from example 2 where

$$t_0 = 0,\ t_1 = \frac{1}{2} \text{ and } t_2 = 1$$

Let $p(t) = 12t^2$ and $q(t) = 2t - 1$

Compute $\langle p, q \rangle$ and $\langle q, q \rangle$

**Solution**

$$\langle p,q\rangle = p(0)\,q(0) + p\left(\tfrac{1}{2}\right) + p(1)\,q(1)$$
$$= (0)(-1) + (3)(0) + (12)(1) = 12$$

$$\langle q,q\rangle = \left[q(0)\right]^2 + \left[q\left(\tfrac{1}{2}\right)\right]^2 + \left[q(1)\right]^2$$
$$= (-1)^2 + (0)^2 + (1)^2 = 2$$

**Norm of a Vector**

Let V be and inner product space with the inner product denoted by $\langle u,v\rangle$ just as in $R^n$ , we define the length or norm of a vector V to be the scalar

$$\|v\| = \sqrt{\langle u,v\rangle} \quad or \quad \|v\|^2 = \langle u,v\rangle$$

1) A unit vector is one whose length is 1.
2) The distance between $u$ & $v$ is $\|u-v\|$ vectors $u$ & $v$ are orthogonal if $\langle u,v\rangle = 0$

**Example 7**

Compute the length of the vectors in example 3.

**Solution**

$$\|p\|^2 = \langle p,p\rangle = \left[p(0)\right]^2 + \left[p\left(\tfrac{1}{2}\right)\right]^2 + \left[p(1)\right]^2$$
$$= (0)^2 + (3)^2 + (12)^2 = 153$$
$$\|p\| = \sqrt{153}$$

In example 3 we found that

$$\langle q,q\rangle = 2$$
$$Hence \quad \|q\| = \sqrt{2}$$

**Example 8**

Let $\Re^2$ have the inner product of example 1 and let x=(1,1) and y=(5,-1)

a) Find $\|x\|$ , $\|y\|$ $and$ $|\langle x,y\rangle|^2$ b) Describe all vectors $(z_1, z_2)$ that are orthogonal to y.

**Solution**

**a)** We have x=(1,1) and y=(5,-1)

And $\langle x,y\rangle = 4x_1y_1 + 5x_2y_2$

$$\|x\| = \sqrt{\langle x, x\rangle} = \sqrt{4(1)(1) + 5(1)(1)}$$
$$= \sqrt{4+5} = \sqrt{9} = 3$$
$$\|y\| = \sqrt{\langle y, y\rangle} = \sqrt{4(5)(5) + 5(-1)(-1)}$$
$$= \sqrt{100+5} = \sqrt{105}$$

$$|\langle x, y\rangle|^2 = \langle x, y\rangle\langle x, y\rangle$$
$$= [4(1)(5) + 5(1)(-1)]^2$$
$$= [20-5]^2$$
$$= [15]^2 = 225$$

**b)** All vectors $z = (z_1, z_2)$ orthogonal to y=(5,-1)

$< y, z >= 0$

$4(5)(z_1) + 5(-1)(z_2) = 0$

$20z_1 - 5z_2 = 0$

$4z_1 - z_2 = 0$

$$\begin{bmatrix} 4 & -1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = 0$$

So all multiples of $\begin{bmatrix} 4 \\ -1 \end{bmatrix}$ are orthogonal to y.

**Example 9**

Le V be $P_4$ with the inner product in example 2 involving evaluation of polynomials at -2,-1,0,1,2 and view $P_2$ as a subspace of V. Produce an orthogonal basis for $P_2$ by applying the Gram Schmidt process to the polynomials $1, t \& t^2$.

**Solution**

Given polynomials **1** ***t*** $\boldsymbol{t^2}$ at -2,-1, 0, 1 and 2

Polynomial: **1** ***t*** $\boldsymbol{t^2}$

Vector of values: $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ 0 \\ 1 \\ 4 \end{bmatrix}$

The inner product of two polynomials in ***V*** equals the (standard) inner product of their corresponding vectors in $\boldsymbol{R^5}$. Observe that ***t*** is orthogonal to the constant function **1**. So

take $\boldsymbol{p_0}(t) = 1$ and $\boldsymbol{p_1}(t) = t$. For $p_2$, use the vectors in $\boldsymbol{R^5}$ to compute the projection of $\boldsymbol{t^2}$ onto Span $\{\boldsymbol{p_0}, \boldsymbol{p_1}\}$:

$$\langle t^2, p_0 \rangle = \langle t^2, 1 \rangle = 4+1+0+1+4 = 10$$

$$\langle p_0, p_0 \rangle = 5$$

$$\langle t^2, p_1 \rangle = \langle t^2, t \rangle = -8+(-1)+0+1+8 = 0$$

The orthogonal projection of $\boldsymbol{t^2}$ onto Span $\{\boldsymbol{1}, \boldsymbol{t}\}$ is $\frac{10}{5} p_0 + 0 p_1$. Thus

$$p_2(t) = t^2 - 2p_0(t) = t^2 - 2$$

An orthogonal basis for the subspace $\boldsymbol{P_2}$ of $\boldsymbol{V}$ is:

Polynomial: $\boldsymbol{p_0}$ $\boldsymbol{p_1}$ $\boldsymbol{p_2}$

Vector of values: $\begin{bmatrix}1\\1\\1\\1\\1\end{bmatrix}, \begin{bmatrix}-2\\-1\\0\\1\\2\end{bmatrix}, \begin{bmatrix}2\\-1\\-2\\-1\\2\end{bmatrix}$

**Best Approximation in Inner Produce Spaces**

A common problem in applied mathematics involves a vector space V whose elements are functions. The problem is to approximate a function $f$ in V by a function $g$ from a specified subspace W of V. The "closeness" of the approximation of $f$ depends on the way $\|f-g\|$ is defined. We will consider only the case in which the distance between $f$ and $g$ is determined by an inner product. In this case the best approximation to $f$ by functions in W is the orthogonal projection of $f$ onto the subspace W.

**Example 10**

Let V be $P_4$ with the inner product in example 5 and let $P_0, P_1 \& P_2$ be the orthogonal basis for the subspace $P_2$, find the best approximation to $p(t) = 5 - \frac{1}{2}t^4$ by polynomials in $P_2$.

**Solution:**

The values of $\boldsymbol{p_0}$, $\boldsymbol{p_1}$, and $\boldsymbol{p_2}$ at the numbers – 2, –1, 0, 1, and 2 are listed in $\boldsymbol{R^5}$ vectors in

Polynomial: $\boldsymbol{p_0}$ $\boldsymbol{p_1}$ $\boldsymbol{p_2}$

Vector of values: $\begin{bmatrix}1\\1\\1\\1\\1\end{bmatrix}, \begin{bmatrix}-2\\-1\\0\\1\\2\end{bmatrix}, \begin{bmatrix}2\\-1\\-2_1\\2\end{bmatrix}$

The corresponding values for $\boldsymbol{p}$ are: $-3, 9/2, 5, 9/2$, and $-3$.

We compute

$$\langle p, p_0\rangle = 8 \qquad \langle p, p_1\rangle = 0 \qquad \langle p, p_2\rangle = -31$$
$$\langle p_0, p_0\rangle = 5, \qquad \langle p_2, p_2\rangle = 14$$

Then the best approximation in $\boldsymbol{V}$ to $\boldsymbol{p}$ by polynomials in $\boldsymbol{P_2}$ is

$$\hat{p} = proj_{p_2} p = \frac{\langle p, p_0\rangle}{\langle p_0, p_0\rangle} p_0 + \frac{\langle p, p_1\rangle}{\langle p_1, p_1\rangle} p_1 + \frac{\langle p, p_2\rangle}{\langle p_2, p_2\rangle} p_2$$
$$= \tfrac{8}{5} p_0 + \tfrac{-31}{14} p_2 = \tfrac{8}{5} - \tfrac{31}{14}(t^2 - 2).$$

This polynomial is the closest to $\boldsymbol{P}$ of all polynomials in $\boldsymbol{P_2}$, when the distance between polynomials is measured only at –2, –1, 0, 1, and 2.

**Cauchy – Schwarz Inquality**

$$For\ all \quad u, v\ in\ V$$
$$|\langle u, v\rangle| \le \|u\|\,\|v\|$$

**Triangle Inequality**

$$For\ all\ \ u, v\ in\ V$$
$$\|u+v\| \le \|u\| + \|v\|$$

**Proof**

$$\|u+v\|^2 = \langle u+v, u+v\rangle$$
$$= \langle u, u\rangle + 2\langle u, v\rangle + \langle v, v\rangle$$
$$\le \|u\|^2 + 2|\langle u, v\rangle| + \|v\|^2$$
$$\le \|u\|^2 + 2\|u\|\|v\| + \|v\|^2$$
$$\|u+v\|^2 = \left(\|u\| + \|v\|\right)^2$$

$$\Rightarrow \quad \|u+v\| = \|u\| + \|v\|$$

**Inner product for $C[a,b]$**

Probably the most widely used inner product space for applications is the vector space $C[a,b]$ of all continuous functions on an interval $a \le t \le b$, with an inner product that will describe.

**Example 11**

For $f$, $g$ in $C[a,b]$, set

$$\langle f,g\rangle = \int_b^a f(t)\, g(t)\, dt$$

Show that it defines an inner product on $C[a,b]$.

**Solution**

Inner product Axioms 1 to 3 follow from elementary properties of definite integrals

1. $\langle f,g\rangle = \langle g, f\rangle$
2. $\langle f+h,g\rangle = \langle f,g\rangle + \langle h,g\rangle$
3. $\langle cf,g\rangle = c\langle f,g\rangle$

For Axiom 4, observe that

$$\langle f, f\rangle = \int_a^b [f(t)]^2 dt \ge 0$$

The function $[\boldsymbol{f}(t)]^2$ is continuous and nonnegative on $[a, b]$. If the definite integral of $[\boldsymbol{f}(t)]^2$ is zero, then $[\boldsymbol{f}(t)]^2$ must be identically zero on $[a, b]$, by a theorem in advanced calculus, in which case $\boldsymbol{f}$ is the zero function. Thus $\langle \boldsymbol{f}, \boldsymbol{f}\rangle = 0$ implies that $\boldsymbol{f}$ is the zero function of $[a, b]$.

So $\langle \boldsymbol{f}, \boldsymbol{g}\rangle = \int_a^b \boldsymbol{f}(t)\boldsymbol{g}(t)dt$ defines an inner product on $\boldsymbol{C}[a, b]$.

**Example 12**

Compute $\langle f,g\rangle$ where $f(t) = 1-3t^2$ *and* $g(t) = t-t^3$ *on* $v = C[0,1]$.

**Solution**

Let V be the space $C[a,b]$ with the inner product

$$\langle f,g\rangle = \int_b^a f(t)\, g(t)\, dt$$

$$f(t) = 1-3t^2 \quad , \quad g(t) = t-t^3$$

$$\langle f,g\rangle = \int_0^1 (1-3t^2)\left(t-t^3\right)dt$$

$$= \int_0^1 \left(3t^5 - 4t^3 + t\right)dt$$

$$= \left|\frac{1}{2}t^6 - t^4 + \frac{1}{2}t^2\right|_0^1$$

$$= 0$$

**Example 13**

Let V be the space $C[a,b]$ with the inner product $\langle f,g\rangle = \int_b^a f(t)\,g(t)\,dt$

Let W be the subspace spanned by the polynomials $P_1(t)=1$, $P_2(t)=2t-1$ & $P_3(t)=12t^2$

Use the Gram – Schmidt process to find an orthogonal basis for W.

**Solution**

Let $\boldsymbol{q_1} = \boldsymbol{p_1}$, and compute

$$\langle \boldsymbol{p}_2, \boldsymbol{q}_1 \rangle = \int_0^1 (2t-1)(1)dt = (t^2-t)\Big|_0^1 = 0$$

So $\boldsymbol{p_2}$ is already orthogonal to $\boldsymbol{q_1}$, and we can take $\boldsymbol{q_2} = \boldsymbol{p_2}$. For the projection of $\boldsymbol{p_3}$ onto $\boldsymbol{W_2}$ = Span $\{\boldsymbol{q_1}, \boldsymbol{q_2}\}$, we compute

$$\langle \boldsymbol{p_3}, \boldsymbol{q_1} \rangle = \int_0^1 12t^2 \cdot 1 dt = 4t^3\Big|_0^1 = 4$$

$$\langle \boldsymbol{q_1}, \boldsymbol{q_1} \rangle = \int_0^1 1 \cdot 1 dt = t\Big|_0^1 = 1$$

$$\langle \boldsymbol{p_3}, \boldsymbol{q_2} \rangle = \int_0^1 12t^2(2t-1)dt = \int_0^1 (24t^3 - 12t^2)dt = 2$$

$$\langle \boldsymbol{q}_2, \boldsymbol{q}_2 \rangle = \int_0^1 (2t-1)^2 dt = \frac{1}{6}(2t-1)^3\Big|_0^1 = \frac{1}{3}$$

Then $$\text{proj}_{w_2} \boldsymbol{p}_3 = \frac{\langle \boldsymbol{p_3}, \boldsymbol{q_1} \rangle}{\langle \boldsymbol{q_1}, \boldsymbol{q_1} \rangle} \boldsymbol{q}_1 + \frac{\langle \boldsymbol{p_3}, \boldsymbol{q_2} \rangle}{\langle \boldsymbol{q_2}, \boldsymbol{q_2} \rangle} \boldsymbol{q}_2 = \frac{4}{1}\boldsymbol{q}_1 + \frac{2}{1/3}\boldsymbol{q}_2 = 4\boldsymbol{q}_1 + 6\boldsymbol{q}_2$$

And $$\boldsymbol{q_3} = \boldsymbol{p_3} - \text{proj}_{w_2} \boldsymbol{p_3} = \boldsymbol{p_3} - 4\boldsymbol{q_1} - 6\boldsymbol{q_2}$$

As a function, $\boldsymbol{q_3}(t) = 12\boldsymbol{t^2} - \boldsymbol{4} - 6(2\boldsymbol{t} - \boldsymbol{1}) = 12\boldsymbol{t^2} - 12\boldsymbol{t} + \boldsymbol{2}$. The orthogonal basis for the subspace $\boldsymbol{W}$ is $\{\boldsymbol{q_1}, \boldsymbol{q_2}, \boldsymbol{q_3}\}$

**Exercises**

Let $\Re^2$ have the inner product of example 1 and let x=(1,1) and y=(5,-1)

a) Find $\|x\|$, $\|y\|$ *and* $|\langle x,y\rangle|^2$ b) Describe all vectors $(z_1, z_2)$ that are orthogonal to y.

**2)** Let $\Re^2$ have the inner product of Example 1. Show that the Cauchy-Shwarz inequality holds for x=(3,-2) and y=(-2,1)

Exercise 3-8 refer to $P_2$ with the inner product given by evaluation at -1,0 and 1 in example 2.

**3)** Compute $\langle p,q\rangle$ where p(t)= 4+t q(t) = 5-4t$^2$

**4)** Compute $\langle p,q\rangle$ where p(t)= 3t - t$^2$ q(t) = 3 + t$^2$

**5)** Compute $\|P\|$ *and* $\|q\|$ for p and q in exercise 3.

**6)** Compute $\|P\|$ *and* $\|q\|$ for p and q in exercise 4.

**7)** Compute the orthogonal projection of q onto the subspace spanned by p, for p and q in Exercise 3.

**8)** Compute the orthogonal projection of q onto the subspace spanned by p, for p and q in Exercise 4.

**9)** Let $P^3$ have the inner product given by evaluation at -3,-1,1, and 3. Let $p_o(t)=1$ , $p_1(t)=t$, *and* $p_2(t)=t^2$

a)Computer the orthogonal projection of $P_2$ on to the subspace spanned by $P_0$ and $P_1$. b) Find a polynomial q that is orthogonal to $P_0$ and $P_1$ such tha $\{p_0, p_1, q\}$ is an orthogonal basis for span $\{p_0, p_1, q\}$ . Scale the polynomial q so that its vector of values at (-3,-1,1,3) is (1,-1,-1,1)

**10)** Let $P^3$ have the inner product given by evaluation at -3,-1,1, and 3. Let $p_o(t)=1$ , $p_1(t)=t$, *and* $p_2(t)=t^2$

Find the best approximation to $p(t)=t^3$ by polynomials in Span $\{p_0, p_1, q\}$.

**11)** Let $p_0, p_1, p_2$ be the orthogonal polynomials described in example 5, where the inner product on $P_4$ is given by evaluation at -2, -1, 0, 1, and 2. Find the orthogonal projection of $t^3$ onto Span $\{p_0, p_1, p_2\}$

**12)** Compute $\langle f, g\rangle$ where $f(t)=1-3t^2$ *and* $g(t)=t-t^3$ *on* $v=C[0,1]$.

**13)** Compute $\langle f, g\rangle$ where $f(t)=5t-3$ *and* $g(t)=t^3-t^2$ *on* $v=C[0,1]$.

**14)** Compute $\|f\|$ for f in exercise 12.

**15)** Compute $\|g\|$ for g in exercise 13.

**16)** Let V be the space C[-2,2] with the inner product of Example 7. Find an orthogonal basis for the subspace spanned by the polynomials$1, t, t^2$ .

**17)** Let $u=\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $v=\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be two vectors in $R^2$. Show that $\langle u, v\rangle = 2u_1v_1 + 3u_2v_2$ defines an inner product.

# Lecture 44
# Application of inner product spaces

**Definition**

An inner product on a vector space *V* is a function that associates to each pair of vectors *u* and *v* in *V*, a real number $\langle u, v \rangle$ and satisfies the following axioms, for all *u*, *v*, *w* in *V* and all scalars *c*:

1. $\langle u, v \rangle = \langle v, u \rangle$
2. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
3. $\langle cu, v \rangle = c \langle u, v \rangle$
4. $\langle u, u \rangle \geq 0$ *and* $\langle u, u \rangle = 0$ *iff* $u = 0$.

A vector space with an inner product is called an inner product space.

**Least Squares Lines**

The simplest relation between two variables x and y is the linear equation $y = \beta_0 + \beta_1 x$. Often experimental data produces points $(x_1, y_1), \ldots, (x_n, y_n)$ that when graphed, seem to lie close to a line. Actually we want to determine the parameters $\beta_0$ and $\beta_1$ that make the line as "close" to the points as possible. There are several ways to measure how close the line is to the data. The usual choice is to add the squares of the residuals. The least squares line is the line $y = \beta_0 + \beta_1 x$ that minimizes the sum of the squares of the residuals.

If the data points are on the line, the parameters $\beta_0$ and $\beta_1$ would satisfy the equations

| *predicted value* | *Observed value* |
|---|---|
| $\beta_0 + \beta_1 x_1$ | $= y_1$ |
| $\beta_0 + \beta_1 x_2$ | $= y_2$ |
| . | . |
| . | . |
| . | . |
| $\beta_0 + \beta_1 x_n$ | $= y_n$ |

We can write this system as

$$X\beta = y$$

$$\text{Where } X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \\ 1 & x_n \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad y = \begin{bmatrix} y_0 \\ y_1 \\ \cdot \\ \cdot \\ \cdot \\ y_n \end{bmatrix}$$

Computing the least-squares solution of $X\beta = y$ is equivalent to finding the $\beta$ that determines the least-squares line.

**Example 1**

Find the equation $y = \beta_0 + \beta_1 x$ of the least-squares line that best fits the data points (2, 1), (5, 2), (7, 3), (8, 3).

**Solution**

$$X\beta = y$$

$$\text{Here } X = \begin{bmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix}$$

For the least-squares solution of $x\beta = y$, obtain the normal equations(with the new notation) :

$$X^T X\hat{\beta} = X^T y$$

i.e, compute

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 5 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{bmatrix} = \begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 5 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 9 \\ 57 \end{bmatrix}$$

The normal equations are

$$\begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix}\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 9 \\ 57 \end{bmatrix}$$

Hence,

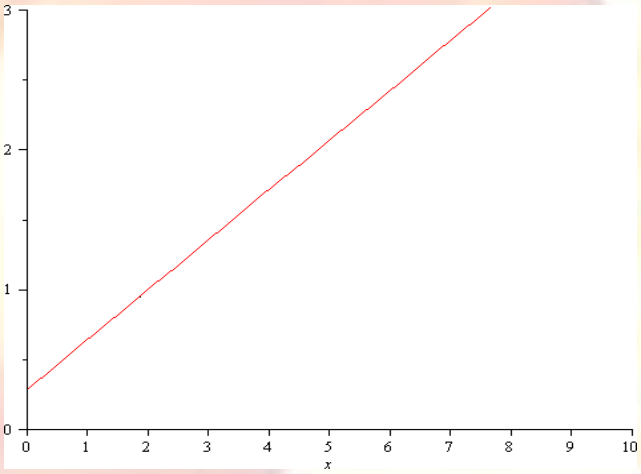
$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix}^{-1}\begin{bmatrix} 9 \\ 57 \end{bmatrix}$$

$$= \frac{1}{84}\begin{bmatrix} 142 & -22 \\ -22 & 4 \end{bmatrix}\begin{bmatrix} 9 \\ 57 \end{bmatrix}$$

$$= \frac{1}{84}\begin{bmatrix} 24 \\ 30 \end{bmatrix} = \begin{bmatrix} 2/7 \\ 5/14 \end{bmatrix}$$

Thus, the least -squares line has the equation

$$y = \frac{2}{7} + \frac{5}{14}x$$




**Weighted Least-Squares**

Let y be a vector of $n$ observations, $y_1, y_2, ..., y_n$ and suppose we wish to approximate y by a vector $y\hat{}$ that belongs to some specified subspace of $\mathbb{R}^n$ (as discussed previously that $\hat{y}$ is written as Ax so that $\hat{y}$ was in the column space of A).Now suppose the approximating vector $y\hat{}$ is to be constructed from the columns of matrix A. Then we find an $\hat{x}$ that makes $Ax\hat{} = y\hat{}$ as close to y as possible. So that measure of closeness is the weighted error

$$\|Wy - Wy\hat{}\|^2 = \|Wy - WAy\hat{}\|^2$$

Where W is the diagonal matrix with (positive) $w_1,..., w_n$ on its diagonal, that is

$$W = \begin{bmatrix} w_1 & 0 & \dots & 0 \\ 0 & w_2 & & \\ . & & . & . \\ . & & . & . \\ . & & . & . \\ 0 & & \dots & w_n \end{bmatrix}$$

Thus, $\hat{x}$ is the ordinary least-squares solution of the equation

$$WAx = Wy$$

The normal equation for the weighted least-squares solution is

$$(WA)^T WAX = (WA)^T Wy$$

**Example 2**

Find the least squares line $y = \beta_0 + \beta_1 x$ that best fits the data (–2, 3), (–1, 5), (0, 5), (1, 4), (2, 3). Suppose that the errors in measuring the *y*-values of the last two data points are greater than for the other points. Weight this data half as much as the rest of the data.

**Solution**

*Write* $X, \beta$ *and* $y$

$$X = \begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, y = \begin{bmatrix} 3 \\ 5 \\ 5 \\ 4 \\ 3 \end{bmatrix}$$

For a weighting matrix, choose W with diagonal entries 2 , 2 , 2 , 1 and 1.
Left-multiplication by W scales the rows of X and y:

$$WX = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$WX = \begin{bmatrix} 2 & -4 \\ 2 & -2 \\ 2 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, Wy = \begin{bmatrix} 6 \\ 10 \\ 10 \\ 4 \\ 3 \end{bmatrix}$$

For normal equation, compute

$$(WX)^T WX = \begin{bmatrix} 14 & -9 \\ -9 & 25 \end{bmatrix}, \text{ and } (WX)^T Wy = \begin{bmatrix} 59 \\ -34 \end{bmatrix}$$

And solve

$$\begin{bmatrix} 14 & -9 \\ -9 & 25 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 59 \\ -34 \end{bmatrix}$$

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 14 & -9 \\ -9 & 25 \end{bmatrix}^{-1} \begin{bmatrix} 59 \\ -34 \end{bmatrix}$$

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \frac{1}{269} \begin{bmatrix} 25 & 9 \\ 9 & 14 \end{bmatrix} \begin{bmatrix} 59 \\ -34 \end{bmatrix}$$

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \frac{1}{269} \begin{bmatrix} 25 & 9 \\ 9 & 14 \end{bmatrix} \begin{bmatrix} 59 \\ -34 \end{bmatrix}$$

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \frac{1}{269} \begin{bmatrix} 1169 \\ 55 \end{bmatrix} = \begin{bmatrix} 4.3 \\ 0.2 \end{bmatrix}$$

Therefore, the solution to two significant digits is $\beta_0 = 4.3 \text{ and } \beta_1 = 0.20$.

Hence the required line is $y = 4.3 + 0.2x$

In contrast, the ordinary least-squares line for this data can be found as:

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 5 \\ 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 20 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 20 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}^{-1} \begin{bmatrix} 20 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 20 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 200 \\ -5 \end{bmatrix} = \begin{bmatrix} 4.0 \\ -0.1 \end{bmatrix}$$

Hence the equation of least-squares line is

$$y = 1.0 - 0.1x$$

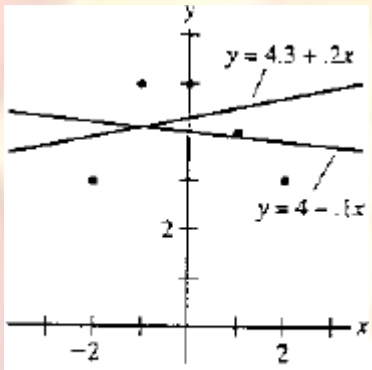



FIGURE 1 Weighted and ordinary least-squares lines.

**What Does Trend Analysis Mean?**

An aspect of technical analysis that tries to predict the future movement of a stock based on past data. Trend analysis is based on the idea that what has happened in the past gives traders an idea of what will happen in the future.

**Linear Trend**

A first step in analyzing a time series, to determine whether a linear relationship provides a good approximation to the long-term movement of the series computed by the method of semi averages or by the method of least squares.

**Note**

The simplest and most common use of trend analysis occurs when the points $t_0, t_1, \ldots, t_n$ can be adjusted so that they are evenly spaced and sum to zero.

**Example**

Fit a quadratic trend function to the data (-2,3), (-1,5), (0,5), (1,4) and (2,3)

**Solution**

The t-coordinates are suitably scaled to use the orthogonal polynomials found in Example 5 of the last lecture. We have

Polynomial: $p_0$ $p_1$ $p_2$ data: g

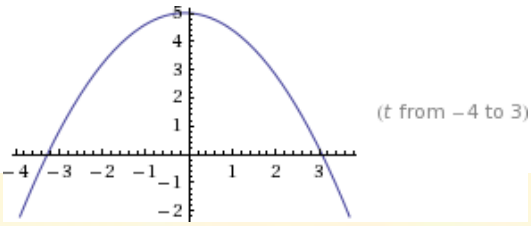
*Vector of values*: $\begin{bmatrix}1\\1\\1\\1\\1\end{bmatrix}, \begin{bmatrix}-2\\-1\\0\\1\\2\end{bmatrix}, \begin{bmatrix}2\\-1\\-2\\-1\\2\end{bmatrix}, \begin{bmatrix}3\\5\\5\\4\\3\end{bmatrix}$

$$\hat{p} = \frac{< g, p_0 >}{< p_0, p_0 >} p_0 + \frac{< g, p_1 >}{< p_1, p_1 >} p_1 + \frac{< g, p_2 >}{< p_2, p_2 >} p_2$$

$$= \frac{20}{5} p_0 - \frac{1}{10} p_1 - \frac{7}{14} p_2$$

*and* $\hat{p}(t) = 4 - 0.1t - 0.5(t^2 - 2)$

Since, the coefficient of $p_2$ is not extremely small, it would be reasonable to conclude that the trend is at least quadratic.




Above figure shows that approximation by a quadratic trend function

**Fourier series**

If f is a $2\pi$ -periodic function then

$$f(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} (a_m \cos mt + b_m \sin mt)$$

is called Fourier series of f where

$$a_m = \frac{1}{\pi} \int_0^{2\pi} f(t) \cos mt \, dt \qquad \text{and}$$

$$b_m = \frac{1}{\pi} \int_0^{2\pi} f(t) \sin mt \, dt$$

**Example**

Let $C[0, 2\pi]$ has the inner product

$$< f, g >= \int_0^{2\pi} f(t)\, g(t)\, dt$$

and let m and n be unequal positive integers. Show that $\cos mt$ and $\cos nt$ are orthogonal.

**Solution**

When $m \neq n$

$$< \cos mt, \cos nt >= \int_0^{2\pi} \cos mt \cos nt \, dt$$

$$= \frac{1}{2} \int_0^{2\pi} [\cos(mt + nt) + \cos(mt - nt)\, dt$$

$$= \frac{1}{2}\left[\frac{\sin(mt+nt)}{m+n} + \frac{\sin(mt-nt)}{m-n}\right]_0^{2\pi}$$
$$= 0$$

**Example**

Find the nth-order Fourier approximation to the function

$f(t) = t$ on the interval$[0, 2\pi]$.

**Solution**

We compute

$$\frac{a_0}{2} = \frac{1}{2}\cdot\frac{1}{\pi}\int_0^{2\pi} t\,dt = \frac{1}{2\pi}\left[\frac{1}{2}t^2\Big|_0^{2\pi}\right] = \pi$$

and for k>0, using integration by parts,

$$a_k = \frac{1}{\pi}\int_0^{2\pi} t\cos kt\,dt = \frac{1}{\pi}\left[\frac{1}{k^2}\cos kt + \frac{t}{k}\sin kt\right]_0^{2\pi} = 0$$

$$b_k = \frac{1}{\pi}\int_0^{2\pi} t\sin kt\,dt = \frac{1}{\pi}\left[\frac{1}{k^2}\sin kt - \frac{t}{k}\cos kt\right]_0^{2\pi} = -\frac{2}{k}$$

Thus, the nth-order Fourier approximation of $f(t) = t$ is

$$\pi - 2\sin t - \sin 2t - \frac{2}{3}\sin 3t - \cdots - \frac{2}{n}\sin nt$$

The norm of the difference between f and a Fourier approximation is called the mean square error in the approximation.
It is common to write

$$f(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty}(a_m\cos mt + b_m\sin mt)$$

This expression for f (t) is called the Fourier series for f on$[0, 2\pi]$. The term $a_m\cos mt$, for example, is the projection of f onto the one-dimensional subspace spanned by$\cos mt$.

**Example**

Let $q_1(t) = 1,\ q_2(t) = t, and\ q_3(t) = 3t^2 - 4$. Verify that $\{q_1, q_2, q_3\}$ is an orthogonal set in C{-2,2] with the inner product

$$< f, g >= \int_a^b f(t)\, g(t)\, dt$$

**Solution:**

$$< q_1, q_2 >= \int_{-2}^{2} 1.t\, dt = \frac{1}{2} t^2 \Big|_{-2}^{2} = 0$$

$$< q_1, q_3 >= \int_{-2}^{2} 1.(3t^2 - 4)\, dt = (t^2 - 4t)\Big|_{-2}^{2} = 0$$

$$< q_2, q_3 >= \int_{-2}^{2} t.(3t^2 - 4)\, dt = (\frac{3}{4} t^4 - 2t^2)\Big|_{-2}^{2} = 0$$

**Exercise**

1. Find the equation $y = \beta_0 + \beta_1 x$ of the least-squares line that best fits the data points (0, 1), (1, 1), (2, 2), (3, 2).
2. Find the equation $y = \beta_0 + \beta_1 x$ of the least-squares line that best fits the data points (-1, 0), (0, 1), (1, 2,),(2, 4).
3. Find the least-squares line $y = \beta_0 + \beta_1 x$ that best fits the data (-2, 0), (-1, 0), (0, 2,),(1, 4),(2, 4), assuming that the first and last data points are less reliable. Weight them half as much as the three interior points.

4: To make a trend analysis of six evenly spaced data points, one can use orthogonal polynomials with respect to evaluation at the points t=-5, -3, -1, 1, 3and 5

(a). Show that the first three orthogonal polynomials are

$$p_0(t) = 1, \qquad p_1(t) = t, \qquad and \qquad p_2(t) = \frac{3}{8} t^2 - \frac{35}{8}$$

(b) Fit a quadratic trend function to the data (-5, 1), (-3, 1), (-1, 4), (1, 4), (3, 6), (5, 8)

5: For the space $C[0, 2\pi]$ with the inner product defined by

$$< f, g >= \int_0^{2\pi} f(t)\, g(t)\, dt$$

(a) Show that $\sin mt$ and $\sin nt$ are orthogonal when $m \neq n$

(b) Find the third–order Fourier approximation to $f(t) = 2\pi - t$

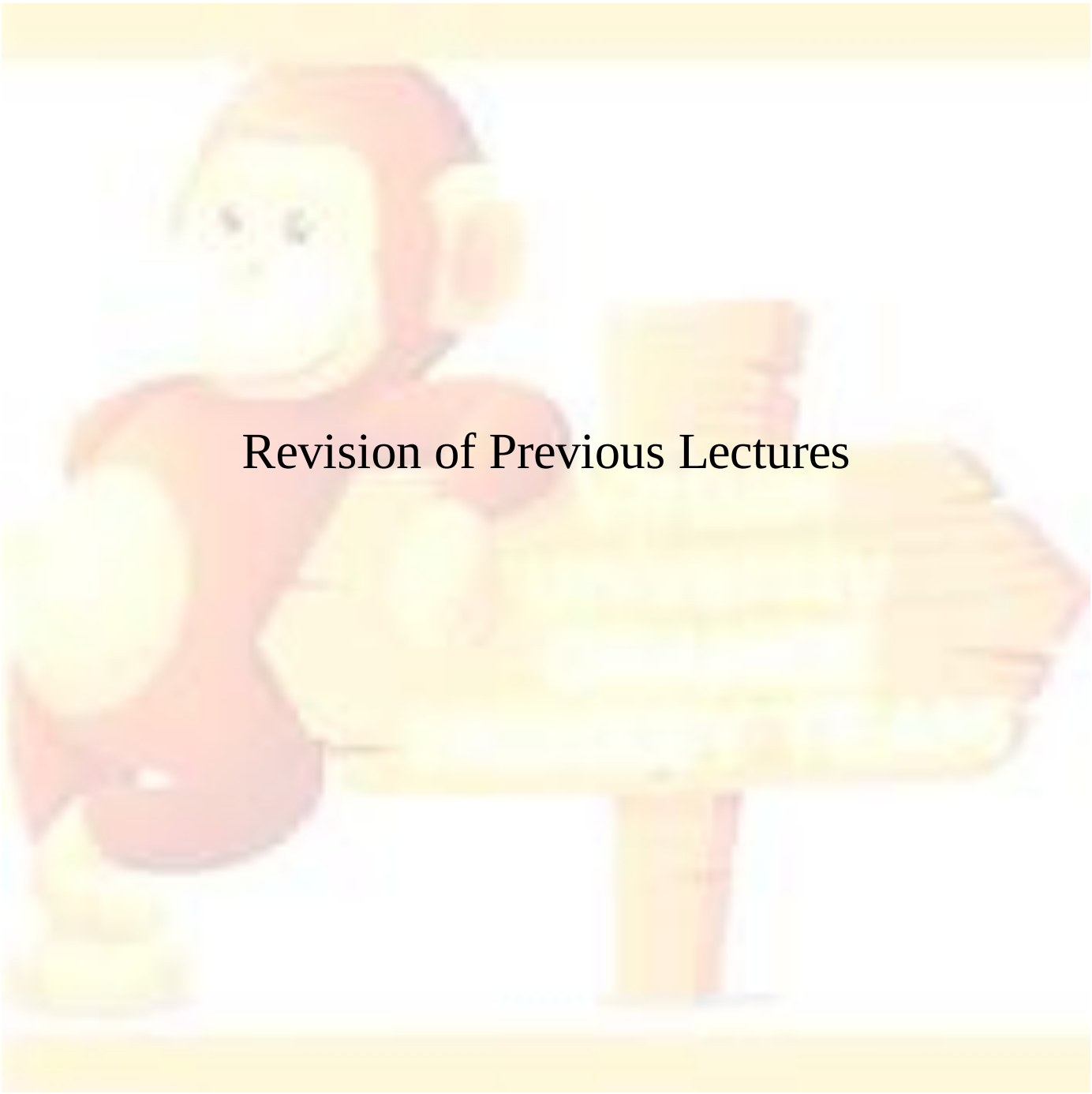
(c) Find the third order Fourier approximation to $\cos^3 t$, without performing any

integration calculations.

6: Find the first-order and third order Fourier approximations to

$$f(t) = 3 - 2\sin t + 5\sin 2t - 6\cos 2t$$





# Revision of Previous Lectures

For any subspace W of a vector space V, which one is not the axiom for subspace.

0 must be in W.
**For all u, v in W and u – v must be in W.**
For all u, v in W and u.v must be in W.
For any scalar k and u in W then k.u in W.

Which one is not the axiom for vector space?
0 + u = u
**0.u = u**
1.u = u
u + v = v + u

The Gauss-Seidel method is applicable to strictly diagonally dominant matrix.
**TRUE**
FALSE

By using determinants, we can easily check that the solution of the given system of linear equation exits and it is unique.

At what condition det(AB)=(detA)(detB) is possible?
**When A and B are n x n matrices**
When A is a row matrix
When A and B are m x n matrices
When B is a column matrix

For any 3x3 matrix A where det (A) = 3, then det (2A) = _____.
24
20
15
**6**

If a multiple of one row of a square matrix A is added to another row to produce a matrix B, then which of the following condition is true?
**detB = detA**
detB = k detA
detA detB = 0
detA detB = detA

The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its main diagonal.
**TRUE**

FALSE

While using the Cramer's rule, if determinant D = 0, and other determinant is not zero then how many solutions are there?
Many solutions
**No solution**
Two solutions
One solution

Which of the following is all permutations of {1,2}?
**(1,2,2,1)**

Question # 1 of 10 ( Start time: 09:52:17 PM ) Total Marks: 1
By using determinants, we can easily check that the solution of the given system of linear equation exits and it is unique.
Select correct option:

FALSE
**TRUE**

Question # 2 of 10 ( Start time: 09:53:11 PM ) Total Marks: 1
If a multiple of one row of a square matrix A is added to another row to produce a matrix B, then which of the following condition is true?
Select correct option:

**detB = k detA**
detB = detA
detA detB = 0
detA detB = detA

Question # 3 of 10 ( Start time: 09:54:09 PM ) Total Marks: 1
At what condition the Cramer's formula is valid for linear systems?
Select correct option:

**When matrix is n x n**
When det(A) is equal to zero
When matrix is m x n
When det(A) in not equal to zero

Question # 4 of 10 ( Start time: 09:54:44 PM ) Total Marks: 1
A matrix has not the same determinant if we add a multiple of a column to another column.

Select correct option:

**TRUE**
FALSE

Question # 5 of 10 ( Start time: 09:55:30 PM ) Total Marks: 1
The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its main diagonal.
Select correct option:

**TRUE**
FALSE

Question # 6 of 10 ( Start time: 09:56:10 PM ) Total Marks: 1
Which of the following is the volume of the parallelepiped determined by the columns of A where A is a 3 x 3 matrix?
Select correct option:

**|det A|**
[A]
det A
A^(-1) ,that is inverse of A

Question # 7 of 10 ( Start time: 09:57:25 PM ) Total Marks: 1
For any 3x3 matrix A where det (A) = 3, then det (2A) = _____.
Select correct option:

24
20
15
**6**

Question # 8 of 10 ( Start time: 09:58:24 PM ) Total Marks: 1
Which one is not the axiom for vector space?
Select correct option:

0 + u = u
**0.u = u**
1.u = u
u + v = v + u
2

Question # 9 of 10 ( Start time: 09:58:58 PM ) Total Marks: 1
Which of the following is NOT the axiom for vector space where u, v, w in V are set of vectors and l, m, n are scalars?
Select correct option:

u + (v + w) = (u + v) + w
**u.v =v.u**
l (u + v)= l u + l v
(l +m) u= I u+ m u

Question # 10 of 10 ( Start time: 09:59:48 PM ) Total Marks: 1
If two rows or columns of a square matrix are identical, then det (A)wil be _____.
Select correct option:

**zero**
non zero
one
positive

Question # 1 of 10 ( Start time: 10:30:38 PM ) Total Marks: 1
If A is strictly diagonally dominant, then A is _____.
Select correct option:

**invertible**
singular
symmetric
scalar

Question # 2 of 10 ( Start time: 10:31:17 PM ) Total Marks: 1
The Gauss-Seidel method is applicable to strictly diagonally dominant matrix.
Select correct option:

**TRUE**
FALSE

Question # 3 of 10 ( Start time: 10:32:00 PM ) Total Marks: 1
If the absolute value of each diagonal entry exceeds the sum of the absolute values of the other entries in the same row then a matrix A is called:
Select correct option:

invertible
**strictly diagonally dominant**
diagonally
scalar

Question # 4 of 10 ( Start time: 10:33:26 PM ) Total Marks: 1
The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its main diagonal.
Select correct option:

**TRUE**
FALSE

Question # 5 of 10 ( Start time: 10:33:52 PM ) Total Marks: 1
Which one is not the axiom for vector space?
Select correct option:

0 + u = u
**0.u = u**
1.u = u
u + v = v + u

Question # 6 of 10 ( Start time: 10:34:16 PM ) Total Marks: 1
Let W = {(x, y) such that x, y in R and x = y}. Is W a vector subspace of plane.
Select correct option:

**YES**
NO

Question # 7 of 10 ( Start time: 10:34:58 PM ) Total Marks: 1
If A is a triangular matrix, then det(A) is the product of the entries on the _____.
Select correct option:

**main diagonal of A**
first two rows of A
diagonal of A
first two columns of A

Question # 8 of 10 ( Start time: 10:35:59 PM ) Total Marks: 1
By using determinants, we can easily check that the solution of the given system of linear equation exits and it is unique.
Select correct option:

FALSE
**TRUE**

Question # 9 of 10 ( Start time: 10:36:55 PM ) Total Marks: 1
If a matrix A is invertible than adj(A) is also invertible.
Select correct option:

**TRUE**
FALSE

Question # 10 of 10 ( Start time: 10:37:57 PM ) Total Marks: 1
If all the entries of a row or a column of a square matrix are zero, then det (A) will be _____.
Select correct option:

**zero**
infinity
one
non zero

/w EPDw UKMTY2
/w EWBgLTm9GL

BC090201047 : Muhmmad Asif

Time Left 57 sec(s)

Quiz Start Time: 08:32 PM

Question # 3 of 10 ( Start time: 08:34:27 PM ) Total Marks: 1

A matrix A and its transpose have the same determinant.

Select correct option:

TRUE

FALSE

Click here to Save Answer & Move to Next Question

/w EPDw UKMTY2
/w EWCgKEw r6D

BC090201047 : Muhmmad Asif

Time Left 48 sec(s)

Quiz Start Time: 08:32 PM

Question # 5 of 10 ( Start time: 08:36:08 PM ) Total Marks: 1

If a system of equations is solved using the Jacobi's method , then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?

Select correct option:

- All of its entries on the diagonal must be zero.
- All of its entries below the diagonal must be zero.
- All of its entries above the diagonal must be zero.
- All of its entries below and above the diagonal must

Click here to Save Answer & Move to Next Question

/w EPDw UKMTY2
/w EWCgLzyYf+C

BC090201047 : Muhmmad Asif

Time Left 66 sec(s)

Quiz Start Time: 08:32 PM

Question # 6 of 10 ( Start time: 08:37:37 PM ) Total Marks: 1

If A is a square matrix, then the Minor of entry ith row and jth column is to be the determinant of the sub matrix that remains when the ith row and jth column of A are:

Select correct option:

- added
- deleted

multiplied

divided

Click here to Save Answer & Move to Next Question

/w EPDw UKMTY2

/w EWCgLH9+aFt

BC090201047 : Muhmmad Asif

Time Left 36 sec(s)

Quiz Start Time: 08:32 PM

Question # 7 of 10 ( Start time: 08:38:54 PM ) Total Marks: 1

If M=[3] then which of the following is the determinant of the matrix M?

Select correct option:

1

[1]

3

[3]

Click here to Save Answer & Move to Next Question

# MTH501 SOLVED MCQS BY $HINING $TAR

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**For which of the matrix, the Gauss-Seidel method is applicable**

/w EPDw UKMTY2

/w EWCgKH3vLPI

BC090201047 : Muhmmad Asif

Time Left 71 sec(s)

Quiz Start Time: 08:32 PM

Question # 9 of 10 ( Start time: 08:41:21 PM ) Total Marks: 1

Which of the following is all permutations of {1,2}?

Select correct option:

- (1,2)
- (2,1)
- (1,2,2,1)
- (1,2) and (2,1)

Click here to Save Answer & Move to Next Question

/w EPDw UKMTY2

/w EWCgLN0IuJD

BC090201047 : Muhmmad Asif

Time Left 56 sec(s)

Quiz Start Time: 08:32 PM

Question # 10 of 10 ( Start time: 08:42:29 PM ) Total Marks: 1


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# MTH501 SOLVED MCQS BY $HINING $TAR

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If M is a square matrix having two rows equal then which of the following about the determinant of the matrix is true?

Select correct option:

- det (M) is not equal to '1'
- det (M)=1flinearly dependent
- det (M) is not equal to '0'
- det (M)=0

Click here to Save Answer & Move to Next Question

Time Left 37 sec(s)

Quiz Start Time: 05:46 PM

Question # 1 of 10 ( Start time: 05:46:20 PM )

If a system of equations is solved using the Gauss-Seidel method, then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?

Select correct option:

- All of its entries on the diagonal must be zero.
- All of its entries below the diagonal must be zero.

All of its entries above the diagonal must be zero.

true

All of its entries below and above the diagonal must

Time Left 81 sec(s)

Quiz Start Time: 05:46 PM

Question # 3 of 10 ( Start time: 05:48:07 PM )

Let W = {(1, y) such that y in R}. Is W a vector subspace of plane.

Select correct option:

YES

NO

true

Time Left 89 sec(s)

Quiz Start Time: 05:46 PM

Question # 4 of 10 ( Start time: 05:48:39 PM )

Which of the following is the volume of the parallelepiped determined by the columns of A where A is a 3 x 3 matrix?

Select correct option:

- |det A| true
- [A]
- det A
- A^(-1) ,that is inverse of A

Time Left 89 sec(s)

Quiz Start Time: 05:46 PM

Question # 5 of 10 ( Start time: 05:50:05 PM )

At what condition the Cramer's rule fails?

Select correct option:

- When the determinant of the coefficient matrix is not
- When matrix is n x n
- When the determinant of the coefficient matrix is zer true

The coefficient matrix A is not invertible

Time Left 74 sec(s)

Quiz Start Time: 05:46 PM

Question # 6 of 10 ( Start time: 05:50:21 PM )

All the lines those passes through origin are not the subspace of a plane.

Select correct option:

FALSE

true

TRUE

Time Left 79 sec(s)

Quiz Start Time: 05:46 PM

Question # 7 of 10 ( Start time: 05:50:55 PM )

A matrix A and its transpose have the same determinant.

Select correct option:

TRUE

true

FALSE

Time Left 83 sec(s)

Quiz Start Time: 05:46 PM

Question # 8 of 10 ( Start time: 05:51:08 PM )

Cramer's rule is a formula for solving systems of equations by _____.

Select correct option:

matrix inversion

determinants

true

comparison

substitutions

Time Left 83 sec(s)

Quiz Start Time: 05:46 PM

Question # 9 of 10 ( Start time: 05:51:30 PM )

If a system of equations is solved using the Jacobi's method , then which of the following is the

most appropriate answer about the matrix M that is derived from the coefficient matrix ?
Select correct option:

- All of its entries on the diagonal must be zero.
- All of its entries below the diagonal must be zero.
- All of its entries above the diagonal must be zero.
- All of its entries below and above the diagonal must

Time Left 87 sec(s)

Quiz Start Time: 05:46 PM

Question # 10 of 10 ( Start time: 05:52:59 PM )

Which of the following is all permutations of {1,2}?
Select correct option:

- (1,2)
- (2,1)

(1,2,2,1)

(1,2) and (2,1)

true

**Consider a system of linear equations A x =b where A is a3 ×3 matrix having 3 pivot positions, then which statement is false about the system Ax =b**

**(a) System has unique solution.**

**(b) Rank of the matrix is 3.**

**(c) There is only one free variable in solution of that system.**

**(d) The associated homogeneous system Ax =0 has only trivial system.**

**If a finite set S of non zero vectors span a vector space V, then some subset of S is a basis for V.**

1. **True**
2. **false**

---

Angle between the vectors $x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ & $y = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ is,

(a) $\frac{\pi}{2}$

(b) $\pi$

(c) 0

(d) $\frac{\pi}{3}$

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If $\lambda$ is an eigenvalue of the invertible matrix A then an eigenvalue of $A^{-1}$ will be

(a) $\lambda^{-1}$

(b) $\frac{1}{\lambda^2}$

(c) $\lambda^2$

(d) Can't be determined.

**D**

**If rank of a3 x 5 matrix is 3 then dimension of its Null space is**

**(a) 0**

**(b) 3**

**(c) 2**

**(d) We can't say anything**

If $\{v_1, v_2, v_3, \ldots, v_n\}$ be the orthogonal set of vectors then which statement(s) must be true.

I. $v_i \cdot v_j = 0$ for all.

II. $v_i \cdot v_i > 0$ for all $i$ and $v_i \neq 0$.

III. Set$\{c_1 v_1, c_2 v_2, c_3 v_3, \ldots, c_n v_n\}$

1 I only.
2 I and II only.
3 II and III only
4 All of the three.

**1**

**If matrix A has zero as an eigenvalue then which statement(s) about A must be true.**

**I. Matrix A is not invertible.**

**II. Matrix A will also have an eigenvalue 2.**

**III. Matrix is diagonalizable.**

**1 II and III only.**

**2 I only. (true)**

**3 II and III only.**

**4 All three.**

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A Linear transformation $T:R^2 \rightarrow R^2$ is such that $T[e_1]=\begin{bmatrix}1\\1\end{bmatrix}$ & $T[e_2]=\begin{bmatrix}0\\-1\end{bmatrix}$ then $T\begin{bmatrix}1\\-3\end{bmatrix}$ is,

1 $\begin{bmatrix}1\\2\end{bmatrix}$

2 $\begin{bmatrix}1\\-3\end{bmatrix}$

3 $\begin{bmatrix}3\\5\end{bmatrix}$

4 $\begin{bmatrix}1\\4\end{bmatrix}$

A linear transformation T : $R^2 \rightarrow R^2$ defined by $T(x)=\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}-x_2\\x_1\end{bmatrix}$, $u=\begin{bmatrix}4\\1\end{bmatrix}$, $v=\begin{bmatrix}2\\3\end{bmatrix}$ then the image of u+v is

- $\begin{bmatrix}4\\6\end{bmatrix}$
- $\begin{bmatrix}4\\-6\end{bmatrix}$
- $\begin{bmatrix}-4\\6\end{bmatrix}$
- $\begin{bmatrix}2\\4\end{bmatrix}$

In the set $S=\{\begin{bmatrix}x-2t\\x+t\\3t\end{bmatrix}$ ;x , t in R $\}$ then its dimension is

- **1**
- **2**
- **0**
- **3**

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Let $A = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix}$ and $u = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$, then u is in

- Column space of A
- Null space of A
- Row space of A
- None of them

Let B = { $b_1$ , $b_2$ } and C={ $c_1$, $c_2$ } for a vector space V such that $b_1 = 4c_1 + c_2$ and $b_2 = -6c_1 + c_2$ . Suppose $x = 3b_1 + b_2$ and $[x]_B = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ then $[x]_C$ will be

- $\begin{bmatrix} 6 \\ 4 \end{bmatrix}$
- $\begin{bmatrix} -4 \\ 6 \end{bmatrix}$
- $\begin{bmatrix} -6 \\ 4 \end{bmatrix}$
- $\begin{bmatrix} -6 \\ -4 \end{bmatrix}$

Consider a basis $B = \{ b_1 , b_2 \}$ for $R^2$ where $b_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and suppose x is in $R^2$ has the Coordinate vector $[x]_B = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ then the value x is

- $\begin{bmatrix} -1 \\ 6 \end{bmatrix}$
- $\begin{bmatrix} -6 \\ 1 \end{bmatrix}$
- $\begin{bmatrix} -1 \\ -6 \end{bmatrix}$
- $\begin{bmatrix} 1 \\ 6 \end{bmatrix}$

In the matrix equation $Ax = b$ if the order of $A$ is $4 \times 3$ and of $b$ is $4 \times 1$ then the order of the column vector $x$ is

**1 x 3**
**3 x 1**
**3 x 3**
**4 x 1**

# MTH501 SOLVED MCQS BY $HINING $TAR

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Transpose of row vectors of a $3 \times 3$ identity matrix

- Span $\mathbb{R}^4$
- are basis for any subspace of $\dim \geq 4$
- are linearly dependent
- are linearly independent

**Determinant of a non-invertible(singular) matrix always**

- **vanish**
- **unity**
- **non zero negative**
- **non zero positive**

**Rank of a zero matrix of any order is**

- **zero**
- **three**
- **four**
- **nine**

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Roots of characteristic equation; $(\lambda+1)^2=0$ are of multiplicity

► **Zero**

► **One**

► **Two**

► **Three**

If there are two vectors parallel to $x$ and $y-axes$ respectively then both vectors

► **Are orthogonal**

►**Having their inner product zero**

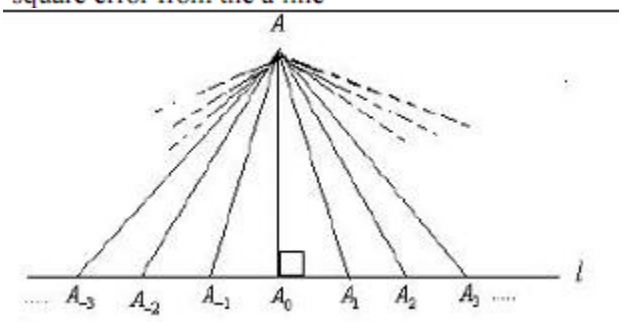
► **Can span a subspace while both passing through the origin**

► **All above statements are equivalent**

# MTH501 SOLVED MCQS BY $HINING $TAR

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Which path in the following figure can give the least square distance and its corresponding least square error from the a line $l$




- $AA_{-1}$ or any path to the left of $A_{-1}$
- $AA_1$ or any path to the right of $A_1$
- Unique path $AA_0$ for which $AA_0$ is orthogonal to line $l$
- None of these

Linear equation $0x + 0y = 5$ has

- **Infinite many solutions**
- **Empty solution**
- **Unique, non-trivial solution**

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► **Unique, trivial solution**

$$T\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_1 \\ -x_2 \end{bmatrix}$$

is an example of transformation from

► $\mathbb{R}^3 \to \mathbb{R}^2$

► $\mathbb{R}^2 \to \mathbb{R}^3$

► $\mathbb{R}^2 \to \mathbb{R}^2$

► $\mathbb{R}^3 \to \mathbb{R}^3$

Dimension of zero vector space $\{0\}$ is

► **Not defined**

►**One**

► **Zero**

► **Arbitrary**

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

As the rank of a matrix is number of its pivot columns then the rank of is

► 4

► 2

► 1

► Inconclusive

If there exists a matrix for which characteristic equation is $\lambda^3 = 1$ then the roots of the equation are

► $1, \frac{-1 \pm \sqrt{-3}}{2}$

► 1

► $\pm 1$

► $\varphi = \{\}$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Distance between the vector and its reflection about $x_2 -$ axis is

► Zero

- 4
- $\sqrt{2}$
- 2

Vector $u$ is not an eigen vector of a matrix $A$ if

- There exists $\lambda$ such that $Au = \lambda u$
- There does not exist $\lambda$ such that $Au = \lambda u$
- $Au = 0$
- $Au = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Linear equation $0x + 0y = 5$ has

- **Infinite many solutions**
- **Empty solution**
- **Unique, non-trivial solution**
- **Unique, trivial solution**

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**Question No: 1 ( Marks: 2 ) - Please choose one**

**The characteristics polynomial for the matrix A= $\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$ is**

- $\lambda^2 - 4\lambda - 45$
- $\lambda^2 + 4\lambda - 45$
- $\lambda^2 + 4\lambda + 45$
- $\lambda^2 - 4\lambda + 45$

**Question No: 2 ( Marks: 2 ) - Please choose one**

**Let A=PA$P^{-1}$ then what will be $A^4$ where A= $\begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix}$ , D= $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$**

- $\begin{bmatrix} 226 & -525 \\ 90 & -209 \end{bmatrix}$
- $\begin{bmatrix} -525 & -226 \\ 90 & -209 \end{bmatrix}$

- $\begin{bmatrix} 266 & -525 \\ -209 & 90 \end{bmatrix}$
- **None of the other**

**Question No: 3 ( Marks: 2 ) - Please choose one**

**The origin of the dynamical system $X_{k+1}$ =A $X_k$ for the matrix A= $\begin{bmatrix} .8 & .3 \\ -.4 & 1.5 \end{bmatrix}$ is**

- **Attractor**
- **Repellor**
- **Saddle point**
- **None of them**

**Question No: 4 ( Marks: 2 ) - Please choose one**

The distance between the vectors u=(7,1) and v=(3,2)is

- $\sqrt{-17}$
- $\sqrt{17}$

- ▶ $\sqrt{7}$
- ▶ $\sqrt{-7}$

**Question No: 5 ( Marks: 2 ) - Please choose one**

$$\begin{bmatrix} 3 \\ -2 \\ 1 \\ 3 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 3 \\ -3 \\ 4 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 8 \\ 7 \\ 0 \end{bmatrix}$$

**The vectors , , are**

- ▶ **Parallel**
- ▶ **Orthogonal**
- ▶ **Not Orthogonal**
- ▶ **Not parallel**

**Question No: 1 ( Marks: 1 ) - Please choose one**

---

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$$

**Reduced echelon form of the matrix is**

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

▶

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

▶

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$$

▶

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

▶

**Question No: 2 ( Marks: 1 ) - Please choose one**

---

**A matrix that results from applying a single elementary row operation to an identity matrix is called**

▶ **Invertible matrix**

▶ **Singular matrix**

▶ **Scalar matrix**

► **Elementary matrix**

**Question No: 3 ( Marks: 1 ) - Please choose one**

---

**For an n×n matrix $(A^t)^t$ =**

► **$A^t$**

► **A**

► **$A^{-1}$**

► **$(A^{-1})^{-1}$**

**Question No: 4 ( Marks: 1 ) - Please choose one**

---

**What is the largest possible number of pivots a 4×6 matrix can have?**

► 4 (true)

► 6

► 10

► 0

**Question No: 5 ( Marks: 1 ) - Please choose one**

---

**The characteristic polynomial of a 5×5 matrix is $\lambda^5 - 4\lambda^4 - 45\lambda^3 = 0$ ,the eigenvalues are**

► **0,-5, 9**

► **0,0,0,5,9**

► **0,0,0,-5,9 (true)**

► **0,0,5,-9**

**Question No: 6 ( Marks: 1 ) - Please choose one**

---

**Find the characteristic equation of the given matrix**

$$\begin{bmatrix} 6 & 8 & 5 & 4 \\ 0 & 2 & -8 & 7 \\ 0 & 0 & 9 & 6 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

- $(6-\lambda)(2-\lambda)(9-\lambda)=0$
- $(6-\lambda)(8-\lambda)(5-\lambda)(4-\lambda)=0$
- $(6-\lambda)^2(2-\lambda)(9-\lambda)=0$
- $(6-\lambda)(6-\lambda)(7-\lambda)(4-\lambda)=0$

**Question No: 7 ( Marks: 1 ) - Please choose one**

---

**A is diagonalizable if $A=PDP^{-1}$ Where**

- **D is any matrix and P is an invertible matrix**
- **D is a diagonal matrix and P is any matrix**
- **D is a diagonal matrix and P is invertible matrix**
- **D is a invertible matrix and P is any matrix**

**Question No: 8 ( Marks: 1 ) - Please choose one**

---

**The inverse of an invertible lower triangular matrix is**

- ► **lower triangular matrix**
- ► **upper triangular matrix**
- ► **diagonal matrix**

**Question No: 9 ( Marks: 1 ) - Please choose one**

---

**If P is a parallelepiped in $R^3$, then**
**{volume of T (P)} = |detA|. {volume of P}**

- ► **Where T is determined by a $2\times2$ matrix A**
- ► **Where T is determined by a $2\times3$ matrix A**
- ► **Where T is determined by a $3\times3$ matrix A**
- ► **Where T is determined by a $3\times2$ matrix A**

**Question No: 10 ( Marks: 1 ) - Please choose one**

---

**Let A be a $n \times m$ matrix of rank $r$ then row space of A has dimension**

- $m-r$
- $n-r$
- $r$
- $nm$

**Question No: 11 ( Marks: 1 ) - Please choose one**

---

**The dimension of the vector space $P_4$ is**

- 4
- 3
- 5
- 1

**Question No: 12 ( Marks: 1 ) - Please choose one**

---

**Let** $u=(3,-2), v=(4,5)$ **.For the weighted Euclidean inner product** $\langle u,v\rangle = 4u_1v_1 + 5u_2v_2$

$\langle v,u\rangle =$

- ► 2
- ► **-2**
- ► 3
- ► -3

**Question No: 13 ( Marks: 1 ) - Please choose one**

---

**Let A be** $n\times n$ **matrix whose entries are real. If** $\lambda$ **is an eigenvalue of A with x a corresponding eigenvector in** $\mathbb{C}^n$**, then**

- ► $A\bar{x} = \lambda\bar{x}$
- ► $A\bar{x} = \bar{\lambda}\bar{x}$
- ► $A\bar{x} = \bar{\lambda}x$

▶ $A\bar{x} = \lambda^{-1}\bar{x}$

**Question No: 14 ( Marks: 1 ) - Please choose one**

---

$$A = \begin{bmatrix} 1.25 & -.75 \\ -.75 & 1.25 \end{bmatrix}$$

**Suppose that has eigenvalues 2 and 0.5 .Then origin is a**

▶ **Saddle point**

▶ **Repellor**

▶ **Attractor**

**Question No: 15 ( Marks: 1 ) - Please choose one**

---

**Which one is the numerical method used for approximation of dominant eigenvalue of a matrix.**

▶ **Power method**

▶ **Jacobi's method**

▶ **Guass Seidal method**

► Gram Schmidt process

**Question No: 16 ( Marks: 1 ) - Please choose one**

---

**The matrix equation $A^T A\hat{x} = A^T b$ represents a system of linear equations commonly referred to as the**

► normal equations for $x$

► normal equations for $\hat{x}$

► normal equations for $A$

► normal equations for $b$

**Question No: 17 ( Marks: 1 ) - Please choose one**

---

**Let $A$ have eigenvalues 2, 5, 0,-7, and -2. Then the dominant eigenvalue for A is**

► $\lambda = 5$

- $\lambda = 0$
- $\lambda = -7$
- $\lambda = 2$

**Question No: 18 ( Marks: 1 ) - Please choose one**

---

**If W is a subspace of $\mathbb{R}^m$, then the transformation $T : \mathbb{R}^m \to W$ that maps each vector x in $\mathbb{R}^m$ into its orthogonal x in W is called the orthogonal projection of**

- $\mathbb{R}^m$ **in** $\mathbb{R}^m$
- $\mathbb{R}^m$ **in W**
- $\mathbb{R}^m$ **in x**

**Question No: 19 ( Marks: 1 ) - Please choose one**

---

**If** $V = \mathbb{R}^n$ , $B = \{b_1, b_2\}$ **and** $C = \{c_1, c_2\}$ **then row reduction of** $\begin{bmatrix} c_1 & c_2 & b_1 & b_2 \end{bmatrix}$ **to** $\begin{bmatrix} I & P \end{bmatrix}$

**Produces a matrix P that satisfies**

- ▶ $[x]_B = P[x]_B$ **for all x in V**
- ▶ $[x]_C = P[x]_B$ **for all x in V**
- ▶ $[x]_B = P[x]_C$ **for all x in V**
- ▶ $[x]_B = [x]_C$ **for all x in V**

**Question No: 20 ( Marks: 1 ) - Please choose one**

---

**The Casorati matrix for the signals *1*$^k$, *(-2)*$^k$ and *3*$^k$ is**

$$\begin{bmatrix} 1^k & (-2)^k & 3^0 \\ 1^{k+1} & (-2)^{k+1} & 3^1 \\ 1^{k+2} & (-2)^{k+2} & 3^2 \end{bmatrix}$$

- ▶

$$\begin{bmatrix} 1^{k} & (-2)^{k} & 3^{k} \\ 1^{k+1} & (-2)^{k+1} & 3^{k+1} \\ 1^{k+2} & (-2)^{k+2} & 3^{k+2} \end{bmatrix}$$

▶

$$\begin{bmatrix} 1^{0} & (-2)^{k} & 3^{k} \\ 1^{1} & (-2)^{k+1} & 3^{k+1} \\ 1^{2} & (-2)^{k+2} & 3^{k+2} \end{bmatrix}$$

▶

Which statement about the set S is false where S = {(1, 1, 3), (2, 3, 7),(2, 2, 6)}
(a) The set S contain an element which is solution of the equation 5x – y- z = 0
**(b) The Set S is linearly independent.**
(c) The set S contain two elements which are multiple of each other.
(d) The Set S is linearly dependent.

How many subspaces $R_2$ have?
(a) only two: {0} and $R_2$
(b) Only four: {0} *x*- axis and *y* -axis and $R_2$
**(c) Infinitely many.**
(d) None of the above.

The set of vectors {(5,0,0), (7,2,-6), (9,4,-8)} is,

- a) Linearly independent
- **b) Linearly dependent**
- c) Basis of $R^3$

**Question No: 1 ( Marks: 1 ) - Please choose one**

If A is a $2\times 2$ matrix, the area of the parallelogram determined by the columns of A is

▶ $A^{-1}$
▶ **det A**

▶ adj A

**Question No: 2 ( Marks: 1 ) - Please choose one**

Cramer's rule leads easily to a general formula for

► **the inverse of an $n\times n$ matrix A**

► the adjugate of an $n\times n$ matrix A

► the determinant of an $n\times n$ matrix A

**Question No: 3 ( Marks: 1 ) - Please choose one**

The transpose of an lower triangular matrix is

► lower triangular matrix
► **upper triangular matrix**

► diagonal matrix

**Question No: 4 ( Marks: 1 ) - Please choose one**

The transpose of an upper triangular matrix is

► **lower triangular matrix**
► upper triangular matrix

► diagonal matrix

**Question No: 5 ( Marks: 1 ) - Please choose one**

Let A be a square matrix of order $3\times 3$ with $\det(A)=21$, then $\det(2A)=$

► **168**
► 186
► 21
► 126

**Question No: 6 ( Marks: 1 ) - Please choose one**

A basis is a linearly independent set that is as large as possible.

► **True**
► False

**Question No: 7 ( Marks: 1 ) - Please choose one**

Let A be an $m \times n$ matrix. If for each b in $\mathbb{R}^m$ the equation Ax=b has a solution then

► **A has pivot position in only one row (may be this option is true)**

► Columns of A span $\mathbb{R}^m$
► Rows of A span $\mathbb{R}^m$

**Question No: 8 ( Marks: 1 ) - Please choose one**

Reduced echelon form of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$ is

$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$

► $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$

► $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

►

**Question No: 22 ( Marks: 3 )**

find that is invertible or not $T(X_1,X_2)=T(6X_1+8X_2 , 5X_1-8X_2)$

**Question No: 23 ( Marks: 3 )**

Find the volume of parallelogram of the vertices (1,2,4) (2,4,-7) and (-1,-3,20

**Question No: 24 ( Marks: 2 )**

Which of the following is true? If V is a vector space over the field F.(justify your answer)

(a) $\{x+y / x \in V, y \in V\} = V$

(b) $\{x+y / x \in V, y \in V\} = V \times V$

(c) $\{\lambda v / v \in V, \lambda \in F\} = F \times V$

**Question No: 25 ( Marks: 5 )**

Let $\begin{bmatrix}1\\0\\-2\end{bmatrix}, \begin{bmatrix}3\\2\\-4\end{bmatrix}, \begin{bmatrix}-3\\-5\\1\end{bmatrix}$. is this in $R^3$ or not?

**Question No: 26 ( Marks: 5 )**

Justify that $A^2=I$ if $A=\begin{bmatrix}1 & 0\\3 & -1\end{bmatrix}$, if and only $M^2=I$. justify your answer by portioned matrix of M

$M=\begin{bmatrix}1 & 0 & 0 & 0\\3 & -1 & 0 & 0\\1 & 0 & -1 & 0\\0 & 1 & -3 & 1\end{bmatrix}$

**Question No: 1 ( Marks: 1 ) - Please choose one**

If A is a $2\times 2$ matrix, the area of the parallelogram determined by the columns of A is

► $A^{-1}$
► **det A**

► adj A

**Question No: 2 ( Marks: 1 ) - Please choose one**

Cramer's rule leads easily to a general formula for

► **the inverse of an $n \times n$ matrix A**

► the adjugate of an $n \times n$ matrix A

► the determinant of an $n \times n$ matrix A

**Question No: 3 ( Marks: 1 ) - Please choose one**

The transpose of an upper triangular matrix is

► lower triangular matrix
► upper triangular matrix

► **diagonal matrix**

**Question No: 4 ( Marks: 1 ) - Please choose one**

Let A be a square matrix of order $3 \times 3$ with $\det(A) = 21$, then $\det(2A) =$

► **168**
► 186
► 21
► 126

**Question No: 5 ( Marks: 1 ) - Please choose one**

A basis is a linearly independent set that is as large as possible.

► **True**

► False

**Question No: 6 ( Marks: 1 ) - Please choose one**

Col

A is all of $\mathbb{R}^m$ if and only if

► the equation $Ax = 0$ has a solution for each b in $\mathbb{R}^m$

► **the equation $Ax = b$ has a solution for each b in $\mathbb{R}^m$**

► the equation $Ax = b$ has a solution for a fixed b in $\mathbb{R}^m$.

**Question No: 7 ( Marks: 1 ) - Please choose one**

If

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

and , then the partitions of A and B

► are not conformable for block multiplication

► **are conformable for AB block multiplication**

► are not conformable for BA block multiplication

**Question No: 8 ( Marks: 1 ) - Please choose one**

Two

vectors are linearly dependent if and only if they lie

► on a line parallel to x-axis

► **on a line through origin**

► on a line parallel to y-axis

**Question No: 9 ( Marks: 1 ) - Please choose one**

The equation x = p + t v describes a line

- ► through v parallel to p
- ► through p parallel to v
- ► **through origin parallel to p**

**Question No: 10 ( Marks: 1 ) - Please choose one**

Let A be an $m \times n$ matrix. If for each b in $\mathbb{R}^m$ the equation Ax=b has a solution then

- ► **A has pivot position in only one row**
- ► Columns of A span $\mathbb{R}^m$
- ► Rows of A span $\mathbb{R}^m$

**Question No: 11 ( Marks: 1 ) - Please choose one**

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 8 \\ 2x_2 - 7x_3 &= 0 \\ -4x_1 + 3x_2 + 9x_3 &= -6 \end{aligned}$$

Given the system the augmented matrix for the system is

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -7 \\ -4 & 3 & 9 \end{bmatrix}$$

- ►
$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -7 & 8 \\ -4 & 3 & 9 & -6 \end{bmatrix}$$
- ►

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ -4 & 5 & 9 \end{bmatrix}$$

▶

▶ $\begin{bmatrix} 1 & -2 & 1 & 8 \\ 0 & 2 & -7 & 0 \\ -4 & 3 & 9 & -6 \end{bmatrix}$

**Question No: 12 ( Marks: 1 ) - Please choose one**

---

Consider the linear transformation $T$ such that $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ is the matrix of linear transformation then $T \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$ is

$$\begin{bmatrix} 10 \\ 4 \\ 2 \end{bmatrix}$$

▶

$$\begin{bmatrix} 1 \\ 0 \\ 9 \end{bmatrix}$$

▶

$$\begin{bmatrix} 10 \\ 4 \\ 1 \end{bmatrix}$$

▶

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \end{bmatrix}$$

►

**Question No: 13 ( Marks: 1 ) - Please choose one**

---

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 5 \qquad \begin{vmatrix} a & b & c \\ 3d & 3e & 3f \\ g & h & i \end{vmatrix}$$

If then will be

► 15
► **45**
► 135
► 60

**Question No: 14 ( Marks: 1 ) - Please choose one**

---

For an n×n matrix $(A^t)^t$ =

► $A^t$

► **A**

► $A^{-1}$

► $(A^{-1})^{-1}$

**Question No: 15 ( Marks: 1 ) - Please choose one**

---

Each Linear Transformation T from $R^n$ to $R^m$ is equivalent to multiplication by a matrix A of order

► m´n

- ▶ **n´m**
- ▶ n´n
- ▶ m´m

**Question No: 16 ( Marks: 1 ) - Please choose one**

---

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$$

Reduced echelon form of the matrix is

- $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$
- ▶ $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$
- ▶ $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$
- ▶

**Question No: 17 ( Marks: 2 )**

---

Find vector and parametric equations of the plane that passes through the origin of $\boldsymbol{R^3}$ and is parallel to the vectors $\boldsymbol{v_1}$ = (1, 2, 5) and $\boldsymbol{v_2}$ = (5, 0, 4).

**Question No: 18 ( Marks: 2 )**

---

Which of the following is true? If V is a vector space over the field F.(justify your answer)

(a) $\{x+y / x \in V, y \in V\} = V$

(b) $\{x+y / x \in V, y \in V\} = V \times V$

(c) $\{\lambda v / v \in V, \lambda \in F\} = F \times V$

**Question No: 19 ( Marks: 3 )**

Let $v_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix}, and\ y = \begin{bmatrix} h \\ -3 \\ -5 \end{bmatrix}.$

For what value(s) of h is ***y*** in the plane generated by $v_1$ and $v_2$?

**Question No: 20 ( Marks: 5 )**

With T defined by T(x)= Ax , find a vector x whose image under T is b, and determine whether x is unique.

$$\begin{bmatrix} 1 & -5 & -7 \\ -3 & 7 & 5 \end{bmatrix}, b = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

**Question No: 21 ( Marks: 10 )**

Given A and b , write the augmented matrix for the linear system that corresponds to the matrix equation $Ax = b$ .Then solve the system and write the solution as a vector.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ -3 & -1 & 2 \\ 0 & 5 & 3 \end{bmatrix} \qquad b = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

**Question No: 1 ( Marks: 1 ) - Please choose one**

If for a linear transformation the equation T(x)=0 has only the trivial solution then T is

- one-to-one
- onto

**Question No: 2 ( Marks: 1 ) - Please choose one**

Which one of the following is an elementary matrix?

- $\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 1 \\ 0 & -3 & -3 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 2 & -3 \end{bmatrix}$
- $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$
-

**Question No: 3 ( Marks: 1 ) - Please choose one**

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and let k be a scalar .A formula that relates det kA to k and det A is

- det kA= k det A
- det kA = det (k+A)

► **det k A = $k^2$ det A**

► **det kA = det A**

**Question No: 4 ( Marks: 1 ) - Please choose one**

---

**The equation x = p + t v describes a line**

► **through v parallel to p**

► **through p parallel to v**

► **through origin parallel to p**

**Question No: 5 ( Marks: 1 ) - Please choose one**

---

**Determine which of the following sets of vectors are linearly dependent.**

► $v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

► $v_1 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$

► $v_1 = \begin{bmatrix} 5 \\ 2 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 10 \\ 4 \\ 6 \end{bmatrix}$

►

**Question No: 6 ( Marks: 1 ) - Please choose one**

---

**Every linear transformation is a matrix transformation**

- ► **True**
- ► **False**

**Question No: 7 ( Marks: 1 ) - Please choose one**

---

**A null space is a vector space.**

- ► **True**
- ► **False**

**Question No: 8 ( Marks: 1 ) - Please choose one**

---

**If two row interchanges are made in succession, then the new determinant**

- ► **equals to the old determinant**
- ► **equals to -1 times the old determinant**

**Question No: 9 ( Marks: 1 ) - Please choose one**

---

**The determinant of A is the product of the pivots in any echelon form U of A , multiplied by $(-1)^r$, Where r is**

- ► **the number of rows of A**
- ► **the number of row interchanges made during row reduction from A to U**
- ► **the number of rows of U**

► the number of row interchanges made during row reduction U to A

**Question No: 10 ( Marks: 1 ) - Please choose one**

If A is invertible, then $\det(A)\det(A^{-1})=1$.

► True
► False

**Question No: 11 ( Marks: 1 ) - Please choose one**

A square matrix $A=[a_{ij}]$ is lower triangular if and only if $a_{ij}=0$ for

► $i>j$
► $i<j$
► $i\leq j$
► $i=j$

**Question No: 12 ( Marks: 1 ) - Please choose one**

The product of upper triangular matrices is

► lower triangular matrix

► upper triangular matrix
► diagonal matrix

**Question No: 13 ( Marks: 1 ) - Please choose one**

**The matrix multiplication is associative**

- ► **True**
- ► **False**

**Question No: 14 ( Marks: 1 ) - Please choose one**

**We can add the matrices of _____.**

- ► **same order**
- ► **same number of columns.**
- ► **same number of rows**
- ► **different order**

**Question No: 15 ( Marks: 1 ) - Please choose one**

By solving system of equations with iterative method, we stop the process when the entries in two successive iterations are _____.

- ► **repeat**
- ► **large difference**
- ► **different**

**Question No: 16 ( Marks: 1 ) - Please choose one**

Jacobi's Method is _____ converges to solution than Gauss Siedal Method.

- ► **slow**
- ► **fast**
- ► **better**

**Question No: 17 ( Marks: 1 ) - Please choose one**

A system of linear equations is said to be homogeneous if it can be written in the form _____.

- ► AX = B
- ► AX = 0
- ► AB = X

- $X = A^{-1}$

**Question No: 18 ( Marks: 1 ) - Please choose one**

The row reduction algorithm applies only to augmented matrices for a linear system.

- **True**
- **False**

**Question No: 19 ( Marks: 1 ) - Please choose one**

Whenever a system has no free variable, the solution set contains many solutions.

- **True**
- **False**

**Question No: 20 ( Marks: 1 ) - Please choose one**

Which of the following is not a linear equation?

- $x_1 + 4x_2 + 1 = x_3$
- $x_1 = 1$
- $x_1 + 4x_2 - \sqrt{2}x_3 = \sqrt{4}$
- $x_1 + 4x_1x_2 - \sqrt{2}x_3 = \sqrt{4}$

**Question No: 21 ( Marks: 2 )**

If a square idempotent matrix A is non singular then show that A is equal to the identity matrix I.

**Question No: 22 ( Marks: 2 )**

---

$$v_1 = \begin{bmatrix} 7 \\ 4 \\ -9 \\ -5 \end{bmatrix}, v_2 = \begin{bmatrix} 4 \\ -7 \\ 2 \\ 5 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ -5 \\ 3 \\ 4 \end{bmatrix}$$

Let , and $H = span\{v_1, v_2, v_3\}$ .It can be verified that $v_1 - 3v_2 + 5v_3 = 0.$ Use this information to find a basis for H.

**Question No: 23 ( Marks: 3 )**

---

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{vmatrix}$$

Find

**Question No: 24 ( Marks: 3 )**

---

Determine bases for the plane 3x – 2y + 5z = 0 as a subspace of $R^3$

**Question No: 25 ( Marks: 5 )**

---

$$\begin{bmatrix} I & 0 \\ A & I \end{bmatrix}$$

Show that is invertible and find its inverse.

**Question No: 26 ( Marks: 5 )**

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---

Find the condition for r and s such that the vectors (r,2,s) , (r+1,2,1) and (3,s,1) are linear dependent.

Matrix $\begin{bmatrix}3^T\\1\end{bmatrix}$ is an example of

► Non-Singular matrix
► Square matrix
► Column vector
► Row vector

Standard matrix for transformation T(x1, x2 ) = (−x1 + x2 , x1 − x2 )is

- $\begin{bmatrix}-1 & 1\\1 & -1\end{bmatrix}$
- $\begin{bmatrix}1 & -1\\-1 & 1\end{bmatrix}$
- $\begin{bmatrix}-1 & 1\\-1 & 1\end{bmatrix}$
- None of these

Matrix $\begin{matrix}a & b\\c & d\end{matrix}$ is singular if

► Ad=bc
► ad ≠ bc
► ad − bc = 1
► None of these

## $$ QUIZ....

All the lines those passes through origin are not the subspace of a plane.

**Select correct option:**

FALSE

CORRECT

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TRUE

Click here to Save Answer & Move to Next Question

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Time Left 17 sec(s)

**Quiz Start Time: 09:54 PM**

**Question # 2 of 10 ( Start time: 09:55:53 PM )** **Total Marks:** 1

If a system of equations is solved using the Gauss-Seidel method, then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?

**Select correct option:**

All of its entries on the diagonal must be zero.

All of its entries below the diagonal must be zero.

All of its entries above the diagonal must be zero.

CORRECT

All of its entries below and above the diagonal must

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Why inverse of the matrix A= [1 2] is NOT possible?

**Select correct option:**

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- Because it is a square matrix.
- Because it is a zero matrix.
- Because it is an identity matrix.
- Because it is a rectangular matrix.

CORRECT

Let W = {(1, y) such that y in R}. Is W a vector subspace of plane.

**Select correct option:**

- YES
- NO

CORRECT

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Time Left 7 sec(s)

**Quiz Start Time: 09:54 PM**

**Question # 5 of 10 ( Start time: 10:00:20 PM )** **Total Marks:** 1

If M is a square matrix having two rows equal then which of the following about the determinant of the matrix is true?

**Select correct option:**

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- det (M) is not equal to 1'
- det (M)=1
- det (M) is not equal to 0'
- det (M)=0

CORRECT

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Time Left 12 sec(s)

**Quiz Start Time: 09:54 PM**

**Question # 6 of 10 ( Start time: 10:01:48 PM )** **Total Marks:** 1

If a system of equations is solved using the Jacobi's method , then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?

**Select correct option:**

- All of its entries on the diagonal must be zero.
- All of its entries below the diagonal must be zero.

All of its entries above the diagonal must be zero.

All of its entries below and above the diagonal must

CORRECT

Click here to Save Answer & Move to Next Question

Which of the following is the volume of the parallelepiped determined by the columns of A where A is a 3 x 3 matrix?

**Select correct option:**

|det A|

CORRECT

[A]

det A

A^(-1) ,that is inverse of A

Click here to Save Answer & Move to Next Question

If all the entries of a row or a column of a square matrix are zero, then det (A) will be _____.

**Select correct option:**

zero

CORRECT

- infinity
- one
- non zero

If both the Jacobi and Gauss-Seidel sequences converge for the solution of Ax=b, for any initial x(0), then which of the following is true about both the solutions?

**Select correct option:**

- No solution
- Unique solution

CORRECT

- Different solutions
- Infinitely many solutions

Click here to Save Answer & Move to Next Question

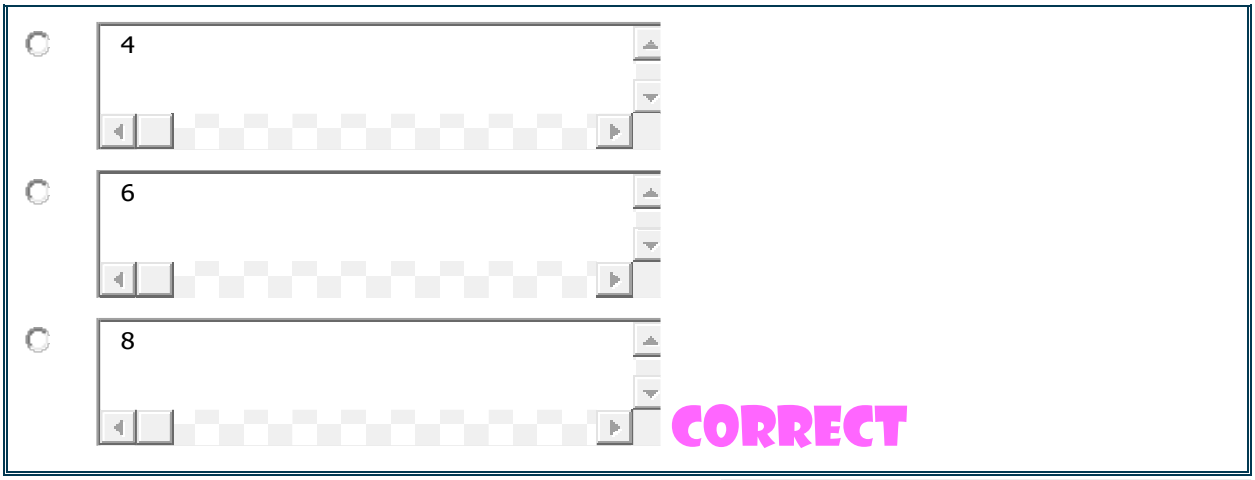
How many different permutations are there in the set of integers {1,2,3}?

**Select correct option:**

- 2

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## LATEST 9th Dec 2011 Paper:

**Q=26** **Let w be the set of all vectors of the form** $\begin{bmatrix} 5b+2c \\ b \\ c \end{bmatrix}$ **,where ' b' and 'c' are arbitrary scalars.show that w is a subspace of $R^3$.**

**SOLUTION:**

W is a subspace of $R^3$ as

$$W = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} : a = 5b + 2c, where\, b\, and\, c\, are\, the\, arbitrary \right\}$$

Reference to the scalars b and c "arbitrary" means that the scalars can be any real numbers.

**Q=25** **compute AB using block multiplication ,where** $A = \left(\begin{array}{cc|c} 1 & 2 & 1 \\ 3 & 4 & 0 \\ \hline 0 & 0 & 2 \end{array}\right)$ **and**

$$B = \begin{pmatrix} 1 & 2 & 3 & 1 \\ 4 & 5 & 6 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \times \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$$

$$A_{11} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, A_{12} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, A_{21}\begin{bmatrix} 0 & 0 \end{bmatrix}, A_{22}\begin{bmatrix} 2 \end{bmatrix}$$

$$B_{11} = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}, B_{12} = \begin{pmatrix} 3 & 1 \\ 6 & 1 \end{pmatrix}, B_{21}\begin{bmatrix} 0 & 0 \end{bmatrix}, B_{22} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$AB = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}$$

$$A_{11}B_{11} + A_{12}B_{21} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}\begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$= \begin{pmatrix} 9 & 12 \\ 19 & 26 \end{pmatrix}$$

$$A_{11}B_{12} + A_{12}B_{22} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 3 & 1 \\ 6 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}\begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$= \begin{pmatrix} 15 & 3 \\ 33 & 7 \end{pmatrix} + \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$A_{21}B_{11} + A_{22}B_{21} = \begin{bmatrix} 0 & 0 \end{bmatrix}\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} + \begin{bmatrix} 2 \end{bmatrix}\begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} + \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$A_{21}B_{12} + A_{22}B_{22} = \begin{bmatrix} 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 \end{bmatrix}\begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 2 \end{bmatrix}$$

$$AB = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix} = \begin{pmatrix} 9 & 12 & 15 & 3 \\ 19 & 26 & 33 & 7 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \text{..Answer}$$

**Q =24 let W be the set of all vectors of the form** $\begin{pmatrix} 3a + 4b \\ 2a + 6b \\ 7a + 9b \end{pmatrix}$ **, where 'a' and 'b' are**

**arbitrary scalars.find the vectors $\vec{U}$ and $\vec{V}$ such that w= span ($\vec{U}$, $\vec{V}$ ) ……3 marks**

Answer.

$$= a\begin{pmatrix}3\\2\\7\end{pmatrix} + b\begin{pmatrix}4\\6\\9\end{pmatrix}$$

therefore, $\{u,v\} = \left\{\begin{pmatrix}3\\2\\7\end{pmatrix}, \begin{pmatrix}4\\6\\9\end{pmatrix}\right\}$ are the vectors in w such that w=span $\{u,v\}$

**Q=23 find the increase of the matrix.** $A = \begin{pmatrix}1 & 0 & 0\\3 & 1 & 0\\0 & 0 & 1\end{pmatrix}$ **using the inversion**

algorithm…..3marks

answer 23

$$A = \begin{pmatrix}1 & 0 & 0\\3 & 1 & 0\\0 & 0 & 1\end{pmatrix}$$

$$AI = \left(\begin{array}{ccc|ccc}1 & 0 & 0 & 1 & 0 & 0\\3 & 1 & 0 & 0 & 1 & 0\\0 & 0 & 1 & 0 & 0 & 1\end{array}\right)$$

$R_2 \leftarrow R_2 - 3R_1$

$$\left(\begin{array}{ccc|ccc}1 & 0 & 0 & 1 & 0 & 0\\0 & 1 & 0 & -3 & 1 & 0\\0 & 0 & 1 & 0 & 0 & 1\end{array}\right)$$

therefore $A^{-1} = \begin{pmatrix}1 & 0 & 0\\-3 & 1 & 0\\0 & 0 & 1\end{pmatrix}$

**Q=22 = determine whether the set v = {(x,y,z)| x,y,z ∈ R and x,y =11} is a subspace of $R^3$ or not….2**

augmented matrix is $\left(\begin{array}{cc|c} 4 & 3 & 4 \\ 5 & 8 & 6 \end{array}\right)$

$R_2 \leftarrow 4R_2 - 5R_1$ $\left(\begin{array}{cc|c} 4 & 3 & 4 \\ 0 & 17 & 4 \end{array}\right)$

$R_1 \leftarrow 17R_1 - 3R_1$ $\left(\begin{array}{cc|c} 68 & 0 & 56 \\ 0 & 17 & 4 \end{array}\right)$

From above matrix it is clear that in 2x2 system, no free variable exists, neither there is any inconsistency such as sum of zero coefficients equal to non-zero real number. therefore, given system has unique solution

**Q=21 show that the following system of linear equations has unique solutions.**
**$4x_1 + 3x_2 = 4$**
**$5x_1 + 8x_2 = 6$**

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Time Left 9 sec(s)

**Quiz Start Time: 11:11 PM**

**Question # 1 of 10 ( Start time: 11:11:37 PM )** **Total Marks:** 1

Let t be any m x n matrix with orthonormal columns and v be any vector then ||t . v || = _____.

**Select correct option:**

|| v ||

| v |

Correct

v

t . || v ||

Click here to Save Answer & Move to Next Question

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/wEWBgLD3NfSD

Time Left 69 sec(s)

**Quiz Start Time: 11:11 PM**

**Question # 3 of 10 ( Start time: 11:14:36 PM )** **Total Marks:** 1

If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

**Select correct option:**

TRUE

Correct

FALSE

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Time Left 19 sec(s)

**Quiz Start Time: 11:11 PM**

**Question # 4 of 10 ( Start time: 11:15:21 PM )** **Total Marks:** 1

If A^2 (A square) is the _____, then the only eigen value of A is zero.

**Select correct option:**

- zero matrix — Correct
- scalar matrix
- symmetric matrix
- identity matrix

Click here to Save Answer & Move to Next Question

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Time Left 9 sec(s)

**Quiz Start Time: 11:11 PM**

**Question # 5 of 10 ( Start time: 11:16:45 PM )** **Total Marks:** 1

Let t be any m x n matrix with orthonormal columns and v be any vector then ||t . v || = t . || v ||.

**Select correct option:**


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TRUE

FALSE

Correct

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If a square matrix has orthonormal columns, then it also has _____ rows.

**Select correct option:**

orthonormal

Correct

orthogonal

Click here to Save Answer & Move to Next Question

If x is _____ to both u and v, then x must be orthogonal to u – v.

**Select correct option:**

orthogonal

Correct

orthonormal

If a _____ matrix has orthonormal columns, then it also has orthonormal rows.

**Select correct option:**

- square
- singular
- diagonal
- rectangular

Correct

For any vectors u and v, the length of vector u – v will be || u – v ||.

**Select correct option:**

- TRUE
- FALSE

Correct

n x n matrix A is invertible if and only if _____ is not an eigen value of A.

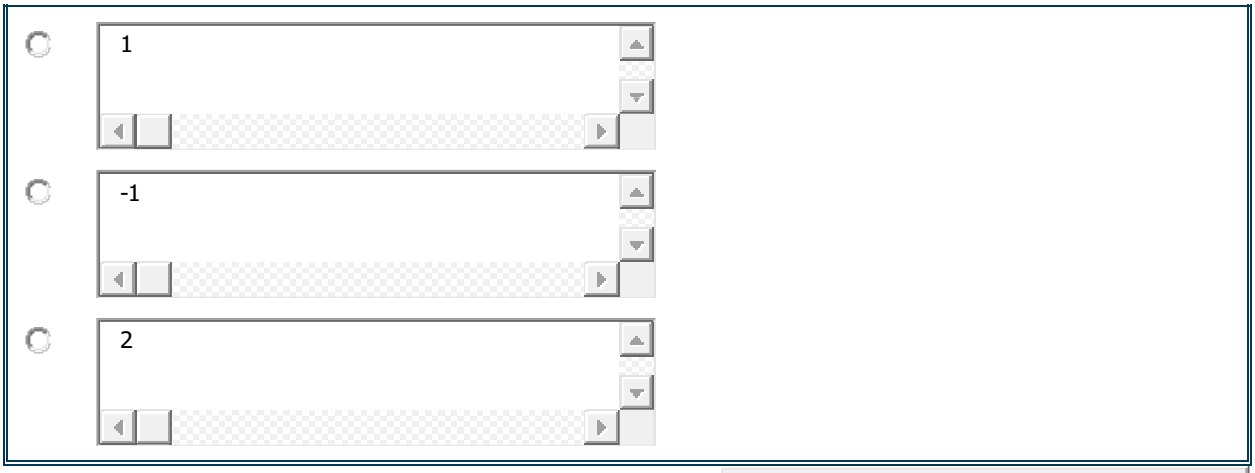
**Select correct option:**

- 0

Correct

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1. If w is the orthogonal projection of a vector v in R^n onto a subspace W of

R^n then w is orthogonal to v

False

========

2. Let W be a subspace of R^n and v be a vector in R^n. Among all vectors in W,

the vector closet to v is the orthogonal projection of v onto W

False

=========

3. The set of all vectors in R^n orthogonal to one fixed vector is a subspace

of R^n

True

================

4. If W is a subspace of R^n, then W and W + have no vectors in common

False

=========

5. If a square matrix has orthonormal columns, then it also has orthonormal

```
rows

        True
```

AR Lucky Team

# MTH501 Assignment Spring 2023

# Section: Saima Shafi




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# AR Lucky Team

**Question:**

To determine whether the vectors

$$v_1 = \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \text{ form a basis for } R^3,$$

We need to check two conditions:

**1. Linear Independence:**

The vectors $v_1$, $v_2$ and $v_3$ must be linearly independent, meaning that no vector can be written as a linear combination of the other vectors.

**2. Spanning:**

The vectors $v_1$, $v_2$ and $v_3$ must span $R^3$, meaning that every vector in $R^3$ can be expressed as a linear combination of $v_1$, $v_2$ and $v_3$.
Let's analyze these conditions:

**1) Linear Independence:**

To check for linear independence, we can set up an augmented matrix with the vectors as its columns and perform row reduction to see if the system has a unique solution. If the system has a unique solution (i.e., the row-reduced form has no free variables), then the vectors are linearly independent.

The augmented matrix [ $v_1$ $v_2$ $v_3$ ] is as follows:

$$\begin{bmatrix} 5 & 1 & 1 \\ 3 & 2 & 1 \\ 4 & 5 & 2 \end{bmatrix}$$

Applying row reduction, we can transform the matrix to its row-reduced echelon form:

$$\begin{bmatrix} 5 & 1 & 1 \\ 3 & 2 & 1 \\ 4 & 5 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 & -1 \\ 3 & 2 & 1 \\ 4 & 5 & 2 \end{bmatrix} R_1 - R_3$$

$$\begin{bmatrix} 1 & -4 & -1 \\ 0 & 14 & 4 \\ 0 & 21 & 6 \end{bmatrix} R_2 - 3R_1, R_3 - 4R_1$$

$$\begin{bmatrix} 1 & -4 & -1 \\ 0 & 7 & 2 \\ 0 & 7 & 2 \end{bmatrix} \frac{1}{2}R_2, \frac{1}{3}R_3$$

$$\begin{bmatrix} 1 & -4 & -1 \\ 0 & 1 & \frac{2}{7} \\ 0 & 0 & 0 \end{bmatrix} R_3 - R_2, \frac{1}{7}R_2$$

The row-reduced echelon form has a row of zeros, indicating that the system has a free variable.
Therefore, the vectors $v_1$, $v_2$ and $v_3$, are linearly dependent.

**2) Spanning:**

Since the vectors $v_1$, $v_2$ and $v_3$, are linearly dependent, they cannot span $R^3$. A set of vectors can only form a basis for a vector space if the vectors are linearly independent and span the entire space. In this case, since the vectors are linearly dependent, they cannot form a basis for $R^3$.

Therefore, the vectors $v_1 = \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ *do not* form a basis for $R^3$.


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CHE301, CS101, CS201, CS202, CS205, CS206, CS301, CS302, CS304, CS311, CS312, CS315, CS401, CS402, CS403, CS405, CS407, CS408, CS410, CS411, CS432, CS435, CS501, CS502, CS504, CS506, CS507, CS508, CS601, CS602, CS603, CS604, CS605, CS606, CS607, CS609, CS610, CS611, CS614, CS615

ECO401, ECO402, ECO403, ECO404, ECO501, ECO601, ECO603, ECO606, ECO615, EDU101, EDU201, EDU301, EDU303, EDU304, EDU305, EDU401, EDU402, EDU403, EDU404, EDU405, EDU406, EDU410, EDU411, EDU430, EDU431, EDU501, EDU505, EDU510, EDU512, EDU515, EDU516, EDU601, EDU602, EDU603, EDU604, EDU654, EDU705, EDU712, ENG001, ENG101, ENG201, ENG301, ENG501, ENG502, ENG503, ENG504, ENG505, ENG506, ENG507, ENG508, ENG509, ENG510, ENG511, ENG512, ENG513, ENG515, ENG516, ENG518, ENG519, ETH202

FIN611, FIN621, FIN622, FIN623, FIN625, FIN630, FIN701

GSC101, GSC201

HRM613, HRM617, HRM624, HRM626, HRM627, HRM713

ISL201, IT430

MCD401, MCD402, MCD403, MCD404, MCD501, MCD502, MCD503, MCD504, MCM101, MCM301, MCM304, MCM310, MCM311, MCM401, MCM404, MCM411, MCM501, MCM511, MCM514, MCM515, MCM516, MCM604, MCM610, MGMT611, MGMT614, MGMT615, MGMT617, MGMT622, MGMT623, MGMT625, MGMT627, MGMT628, MGMT629, MGMT630, MGMT631, MGMT715, MGMT727, MGMT731, MGT101, MGT111, MGT201, MGT211, MGT301, MGT401, MGT402, MGT404, MGT411, MGT501, MGT502, MGT503, MGT504, MGT510, MGT513, MGT520, MGT522, MGT601, MGT602, MGT603, MGT604, MGT610, MGT611, MGT612, MGT613, MGT621, MGT703, MGT705, MKT501, MKT529, MKT530, MKT603, MKT610, MKT611, MKT621, MKT624, MKT625, MKT626, MKT627, MKT630

MTH001, MTH100, MTH101, MTH102, MTH201, MTH202, MTH301, MTH302, MTH303, MTH401, MTH501, MTH601, MTH603, MTH621, MTH622, MTH631, MTH632, MTH633, MTH634, MTH641

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PAK301, PAK302, PHY101, PHY301, PSC201, PSC401, PSY101, PSY401, PSY403, PSY404, PSY405, PSY406, PSY407, PSY408, PSY409, PSY502, PSY504, PSY510, PSY511, PSY512, PSY513, PSY514, PSY610, PSY631, PSY632

SOC101, SOC301, SOC302, SOC401, SOC402, SOC403, SOC601, SOC603, STA100, STA301, STA621, STA630, STA631, STA632, STA642, STA643, STA644, STA730,

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# Current Paper of MTH 501...DATE 11 may 2012 …

**MASOOM FAIRY**

**Total 26 Questions:**
**20 MCQ'S and 6 Subjective.**
**All Subjective and MCQ'S are New Not any one from Past papers:**
**Subjective Questions are there:**

**Q: 21: If A=B, then determine the Value of x and y, Where** $A=\begin{bmatrix}1 & Y+2\\ X+2 & 3\end{bmatrix}$ and $B=\begin{bmatrix}1 & 2\\ 4 & 3\end{bmatrix}$.

**MARKS: 2:**

**Q: 22: Let** $A=\begin{bmatrix}1&2&3&4\\2&3&4&5\\2&4&6&8\end{bmatrix}\square\begin{bmatrix}1&0&-1&-2\\0&1&2&3\\0&0&0&0\end{bmatrix}$, Solutions of Ax=0 in term of free Variables are

$x_1 = x_3 + 2x_4$

$x_2 = -2x_3 - 3x_4$

$x_3 = free\ of\ \text{var}iable$

$x_4 = free\ of\ \text{var}iable$

Masoom fairy

**MARKS: 2:**

**Q: 23: Determine the matrix is block upper triangular, Block lower triangular or Block diagonal.**

$$B=\left[\begin{array}{cccc}1&0&0&0\\ \hline 2&3&4&0\\ 2&1&2&0\\ \hline 3&2&0&9\end{array}\right].$$

**Q: 24: Determine the Value "s" for which the following system of Linear equation does not has unique solution:**

$$Sx_1 + 2x_2 = 10$$
$$5x_1 + 7x_2 = 20 \qquad ;'s' \text{ is parameter}$$

**Q: 25: Compute the following by using product rule for**

$$(AB)_{23}, (AB)_{13} \text{ and } (AB)_{12};$$

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 2 & 3 \\ 3 & 5 & 8 \\ 2 & 1 & 3 \end{bmatrix}.$$

**MARKS: 5:**

**Q: 26: From Lecture number 22.**

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**FINALTERM EXAMINATION**

Fall 2008

MTH501- Linear Algebra

Ref No: 417476

Time: 120 min

Marks: 70

**Student Info**

| | |
|---|---|
| Student ID: | |
| Center: | |
| Exam Date: | |

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| **Q No.** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **Total** |
|---|---|---|---|---|---|---|---|---|---|
| **Marks** | | | | | | | | | |
| **Q No.** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **16** | |
| **Marks** | | | | | | | | | |
| **Q No.** | **17** | **18** | **19** | **20** | **21** | **22** | **23** | **24** | |
| **Marks** | | | | | | | | | |
| **Q No.** | **25** | **26** | **27** | **28** | **29** | **30** | **31** | | |
| **Marks** | | | | | | | | | |

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**Question No: 1 ( Marks: 1 ) - Please choose one**

---

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$$

Reduced echelon form of the matrix is

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

▶

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

▶

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$$

▶

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

▶

**Question No: 2 ( Marks: 1 ) - Please choose one**

---

A matrix that results from applying a single elementary row operation to an identity matrix is called

▶ **Invertible matrix**

▶ Singular matrix

▶ Scalar matrix

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► Elementary matrix

**Question No: 3 ( Marks: 1 ) - Please choose one**

---

For an n×n matrix $(A^t)^t$ =

► $A^t$

► **A**

► $A^{-1}$

► $(A^{-1})^{-1}$

**Question No: 4 ( Marks: 1 ) - Please choose one**

---

What is the largest possible number of pivots a 4×6 matrix can have?

► 4

► 6

► 10

► 0

**Question No: 5 ( Marks: 1 ) - Please choose one**

---

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The characteristic polynomial of a 5×5 matrix is $\lambda^5 - 4\lambda^4 - 45\lambda^3 = 0$ ,the eigenvalues are

- 0,-5, 9
- 0,0,0,5,9
- 0,0,0,-5,9
- 0,0,5,-9

**Question No: 6 ( Marks: 1 ) - Please choose one**

---

Find the characteristic equation of the given matrix

$$\begin{bmatrix} 6 & 8 & 5 & 4 \\ 0 & 2 & -8 & 7 \\ 0 & 0 & 9 & 6 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

- $(6-\lambda)(2-\lambda)(9-\lambda)=0$
- $(6-\lambda)(8-\lambda)(5-\lambda)(4-\lambda)=0$
- $(6-\lambda)^2(2-\lambda)(9-\lambda)=0$
- $(6-\lambda)(6-\lambda)(7-\lambda)(4-\lambda)=0$

**Question No: 7 ( Marks: 1 ) - Please choose one**

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---

A is diagonalizable if $A = PDP^{-1}$ Where

- D is any matrix and P is an invertible matrix
- D is a diagonal matrix and P is any matrix
- D is a diagonal matrix and P is invertible matrix
- D is a invertible matrix and P is any matrix

**from book**
A square matrix *A* is said to be diagonalizable if *A* is similar to a diagonal matrix. i.e. if *A* = $^{-1}$*PDP* for some invertible matrix *P* and some diagonal matrix *D*.

**Question No: 8 ( Marks: 1 ) - Please choose one**

---

The inverse of an invertible lower triangular matrix is

- lower triangular matrix
- upper triangular matrix
- diagonal matrix

The standard operations on triangular matrices conveniently preserve the triangular form: the sum and product of two upper triangular matrices is again upper triangular. The inverse of an upper triangular matrix is also upper triangular, and of course we can multiply an upper triangular matrix by a constant and it will still be upper triangular. This means that the upper triangular matrices form a subalgebra of the ring of square matrices for any given size. The analogous result holds for lower triangular matrices. Note, however, that the product of a *lower* triangular with an *upper* triangular matrix does *not* preserve triangularity.

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**http://en.wikipedia.org/wiki/Triangular_matrix**

**Question No: 9 ( Marks: 1 ) - Please choose one**

---

If P is a parallelepiped in $R^3$, then
{volume of T (P)} = |detA|. {volume of P}

- Where T is determined by a $2\times2$ matrix A
- Where T is determined by a $2\times3$ matrix A
- **Where T is determined by a $3\times3$ matrix** A not conf.
- Where T is determined by a $3\times2$ matrix A

**Question No: 10 ( Marks: 1 ) - Please choose one**

---

Let A be a $n\times m$ matrix of rank $r$ then row space of A has dimension

- $m-r$
- $n-r$
- $r$
- $nm$

*Main article: Rank (linear algebra)*

The dimension of the row space is called the **rank** of the matrix. This is the same as the maximum number of linearly independent rows that can be chosen from the matrix. For example, the 3 × 3 matrix in the example above has rank two.

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The rank of a matrix is also equal to the dimension of the column space. The dimension of the null space is called the **nullity** of the matrix, and is related to the rank by the following equation:

$$\operatorname{rank}(A) + \operatorname{nullity}(A) = n,$$

where *n* is the number of columns of the matrix *A*. The equation above is known as the rank-nullity theorem.

[edit]

**Question No: 11 ( Marks: 1 ) - Please choose one**

---

The dimension of the vector space $P_4$ is

► 4

► 3

► 5

► 1

The dimension of the vector space P4 is 4. FALSE It's 5.

**Question No: 12 ( Marks: 1 ) - Please choose one**

---

Let $u = (3,-2), v = (4,5)$ .For the weighted Euclidean inner product $\langle u,v\rangle = 4u_1v_1 + 5u_2v_2$ $\langle v,u\rangle =$

► 2

► -2

► 3

► -3

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**Question No: 13 ( Marks: 1 ) - Please choose one**

---

Let A be $n \times n$ matrix whose entries are real. If $\lambda$ is an eigenvalue of A with x a corresponding eigenvector in $\mathbb{C}^n$, then

- $A\bar{x} = \lambda\bar{x}$
- $A\bar{x} = \bar{\lambda}\bar{x}$
- $A\bar{x} = \bar{\lambda}x$
- $A\bar{x} = \lambda^{-1}\bar{x}$

**Question No: 14 ( Marks: 1 ) - Please choose one**

---

$$A = \begin{bmatrix} 1.25 & -.75 \\ -.75 & 1.25 \end{bmatrix}$$

Suppose that has eigenvalues 2 and 0.5 .Then origin is a

- Saddle point
- Repellor
- Attractor

**Question No: 15 ( Marks: 1 ) - Please choose one**

---

Which one is the numerical method used for approximation of dominant eigenvalue of a matrix.

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- Power method
- Jacobi's method
- Guass Seidal method
- Gram Schmidt process

**Question No: 16 ( Marks: 1 ) - Please choose one**

---

The matrix equation $A^T A\hat{x} = A^T b$ represents a system of linear equations commonly referred to as the

- normal equations for $x$
- normal equations for $\hat{x}$
- normal equations for $A$
- normal equations for $b$

**Question No: 17 ( Marks: 1 ) - Please choose one**

---

Let $A$ have eigenvalues 2, 5, 0,-7, and -2. Then the dominant eigenvalue for A is

- $\lambda = 5$
- $\lambda = 0$
- $\lambda = -7$

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- $\lambda = 2$

**Question No: 18 ( Marks: 1 ) - Please choose one**

---

If W is a subspace of $\mathbb{R}^m$, then the transformation $T : \mathbb{R}^m \to W$ that maps each vector x in $\mathbb{R}^m$ into its orthogonal x in W is called the orthogonal projection of

- $\mathbb{R}^m$ in $\mathbb{R}^m$
- $\mathbb{R}^m$ in W
- $\mathbb{R}^m$ in x

**Question No: 19 ( Marks: 1 ) - Please choose one**

---

If $V = \mathbb{R}^n$, $B = \{b_1, b_2\}$ and $C = \{c_1, c_2\}$ then row reduction of $\begin{bmatrix} c_1 & c_2 & b_1 & b_2 \end{bmatrix}$ to $\begin{bmatrix} I & P \end{bmatrix}$ Produces a matrix P that satisfies

- $[x]_B = P[x]_B$ for all x in V
- $[x]_C = P[x]_B$ for all x in V
- $[x]_B = P[x]_C$ for all x in V
- $[x]_B = [x]_C$ for all x in V

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**Question No: 20 ( Marks: 1 ) - Please choose one**

---

The Casorati matrix for the signals $1^k$, $(-2)^k$ and $3^k$ is

$$\begin{bmatrix} 1^k & (-2)^k & 3^0 \\ 1^{k+1} & (-2)^{k+1} & 3^1 \\ 1^{k+2} & (-2)^{k+2} & 3^2 \end{bmatrix}$$

▶

$$\begin{bmatrix} 1^k & (-2)^k & 3^k \\ 1^{k+1} & (-2)^{k+1} & 3^{k+1} \\ 1^{k+2} & (-2)^{k+2} & 3^{k+2} \end{bmatrix}$$ not conf.

▶

$$\begin{bmatrix} 1^0 & (-2)^k & 3^k \\ 1^1 & (-2)^{k+1} & 3^{k+1} \\ 1^2 & (-2)^{k+2} & 3^{k+2} \end{bmatrix}$$

▶

**Question No: 21 ( Marks: 2 )**

---

Find the characteristic polynomial and all eigenvalues of the given matrix

$$\begin{bmatrix} a & a \\ a & a \end{bmatrix}$$

**Question No: 22 ( Marks: 2 )**

---

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Write the Fourier coefficients $a_k \text{ and } b_k$ to the function $f(t) = 2\pi - t$ on the interval $[0, 2\pi]$.

**Question No: 23 ( Marks: 2 )**

---

The matrix A is followed by a sequence $\{x_k\}$ produced by the power method. Use these data to estimate the largest eigenvalue of A.

$$A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0.25 \end{bmatrix}, \begin{bmatrix} 1 \\ 0.3158 \end{bmatrix}, \begin{bmatrix} 1 \\ 0.3298 \end{bmatrix}, \begin{bmatrix} 1 \\ 0.3326 \end{bmatrix}$$

**Question No: 24 ( Marks: 3 )**

---

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

If then find an invertible matrix P such that $P^{-1}AP = D(\textit{diagonal matrix})$

**Question No: 25 ( Marks: 3 )**

---

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & -\frac{2}{3} \\ 0 & \frac{1}{3} \end{bmatrix}$$

Check whether the matrix has orthonormal columns or not?

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**Question No: 26 ( Marks: 3 )**

---

If A is a $6\times4$ matrix, what is the smallest possible dimension of Null A?

**Question No: 27 ( Marks: 5 )**

---

Assume that the matrix ***A*** is row equivalent to ***B***. Without calculations, list rank ***A*** and dim Nul ***A***. Then find bases for Col ***A*** and Row ***A.***

$$A=\begin{bmatrix}1 & 2 & -2 & 3\\ 2 & 5 & 0 & -7\\ 3 & 7 & -2 & -4\end{bmatrix} \qquad B=\begin{bmatrix}1 & 2 & -2 & 3\\ 0 & 1 & 4 & -13\\ 0 & 0 & 0 & 0\end{bmatrix}$$

**Question No: 28 ( Marks: 5 )**

---

$$\begin{bmatrix}a & -b\\ b & a\end{bmatrix}$$

Find an invertible matrix P and a matrix C of the form such that the given matrix A has the form $A=PCP^{-1}$.

$$A=\begin{bmatrix}1 & 5\\ -2 & 3\end{bmatrix}$$

with eigenvalue $\lambda=2\pm3i$ and eigenvector

$v=\begin{bmatrix}1+3i\\ 2\end{bmatrix}$ *which corresponds the eigenvalue* $\lambda=2-3i$

**Question No: 29 ( Marks: 5 )**

---

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$$u=\begin{bmatrix}2\\-5\\-1\end{bmatrix}, v=\begin{bmatrix}-7\\-4\\6\end{bmatrix}$$

Let .Compute and compare $u.v, \|u\|^2, \|v\|^2 \text{ and } \|u+v\|^2$. Do not use Pythagorean Theorem.

**Question No: 30 ( Marks: 10 )**

---

$$A=\begin{bmatrix}1&1\\1&2\\1&3\end{bmatrix}, b=\begin{bmatrix}0\\0\\6\end{bmatrix}$$

Find a least squares solution of the inconsistent system $Ax = b$ where .

**Question No: 31 ( Marks: 10 )**

---

Determine whether the signals $1^k, 2^k$, and $(-2)^k$ are the solutions of the difference equation $y_{k+3} - y_{k+2} - 4y_{k+1} + 4y_k = 0$.

The null space of A is the solution set of the equation
Ax = 0. TRUE
The null space of an m _ n matrix is in $R_m$. False. It's $R_n$
The column space of A is the range of the mapping x 7! Ax.
TRUE
If the equation Ax = b is consistent, then Col A is $R_m$.
FALSE must be consistent for all b
The kernel of a linear transformation is a vector space. TRUE
To show this we show it is a subspace

Col A is the set of a vectors that can be written as Ax for
some x. TRUE Remember that Ax gives a linear combination
of columns of A using x entries as weights.

The null space is a vector space. TRUE

The column space of an m _ n matrix is in $R_m$ TRUE

Col A is the set of all solutions of Ax = b. FALSE It is the set

of all b that have solutions.

Nul A is the kernel of the mapping x 7! Ax. TRUE

The range of a linear transformation is a vector space. TRUE
It's a subspace(check), thus vector space.

The set of all solutions of a homogenous linear di_erential equation is the kernel of a linear transformation. TRUE

A single vector is itself linearly dependent. FALSE unless it in the zero vector

If H =Spanfb$_1$; : : : ; b$_n$g then fb$_1$; : : : ; b$_n$g is a basis for H. FALSE They may not be linearly independent.

The columns of an invertible n _ n matrix form a basis for R$_n$ TRUE They are linerly independent and span R$_n$. (why?)

A basis is a spanning set that is as large as possible. FALSE It is is too large, then it is no longer linearly independent.

In some cases, the linear dependence relations among the columns of a matrix can be a_ected by certain elementary row operations on the matrix. FALSE They are not a_ected.

A linearly independent set in a subspace H is a basis for H. FALSE It may not span.

If a _nite set S of nonzero vectors spans a vector space V, the some subset is a basis for V. TRUE by Spanning Set Theorem
A basis is a linearly independent set that is as large as possible. TRUE

The standard method for producing a spanning set for Nul A, described in this section, sometimes fails to produce a basis. FALSE It NEVER fails!!!

If B is an echelon form of a matrix A, then the pivot columns of B form a basis for Col A. FALSE Must look at corresponding columns in A.

The number of pivot columns of a matrix equals the dimension of its column space. TRUE Remember these

columns and linearly independent and span the column space.
A plane in $R_3$ is a two dimensional subspace of $R_3$. FALSE unless the plane is through the origin.

The dimension of the vector space $P_4$ is 4. FALSE It's 5.

If dimV = n and S is a linearly independent set in V, then S is a basis for V. FALSE S must have exactly n elements.

If a set $fv_1 : : : v_ng$ spans a _nite dimensional vector space V and if T is a set of more than n vectors in V, then T is linearly dependent. TRUE The number of linearly independent vectors that span a set is unique.

$R_2$ is a two dimensional subspace of $R_3$. FALSE Not a subset, as before.

The number of variables in the equation Ax = 0 equals the dimension of Nul A. FALSE It's the number of free variables.

A vector space is in_nite dimensional is it is spanned by an in_nite set. FALSE It must be impossible to span it by a _nite set.

If dim V = n and if S spans V. then S is a basis for V.
FALSE S must have exactly n elements or be noted as linearly independent.

The only three dimensional subspace of $R_3$ is $R_3$ itself. TRUE
If spanned by three vectors must be all of $R_3$.